

Optimal Transport for Machine Learners

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Abstract

Optimal transport provides a geometric way to compare probability measures by optimizing how mass is matched, split, regularized and moved. This book follows that progression. It starts from finite assignment and the Monge map viewpoint, passes to Kantorovich couplings and dual potentials, and then explains the algorithmic ideas that make transport usable: linear programming, semi-discrete cells, Sinkhorn scaling and low-dimensional projections. The same objects are then reused as a geometry of measures, giving Wasserstein distances, barycenters, gradient flows, dynamic formulations and Gaussian/Bures formulas. The final chapters emphasize the variants most relevant to machine learning: divergences and adversarial losses, entropic and unbalanced relaxations, robust or spectral ground geometries, Gromov and quantum extensions, and transport-based views of generative models, mean-field networks and attention dynamics.

Guide to the Literature and Scope

Several books already cover optimal transport from complementary viewpoints. The two-volume monograph of Rachev and Rüschendorf [143, 144] gives a broad probabilistic treatment of mass transportation and its applications. Villani's books [170, 171] are the standard references for the modern mathematical theory, from Kantorovich duality to curvature, concentration and geometric analysis. Santambrogio's text [154] offers a concise applied-mathematics route through the same foundations, with a strong emphasis on PDEs and variational arguments. Ambrosio, Gigli and Savaré [6] develop the metric-space theory of gradient flows that underlies the dynamical part of the subject.

On the computational side, Peyré and Cuturi [138] provide the reference account of numerical OT, entropic regularization and applications in data sciences. Galichon's book [69] explains the economic and matching-theoretic viewpoint, while the statistical theory of OT is developed in the recent monograph of Weed, Niles-Weed and Rigollet [174]. Recent surveys complement these books by emphasizing scalable algorithms and machine-learning applications [99, 126], as well as the role of OT in imaging and graphics [29]. These references remain the natural places to find exhaustive proofs, historical details and specialized variants.

The aim here is different and more selective. The book keeps the core mathematics explicit, but organizes it around the questions that repeatedly arise in machine learning: how to compare singular empirical measures, how to compute differentiable transport losses, how regularization changes optimization and statistics, how dual potentials become discriminators, and how transport geometry produces flows of particles, neurons and tokens. The intended contribution is therefore not a replacement for the references above, but a compact bridge between rigorous OT and the geometric intuitions needed to use it in modern ML.

1 Optimal Matching between Point Clouds

This opening section isolates the simplest form of optimal transport: pairing two finite point clouds. The stakes are algorithmic and geometric at once: one sees the combinatorial nature of transport, the special simplicity of the line, and the first hints that convex relaxation will be necessary in higher dimension. Classical assignment algorithms such as the Hungarian method and auction methods [103, 18] provide the computational backdrop, while the geometric examples prepare the Kantorovich relaxation.

1.1 Monge Problem for Discrete Points

This subsection formulates matching as Monge's deterministic transport problem on two equally weighted clouds. The one-dimensional case is a transparent reference case where the optimal map can be read off by sorting.

Matching Problem Given a cost matrix $(C_{i,j})_{i \in \llbracket n \rrbracket, j \in \llbracket m \rrbracket}$ and assuming $n = m$, the optimal assignment problem aims to find a bijection σ within the set $\text{Perm}(n)$ of permutations of n elements that solves

$$\min_{\sigma \in \text{Perm}(n)} \frac{1}{n} \sum_{i=1}^n C_{i, \sigma(i)}. \quad (1)$$

One could naively evaluate the cost function above using all permutations in the set $\text{Perm}(n)$. However, this set has size $n!$, which becomes enormous even for small values of n . In general, the optimal σ is not unique.

1D Case In 1D, for convex cost, the matching defines a monotonic map.

Proposition 1 (Monotone matching on the line). *Assume that the points $(x_i)_i$ and $(y_j)_j$ are pairwise distinct. If the cost is of the form $C_{i,j} = h(x_i - y_j)$, where $h : \mathbb{R} \rightarrow \mathbb{R}^+$ is strictly convex (for example, $C_{i,j} = |x_i - y_j|^p$ for $p > 1$), then any optimal σ defines a strictly increasing map $x_i \mapsto y_{\sigma(i)}$ (and thus is unique), i.e.,*

$$\forall (i, i'), \quad (x_i - x_{i'})(y_{\sigma(i)} - y_{\sigma(i')}) > 0.$$

Proof. Indeed, if this property is violated, i.e., there exists (i, i') such that $(x_i - x_{i'})(y_{\sigma(i)} - y_{\sigma(i')}) < 0$, then one can define a permutation $\tilde{\sigma}$ by swapping the match, i.e., $\tilde{\sigma}(i) = \sigma(i')$ and $\tilde{\sigma}(i') = \sigma(i)$, yielding a strictly better cost, as proved in the following fact. Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be strictly convex and let $x < x'$ and $y < y'$. Then

$$h(x - y) + h(x' - y') < h(x - y') + h(x' - y).$$

We set the gap $d := y' - y > 0$ and define for every $s \in \mathbb{R}$

$$D(s) := \frac{h(s) - h(s - d)}{d} \quad \text{and} \quad \Delta := h(x - y') + h(x' - y) - h(x - y) - h(x' - y') = d(D(x' - y) - D(x - y)).$$

Because h is strictly convex, D is strictly increasing. Since $x - y < x' - y$, monotonicity yields $D(x - y) < D(x' - y)$, that is $\Delta > 0$. $\square$

This property extends by continuity to convex (not strictly convex) costs such as $|x - y|$, but in this case, the matching is not necessarily unique. Note that if h is concave instead of being convex, then the behavior is entirely different, and the optimal match tends to exchange the positions. In this case, there exists an $O(n^2)$ algorithm.

Therefore, the algorithm to compute an optimal transport is to sort the points, i.e., find some pair of permutations σ_X, σ_Y such that

$$x_{\sigma_X(1)} \leq x_{\sigma_X(2)} \leq \dots \quad \text{and} \quad y_{\sigma_Y(1)} \leq y_{\sigma_Y(2)} \leq \dots$$

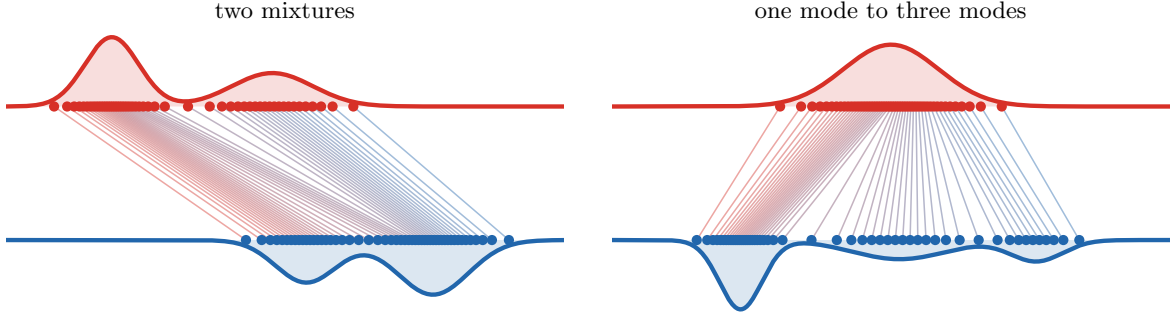


Figure 1: One-dimensional optimal matching by quantile sorting. The red and blue curves are smooth laws used to generate equal-weight empirical measures; the dots are inverse-CDF samples at common quantile levels. The monotone assignment connects equal ranks, both for two Gaussian mixtures and for the transport from one central Gaussian toward a three-mode target law.

and then an optimal match is mapping $x_{\sigma_X(k)} \mapsto y_{\sigma_Y(k)}$, i.e., an optimal transport is $\sigma = \sigma_Y \circ \sigma_X^{-1}$. The total computational cost is thus $O(n \log(n))$, using, for instance, the quicksort algorithm.

Note that if $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is an increasing map, one can apply this technique to costs of the form $h(|\varphi(x) - \varphi(y)|)$ with a change of variable. A typical application is grayscale histogram equalization. The empirical cumulative distribution of the luminance values of an image is transported to a prescribed target histogram, often a uniform or reference-image histogram. In one dimension, the monotone rearrangement above gives the exact transport map, so the operation is both computationally simple and geometrically faithful: it matches distributions of intensities rather than individual pixels.

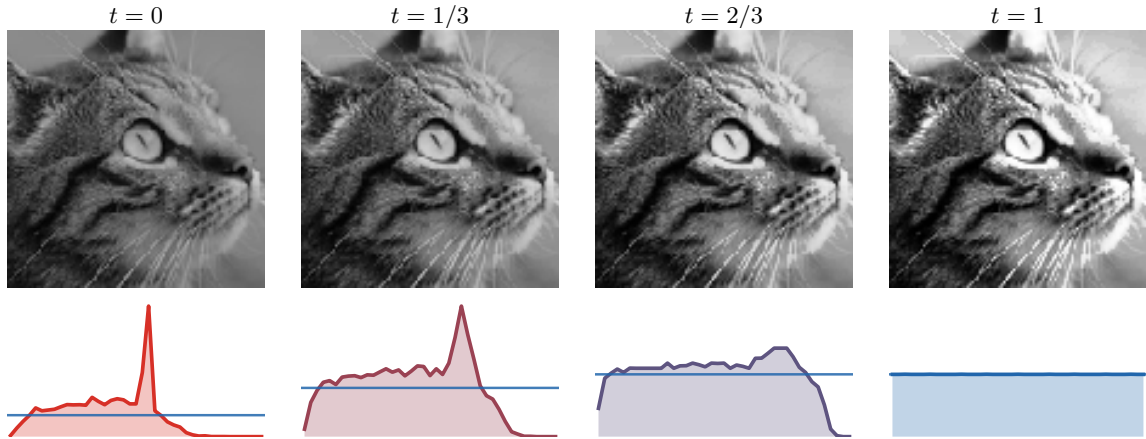


Figure 2: Histogram equalization as one-dimensional Monge transport on pixel intensities. The map is the monotone rearrangement $T = Q_\beta \circ F_\alpha$, using the cumulative and quantile functions defined later in Definition 4. The images are interpolated pointwise by $I_t = (1 - t)I + tT(I)$, and the histograms show the empirical laws of these interpolated intensities; they are not binwise linear interpolations of histograms. This is the one-dimensional version of McCann displacement interpolation, discussed in Section 2.3.

Note that if h is strictly convex, then all optimal assignments are increasing, and if the points are all distinct, this increasing map is unique. If h is not strictly convex, for instance $c(x, y) = |x - y|$, non-increasing optimal assignments can also exist. This happens, for example, in the book-shifting problem with overlapping uniform distributions, where the mass in the intersection can stay fixed.

This efficient strategy to compute the OT in 1-D does not extend to higher dimensions. In 2-D, as already

noted by Monge, one has the following property.

Proposition 2 (Non-crossing optimal matchings). *In dimension 2, for $c(x, y) = \|x - y\|$, if σ is an optimal assignment, then segments $[x_i, y_{\sigma(i)}]$ cannot cross.*

Proof. If two segments $[x_i, y_{\sigma(i)}]$ and $[x_j, y_{\sigma(j)}]$ cross at an interior point z , then the triangle inequality gives

$$\|x_i - y_{\sigma(j)}\| + \|x_j - y_{\sigma(i)}\| < \|x_i - y_{\sigma(i)}\| + \|x_j - y_{\sigma(j)}\|.$$

The assignment obtained by swapping $(\sigma(i), \sigma(j))$ therefore has a strictly smaller cost, which contradicts optimality. $\square$

This property alone is, however, not enough to lead to an efficient algorithm. Non-crossing is only a necessary local test, not a compact certificate of optimality. For instance, if n sources and n targets are placed alternately on the boundary of a convex polygon, the number of non-crossing perfect matchings is the Catalan number

$$C_n = \frac{1}{n+1} \binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n^{3/2}}}.$$

Thus even after forbidding crossings, an exhaustive search remains exponential. The two-segment swap in the proof above is nevertheless useful: it explains why a crossing matching cannot be optimal, but it does not select among the exponentially many planar matchings that survive this local test.

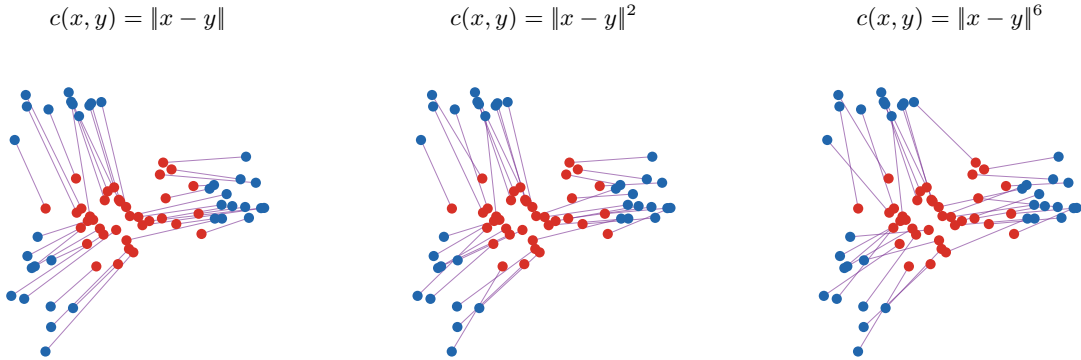


Figure 3: Optimal assignments between the same two point clouds for three powers of the Euclidean distance. The source is a compact central Gaussian cloud and the target is a three-mode Gaussian mixture; this canonical geometry is reused in later coupling and regularization figures. The feasible set is unchanged, but increasing p penalizes the longest edges more strongly and changes the global organization of the permutation.

The strict assignment model is also tied to equal cardinalities and equal weights. As soon as the target resolution changes or the weights are not uniform, a permutation no longer describes the feasible transports. One instead needs a nonnegative transport matrix with prescribed row and column sums, as illustrated in Figure 4; this is the finite-dimensional Kantorovich relaxation developed in Section 3.

1.2 Matching Algorithms

This subsection briefly locates matching within classical combinatorial optimization. Its main point is that efficient algorithms exist, but their cleanest analysis is obtained only after introducing the linear-programming viewpoint.

Efficient algorithms exist to solve the optimal matching problem. The most well-known are the Hungarian method and auction algorithms [103, 17, 18]. Auction algorithms use prices on the target points: each source

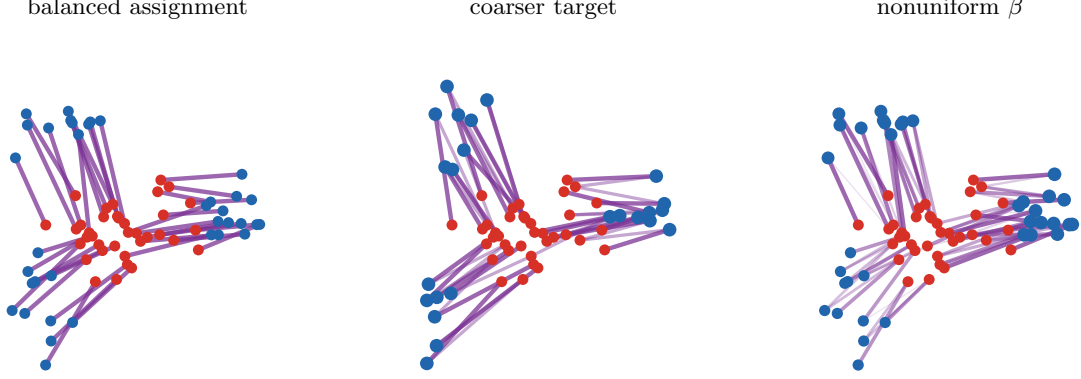


Figure 4: From assignments to transport plans, using the same central-source and three-mode-target geometry as Figure 3. In the balanced equal-weight case, each source atom is matched to one target atom. With a coarser target or nonuniform target weights, the coupling matrix can merge or split mass; segment thickness and opacity encode its nonzero entries, and target marker areas encode the prescribed target masses.

bids for the target with largest reduced profit, the target price is increased, and the process terminates when the ε -complementary slackness conditions are satisfied. For integer costs, choosing $\varepsilon < 1/n$ gives an exact optimum after a finite number of bids [18]. Section 5.2 revisits this algorithm after Kantorovich duality and explains why it is a dual price method, parallel in spirit to Sinkhorn scaling.

Hungarian primal-dual method. The Hungarian method is best understood as a certificate-building algorithm for the assignment linear program. It maintains a partial matching M and dual prices (u_i, v_j) satisfying

$$u_i + v_j \leq C_{i,j} \quad \forall i, j.$$

The equality graph $E(u, v) = \{(i, j) : u_i + v_j = C_{i,j}\}$ contains the edges whose reduced cost is zero. The algorithm only augments M along alternating paths made of equality edges. Starting from an unmatched source, it grows an alternating tree with source set S and target set T . If the tree reaches an unmatched target, the matching is augmented along the path. If no such edge exists, the dual variables are shifted by the smallest slack

$$\delta = \min_{i \in S, j \notin T} (C_{i,j} - u_i - v_j), \quad u_i \leftarrow u_i + \delta \ (i \in S), \quad v_j \leftarrow v_j - \delta \ (j \in T).$$

This update preserves all inequalities $u_i + v_j \leq C_{i,j}$, keeps the current alternating tree tight, and creates at least one new equality edge leaving S . Maintaining these slacks incrementally gives the standard $O(n^3)$ implementation for an $n \times n$ assignment problem.

Proposition 1 (Correctness and complexity of the Hungarian primal-dual method). *Assume the Hungarian method terminates with a perfect matching σ contained in the equality graph*

$$E(u, v) = \{(i, j) : u_i + v_j = C_{i,j}\},$$

where (u, v) is dual feasible, i.e. $u_i + v_j \leq C_{i,j}$ for all (i, j) . Then σ is an optimal assignment. Moreover, the usual Hungarian updates terminate after finitely many augmentations. With maintained slacks, the method uses $O(n^3)$ arithmetic operations.

Proof. For any permutation τ , dual feasibility gives

$$\sum_i C_{i,\tau(i)} \geq \sum_i (u_i + v_{\tau(i)}) = \sum_i u_i + \sum_j v_j.$$

This is the weak duality lower bound. If σ is contained in the equality graph, then

$$\sum_i C_{i,\sigma(i)} = \sum_i u_i + \sum_j v_j,$$

so the primal cost of σ reaches the dual lower bound and is optimal.

It remains to justify finite termination and the complexity bound. At each successful augmentation, the matching cardinality increases by one, so there are at most n augmentations. During one augmentation phase, the algorithm grows an alternating tree in the equality graph. If no augmenting path is available, the dual update uses the smallest slack of an edge leaving the current tree. For edges inside the tree, adding δ to source labels and subtracting δ from target labels preserves tightness; for edges from S to T^c , the definition of δ preserves feasibility and makes at least one new edge tight; all other inequalities are unchanged or become looser. Thus the reachable sets strictly grow between two failed augmentation attempts, and they can grow at most n times within one phase. If the current slacks $\min_{i \in S} (C_{i,j} - u_i - v_j)$ are updated when a source enters S , each tree expansion costs $O(n)$. A phase therefore costs $O(n^2)$, and the n phases give $O(n^3)$ operations. Hence the method reaches a perfect optimal matching. $\square$

Network-flow viewpoint and applications. Their derivation and analysis are greatly simplified by introducing the Kantorovich relaxation and its associated dual problem. The transportation simplex and network-flow viewpoints go back to Dantzig’s transportation linear program and modern minimum-cost-flow algorithms [54, 19, 130]. These finite matching and assignment ideas also underlie color transfer between images. Grayscale equalization is one-dimensional, but transferring a full color palette requires transporting empirical measures in a three-dimensional color space, for instance RGB or Lab. Early color-transfer methods use affine statistics or iterated one-dimensional projections [146, 141]; replacing these projections by a genuine three-dimensional OT coupling gives a more intrinsic way to match colors while preserving the geometry of the palette [142].

2 Monge Problem between Measures

The goal of this section is to pass from finite matching to transport between arbitrary probability laws. The central stakes are to define measures, push-forwards and Monge maps carefully enough that the discrete picture survives, while exposing why deterministic maps can fail to exist. Monge’s original formulation [125] and modern treatments [170, 171, 154, 143] are the conceptual background for this transition.

The presentation of the previous section could only handle two sets with the same number of points. To relax this to a more general setting, one needs to consider probability distributions, so that the points are weighted by masses.

2.1 Measures

Measures are the language that lets point clouds, densities and singular objects be handled uniformly. We only recall the facts needed later: integration, total variation, densities and probabilistic laws.

Histograms We use the terms histogram and probability vector interchangeably for any element $\mathbf{a} \in \Sigma_n$ of the probability simplex

$$\Sigma_n := \left\{ \mathbf{a} \in \mathbb{R}_+^n ; \sum_{i=1}^n a_i = 1 \right\}.$$

Discrete measure, empirical measure A discrete measure with weights $\mathbf{a}$ and locations $x_1, \dots, x_n \in \mathcal{X}$ reads

$$\alpha = \sum_{i=1}^n a_i \delta_{x_i} \tag{2}$$

where δ_x is the Dirac mass at position x , intuitively a unit of mass infinitely concentrated at location x . Such a measure is a probability measure if, additionally, $a \in \Sigma_n$, and more generally a positive measure if all weights a_i are nonnegative. An “empirical” probability distribution is uniform on a point cloud, i.e. $\alpha = \frac{1}{n} \sum_i \delta_{x_i}$. In practice, in many applications, it is useful to be able to manipulate both the positions x_i (“Lagrangian” discretization) and the weights a_i (“Eulerian” discretization). Lagrangian modification is usually more powerful (because it leads to adaptive discretization) but it breaks the convexity of most problems.

General measures We consider Borel measures $\alpha \in \mathcal{M}(\mathcal{X})$ on a metric space $(\mathcal{X}, d)$, meaning that $\alpha(A)$ is defined for every Borel set A (obtained from open sets by countable unions, countable intersections and complements). Unless otherwise stated, the measures are finite. A Dirac measure δ_x is then defined as $\delta_x(A) = 1$ if $x \in A$ and 0 otherwise, and this extends by linearity for discrete measures of the form (2) as

$$\alpha(A) = \sum_{x_i \in A} a_i$$

We denote $\mathcal{M}_+(\mathcal{X})$ the subset of all positive measures on $\mathcal{X}$, i.e. $\alpha(A) \geq 0$ (and $\alpha(\mathcal{X}) < +\infty$ for the measure to be finite). The set of probability measures is denoted $\mathcal{M}_+^1(\mathcal{X})$, which means that any $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ is positive, and that $\alpha(\mathcal{X}) = 1$.

Definition 1 (Support of a measure). *The support $\text{supp}(\alpha)$ of a Borel measure α on a metric space $\mathcal{X}$ is the smallest closed set of full α -mass. Equivalently, $x \in \text{supp}(\alpha)$ if every open ball centered at x has positive α -mass.*

Radon measures Using Lebesgue integration, a Borel measure can be used to compute the integral of measurable functions (i.e. such that level sets $\{x ; f(x) < t\}$ are Borel sets), and we denote this pairing as

$$\langle f, \alpha \rangle := \int f(x) d\alpha(x).$$

Integration of such a measurable f against a discrete measure α computes a sum

$$\int_{\mathcal{X}} f(x) d\alpha(x) = \sum_{i=1}^n a_i f(x_i).$$

This applies in particular to continuous test functions, which are Borel measurable. Integration against a finite measure on a compact space thus defines a continuous linear form $f \mapsto \int f d\alpha$ on the Banach space of continuous functions $(\mathcal{C}(\mathcal{X}), \|\cdot\|_\infty)$, indeed $|\int f d\alpha| \leq \|f\|_\infty |\alpha|(\mathcal{X})$. On compact spaces, the converse is true: every continuous linear form on $\mathcal{C}(\mathcal{X})$ is represented by integration against a finite signed Radon measure. This is the Riesz–Markov–Kakutani representation theorem [149, 26]. It identifies $\mathcal{M}(\mathcal{X})$ with the Banach dual of $\mathcal{C}(\mathcal{X})$, and this duality pairing $\langle f, \alpha \rangle$ will be crucial for the convex duality arguments used later.

Relative densities. A measure α which is a weighting of another reference measure λ is said to have a density, denoted $d\alpha(x) = \rho_\alpha(x) d\lambda(x)$, or $\rho_\alpha = \frac{d\alpha}{d\lambda}$. On $\mathbb{R}^d$ the reference λ is often Lebesgue measure dx . This means that

$$\forall h \in \mathcal{C}(\mathcal{X}), \quad \int_{\mathcal{X}} h(x) d\alpha(x) = \int_{\mathcal{X}} h(x) \rho_\alpha(x) d\lambda(x).$$

Total variation norm. The norm inherited from the duality $\mathcal{M}(\mathcal{X}) = \mathcal{C}(\mathcal{X})^*$ is the total variation norm. We use the notation

Definition 2 (Total variation). *For a finite signed Radon measure α on a compact space $\mathcal{X}$,*

$$\|\alpha\|_{\text{TV}} := \sup_{f \in \mathcal{C}(\mathcal{X})} \{ \langle f, \alpha \rangle ; \|f\|_\infty \leq 1 \}.$$

This formula is useful because it computes the norm of a measure as the operator norm of the corresponding linear form on $\mathcal{C}(\mathcal{X})$.

The same norm also has a direct measure-theoretic expression. The absolute value of a signed measure is

$$|\alpha|(A) := \sup_{A=\cup_i B_i} \sum_i |\alpha(B_i)|,$$

where the supremum is over finite or countable measurable partitions of A . Thus, if $\alpha = \sum_i a_i \delta_{x_i}$ with distinct atoms, $|\alpha| = \sum_i |a_i| \delta_{x_i}$; if $d\alpha(x) = \rho(x)d\lambda(x)$, then $d|\alpha|(x) = |\rho(x)|d\lambda(x)$.

Proposition 2 (Dual and measure definitions of total variation). *For a finite signed Radon measure α on a compact space $\mathcal{X}$,*

$$\|\alpha\|_{\text{TV}} = |\alpha|(\mathcal{X}).$$

Consequently $(\mathcal{M}(\mathcal{X}), \|\cdot\|_{\text{TV}})$ is isometrically the Banach dual of $(\mathcal{C}(\mathcal{X}), \|\cdot\|_{\infty})$.

Proof. The inequality $\|\alpha\|_{\text{TV}} \leq |\alpha|(\mathcal{X})$ follows from

$$\left| \int f d\alpha \right| \leq \int |f| d|\alpha| \leq \|f\|_{\infty} |\alpha|(\mathcal{X}).$$

For the reverse inequality, write the Jordan decomposition $\alpha = \alpha^+ - \alpha^-$, so that $|\alpha| = \alpha^+ + \alpha^-$. The measurable sign $s = \frac{d\alpha}{d|\alpha|}$ takes values in $\{-1, 1\}$ outside a $|\alpha|$ -null set and satisfies $d\alpha = s d|\alpha|$. By regularity of Radon measures on compact spaces, s can be approximated in $L^1(|\alpha|)$ by continuous functions f_k with $\|f_k\|_{\infty} \leq 1$. Hence $\int f_k d\alpha \rightarrow \int s d\alpha = |\alpha|(\mathcal{X})$, which proves the equality. The final statement is the Riesz–Markov–Kakutani representation theorem with this norm. $\square$

For two absolutely continuous measures $d\alpha = \rho_{\alpha}d\lambda$ and $d\beta = \rho_{\beta}d\lambda$, this gives the concrete formula

$$\|\alpha - \beta\|_{\text{TV}} = \int_{\mathcal{X}} |\rho_{\alpha}(x) - \rho_{\beta}(x)| d\lambda(x).$$

For two discrete measures written on the same union of supports, $\alpha = \sum_k a_k \delta_{z_k}$ and $\beta = \sum_k b_k \delta_{z_k}$, with missing coefficients set to zero,

$$\|\alpha - \beta\|_{\text{TV}} = \sum_k |a_k - b_k|.$$

Probabilistic interpretation. Radon probability measures represent the laws of random variables. A random variable with values in $\mathcal{X}$ is a measurable map $X : \Omega \rightarrow \mathcal{X}$ from an abstract probability space $(\Omega, \mathbb{P})$. Its law is the Radon probability measure α defined by

$$\alpha(A) = \mathbb{P}(\{\omega \in \Omega ; X(\omega) \in A\}) \quad \text{for Borel sets } A \subset \mathcal{X}.$$

Integrals with respect to this law are expectations:

$$\int_{\mathcal{X}} f(x) d\alpha(x) = \mathbb{E}[f(X)].$$

2.2 Push Forward

Push-forwards encode how maps move mass. This short subsection is the bridge between deterministic maps and linear operations on measures.

For some continuous map $T : \mathcal{X} \rightarrow \mathcal{Y}$, we define the pushforward operator $T_{\#} : \mathcal{M}(\mathcal{X}) \rightarrow \mathcal{M}(\mathcal{Y})$. For a Dirac mass, one has $T_{\#} \delta_x = \delta_{T(x)}$, and this formula is extended to arbitrary measures by linearity. In some sense, moving from T to $T_{\#}$ is a way to linearize any map at the price of moving from a (possibly) finite-dimensional space $\mathcal{X}$ to the infinite-dimensional space $\mathcal{M}(\mathcal{X})$, and this idea is central to many convex

relaxation methods, most notably Lasserre's relaxation. For discrete measures (2), the pushforward operation consists simply in moving the positions of all the points in the support of the measure

$$T_{\#}\alpha := \sum_i a_i \delta_{T(x_i)}.$$

For more general measures, for instance for those with a density, the notion of push-forward plays a fundamental role in describing spatial modifications of probability measures. The formal definition reads as follows.

Definition 3 (Push-forward). *For $T : \mathcal{X} \rightarrow \mathcal{Y}$, the push forward measure $\beta = T_{\#}\alpha \in \mathcal{M}(\mathcal{Y})$ of some $\alpha \in \mathcal{M}(\mathcal{X})$ satisfies*

$$\forall h \in \mathcal{C}(\mathcal{Y}), \quad \int_{\mathcal{Y}} h(y) d\beta(y) = \int_{\mathcal{X}} h(T(x)) d\alpha(x). \quad (3)$$

Equivalently, for any measurable set $B \subset \mathcal{Y}$, one has

$$\beta(B) = \alpha(\{x \in \mathcal{X} ; T(x) \in B\}). \quad (4)$$

Note that $T_{\#}$ preserves positivity and total mass, so that if $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ then $T_{\#}\alpha \in \mathcal{M}_+^1(\mathcal{Y})$.

Proposition 3 (Push-forward formula for densities). *Let α and β have densities ρ_{α} and ρ_{β} on open subsets of $\mathbb{R}^d$. Assume that T is a C^1 diffeomorphism and that $\beta = T_{\#}\alpha$. Then, for all x ,*

$$\rho_{\alpha}(x) = |\det(T'(x))| \rho_{\beta}(T(x)). \quad (5)$$

Equivalently, for $y = T(x)$,

$$\rho_{\beta}(y) = \rho_{\alpha}(T^{-1}y) \frac{1}{|\det T'(T^{-1}y)|}.$$

Proof. Explicitly doing the change of variable $y = T(x)$, so that $dy = |\det(T'(x))|dx$ in formula (3) for measures with densities $(\rho_{\alpha}, \rho_{\beta})$ on $\mathbb{R}^d$ (assuming T is smooth and a bijection), one has for all $h \in \mathcal{C}(\mathcal{Y})$

$$\begin{aligned} \int_{\mathcal{Y}} h(y) \rho_{\beta}(y) dy &= \int_{\mathcal{Y}} h(y) d\beta(y) = \int_{\mathcal{X}} h(T(x)) d\alpha(x) = \int_{\mathcal{X}} h(T(x)) \rho_{\alpha}(x) dx \\ &= \int_{\mathcal{Y}} h(y) \rho_{\alpha}(T^{-1}y) \frac{dy}{|\det(T'(T^{-1}y))|}, \end{aligned}$$

which shows that

$$\rho_{\beta}(y) = \rho_{\alpha}(T^{-1}y) \frac{1}{|\det(T'(T^{-1}y))|}.$$

Since T is a diffeomorphism, one obtains equivalently

$$\rho_{\alpha}(x) = |\det(T'(x))| \rho_{\beta}(T(x))$$

where $T'(x) \in \mathbb{R}^{d \times d}$ is the Jacobian matrix of T (the matrix formed by taking the gradient of each coordinate of T). This implies, denoting $y = T(x)$

$$|\det(T'(x))| = \frac{\rho_{\alpha}(x)}{\rho_{\beta}(y)}.$$

□

Remark 1 (Probabilistic interpretation). The law, i.e. probability distribution, of a random variable X is the push-forward of $\mathbb{P}$ by X , namely $\alpha = X_{\#}\mathbb{P}$. Applying another push-forward $\beta = T_{\#}\alpha$ for $T : \mathcal{X} \rightarrow \mathcal{Y}$, following (3), is equivalent to defining another random variable $Y = T(X)$, namely $\omega \in \Omega \mapsto T(X(\omega)) \in \mathcal{Y}$. Thus β is the law of Y . Drawing a random sample y from Y is thus simply achieved by computing $y = T(x)$ where x is drawn from X .

2.3 Monge Maps

Monge's formulation asks for a deterministic map carrying one probability law to another. It is geometrically intuitive and gives the maps used for interpolation, but it is also fragile because deterministic maps cannot split mass.

Continuous Monge problem. Given $\alpha \in \mathcal{M}_+^1(\mathcal{X})$, $\beta \in \mathcal{M}_+^1(\mathcal{Y})$ and a cost $c : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}_+$, the continuous Monge problem is

$$\mathcal{M}_c(\alpha, \beta) := \inf_{T: \mathcal{X} \rightarrow \mathcal{Y}} \left\{ \int_{\mathcal{X}} c(x, T(x)) d\alpha(x) ; T_{\#}\alpha = \beta \right\}. \quad (6)$$

When $\mathcal{X} = \mathcal{Y}$ and $c(x, y) = d(x, y)^p$, the associated directed Monge value is

$$\tilde{\mathcal{W}}_p(\alpha, \beta) := \mathcal{M}_{d^p}(\alpha, \beta)^{1/p}. \quad (7)$$

The tilde is important: the constraint set can be empty and the value is not a symmetric distance in general. For instance, one point can be mapped to one point, but a single Dirac mass cannot be mapped deterministically to two separated Dirac masses.

Example 1 (Semi-discrete Monge maps). If α has a density and $\beta = \sum_j b_j \delta_{y_j}$ is discrete, a Monge map can exist even though the target is singular: it sends whole regions of $\mathcal{X}$ to the target atoms. For the quadratic cost these regions are Laguerre cells selected by semi-dual weights. Section 6 studies this semi-discrete construction and its numerical algorithms in detail.

One-dimensional quantiles. In one dimension the Monge map is explicit and is the continuous analogue of the sorting rule from Section 1.1.

Definition 4 (Cumulative and quantile functions). For $\alpha \in \mathcal{M}_+^1(\mathbb{R})$, its cumulative distribution function is

$$\mathcal{C}_\alpha(x) := \alpha((-\infty, x]). \quad (8)$$

Its generalized inverse, or quantile function, is

$$\mathcal{C}_\alpha^{-1}(s) := \inf \{x \in \mathbb{R} ; \mathcal{C}_\alpha(x) \geq s\}, \quad s \in (0, 1). \quad (9)$$

Proposition 4 (Monotone rearrangement on the line). Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R})$ and let $p \geq 1$. The coupling

$$\pi^* = (\mathcal{C}_\alpha^{-1}, \mathcal{C}_\beta^{-1})_{\#} \text{Leb}_{[0,1]}$$

minimizes $\int |x - y|^p d\pi(x, y)$ among couplings. If α has no atoms, it is induced by the monotone Monge map

$$T(x) = \mathcal{C}_\beta^{-1}(\mathcal{C}_\alpha(x)).$$

Proof. If U is uniform on $[0, 1]$, then $\mathcal{C}_\alpha^{-1}(U)$ has law α and $\mathcal{C}_\beta^{-1}(U)$ has law β , so the displayed measure is a coupling. Its support is monotone: for $s < t$, both quantile functions satisfy $\mathcal{C}_\alpha^{-1}(s) \leq \mathcal{C}_\alpha^{-1}(t)$ and $\mathcal{C}_\beta^{-1}(s) \leq \mathcal{C}_\beta^{-1}(t)$. If a coupling had two crossed pairs $x < x'$ and $y > y'$ with positive mass, exchanging the targets decreases the cost for strictly convex powers and does not increase it for $p = 1$ by the same two-point inequality used in Proposition 1. Removing all such crossings gives a monotone coupling, and the monotone coupling with prescribed marginals is exactly the quantile coupling. If α has no atoms, $\mathcal{C}_\alpha(X)$ is uniform for $X \sim \alpha$, so the coupling is generated by the stated map. $\square$

Displacement interpolation. If $T_{\#}\alpha = \beta$, the Monge interpolation between α and β is

$$\alpha_t = ((1 - t)\text{Id} + tT)_{\#}\alpha, \quad t \in [0, 1].$$

For the quadratic cost, when T is optimal, this is the McCann displacement interpolation. In one dimension it corresponds exactly to interpolating quantiles,

$$\mathcal{C}_{\alpha_t}^{-1}(s) = (1 - t)\mathcal{C}_\alpha^{-1}(s) + t\mathcal{C}_\beta^{-1}(s).$$

The histogram-equalization figure in Section 1.1 is this construction applied to pixel intensities.

Brenier maps and generalizations.

Theorem 1 (Brenier). *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R}^d)$ have finite second moments, and assume that α is absolutely continuous with respect to Lebesgue measure. For the quadratic cost $c(x, y) = \|x - y\|^2$, there exists a convex function $\varphi : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ such that*

$$T = \nabla \varphi, \quad T_\# \alpha = \beta,$$

and T is the unique optimal Monge map α -almost everywhere. The optimal Kantorovich plan is $(\text{Id}, T)_\# \alpha$.

Proof. Kantorovich duality for the quadratic cost gives an optimal c -concave potential. After subtracting $\|x\|^2/2$, this is equivalent to a convex potential φ whose subgradient contains the support of every optimal plan: if (x, y) belongs to the support, then $y \in \partial \varphi(x)$. Since α is absolutely continuous, Alexandrov's theorem gives differentiability of convex functions α -almost everywhere, hence $y = \nabla \varphi(x)$ for any optimal pair above such an x . Thus every optimal plan is concentrated on the graph of $\nabla \varphi$, which proves existence and uniqueness of the Monge map and identifies the Kantorovich optimizer. $\square$

Extensions to more general costs require a replacement for the gradient-of-convex structure. Under the usual twist condition, for instance when $y \mapsto \nabla_x c(x, y)$ is injective, optimal maps are recovered from c -convex potentials through $\nabla_x c(x, T(x)) = \nabla u(x)$ whenever the derivatives exist. This is the route developed by Gangbo–McCann and subsequent regularity theory for strictly convex and generated costs [72, 171, 154]. The Brenier theorem is the special quadratic case where this relation becomes $T(x) = x - \nabla u(x) = \nabla \varphi(x)$.

Proposition 5 (Polar factorization). *Let $\Omega \subset \mathbb{R}^d$ be endowed with normalized Lebesgue measure λ , and let $u \in L^2(\Omega; \mathbb{R}^d)$. Assume that the law $\nu = u_\# \lambda$ has finite second moment. Then there exist a measure-preserving map $s : \Omega \rightarrow \Omega$ and a convex function φ such that*

$$u = \nabla \varphi \circ s \quad \lambda\text{-a.e.}$$

The Brenier factor $\nabla \varphi$ is unique λ -almost everywhere.

Proof. By Brenier's theorem there is a unique gradient of a convex function $T = \nabla \varphi$ transporting λ to ν . The maps u and T have the same image law. The rearrangement theorem for non-atomic probability spaces gives a measure-preserving map s such that $u = T \circ s$; equivalently, s chooses, with the correct conditional probabilities, preimages of $u(x)$ under T . Uniqueness of the Brenier factor follows from the uniqueness part of Theorem 1. In the Gaussian linear case, with reference $\mathcal{N}(0, \text{Id})$ and $u(x) = Ax$, this is the usual matrix polar decomposition: $A = SO$ with $S = (AA^\top)^{1/2}$ symmetric positive semidefinite, Sx is the Brenier part and Ox is measure preserving. $\square$

Gaussian case. For Gaussian measures, the Brenier map is affine. If

$$\alpha = \mathcal{N}(\mathbf{m}_\alpha, \Sigma_\alpha), \quad \beta = \mathcal{N}(\mathbf{m}_\beta, \Sigma_\beta), \quad \Sigma_\alpha, \Sigma_\beta \succ 0,$$

then

$$T(x) = \mathbf{m}_\beta + A(x - \mathbf{m}_\alpha), \quad A = \Sigma_\alpha^{-1/2} (\Sigma_\alpha^{1/2} \Sigma_\beta \Sigma_\alpha^{1/2})^{1/2} \Sigma_\alpha^{-1/2}. \quad (10)$$

The covariance part of the distance is the Bures distance

$$\mathcal{B}(\Sigma_\alpha, \Sigma_\beta)^2 := \text{tr}(\Sigma_\alpha) + \text{tr}(\Sigma_\beta) - 2 \text{tr} \left((\Sigma_\alpha^{1/2} \Sigma_\beta \Sigma_\alpha^{1/2})^{1/2} \right), \quad (11)$$

and hence

$$\mathcal{W}_2(\alpha, \beta)^2 = \|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \mathcal{B}(\Sigma_\alpha, \Sigma_\beta)^2. \quad (12)$$

Regularity of OT maps.

Proposition 6 (Caffarelli regularity). *Let $\Omega, \Lambda \subset \mathbb{R}^d$ be bounded uniformly convex domains with C^2 boundaries. Let $\alpha = \rho(x)dx$ be supported on Ω and $\beta = \eta(y)dy$ be supported on Λ , with $0 < \lambda \leq \rho, \eta \leq \Lambda_0 < +\infty$. If $\rho, \eta \in C^\alpha$ for some $\alpha \in (0, 1)$, then the Brenier potential φ transporting α to β satisfies $\varphi \in C_{\text{loc}}^{2,\alpha}(\Omega)$. Under the corresponding boundary compatibility and smoothness assumptions, the regularity holds up to the boundary.*

Proof. The potential solves the Monge–Ampère equation

$$\det(\nabla^2 \varphi(x)) = \frac{\rho(x)}{\eta(\nabla \varphi(x))}$$

in the Alexandrov sense, with second boundary condition $\nabla \varphi(\Omega) = \Lambda$. The density bounds and convexity of the domains give strict convexity and localization of sections. Caffarelli’s interior theory then yields $C_{\text{loc}}^{2,\alpha}$ estimates for φ ; the boundary statement follows from the boundary regularity theory under uniform convexity and compatibility assumptions [32, 171]. $\square$

Proposition 7 (Linearization of the Monge–Ampère equation). *Let $\rho_\varepsilon = \rho_0 + \varepsilon r + o(\varepsilon)$ be a smooth perturbation of a positive density on a smooth bounded domain, with $\int r = 0$. If $T_\varepsilon(x) = x + \varepsilon \nabla u(x) + o(\varepsilon)$ transports the reference density ρ_0 to the perturbed density ρ_ε , then, to first order,*

$$-\nabla \cdot (\rho_0 \nabla u) = r.$$

In particular, when ρ_0 is constant, the linearized equation is $-\Delta u = r/\rho_0$.

Proof. The change-of-variables equation for T_ε is

$$\rho_0(x) = \rho_\varepsilon(T_\varepsilon(x)) \det(\nabla T_\varepsilon(x)).$$

Expanding $\rho_\varepsilon(x + \varepsilon \nabla u) = \rho_0(x) + \varepsilon r(x) + \varepsilon \langle \nabla \rho_0, \nabla u \rangle + o(\varepsilon)$ and

$$\det(I + \varepsilon \nabla^2 u) = 1 + \varepsilon \Delta u + o(\varepsilon)$$

gives

$$\rho_0 = \rho_0 + \varepsilon (r + \langle \nabla \rho_0, \nabla u \rangle + \rho_0 \Delta u) + o(\varepsilon) = \rho_0 + \varepsilon (r + \nabla \cdot (\rho_0 \nabla u)) + o(\varepsilon).$$

The first-order term must vanish, which gives the displayed equation. $\square$

Triangular rearrangements.

Proposition 8 (Knothe–Rosenblatt triangular rearrangement). *Let α and β be probability measures on $\mathbb{R}^d$ with positive densities and regular conditional distributions. There is a triangular map*

$$T(x_1, \dots, x_d) = (T_1(x_1), T_2(x_1, x_2), \dots, T_d(x_1, \dots, x_d))$$

such that $T_\# \alpha = \beta$ and each coordinate map is nondecreasing in its last argument.

Proof. Transport the first marginal of α to the first marginal of β by the one-dimensional quantile map. Then, for each value of the first coordinate, transport the conditional law of x_2 given x_1 to the corresponding conditional law of y_2 given $y_1 = T_1(x_1)$ by another one-dimensional quantile map. Iterating this construction over the coordinates gives a triangular measurable map. The chain rule for conditional probabilities shows that the joint law is transported to β . $\square$

This construction transports successively along coordinate axes and is therefore axis-wise. It depends on the ordering of coordinates and is not generally optimal for the quadratic cost. It is nevertheless a useful limiting object: Brenier maps for increasingly anisotropic quadratic costs converge to triangular rearrangements under suitable assumptions [37].

Illustrative Monge maps. The following figures show three concrete push-forwards. In each case, the map is estimated on a small empirical measure, but the visualization is chosen to emphasize the underlying continuum operation: moving colors in RGB space, displacing points between shapes, and representing a target law by a displacement field from a fixed reference.

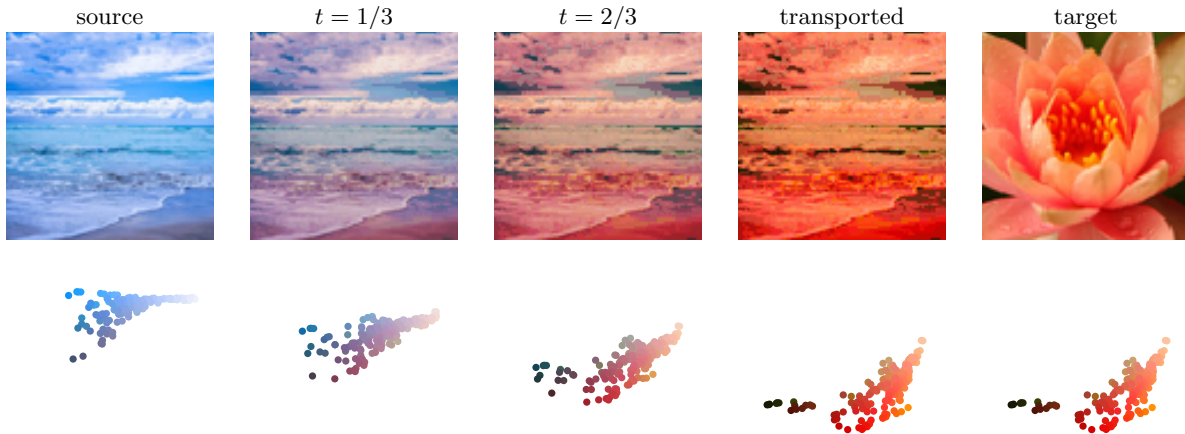


Figure 5: Color transfer as a Monge map in RGB space, from a beach photograph to a flower photograph. The top row applies the palette map to the source image, while the bottom row shows the corresponding empirical color cloud in the RGB cube. Only colors are transported here: pixel locations are left fixed.

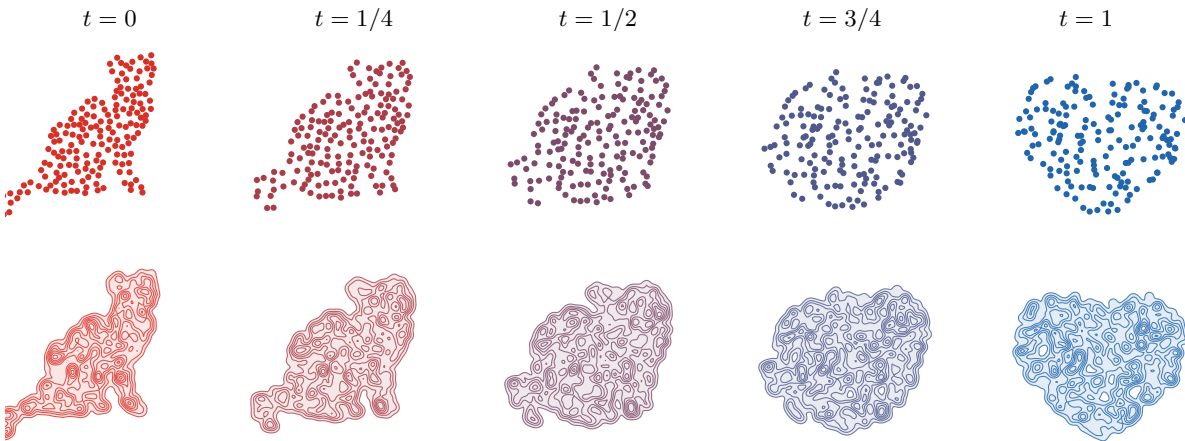


Figure 6: McCann displacement interpolation between a cat silhouette and a heart silhouette. The first row displays transported particles along $T_t(x) = (1 - t)x + tT(x)$; the second row shows kernel-smoothed density estimates computed from a larger cloud, making the evolving support easier to read.

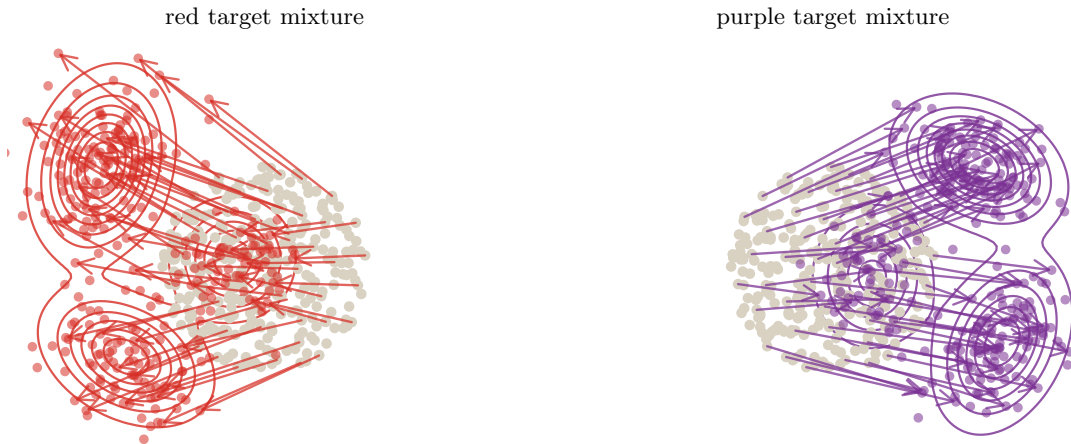


Figure 7: Linearized transport coordinates around a fixed disk reference. The gray points sample the reference law, colored contours show the target density, and the arrows display a farthest-point subsample of the barycentric projection $T_\mu - \text{Id}$ of the quadratic OT map. The full OT map is computed on a denser cloud than the arrows shown here.

3 Kantorovich Relaxation

Kantorovich’s relaxation is the decisive move that turns transport into convex optimization. The section explains how deterministic maps are replaced by couplings, why this fixes infeasibility and symmetry, and how it produces the Wasserstein distances. Historically, this linear-programming viewpoint grew from Kantorovich’s economic planning work [96] and is now the standard foundation of OT [170, 171, 144].

3.1 Discrete Relaxation

The discrete relaxation is the cleanest place to see mass splitting. It replaces permutations by a transportation polytope and reveals the linear-programming structure that algorithms exploit.

Monge’s discrete matching problem is problematic because it cannot be applied when $n \neq m$. One needs to take into account masses (a_i, b_j) to handle this more general situation. Monge’s continuous formulation (6), based on push-forward maps, is also problematic because there may be no transport map T such that $T_\# \alpha = \beta$. This happens, for instance, when α is a single Dirac mass that should be mapped to several Dirac masses. Accordingly, Monge’s formulation is not symmetric with respect to the exchange of α and β : one can map two Dirac masses to a single one, but not the other way around. Moreover, these are non-convex optimization problems, which are difficult to solve numerically.

The key idea of [96] is to relax the deterministic nature of transportation. Instead of requiring each source point x_i to be sent to one and only one target location, Kantorovich allows the mass at x_i to be dispatched across several locations. This moves from deterministic transport maps to probabilistic, or fuzzy, transport plans and allows what is now called mass splitting. This flexibility is encoded, in place of a permutation σ or a map T , by a coupling matrix $P \in \mathbb{R}_+^{n \times m}$, where $P_{i,j}$ describes the amount of mass flowing from x_i to y_j in the formalism of discrete measures $\alpha = \sum_i a_i \delta_{x_i}$, $\beta = \sum_j b_j \delta_{y_j}$. Admissible couplings are only constrained to satisfy conservation of mass

$$U(a, b) := \{P \in \mathbb{R}_+^{n \times m} ; P \mathbf{1}_m = a \quad \text{and} \quad P^\top \mathbf{1}_n = b\}, \quad (13)$$

where we used the following matrix-vector notation

$$P \mathbf{1}_m = \left(\sum_j P_{i,j} \right)_i \in \mathbb{R}^n \quad \text{and} \quad P^\top \mathbf{1}_n = \left(\sum_i P_{i,j} \right)_j \in \mathbb{R}^m.$$

The set of matrices $U(a, b)$ is bounded, defined by $n+m$ equality constraints, and therefore a convex polytope (the convex hull of a finite set of matrices). In one dimension, an ordered coupling can be read as a matrix: rows index source bins, columns index target bins, and the marginal constraints are visible as prescribed row and column sums.

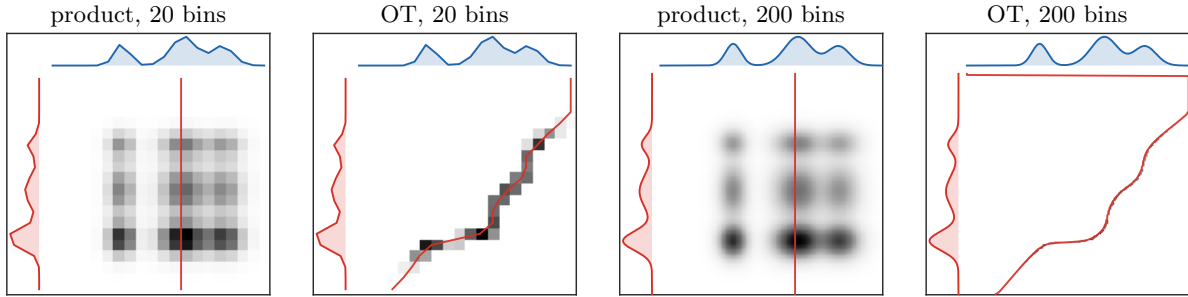


Figure 8: Coupling matrices with their prescribed marginals. The central grayscale image displays P_{ij} ; the red curve on the left is the source marginal a , the blue curve on top is the target marginal b , and the thin red curve inside the matrix is the barycentric projection $i \mapsto \sum_j P_{ij} j / a_i$. The independent product plan is diffuse, whereas the one-dimensional optimal plan concentrates near the monotone quantile correspondence. The dense discretization is the same construction with 200 bins.

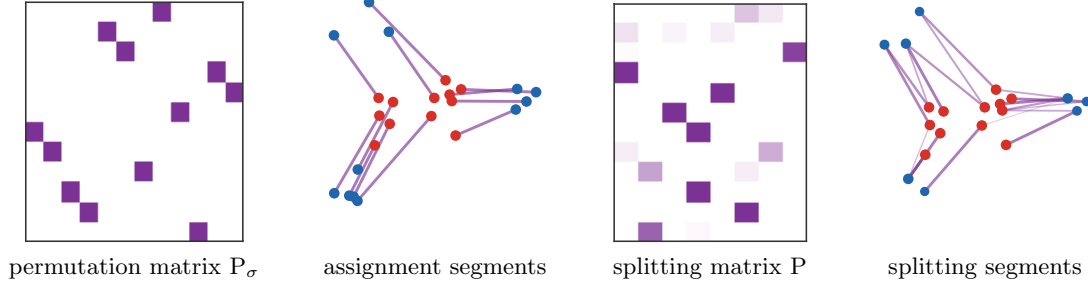


Figure 9: From permutation matrices to splitting couplings. When the two empirical measures have the same number of atoms and uniform weights, an optimal plan can be a permutation matrix. Once target masses are nonuniform, the admissible set $\mathcal{U}(a, b)$ also contains plans in which one source sends mass to several targets and several sources merge into the same target.

Continuous couplings and plans. The same relaxation applies to arbitrary probability measures. A transport plan is a joint probability measure with prescribed marginals.

Definition 5 (Continuous couplings). *For $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ and $\beta \in \mathcal{M}_+^1(\mathcal{Y})$, the set of couplings is*

$$\mathcal{U}(\alpha, \beta) := \left\{ \pi \in \mathcal{M}_+^1(\mathcal{X} \times \mathcal{Y}) ; (P_{\mathcal{X}})_{\#} \pi = \alpha, (P_{\mathcal{Y}})_{\#} \pi = \beta \right\}, \quad (14)$$

where $P_{\mathcal{X}}(x, y) = x$ and $P_{\mathcal{Y}}(x, y) = y$.

The Kantorovich optimal transport value is

$$\mathcal{L}_c(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y). \quad (15)$$

The product measure $\alpha \otimes \beta$ is always feasible, so the relaxation is never empty, unlike the Monge constraint.

Definition 6 (*c-cyclical monotonicity*). *A set $\Gamma \subset \mathcal{X} \times \mathcal{Y}$ is c-cyclically monotone if, for every finite family $(x_i, y_i)_{i=1}^k \subset \Gamma$ and every permutation σ of $\{1, \dots, k\}$,*

$$\sum_{i=1}^k c(x_i, y_i) \leq \sum_{i=1}^k c(x_i, y_{\sigma(i)}).$$

Additionally, whereas the Monge formulation is intrinsically asymmetric, Kantorovich's relaxed formulation is always symmetric, in the sense that a coupling P is in $\mathcal{U}(a, b)$ if and only if $P^{\top}$ is in $\mathcal{U}(b, a)$.

Kantorovich, aiming for economic planning, made a strong simplifying assumption: the cost of transportation should be linear in the amount of transported mass. Under this assumption, denoting $C_{i,j}$ the cost of moving a unit amount of mass from x_i to y_j , Kantorovich's optimal transport problem now reads

$$L_C(a, b) := \min_{P \in \mathcal{U}(a, b)} \langle C, P \rangle := \sum_{i,j} C_{i,j} P_{i,j}. \quad (16)$$

This is a linear program, and as is usually the case with such programs, its solutions are not necessarily unique.

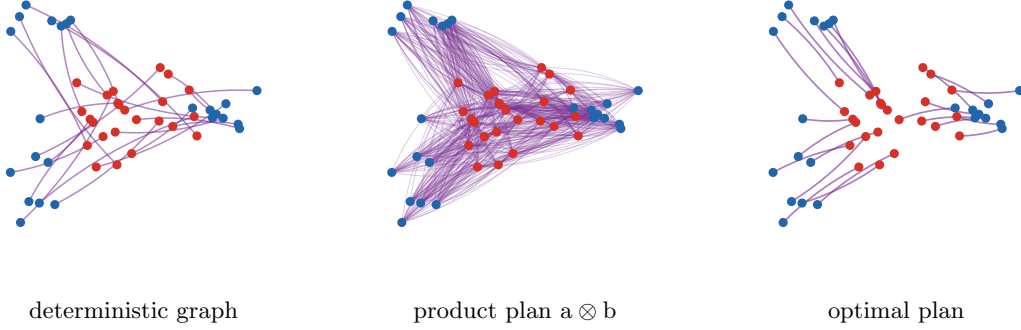


Figure 10: Discrete couplings represented as transport polylines on the canonical point clouds used in the matching section. The deterministic graph is a feasible Monge-type plan, the product plan spreads every source mass over all targets, and the optimal Kantorovich plan minimizes the quadratic transport cost. Line width and opacity encode the transported mass.

Sparse feasible plans. Although the product plan $a \otimes b$ is always feasible, it is dense and therefore a poor initialization for algorithms that exploit sparsity. The transportation polytope always contains sparse feasible points, and linear costs always admit sparse optimizers.

Proposition 9 (Sparse optimal plans). *Assume $a_i > 0$, $b_j > 0$ and $\sum_i a_i = \sum_j b_j = 1$. The linear program (16) admits an optimal coupling with at most $n + m - 1$ nonzero entries.*

Proof. The transportation polytope is compact, so a linear objective attains its minimum at an extreme point. We show that any extreme point has at most $n + m - 1$ positive entries. Let $E = \{(i, j) : P_{ij} > 0\}$ be the support graph on the bipartite vertex set $\{1, \dots, n\} \cup \{1, \dots, m\}$. If this graph contains a cycle, orient the cycle and put alternating signs $+1, -1$ on its edges, obtaining a nonzero matrix H supported on E with zero row and column sums. For sufficiently small $t > 0$, both $P + tH$ and $P - tH$ are nonnegative couplings, and P is their midpoint, contradicting extremality. Thus the support graph is a forest. Since a forest on $n + m$ vertices has at most $n + m - 1$ edges, the claim follows. $\square$

Proposition 10 (North-west corner feasible plan). *For any $a \in \Sigma_n$ and $b \in \Sigma_m$, the following algorithm returns a coupling in $U(a, b)$ with at most $n + m - 1$ nonzero entries. Initialize residuals $r_i = a_i$, $s_j = b_j$ and indices $i = j = 1$. While $i \leq n$ and $j \leq m$, set*

$$P_{ij} = \min(r_i, s_j),$$

replace $r_i \leftarrow r_i - P_{ij}$ and $s_j \leftarrow s_j - P_{ij}$, then increase every index whose residual is zero.

Proof. At each step the algorithm assigns the largest possible mass to the current cell without violating either the current row or current column constraint. The residuals remain nonnegative, and at least one of them becomes zero, so at least one index advances. Hence the procedure terminates after at most $n + m - 1$ assignments unless a final simultaneous row/column exhaustion ends the algorithm earlier. Each row i receives exactly its initial mass a_i before the index moves past it, and each column j receives exactly b_j before the column index moves past it. Therefore the output has row sums a , column sums b , nonnegative entries, and at most one new nonzero per step. In one dimension this is the discrete quantile sweep, hence the discrete analogue of the Knothe–Rosenblatt axis-wise construction of Proposition 8. $\square$

Proposition 11 (One-dimensional weighted sweep). *Let $x_1 \leq \dots \leq x_n$ and $y_1 \leq \dots \leq y_m$ be points on the line, and let $c(x, y) = h(x - y)$ with h convex. The north-west corner plan between the sorted weighted atoms is optimal for (16). Consequently, for unsorted one-dimensional inputs, an optimal plan is obtained in $O(n \log n + m \log m)$ time by sorting and then sweeping the masses once from left to right.*

Proof. The north-west plan is monotone: if $i < i'$ and $j > j'$, it cannot put positive mass on both (i, j) and (i', j') , because the sweep exhausts rows and columns in increasing order. Conversely, any feasible plan with a crossing pair of positive entries can be improved by moving a small mass η from (i, j) and (i', j') to (i, j') and (i', j) . The two marginals are unchanged, and convexity of h gives

$$h(x_i - y_j) + h(x_{i'} - y_{j'}) \geq h(x_i - y_{j'}) + h(x_{i'} - y_j)$$

for $i < i'$ and $j' < j$, with strict inequality for strictly convex h and distinct points. Repeating this uncrossing procedure until no crossing remains yields a monotone optimal plan. There is only one monotone feasible plan with the prescribed sorted marginals, namely the sweep plan: it pairs the leftmost remaining source mass with the leftmost remaining target mass at every step. Sorting costs $O(n \log n + m \log m)$ and the sweep uses at most $n + m - 1$ assignments. $\square$

Optimal matching to optimal transport. Let the marginals be uniform on n points: $\alpha = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$, $\beta = \frac{1}{n} \sum_{i=1}^n \delta_{y_i}$. In this case, there exist solutions to the Kantorovich problem that are permutations. Such a permutation defines, by optimality, a coupling with support $\Gamma := (x_i, y_{\sigma(i)})_i$ that is c -cyclically monotone. The following theorem says this is true for any optimal coupling, not only in the case of a finite set of points.

Theorem 2 (Rockafellar). *For any optimal plan π solving Kantorovich's problem, $\text{supp}(\pi)$ is c -cyclically monotone.*

Proof. We prove the contrapositive. Suppose that $\text{supp}(\pi)$ is not c -cyclically monotone. Then there exist points $(x_i, y_i)_{i=1}^k$ in the support and a permutation σ such that

$$\sum_i c(x_i, y_i) > \sum_i c(x_i, y_{\sigma(i)}).$$

By continuity of c , this strict inequality persists on small neighborhoods $U_i \times V_i$ of the points (x_i, y_i) , chosen disjoint enough so that each has positive π -mass. Let $\eta > 0$ be smaller than all these masses. Removing mass η from each $U_i \times V_i$ and re-inserting the same mass from U_i to $V_{\sigma(i)}$ preserves both marginals, because the operation only permutes target neighborhoods among the same source neighborhoods. The cost strictly decreases by the displayed gap up to the continuity error. This contradicts optimality of π . $\square$

Interpolation induced by a plan. A transport plan also defines an interpolation between measures by moving every active pair (x_i, y_j) along the segment $(1-t)x_i + ty_j$ with mass P_{ij} . When the plan is not induced by a map, one source atom can split into several moving atoms.

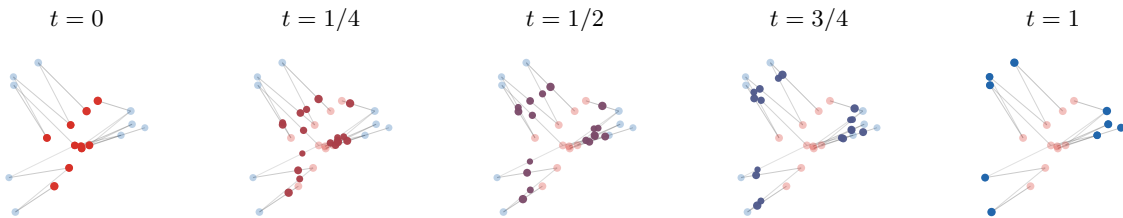


Figure 11: McCann interpolation induced by a non-deterministic transport plan. In every panel, the red and blue endpoint measures are shown with low opacity, thin gray segments display the support $P_{ij} > \text{tol}$ of the coupling, and the moving atoms are colored from red to blue along the interpolation.

Monotonicity. Assume the plan is induced by a measurable map $T : \mathcal{X} \rightarrow \mathcal{Y}$, i.e. $\pi = (\text{Id}, T)_\# \alpha$. For any k points $x_1, \dots, x_k$ in the domain, cyclical monotonicity reads

$$\sum_{i=1}^k c(x_i, T(x_i)) \leq \sum_{i=1}^k c(x_i, T(x_{i+1})), \quad x_{k+1} = x_1.$$

For $c(x, y) = \frac{1}{2}|x - y|^2$, taking $k = 2$, shows that for any x, y ,

$$\langle T(x) - T(y), x - y \rangle \geq 0,$$

so T is a monotone vector field. Brenier's theorem adds that, when α is absolutely continuous, $T = \nabla \varphi$ for a convex potential φ . (The converse fails in dimension $d \geq 2$: a small rotation is monotone yet not a gradient.)

One dimension. In one space dimension, with cost $c(x, y) = |x - y|^p$ for any $p \geq 1$, the two-point inequality becomes

$$|x - T(x)|^p + |y - T(y)|^p \leq |x - T(y)|^p + |y - T(x)|^p,$$

which is equivalent to $T(x) \leq T(y)$ whenever $x < y$. Thus T must be non-decreasing, recovering the classical monotone rearrangement.

3.2 Metric Properties: Wasserstein Distances

The final part of the section proves that OT costs are genuine distances when the ground cost comes from a metric. It also compares Wasserstein convergence with total variation and explains why OT is weak enough to move Dirac masses continuously.

OT defines a distance. An important feature of OT is that it defines a distance between histograms and probability measures as soon as the cost matrix satisfies certain suitable properties. Indeed, OT can be understood as a canonical way to lift a ground distance between points to a distance between histograms or measures. The proof of this result relies on a “gluing lemma”, which we first prove in the discrete case.

Lemma 1 (Discrete gluing lemma). *Given $(a, b, c) \in \Sigma_n \times \Sigma_p \times \Sigma_m$, let $P \in U(a, b)$ and $Q \in U(b, c)$. Then there exists a 3-D tensor coupling $S \in \mathbb{R}_+^{n \times p \times m}$ such that the 2-D marginals satisfy*

$$\sum_k S_{i,j,k} = P_{i,j} \quad \text{and} \quad \sum_i S_{i,j,k} = Q_{j,k}.$$

Note that this implies that the three 1-D marginals of S are (a, b, c) .

Proof. One verifies that

$$S_{i,j,k} = \begin{cases} \frac{P_{i,j} Q_{j,k}}{b_j} & \text{if } b_j \neq 0 \\ 0 & \text{otherwise} \end{cases} \quad (17)$$

is acceptable. Indeed, if $b_j \neq 0$

$$\sum_k S_{i,j,k} = \sum_k \frac{P_{i,j} Q_{j,k}}{b_j} = \frac{P_{i,j}}{b_j} (Q \mathbf{1}_m)_j = \frac{P_{i,j}}{b_j} b_j.$$

If $b_j = 0$, then necessarily $P_{i,j} = 0$ and $\sum_k S_{i,j,k} = 0 = P_{i,j}$. □

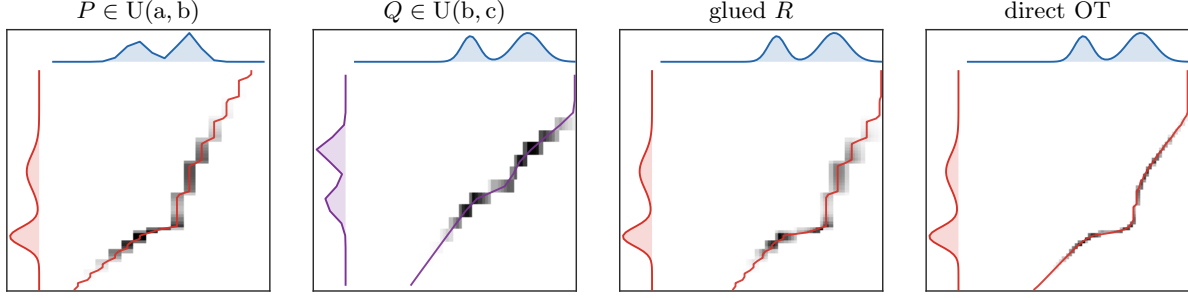


Figure 12: Discrete gluing lemma in matrix form. The first two panels are optimal one-dimensional couplings through an intermediate marginal b . The third panel shows the induced marginal $R = P \text{diag}(1/b)Q$ between a and c ; it is feasible and is the coupling used in the triangle-inequality proof. Because the intermediate marginal is represented on a coarser grid, the glued coupling is more mediated than the direct optimal coupling shown on the right.

Proposition 12 (Discrete Wasserstein distance). *We suppose $n = m$, and that for some $p \geq 1$, $C = D^p = (D_{i,j}^p)_{i,j} \in \mathbb{R}^{n \times n}$ where $D \in \mathbb{R}_+^{n \times n}$ is a distance on $\llbracket n \rrbracket$, i.e.*

1. $D \in \mathbb{R}_+^{n \times n}$ is symmetric;
2. $D_{i,j} = 0$ if and only if $i = j$;
3. $\forall (i, j, k) \in \llbracket n \rrbracket^3, D_{i,k} \leq D_{i,j} + D_{j,k}$.

Then

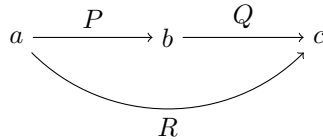
$$W_p(a, b) := L_{D^p}(a, b)^{1/p} \quad (18)$$

(note that W_p depends on D) defines the p -Wasserstein distance on Σ_n , i.e. W_p is symmetric, positive, $W_p(a, b) = 0$ if and only if $a = b$, and it satisfies the triangle inequality

$$\forall a, b, c \in \Sigma_n, \quad W_p(a, c) \leq W_p(a, b) + W_p(b, c).$$

Proof. For symmetry, since D^p is symmetric, we use the fact that if $P \in U(a, b)$ is optimal for $W_p(a, b)$, then $P^\top \in U(b, a)$ is optimal for $W_p(b, a)$. For definiteness, since $C = D^p$ has a null diagonal, $W_p(a, b) = 0$ is achieved by the diagonal coupling $P^* = \text{diag}(a) = \text{diag}(b)$ when $a = b$; by positivity of all off-diagonal elements of D^p , $W_p(a, b) > 0$ whenever $a \neq b$ because any admissible coupling then has a nonzero element outside the diagonal.

To prove the triangle inequality of Wasserstein distances for arbitrary measures, we consider $a, b, c \in \Sigma_n$, and let P and Q be two optimal solutions of the transport problems between a and b , and b and c respectively. We use the gluing Lemma 1 which defines $S \in \mathbb{R}_+^{n \times n}$ with marginals $\sum_k S_{\cdot, \cdot, k} = P$ and $\sum_i S_{i, \cdot, \cdot} = Q$. We define $R = \sum_j S_{\cdot, j, \cdot}$, which is an element of $U(a, c)$.



Note that if one assumes $b > 0$ then $R = P \text{diag}(1/b)Q$.

The triangle inequality follows from

$$\begin{aligned}
W_p(a, c) &= \left(\min_{\tilde{R} \in U(a, c)} \langle \tilde{R}, D^p \rangle \right)^{1/p} \leq \langle R, D^p \rangle^{1/p} \\
&= \left(\sum_{i, k} D_{ik}^p \sum_j S_{i, j, k} \right)^{1/p} \leq \left(\sum_{i, j, k} (D_{ij} + D_{j, k})^p S_{i, j, k} \right)^{1/p} \\
&\leq \left(\sum_{i, j, k} D_{ij}^p S_{i, j, k} \right)^{1/p} + \left(\sum_{i, j, k} D_{j, k}^p S_{i, j, k} \right)^{1/p} \\
&= \left(\sum_{i, j} D_{ij}^p \sum_k S_{i, j, k} \right)^{1/p} + \left(\sum_{j, k} D_{j, k}^p \sum_i S_{i, j, k} \right)^{1/p} \\
&= \left(\sum_{i, j} D_{i, j}^p P_{i, j} \right)^{1/p} + \left(\sum_{j, k} D_{j, k}^p Q_{j, k} \right)^{1/p} = W_p(a, b) + W_p(b, c).
\end{aligned}$$

The first inequality follows from the feasibility of R , the second is the usual triangle inequality for elements in D , and the third comes from Minkowski's inequality. $\square$

Continuous gluing. Proposition 12 generalizes from histogram to arbitrary measures that need not be discrete. For this, one needs the following general gluing lemma.

Lemma 2 (Gluing lemma). *Let $(\alpha, \beta, \gamma) \in \mathcal{M}_+^1(\mathcal{X}) \times \mathcal{M}_+^1(\mathcal{Y}) \times \mathcal{M}_+^1(\mathcal{Z})$ where $(\mathcal{X}, \mathcal{Y}, \mathcal{Z})$ are Polish spaces, i.e. separable topological spaces that can be metrized by complete distances. Given $\pi \in \mathcal{U}(\alpha, \beta)$ and $\xi \in \mathcal{U}(\beta, \gamma)$, then there exists a tensor coupling measure $\sigma \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$ such that*

$$(P_{\mathcal{X}, \mathcal{Y}})_\# \sigma = \pi \quad \text{and} \quad (P_{\mathcal{Y}, \mathcal{Z}})_\# \sigma = \xi$$

where we denoted the projector $P_{\mathcal{X}, \mathcal{Y}}(x, y, z) = (x, y)$ and $P_{\mathcal{Y}, \mathcal{Z}}(x, y, z) = (y, z)$.

Proof. The proof of this fundamental result is involved since it requires using the disintegration of measure (which corresponds to conditional probabilities). The disintegration of measures is applicable because the spaces are Polish. We disintegrate π and ξ against β to obtain two families $(\pi_y)_{y \in \mathcal{Y}}$ and $(\xi_y)_{y \in \mathcal{Y}}$ of probability distributions on $\mathcal{X}$ and $\mathcal{Z}$. These families are defined by the fact that

$$\forall h \in \mathcal{C}(\mathcal{X} \times \mathcal{Y}), \quad \int_{\mathcal{Y}} \left(\int_{\mathcal{X}} h(x, y) d\pi_y(x) \right) d\beta(y) = \int h(x, y) d\pi(x, y).$$

and similarly for ξ . When $\beta = \sum_j b_j \delta_{y_j}$ and $\pi = \sum_{i, j} P_{i, j} \delta_{(x_i, y_j)}$, then this conditional distribution is defined on the support of β as $\pi_{y_j} = \sum_i \frac{P_{i, j}}{b_j} \delta_{x_i}$ (and similarly for ξ). The glued measure is then defined by the conditional-product formula

$$\forall g \in \mathcal{C}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}), \quad \int g(x, y, z) d\sigma(x, y, z) = \int g(x, y, z) d\pi_y(x) d\xi_y(z) d\beta(y).$$

For discrete measures, this matches the definition (17), since $\sigma = \sum_{i, j, k} S_{i, j, k} \delta_{x_i, y_j, z_k}$ where

$$S_{i, j, k} = \frac{P_{i, j}}{b_j} \frac{Q_{j, k}}{b_j} b_j.$$

$\square$

Using this gluing lemma, we can now construct the Wasserstein distance in the general setting of arbitrary distributions on a Polish space.

Proposition 13 (Wasserstein distance on measures). *We assume $\mathcal{X} = \mathcal{Y}$, and that for some $p \geq 1$, $c(x, y) = d(x, y)^p$ where d is a distance on $\mathcal{X}$, i.e.*

- (i) $d(x, y) = d(y, x) \geq 0$;
- (ii) $d(x, y) = 0$ if and only if $x = y$;
- (iii) $\forall (x, y, z) \in \mathcal{X}^3, d(x, z) \leq d(x, y) + d(y, z)$.

Then

$$\mathcal{W}_p(\alpha, \beta) := \mathcal{L}_{d^p}(\alpha, \beta)^{1/p} \quad (19)$$

(note that $\mathcal{W}_p$ depends on d) defines the p -Wasserstein distance on $\mathcal{X}$, i.e. $\mathcal{W}_p$ is symmetric, positive, $\mathcal{W}_p(\alpha, \beta) = 0$ if and only if $\alpha = \beta$, and it satisfies the triangle inequality

$$\forall (\alpha, \beta, \gamma) \in \mathcal{M}_+^1(\mathcal{X})^3, \quad \mathcal{W}_p(\alpha, \gamma) \leq \mathcal{W}_p(\alpha, \beta) + \mathcal{W}_p(\beta, \gamma).$$

Proof. The symmetry follows from the fact that since d is symmetric, if $\pi(x, y)$ is optimal for $\mathcal{L}_{d^p}(\alpha, \beta)$, then $\pi(y, x) \in \mathcal{U}(\beta, \alpha)$ is optimal for $\mathcal{L}_{d^p}(\beta, \alpha)$. If $\mathcal{L}_{d^p}(\alpha, \beta) = 0$, then necessarily an optimal coupling π is supported on the diagonal $\Delta := \{(x, x)\}_x \subset \mathcal{X}^2$. We denote $\lambda(x)$ the corresponding measure on the diagonal, i.e. such that $\int h(x, y) d\pi(x, y) = \int h(x, x) d\lambda(x)$. Then since $\pi \in \mathcal{U}(\alpha, \beta)$ necessarily $\lambda = \alpha$ and $\lambda = \beta$ so that $\alpha = \beta$.

For the triangle inequality, we consider optimal couplings $\pi \in \mathcal{U}(\alpha, \beta)$ and $\xi \in \mathcal{U}(\beta, \gamma)$ and we glue them according to the Lemma 2. We define the composition of the two couplings (π, ξ) as $\rho := (P_{\mathcal{X}, \mathcal{Z}})_\# \sigma$. Note that if π and ξ are couplings induced by two Monge maps $T_\mathcal{X}(x)$ and $T_\mathcal{Y}(y)$, then ρ is itself induced by the Monge map $T_\mathcal{Y} \circ T_\mathcal{X}$, so that this notion of composition of coupling generalizes the composition of maps. The triangular inequality follows from

$$\begin{aligned} \mathcal{W}_p(\alpha, \gamma) &\leq \left(\int_{\mathcal{X} \times \mathcal{Z}} d(x, z)^p d\rho(x, z) \right)^{1/p} = \left(\int_{\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}} d(x, z)^p d\sigma(x, y, z) \right)^{1/p} \\ &\leq \left(\int_{\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}} (d(x, y) + d(y, z))^p d\sigma(x, y, z) \right)^{1/p} \\ &\leq \left(\int_{\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}} d(x, y)^p d\sigma(x, y, z) \right)^{1/p} + \left(\int_{\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}} d(y, z)^p d\sigma(x, y, z) \right)^{1/p} \\ &= \left(\int_{\mathcal{X} \times \mathcal{Y}} d(x, y)^p d\pi(x, y) \right)^{1/p} + \left(\int_{\mathcal{Y} \times \mathcal{Z}} d(y, z)^p d\xi(y, z) \right)^{1/p} = \mathcal{W}_p(\alpha, \beta) + \mathcal{W}_p(\beta, \gamma). \end{aligned}$$

□

Comparison with Monge. This distance $\mathcal{W}_p$ defined through the Kantorovich problem (19) should be contrasted with the directed distance $\tilde{\mathcal{W}}$ obtained using Monge's problem (7). The Kantorovich feasible set is never empty, since it contains the product coupling, although the p -cost may still be infinite without moment assumptions on non-compact spaces. By contrast, Monge's constraint set $\{T; T_\# \alpha = \beta\}$ can be empty. When an optimal Monge map exists, Kantorovich gives the same value by choosing the graph coupling $(\text{Id}, T)_\# \alpha$; in this sense the Kantorovich problem is the convex relaxation of Monge's problem, with much better stability properties.

3.3 Metric Properties: Topology and Applications

This subsection shifts from metric axioms to topology and uses. Wasserstein distances metrize weak convergence under moment control, sit between weak and strong topologies, and provide quantitative estimates in probability and robust optimization.

Convergence in law topology. On a bounded metric space, all $\mathcal{W}_p$ distances define the same topology, although they are not equivalent as distances.

Proposition 14 (Equivalence of Wasserstein distances on compact spaces). *One has for $p \leq q$*

$$\mathcal{W}_p(\alpha, \beta) \leq \mathcal{W}_q(\alpha, \beta) \leq \text{diam}(\mathcal{X})^{\frac{q-p}{q}} \mathcal{W}_p(\alpha, \beta)^{\frac{p}{q}}$$

where $\text{diam}(\mathcal{X}) := \sup_{x,y} d(x, y)$.

Proof. The left inequality follows from Jensen inequality, $\varphi(\int c(x, y) d\pi(x, y)) \leq \int \varphi(c(x, y)) d\pi(x, y)$, applied to any probability distribution π and to the convex function $\varphi(r) = r^{q/p}$ with $c(x, y) = d(x, y)^p$, so that one gets

$$\left(\int d(x, y)^p d\pi(x, y) \right)^{\frac{q}{p}} \leq \int d(x, y)^q d\pi(x, y).$$

The right inequality follows from

$$d(x, y)^q \leq \text{diam}(\mathcal{X})^{q-p} d(x, y)^p.$$

□

The Wasserstein distance $\mathcal{W}_p$ is a weak distance: it compares singular distributions, such as discrete measures, and quantifies spatial shifts between supports. Its topology is studied in detail in [171, 154, 79], while empirical rates are quantified in [57, 66, 173, 27, 28].

Definition 7 (Weak* topology). $(\alpha_k)_k$ converges weakly* to α in $\mathcal{M}_+^1(\mathcal{X})$ (denoted $\alpha_k \rightharpoonup \alpha$) if and only if for any bounded continuous function $f \in \mathcal{C}_b(\mathcal{X})$, $\int_{\mathcal{X}} f d\alpha_k \rightarrow \int_{\mathcal{X}} f d\alpha$. On compact spaces, $\mathcal{C}_b(\mathcal{X}) = \mathcal{C}(\mathcal{X})$, which is why the boundedness is often left implicit there.

Remark 2 (A Riemann-sum weak limit). On $\mathcal{X} = \mathbb{R}$, the empirical measures on a regular grid satisfy

$$\frac{1}{n} \sum_{k=1}^n \delta_{k/n} \rightharpoonup \mathcal{U}_{[0,1]}.$$

Indeed, for every continuous bounded function f ,

$$\frac{1}{n} \sum_{k=1}^n f(k/n) \rightarrow \int_0^1 f(x) dx,$$

which is precisely the convergence of Riemann sums. This convergence is weak but not strong: for every n , the discrete measure and the uniform density are mutually singular, hence their total variation distance is equal to 2.

Remark 3 (Weak convergence for discrete measures). In the special case of a single Dirac, $\delta_{x^{(n)}} \rightharpoonup \delta_x$ is equivalent to $\int f d\delta_{x^{(n)}} = f(x^{(n)}) \rightarrow \int f d\delta_x = f(x)$ for any continuous f . This in turn is equivalent to $x^{(n)} \rightarrow x$. For a fixed number of atoms, if $\alpha_n = \sum_{i=1}^N a_i^{(n)} \delta_{x_i^{(n)}}$ and, after extracting a subsequence and relabeling, $a_i^{(n)} \rightarrow a_i$ and $x_i^{(n)} \rightarrow x_i$, then α_n converges weakly to $\sum_i a_i \delta_{x_i}$, with atoms at identical limits merged. Without a uniform bound on the number of atoms, weak limits of discrete measures can be non-discrete; empirical measures are the standard example.

In terms of random vectors, if $X_n \sim \alpha_n$ and $X \sim \alpha$ (not necessarily defined on the same probability space), weak convergence corresponds to convergence in law of X_n toward X .

Remark 4 (Central limit theorem). The central limit theorem states that if $(X_1, \dots, X_n)$ are i.i.d. random vectors with finite second moments, $\mathbb{E}(X_i) = 0$, and $\mathbb{E}(X_i X_i^\top) = \text{Id}$, then the rescaled average $Z_n := \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i$ converges in law toward a Gaussian $\mathcal{N}(0, \text{Id})$. This means that the measure α_n representing the law of Z_n converges weakly toward the measure α of the centered normalized Gaussian.

Definition 8 (Strong topology). *The simplest distance on Radon measures is the total variation norm, which is the dual norm of the L^∞ norm on $\mathcal{C}(\mathcal{X})$ and whose topology is often called the “strong” topology*

$$\|\alpha - \beta\|_{\text{TV}} := \sup_{\|f\|_\infty \leq 1} \int f d(\alpha - \beta) = |\alpha - \beta|(\mathcal{X})$$

where $|\alpha - \beta|(\mathcal{X})$ is the mass of the absolute value of the difference measure. When $\alpha - \beta = \rho dx$ has a density, then $\|\alpha - \beta\|_{\text{TV}} = \int |\rho(x)| dx = \|\rho\|_{L^1(dx)}$ is the L^1 norm associated to dx . When $\alpha - \beta = \sum_i u_i \delta_{z_i}$ is discrete, then $\|\alpha - \beta\|_{\text{TV}} = \sum_i |u_i| = \|u\|_{\ell^1}$ is the discrete ℓ^1 norm.

The following proposition shows that the TV norm can be seen as a Wasserstein distance, but for a “degenerate” 0/1 metric.

Proposition 15 (Total variation as Wasserstein for the discrete metric). *Denoting d the 0/1 distance such that $d(x, x) = 0$ and $d(x, y) = 1$ if $x \neq y$, then*

$$\mathcal{W}_p(\alpha, \beta)^p = \frac{1}{2} \|\alpha - \beta\|_{\text{TV}}.$$

Proof. For the sake of simplicity, we do the proof for discrete measures with weights (a, b) and without loss of generality assume they have the same support $(x_i)_i$ and we denote $D := (d(x_i, x_j))_{i,j}$ which is 0 on the diagonal and one outside. Also since $d^p = d$ we consider $p = 1$. We denote $c_i = \min(a_i, b_i)$. By conservation of mass, for every $P \in U(a, b)$, $P_{i,i} \leq c_i$, thus

$$\langle P, D \rangle = \sum_{i \neq j} P_{i,j} = 1 - \sum_i P_{i,i} \geq 1 - \sum_i c_i.$$

We need to show that this bound is tight, namely to construct $\hat{P} \in U(a, b)$ such that $\text{diag}(\hat{P}) = c$. Let

$$\bar{a} := a - c = (a - b)_+ \geq 0 \quad \text{and} \quad \bar{b} := b - c = (b - a)_+ \geq 0$$

If $\bar{a} = \bar{b} = 0$, then $a = b$ and the diagonal coupling is optimal. Otherwise, one has

$$\frac{\bar{a} \otimes \bar{b}}{\langle \bar{a}, \mathbf{1} \rangle} \in U(\bar{a}, \bar{b})$$

and we remark that $\langle \bar{a}, \mathbf{1} \rangle = \langle \bar{b}, \mathbf{1} \rangle = 1 - \langle c, \mathbf{1} \rangle$. Thus denoting

$$\hat{P} := \text{diag}(c) + \frac{\bar{a} \otimes \bar{b}}{\langle \bar{a}, \mathbf{1} \rangle} \geq 0$$

satisfies

$$\hat{P}\mathbf{1} = c + \bar{a} = a \quad \text{and} \quad \hat{P}^\top \mathbf{1} = c + \bar{b} = b$$

so that $\hat{P} \in U(a, b)$ is a coupling so that $\text{diag}(\hat{P}) = \text{diag}(c)$ since $\text{diag}(\bar{a} \otimes \bar{b}) = 0$. We thus conclude that

$$W_1(a, b) = \langle D, \hat{P} \rangle = \sum_{i,j} \frac{\bar{a}_i \bar{b}_j}{\langle \bar{a}, \mathbf{1} \rangle} = \sum_i \bar{a}_i = \sum_i \bar{b}_i = \frac{1}{2} \sum_i (\bar{a}_i + \bar{b}_i) = \frac{1}{2} \|a - b\|_{\text{TV}}.$$

□

As explained in Remark 3, in the special case of Diracs, $\delta_{x_n} \rightharpoonup \delta_x$ is equivalent to $x_n \rightarrow x$. One can then contrast the strong topology with the Wasserstein distance if $x_n \neq x$,

$$\|\delta_{x_n} - \delta_x\|_{\text{TV}} = 2 \quad \text{and} \quad \mathcal{W}_p(\delta_{x_n}, \delta_x) = d(x_n, x).$$

This shows that for the strong topology, Diracs never converge, while they do converge for the Wasserstein distance. It is a powerful property of the Wasserstein distance: on compact spaces, it metrizes weak convergence.

Proposition 16 (Wasserstein metrizes weak convergence on compact spaces). *If $\mathcal{X}$ is compact, $\alpha_k \rightharpoonup \alpha$ if and only if $\mathcal{W}_p(\alpha_k, \alpha) \rightarrow 0$.*

Proof. For $p = 1$, this is the Kantorovich–Rubinstein metrization theorem: by duality, $\mathcal{W}_1$ is the supremum over 1-Lipschitz test functions, and on a compact metric space this class is compact modulo constants by Arzelà–Ascoli. Thus convergence in $\mathcal{W}_1$ is equivalent to weak convergence. Proposition 14 then shows that all Wasserstein distances $\mathcal{W}_p$ induce the same convergent sequences on compact spaces. Hence weak convergence is equivalent to convergence in $\mathcal{W}_p$ for every $p \geq 1$. $\square$

On non-compact spaces, one needs also to impose convergence of the p -th moments. More precisely, on a Polish metric space, $\mathcal{W}_p(\alpha_k, \alpha) \rightarrow 0$ if and only if $\alpha_k \rightharpoonup \alpha$ and, for some reference point x_0 ,

$$\int d(x, x_0)^p d\alpha_k(x) \longrightarrow \int d(x, x_0)^p d\alpha(x).$$

Note that there exist alternative distances which also metricize weak convergence. The simplest ones are Hilbertian kernel norms, which are detailed in Section 7.1.

On a discrete space, the strong and weak topologies coincide, and the following proposition relates the TV and Wasserstein distances.

Proposition 17 (Comparison with total variation on discrete spaces). *One has*

$$\frac{d_{\min}}{2} \|\alpha - \beta\|_{\text{TV}} \leq \mathcal{W}_1(\alpha, \beta) \leq \frac{d_{\max}}{2} \|\alpha - \beta\|_{\text{TV}} \quad \text{where} \quad \begin{cases} d_{\min} := \inf_{x \neq y} d(x, y) \\ d_{\max} := \sup_{x, y} d(x, y) \end{cases}$$

Proof. We denote $d_0(x, y)$ the distance such that $d_0(x, x) = 0$ and $d_0(x, y) = 1$ for $x \neq y$. One has

$$d_{\min} d_0(x, y) \leq d(x, y) \leq d_{\max} d_0(x, y)$$

so that integrating this against any $\pi \in \mathcal{U}(\alpha, \beta)$ and taking the minimum among those π gives the result using Proposition 15. $\square$

This bound is sharp, as this can be observed by taking $\alpha = \delta_x$ and $\beta = \delta_y$, in which case the bound simply reads if $x \neq y$

$$d_{\min} \leq d(x, y) \leq d_{\max}.$$

This shows that the ratio between the two distances can blow up as $d_{\max}/d_{\min}$ increases. On non-discrete spaces, if $d_{\min} = 0$, then the two distances are not equivalent, in line with the fact that the strong and weak topologies do not coincide.

Distributional robustness and $\mathcal{W}_\infty$. Wasserstein distances are also used to define ambiguity sets around an empirical law. Given samples z_i and $\hat{\alpha}_n = \frac{1}{n} \sum_i \delta_{z_i}$, a distributionally robust optimization (DRO) problem replaces the empirical risk $\frac{1}{n} \sum_i \ell_\theta(z_i)$ by

$$\sup_{\beta: \mathcal{W}_p(\beta, \hat{\alpha}_n) \leq \rho} \int \ell_\theta(z) d\beta(z),$$

or, in Lagrangian penalized form, by $\sup_\beta \int \ell_\theta d\beta - \lambda \mathcal{W}_p(\beta, \hat{\alpha}_n)^p$. The constrained and penalized formulations are linked by the choice of multiplier λ , but are not the same problem for an arbitrary fixed λ . Both ask for performance against distributions that can be reached by transporting the empirical mass within a budget. The radius ρ is expressed in the geometry of the data space, so it can encode feature perturbations, domain shift or model misspecification [62, 24, 73].

The basic computational reason for the popularity of Wasserstein DRO is a dual reformulation. Under the usual upper-semicontinuity and growth assumptions on the loss, one has

$$\sup_{\beta: \mathcal{W}_p(\beta, \hat{\alpha}_n)^p \leq \rho^p} \int \ell_\theta d\beta = \inf_{\lambda \geq 0} \lambda \rho^p + \frac{1}{n} \sum_{i=1}^n \sup_z \{ \ell_\theta(z) - \lambda d(z, z_i)^p \}. \quad (20)$$

Thus the robust risk is an empirical risk in which each sample is replaced by its worst penalized perturbation. For $p = 1$ and an L_θ -Lipschitz loss, the Kantorovich–Rubinstein dual gives the transparent upper bound

$$\sup_{\beta: \mathcal{W}_1(\beta, \hat{\alpha}_n) \leq \rho} \int \ell_\theta d\beta \leq \frac{1}{n} \sum_i \ell_\theta(z_i) + \rho L_\theta,$$

which exhibits Wasserstein robustness as a Lipschitz regularizer. The bound is sharp for worst-case optimization over a full Lipschitz ball of losses, while a fixed loss may not realize all extremal Lipschitz directions.

In machine learning, the inner supremum in (20) has the form of an adversarial perturbation problem: the adversary moves each datum away from z_i but pays a transport penalty. This connects Wasserstein DRO to adversarial training and certified robustness [159, 158]. In sequential decision problems, the same idea places Wasserstein ambiguity sets around transition kernels or rewards, producing robust Bellman operators and robust reinforcement-learning models [175, 176].

The limiting distance

$$\mathcal{W}_\infty(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \operatorname{ess\,sup}_{(x, y) \sim \pi} d(x, y) \quad (21)$$

is the limit of $\mathcal{W}_p(\alpha, \beta)$ as $p \rightarrow \infty$ on bounded spaces, not the limit of the convex programs defining $\mathcal{W}_p^p$. It minimizes the worst displacement rather than an average displacement, and the resulting optimization is no longer a linear convex program because of the essential-supremum objective. This makes $\mathcal{W}_\infty$ less convenient algorithmically, but very natural in robust formulations where one wants to guarantee that every transported sample stays within a prescribed perturbation radius.

Theoretical application: quantitative central limit theorems. The weak topology only says whether laws converge; Wasserstein distances also quantify how fast they converge. The next result is a representative theoretical application: the central limit theorem becomes a rate estimate in $\mathcal{W}_1$. This is useful because $\mathcal{W}_1$ is exactly the supremum over 1-Lipschitz observables, so the bound controls the error of all stable numerical or statistical measurements of the normalized sum at once. Figure 13 illustrates the elementary Bernoulli case.

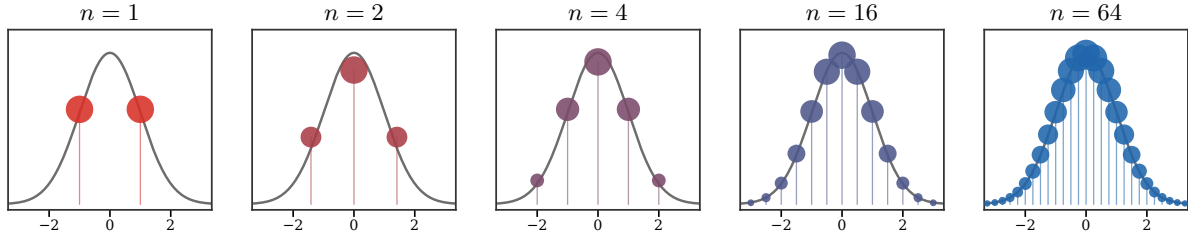


Figure 13: Central-limit theorem for normalized Bernoulli sums. Starting from $\alpha_0 = \frac{1}{2}(\delta_{-1} + \delta_1)$, the law of $Z_n = n^{-1/2} \sum_{i=1}^n X_i$ remains discrete, but its rescaled atom heights approach the standard Gaussian density shown in gray. The Wasserstein Berry–Esseen bound below quantifies this weak convergence by a $\mathcal{W}_1$ distance.

Proposition 18 (Berry–Esseen bound in $\mathcal{W}_1$). *Let $(X_i)_{i=1}^n$ be i.i.d. real random variables such that $\mathbb{E}X_i = 0$, $\mathbb{E}X_i^2 = 1$ and $\mathbb{E}|X_i|^3 < +\infty$. If α_n is the law of $n^{-1/2} \sum_i X_i$ and γ is the standard Gaussian law, then*

$$\mathcal{W}_1(\alpha_n, \gamma) \leq \frac{C \mathbb{E}|X_1|^3}{\sqrt{n}},$$

where C is a universal constant.

Proof. We give the standard Stein-method sketch. By Kantorovich–Rubinstein duality,

$$\mathcal{W}_1(\alpha_n, \gamma) = \sup_{\text{Lip}(h) \leq 1} |\mathbb{E}h(S_n) - \mathbb{E}h(G)|, \quad S_n = n^{-1/2} \sum_i X_i, \quad G \sim \gamma.$$

For each such h , solve Stein’s equation

$$f'_h(x) - xf_h(x) = h(x) - \mathbb{E}h(G).$$

The solution satisfies uniform derivative bounds depending only on the Lipschitz constant of h . Hence

$$\mathbb{E}h(S_n) - \mathbb{E}h(G) = \mathbb{E}[f'_h(S_n) - S_n f_h(S_n)].$$

Expanding $f'_h(S_n)$ and $f_h(S_n)$ by replacing the summands one at a time, the first- and second-order terms cancel because $\mathbb{E}X_i = 0$ and $\mathbb{E}X_i^2 = 1$. The Taylor remainder is bounded by $C \sum_i \mathbb{E}|X_i/\sqrt{n}|^3$, which gives the displayed $n^{-1/2}$ rate. This is the Wasserstein form of the Berry–Esseen theorem; sharper constants and higher-order transport-distance refinements are studied in [25, 147]. $\square$

Applications and implications The practical value of Wasserstein distances is that they provide a geometric loss on probability measures. In barycenter problems, minimizing $\sum_s \lambda_s \mathcal{W}_2^2(\alpha, \beta_s)$ produces an average distribution that respects displacement rather than pointwise density interpolation, which is crucial for shape and image averaging [1, 52, 163]. In registration and morphing, OT penalizes how far mass travels, so it naturally encodes spatial deformations and correspondences [87, 139, 82]. In statistical fitting, one can estimate a parametric measure $\theta \mapsto \alpha_\theta$ from data β by minimizing $\theta \mapsto \mathcal{W}_p(\alpha_\theta, \beta)$; this remains meaningful even when the model and empirical measure are mutually singular, unlike likelihood or KL objectives [33, 68].

4 Sinkhorn

Entropic regularization makes optimal transport smooth, strictly convex and scalable. This section explains why adding entropy produces matrix scaling, how Sinkhorn iterations enforce the marginals, and why this simple algorithm became the computational workhorse of OT. The presentation connects the older matrix-scaling literature [160, 162, 161] with modern entropic OT [50, 138].

4.1 Entropic Regularization for Discrete Measures

Entropy turns a possibly non-unique linear program into a unique smooth problem. The price is bias, but the reward is differentiability and fast scaling algorithms.

Entropic Regularization for Discrete Measures. The idea of the entropic regularization of optimal transport is to use the Shannon-Boltzmann entropy

$$H(P) := - \sum_{i,j} P_{i,j} \log(P_{i,j}),$$

with the convention $0 \log(0) = 0$ as a regularizing function to obtain approximate solutions to the original transport problem (16)

$$L_C^\varepsilon(a, b) := \min_{P \in \mathcal{U}(a, b)} \langle P, C \rangle - \varepsilon H(P). \quad (22)$$

Proposition 19 (Existence and uniqueness of entropic OT). *Assume that a, b are probability histograms and that C is finite. For every $\varepsilon > 0$, problem (22) admits a unique minimizer. If all entries of a and b are positive, then this minimizer is positive on every entry.*

Proof. The transport polytope $U(a, b)$ is non-empty and compact, and the objective is continuous on it with the convention $0 \log 0 = 0$, so a minimizer exists. On the relative interior of the polytope,

$$-\partial^2 H(P) = \text{diag}(1/P_{i,j})$$

is positive definite on every non-zero feasible direction. Hence $-H$ is strictly convex on the polytope and $\langle P, C \rangle - \varepsilon H(P)$ is strictly convex, which implies uniqueness.

If $a_i, b_j > 0$ and a minimizer had $P_{i,j} = 0$, then for small $t > 0$ the perturbation $P_t = (1 - t)P + t a \otimes b$ remains feasible. The directional derivative of the entropic part at $t = 0$ is $-\infty$ because the derivative of $r \log r$ at 0 is $-\infty$ along a positive direction. Thus the objective decreases for small t , contradicting optimality. Therefore the minimizer is strictly positive. $\square$

Smoothing effect. By Proposition 19, problem (22) has a unique optimal solution. This smoothing, beyond providing uniqueness, actually leads to $L_C^\varepsilon(a, b)$ being a smooth function of a, b and C whenever these variables stay in the relative interior of their domains. In finite dimension, this follows from strict convexity and the envelope theorem applied to the dual problem. The effect of the entropy is to act as a barrier function for the positivity constraint. As we will show later, this forces the solution P to be strictly positive on the support of $a \otimes b$. We will also show that as $\varepsilon \rightarrow +\infty$, the solution satisfies $P \rightarrow a \otimes b$.

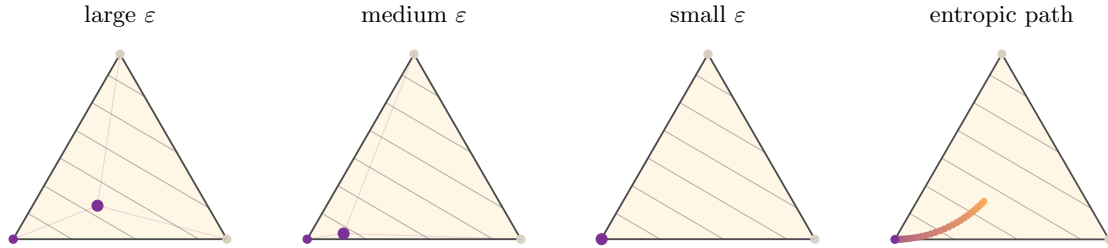


Figure 14: Entropic regularization of a linear objective on a two-dimensional face of the transport polytope. The gray lines are linear objective levels. The first three panels show the unique entropic minimizer for decreasing regularization strengths $\varepsilon = 1.10, 0.35, 0.095$; the last panel shows the continuous path traced by the minimizer as ε decreases. The entropy barrier keeps the solution in the relative interior for every $\varepsilon > 0$, while the path approaches the low-cost vertex.

4.2 Sinkhorn's Algorithm

Sinkhorn's algorithm is alternating normalization of rows and columns. This subsection derives the scaling form of the optimizer and explains why each iteration only needs matrix-vector products.

The underlying matrix-scaling iteration has a long history, including iterative proportional fitting and the work of Sinkhorn and Knopp [160, 162, 161]. Its modern role in OT was transformed by Cuturi's entropic formulation [50]: the algorithm became a practical large-scale tool for machine learning, and also changed the way OT is viewed in ML, from a mostly geometric distance to a differentiable computational primitive.

The following proposition shows that the solution of (22) has a specific form, which can be parameterized using $n + m$ variables. That parameterization is therefore essentially dual, in the sense that a coupling P in $U(a, b)$ has nm variables but $n + m$ constraints.

Proposition 20 (Scaling form of entropic OT). *P is the unique solution to (22) if and only if there exists $(u, v) \in \mathbb{R}_+^n \times \mathbb{R}_+^m$ such that*

$$\forall (i, j) \in \llbracket n \rrbracket \times \llbracket m \rrbracket, \quad P_{i,j} = u_i K_{i,j} v_j \quad \text{where} \quad K_{i,j} := e^{-\frac{c_{i,j}}{\varepsilon}}, \quad (23)$$

and $P \in \mathcal{U}(a, b)$.

Proof. Without loss of generality, we assume $a_i, b_j > 0$ (otherwise, rows or columns with zero mass are fixed to zero and can be removed). By Proposition 19, the minimizer is strictly positive on the remaining support.

We can thus ignore the positivity constraint when introducing two dual variables $f \in \mathbb{R}^n, g \in \mathbb{R}^m$ for each marginal constraint so that the Lagrangian of (22) reads

$$\mathcal{E}(P, f, g) = \langle P, C \rangle + \varepsilon \sum_{i,j} P_{i,j} \log(P_{i,j}) + \langle f, a - P \mathbf{1}_m \rangle + \langle g, b - P^\top \mathbf{1}_n \rangle.$$

Considering first-order conditions (where we ignore the positivity constraint as explained above), we have

$$\frac{\partial \mathcal{E}(P, f, g)}{\partial P_{i,j}} = C_{i,j} + \varepsilon(\log(P_{i,j}) + 1) - f_i - g_j = 0.$$

which results, in an optimal P coupling of the regularized problem, in the expression $P_{i,j} = e^{\frac{f_i + g_j - C_{i,j}}{\varepsilon} - 1}$ which can be rewritten in the form provided in the proposition using non-negative vectors $u_i := e^{f_i/\varepsilon - 1}$ and $v_j := e^{g_j/\varepsilon}$. $\square$

The factorization of the optimal solution exhibited in Equation (23) can be conveniently rewritten in matrix form as $P = \text{diag}(u)K \text{diag}(v)$. u, v must therefore satisfy the following nonlinear equations which correspond to the mass conservation constraints inherent to $U(a, b)$,

$$\text{diag}(u)K \text{diag}(v) \mathbf{1}_m = a, \quad \text{and} \quad \text{diag}(v)K^\top \text{diag}(u) \mathbf{1}_n = b, \quad (24)$$

These two equations can be further simplified, since $\text{diag}(v) \mathbf{1}_m$ is v , and the multiplication of $\text{diag}(u)$ times Kv is

$$u \odot (Kv) = a \quad \text{and} \quad v \odot (K^\top u) = b \quad (25)$$

where $\odot$ corresponds to the entry-wise multiplication of vectors. This problem is known in the numerical analysis community as the matrix scaling problem (see [127] and references therein).

The problem of normalizing a positive matrix K has a long history, from iterative proportional fitting in statistics and economics [102, 177, 70] to modern matrix balancing algorithms [94, 49]. The problem of normalizing a positive matrix K by diagonal scaling is well known, in particular when $n = m$ and a and b are uniform. This corresponds to diagonal scaling toward bistochasticity, which is a very old problem. The previous result shows that there is a unique such scaled matrix P , thanks to the strong convexity of the regularized problem. The remaining question is how to compute this scaled matrix in practice. If some entries of K vanish (equivalently, if the cost matrix C can have infinite values), additional support conditions are needed; here we focus on the strictly positive case.

An intuitive way to try to solve these equations is to solve them iteratively, by modifying first u so that it satisfies the left-hand side of Equation (25) and then v to satisfy its right-hand side. These two updates define Sinkhorn's algorithm

$$u^{(\ell+1)} := \frac{a}{Kv^{(\ell)}} \quad \text{and} \quad v^{(\ell+1)} := \frac{b}{K^\top u^{(\ell+1)}}, \quad (26)$$

initialized with an arbitrary positive vector, for instance $v^{(0)} = \mathbf{1}_m$. The division operator used above between two vectors is to be understood entry-wise. Note that a different initialization will likely lead to a different solution for u, v , since u, v are only defined up to a multiplicative constant (if u, v satisfy (24) then so do $\lambda u, v/\lambda$ for any $\lambda > 0$). The alternating normalization can be read directly on the coupling matrix: a row update enforces the source marginal and generally perturbs the target marginal, while the next column update does the converse.

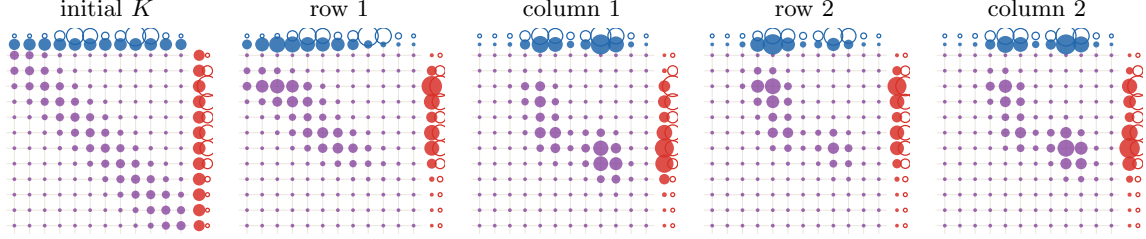


Figure 15: Marginal constraints during Sinkhorn scaling on a twelve-bin one-dimensional problem. Violet circles represent the current coupling matrix; hollow side circles show the prescribed red row marginal and blue column marginal, while filled side circles show the current marginals. Row normalizations align the red marginal and leave a blue defect; column normalizations align the blue marginal and leave a red defect.

The same alternating projection mechanism is clearer on a dense one-dimensional discretization, where the marginal defects appear as continuous side curves.

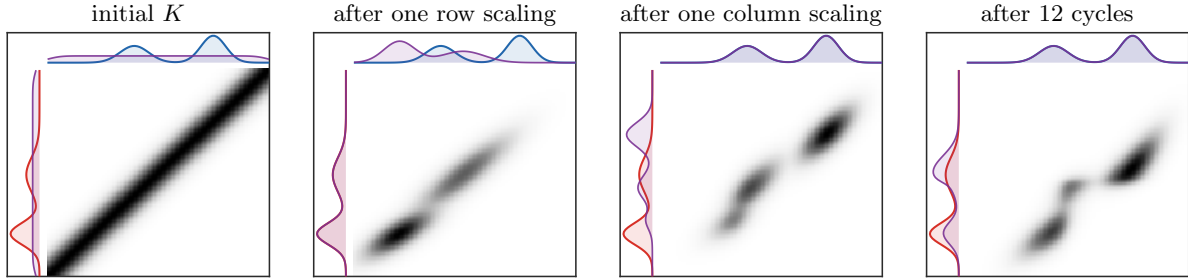


Figure 16: Dense Sinkhorn scaling for one-dimensional Gaussian-mixture marginals. The grayscale matrix is the current coupling. The red and blue side curves are the prescribed source and target marginals, while the violet side curves are the current row and column sums. A row scaling makes the violet curve coincide with the red source marginal, a column scaling makes it coincide with the blue target marginal, and the alternation rapidly stabilizes both sides.

Once both marginals are nearly enforced, the coupling matrix itself stabilizes.

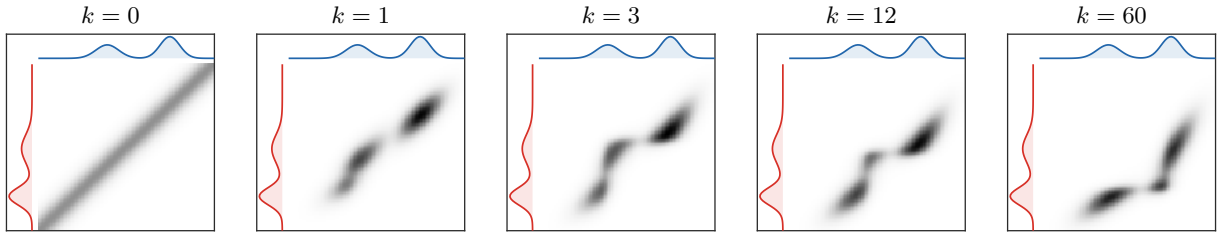


Figure 17: Coupling matrices along Sinkhorn iterations for the same one-dimensional setting as Figure 20, with fixed $\varepsilon = 0.045$. The panels show the matrix after k complete row-column scaling cycles and use a common grayscale normalization. The initial Gibbs kernel is a diagonal-like affinity matrix; scaling bends and reweights it until its row and column sums match the prescribed marginals.

The next paragraphs explain why these iterations converge.

Complexity bounds for Sinkhorn and comparisons with accelerated first-order methods are discussed in [4, 59, 100]. A chief computational advantage of Sinkhorn's algorithm, besides its simplicity, is that the

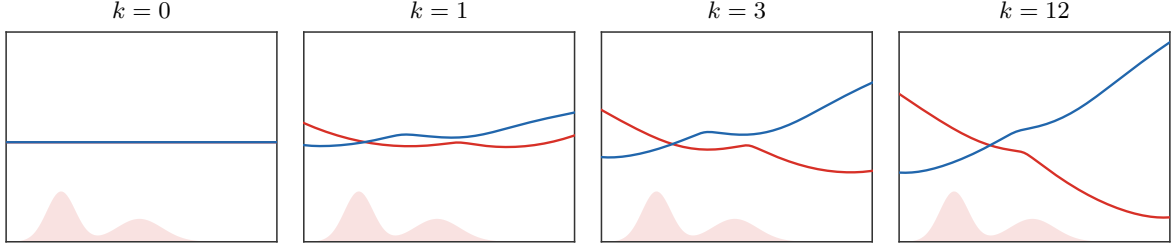


Figure 18: KL-normalized Sinkhorn dual potentials along the scaling iteration for the same one-dimensional Gaussian-mixture setting as Figure 20, with fixed regularization strength $\varepsilon = 0.045$. The faint red and blue silhouettes show the source and target histograms, and the red and blue curves are the logarithmic scaling potentials. All panels share the same axes, making stabilization visible.

only expensive step is multiplication by the Gibbs kernel. Its complexity therefore scales like Cnm , where C is the number of Sinkhorn iterations. Section 8.2 gives a more precise convergence statement: for a fixed regularization strength ε , the entropic dual gap has an $O(1/k)$ bound with constants proportional to $1/\varepsilon$, while Hilbert-metric arguments give eventual linear convergence when the Gibbs kernel is uniformly positive. To approximate the unregularized OT value to accuracy δ , one must also balance the entropic bias, which is typically $O(\varepsilon)$ in finite dimension. Choosing ε proportional to δ and solving the entropic problem to accuracy $O(\delta)$ leads to the familiar iteration scaling of order $1/\delta^2$ for the unregularized value, up to logarithmic and cost-range factors [4, 59].

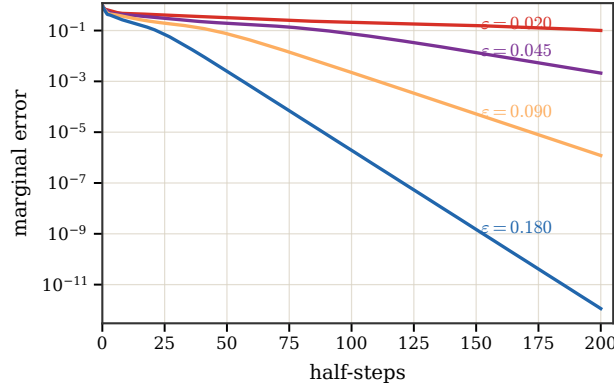


Figure 19: Marginal violation along Sinkhorn half-steps for four values of ε on the same one-dimensional Gaussian-mixture problem. The plotted error is $\frac{1}{2}(\|P_k \mathbf{1} - \mathbf{a}\|_1 + \|P_k^\top \mathbf{1} - \mathbf{b}\|_1)$ on a logarithmic scale. For fixed positive ε , the curves enter a linear regime; smaller ε gives a sharper transport geometry but a more peaked Gibbs kernel and slower scaling.

In many applications, however, one does not need a highly accurate optimizer for the OT subproblem. Downstream performance is often tied more to the geometric bias of the discrepancy than to exact optimality. In such settings, C is often moderate. This should be contrasted with interior point methods, which also operate by introducing a barrier function of the form $-\sum_i \log(P_{i,j})$. For a dense transportation linear program, these algorithms typically have worst-case complexity of order $O(n^6 \log(1/\delta))$ to reach accuracy δ , up to problem-dependent conditioning factors.

The second crucial aspect of Sinkhorn is that matrix-vector multiplication streams extremely well on GPU. Even better, if one is interested in computing many OT problems with a fixed cost matrix C , one can replace many matrix-vector multiplications with matrix-matrix multiplications, so that the computational

gain can be substantial.

Remark 5 (Separable Gaussian kernels on grids). When the samples lie on a Cartesian grid and $c(x, y) = \|x - y\|^2$, the Gibbs kernel is Gaussian and factorizes along coordinates. If the grid has q points per axis in dimension d , so that $N = q^d$ grid points are used, then

$$K(x, y) = \exp\left(-\frac{\|x - y\|^2}{\varepsilon}\right) = \prod_{\ell=1}^d \exp\left(-\frac{(x_\ell - y_\ell)^2}{\varepsilon}\right).$$

Multiplication by K can therefore be applied by successively multiplying along each coordinate direction, equivalently by applying one-dimensional Gaussian kernel operators along the axes. On a periodic or sufficiently padded uniform grid these are literal discrete convolutions. A direct dense one-dimensional multiplication costs $O(q^2)$ on each of the q^{d-1} coordinate lines, and this is repeated for d axes. Hence one Sinkhorn half-step costs

$$O(dq^{d+1}) = O(dN^{1+1/d})$$

instead of $O(N^2)$. With FFT-based or truncated Gaussian convolutions, the same separability can be pushed further, but the simple tensor-product estimate already explains why grid-based Sinkhorn can scale much better than a generic dense coupling.

4.3 Reformulation using relative entropy

The KL formulation identifies Sinkhorn as a projection method. It also prepares the continuous and unbalanced settings, where a reference measure is essential.

A convenient tool to reformulate and “normalize” this discrete entropy (which is crucial to formulate a continuous version of the problem below) is to consider the relative entropy, also called the Kullback–Leibler divergence, which is defined as

$$\text{KL}(\mathbf{P}|\mathbf{Q}) := \sum_{i,j} \mathbf{P}_{i,j} \log\left(\frac{\mathbf{P}_{i,j}}{\mathbf{Q}_{i,j}}\right) - \mathbf{P}_{i,j} + \mathbf{Q}_{i,j}. \quad (27)$$

with the convention $0 \log(0) = 0$ and $\text{KL}(\mathbf{P}|\mathbf{Q}) = +\infty$ if there exists some (i, j) such that $\mathbf{Q}_{i,j} = 0$ but $\mathbf{P}_{i,j} \neq 0$. For the specific case of comparing probability distributions, this further simplifies to

$$\text{KL}(\mathbf{P}|\mathbf{Q}) = \sum_{i,j} \mathbf{P}_{i,j} \log\left(\frac{\mathbf{P}_{i,j}}{\mathbf{Q}_{i,j}}\right).$$

The Shannon-Boltzmann neg-entropy is obtained when considering $\mathbf{Q} = \mathbf{1}_{n \times m}$, i.e.

$$\mathbf{H}(\mathbf{P}) = -\text{KL}(\mathbf{P}|\mathbf{1}_{n \times m}).$$

KL is a particular instance of both a φ -divergence (as defined in Section 7.3) and a Bregman divergence; up to standard affine rescalings, it is the canonical overlap between these two families. This special property is at the heart of the fact that this regularization leads to elegant algorithms and a tractable mathematical analysis.

Proposition 21 (Relative entropy is distance-like). *Let $\mathbf{P}, \mathbf{Q} \in \mathbb{R}_+^{n \times m}$ have the same total mass and assume $\mathbf{Q}_{i,j} > 0$ on the support of $\mathbf{P}$. Then $\text{KL}(\mathbf{P}|\mathbf{Q}) \geq 0$, with equality if and only if $\mathbf{P} = \mathbf{Q}$.*

Proof. Write $\varphi(s) = s \log s - s + 1$. Convexity gives $\varphi(s) \geq \varphi(1) + \varphi'(1)(s - 1) = 0$, and strict convexity gives equality only at $s = 1$. Hence

$$\text{KL}(\mathbf{P}|\mathbf{Q}) = \sum_{i,j} \mathbf{Q}_{i,j} \varphi(\mathbf{P}_{i,j}/\mathbf{Q}_{i,j}) \geq 0,$$

with equality only when $\mathbf{P}_{i,j}/\mathbf{Q}_{i,j} = 1$ for all entries with $\mathbf{Q}_{i,j} > 0$. The support convention rules out positive $\mathbf{P}$ where $\mathbf{Q} = 0$, so equality is equivalent to $\mathbf{P} = \mathbf{Q}$. $\square$

Equivalently, when P and Q have the same total mass, it reads

$$\text{KL}(P|Q) = \sum_{i,j} \varphi(P_{i,j}/Q_{i,j}) Q_{i,j}.$$

where $\varphi(s) = s \log(s)$. For any convex φ such that $\varphi(1) = 0$, one has indeed by Jensen

$$\sum_{i,j} \varphi(P_{i,j}/Q_{i,j}) Q_{i,j} \geq \varphi\left(\sum_{i,j} P_{i,j}/Q_{i,j} Q_{i,j}\right) = \varphi\left(\sum_{i,j} P_{i,j}\right) = \varphi(1) = 0.$$

For instance, one can use as reference measure the tensor product $a \otimes b = (a_i b_j)_{i,j}$ and consider

$$\min_{P \in U(a,b)} \langle P, C \rangle + \varepsilon \text{KL}(P|a \otimes b). \quad (28)$$

Note also that for later, it will be important to have this normalization when considering unbalanced OT.

But the choice of normalization (i.e. reference measure) has no importance for the selection of the optimal P since it only affects the objective by a constant, as shown in the following proposition. In particular, (28) and (22) have the same unique solution.

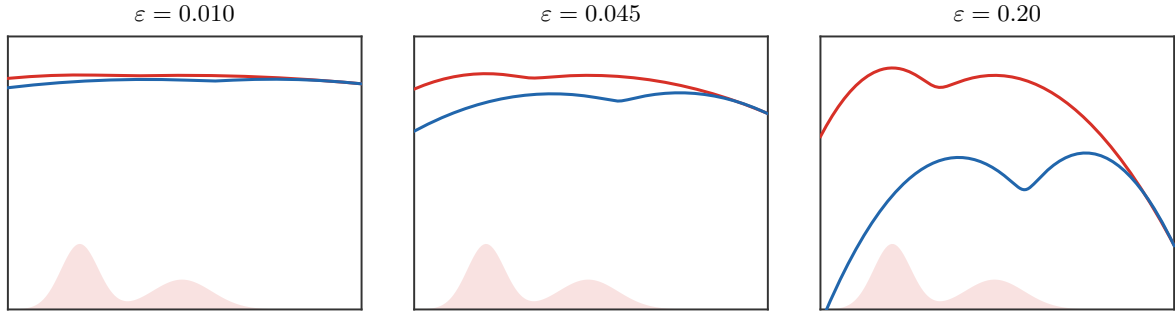


Figure 20: KL-normalized Sinkhorn dual potentials for the same one-dimensional Gaussian-mixture histograms. The faint red and blue silhouettes show the source histogram a and target histogram b . The red and blue curves are the logarithmic scalings $f_i^\varepsilon = \varepsilon \log u_i^\varepsilon$ and $g_j^\varepsilon = \varepsilon \log v_j^\varepsilon$, with gauge $\langle f^\varepsilon, a \rangle = 0$, computed from $P_{i,j} = u_i a_i b_j e^{-C_{i,j}/\varepsilon} v_j$. The squared Euclidean cost is normalized by its median positive entry, and all panels use the same axes; increasing ε turns the hard c -transform geometry into smoother log-sum-exp potentials.

Proposition 22 (Reference measure shift for KL). *For $P \in U(a, b)$, one has*

$$\text{KL}(P|a \otimes b) = \text{KL}(P|a' \otimes b') - \text{KL}(a|a') - \text{KL}(b|b').$$

Consequently, changing the tensor-product reference measure only adds a constant on the transport polytope. In particular, (28) and (22) have the same unique minimizer.

Proof. This follows from

$$\sum_{i,j} P_{i,j} \log \frac{P_{i,j}}{a_i b_j} = \sum_{i,j} P_{i,j} \log \frac{P_{i,j}}{a'_i b'_j} + \sum_{i,j} P_{i,j} \left(\log \frac{a'_i}{a_i} + \log \frac{b'_j}{b_j} \right).$$

□

The choice of using the reference measure $a \otimes b$ is important to deal with situations where the support of a and b can change (so that some coordinate of a or b might vanish), and more importantly in the following section which deals with possibly continuous distributions.

One has the following convergence property.

Proposition 23 (Convergence with ε). *Assume, after removing zero-mass rows and columns, that $\mathbf{a}$ and $\mathbf{b}$ are positive and that $\mathbf{C}$ is finite. The unique solution $\mathbf{P}_\varepsilon$ of (22) converges to the optimal solution with maximal entropy within the set of all optimal solutions of the Kantorovich problem, namely*

$$\mathbf{P}_\varepsilon \xrightarrow[\mathbf{P}]{\varepsilon \rightarrow 0} \operatorname{argmin} \{ -H(\mathbf{P}) ; \mathbf{P} \in \mathcal{U}(\mathbf{a}, \mathbf{b}), \langle \mathbf{P}, \mathbf{C} \rangle = L_C(\mathbf{a}, \mathbf{b}) \} \quad (29)$$

so that in particular

$$L_C^\varepsilon(\mathbf{a}, \mathbf{b}) \xrightarrow{\varepsilon \rightarrow 0} L_C(\mathbf{a}, \mathbf{b}).$$

One has

$$\mathbf{P}_\varepsilon \xrightarrow{\varepsilon \rightarrow \infty} \mathbf{a} \otimes \mathbf{b}. \quad (30)$$

Proof. **Case $\varepsilon \rightarrow 0$.** We consider a sequence $(\varepsilon_\ell)_\ell$ such that $\varepsilon_\ell \rightarrow 0$ and $\varepsilon_\ell > 0$. We denote $\mathbf{P}_\ell$ the solution of (22) for $\varepsilon = \varepsilon_\ell$. Since $\mathcal{U}(\mathbf{a}, \mathbf{b})$ is bounded, we can extract a sequence (that we do not relabel for the sake of simplicity) such that $\mathbf{P}_\ell \rightarrow \mathbf{P}^*$. Since $\mathcal{U}(\mathbf{a}, \mathbf{b})$ is closed, $\mathbf{P}^* \in \mathcal{U}(\mathbf{a}, \mathbf{b})$. We consider any $\mathbf{P}$ such that $\langle \mathbf{C}, \mathbf{P} \rangle = L_C(\mathbf{a}, \mathbf{b})$. Using the equivalent KL-normalized formulation (28), optimality of $\mathbf{P}$ and $\mathbf{P}_\ell$ for their respective optimization problems (for $\varepsilon = 0$ and $\varepsilon = \varepsilon_\ell$) gives

$$0 \leq \langle \mathbf{C}, \mathbf{P}_\ell \rangle - \langle \mathbf{C}, \mathbf{P} \rangle \leq \varepsilon_\ell (\text{KL}(\mathbf{P}|\mathbf{a} \otimes \mathbf{b}) - \text{KL}(\mathbf{P}_\ell|\mathbf{a} \otimes \mathbf{b})). \quad (31)$$

Since KL is continuous, taking the limit $\ell \rightarrow +\infty$ in this expression shows that $\langle \mathbf{C}, \mathbf{P}^* \rangle = \langle \mathbf{C}, \mathbf{P} \rangle$ so that $\mathbf{P}^*$ is a feasible point of (29). Furthermore, dividing by ε_ℓ in (31) and taking the limit shows that $\text{KL}(\mathbf{P}^*|\mathbf{a} \otimes \mathbf{b}) \leq \text{KL}(\mathbf{P}|\mathbf{a} \otimes \mathbf{b})$, which shows that $\mathbf{P}^*$ is a solution of (29). Since the solution $\mathbf{P}_0^*$ to this program is unique by strict convexity of $\text{KL}(\cdot|\mathbf{a} \otimes \mathbf{b})$ on the optimal face, one has $\mathbf{P}^* = \mathbf{P}_0^*$, and the whole sequence is converging.

Case $\varepsilon \rightarrow +\infty$. Evaluating at $\mathbf{a} \otimes \mathbf{b}$ (which belongs to the constraint set $\mathcal{U}(\mathbf{a}, \mathbf{b})$) the energy, one has

$$\langle \mathbf{C}, \mathbf{P}_\varepsilon \rangle + \varepsilon \text{KL}(\mathbf{P}_\varepsilon|\mathbf{a} \otimes \mathbf{b}) \leq \langle \mathbf{C}, \mathbf{a} \otimes \mathbf{b} \rangle + \varepsilon \times 0$$

and since $\langle \mathbf{C}, \mathbf{P}_\varepsilon \rangle \geq 0$, this leads to

$$\text{KL}(\mathbf{P}_\varepsilon|\mathbf{a} \otimes \mathbf{b}) \leq \varepsilon^{-1} \langle \mathbf{C}, \mathbf{a} \otimes \mathbf{b} \rangle \leq \frac{\|\mathbf{C}\|_\infty}{\varepsilon}$$

so that $\text{KL}(\mathbf{P}_\varepsilon|\mathbf{a} \otimes \mathbf{b}) \rightarrow 0$ and thus $\mathbf{P}_\varepsilon \rightarrow \mathbf{a} \otimes \mathbf{b}$ since KL is a valid divergence. $\square$

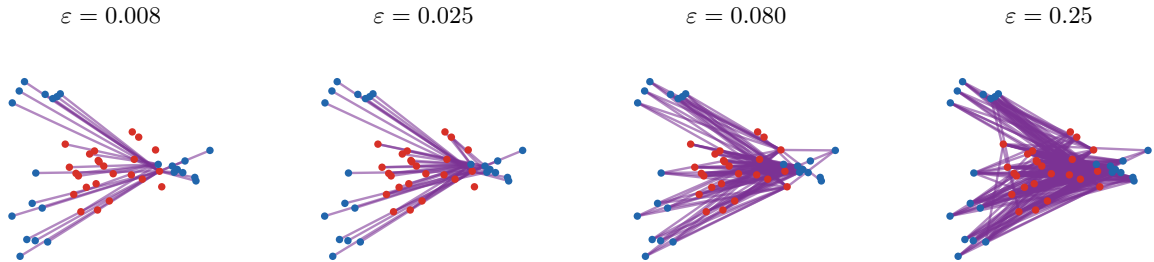


Figure 21: Entropically regularized couplings between the canonical red and blue point clouds for four fixed regularization strengths. The squared Euclidean cost is normalized by its median. Violet segments display the largest visible entries of the computed Sinkhorn plan, with thickness and opacity proportional to transported mass. The computed plans are strictly positive for every $\varepsilon > 0$, but the visible mass pattern evolves from nearly sparse to diffuse as ε increases.

4.4 General Formulation

The continuous entropic problem is the static Schrödinger problem. It gives a probabilistic interpretation of Sinkhorn as the most likely coupling under a noisy reference dynamics.

One can consider arbitrary measures by replacing the discrete entropy with the relative entropy with respect to the product measure $d\alpha \otimes d\beta(x, y) := d\alpha(x)d\beta(y)$, and propose a regularized counterpart to (15) using

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) := \min_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) + \varepsilon \text{KL}(\pi | \alpha \otimes \beta) \quad (32)$$

where the relative entropy is a generalization of the discrete Kullback-Leibler divergence (27)

$$\text{KL}(\pi | \xi) := \int_{\mathcal{X} \times \mathcal{Y}} \log \left(\frac{d\pi}{d\xi}(x, y) \right) d\pi(x, y) + \int_{\mathcal{X} \times \mathcal{Y}} (d\xi(x, y) - d\pi(x, y)), \quad (33)$$

and by convention $\text{KL}(\pi | \xi) = +\infty$ if π does not have a density $\frac{d\pi}{d\xi}$ with respect to ξ . It is important to realize that the reference measure $\alpha \otimes \beta$ chosen in (32) to define the entropic regularizing term $\text{KL}(\cdot | \alpha \otimes \beta)$ plays no specific role (because Proposition 22 still applies in this general setting), only its support matters. This problem is often referred to as the “static Schrödinger problem”, since π is intended to model the most likely coupling between gas particles observed only at two different times (the so-called lazy gas model). The parameter ε controls the temperature of the gas, and particles do not move in a deterministic straight line as in optimal transport for the Euclidean cost, but rather according to a stochastic Brownian bridge. This viewpoint goes back to Schrödinger’s reciprocal problem [157] and is surveyed in modern OT language in [108, 109]; stochastic-control formulations are developed in [42].

Remark 6 (Probabilistic interpretation). If $(X, Y) \sim \pi$ have marginals $X \sim \alpha$ and $Y \sim \beta$, then $\text{KL}(\pi | \alpha \otimes \beta) = \mathcal{I}(X, Y)$ is the mutual information of the couple, which is 0 if and only if X and Y are independent. The entropic problem (32) is thus equivalent to

$$\min_{(X, Y), X \sim \alpha, Y \sim \beta} \mathbb{E}(c(X, Y)) + \varepsilon \mathcal{I}(X, Y).$$

Using a large ε thus enforces the optimal coupling to describe independent variables, while, according to Brenier’s theorem, small ε rather imposes a deterministic dependency between the couple according to a Monge map.

4.5 Bregman Projection View of Sinkhorn

This subsection explains Sinkhorn as alternating Bregman projections. The main message is geometric: each row or column rescaling is the KL projection onto one affine marginal constraint, so convergence follows from the Pythagorean identity for Bregman divergences.

For the sake of simplicity, this section is written for discrete measures, but the analysis carries over to general measures. This analysis is revisited in Section 8.2 using convex duality.

Alternating KL projections. The projection viewpoint explains Sinkhorn as repeated enforcement of one marginal constraint at a time. It is not specific to the entropy, although the KL case is the one where the projections reduce to elementary row and column scalings.

Definition 9 (Bregman divergence). *Let Φ be a differentiable strictly convex function on a convex domain Ω . The Bregman divergence generated by Φ is*

$$D_\Phi(P | Q) := \Phi(P) - \Phi(Q) - \langle \nabla \Phi(Q), P - Q \rangle.$$

For the negative entropy $\Phi(P) = \sum_{i,j} P_{i,j} \log P_{i,j}$ on the positive orthant, one obtains $D_\Phi(P | Q) = \text{KL}(P | Q)$ up to the harmless convention at the boundary.

The main interest of Bregman divergences is that the geometry can encode constraints. A Legendre-type generator Φ blows up, or has an infinite derivative, at the boundary of its domain; for negative entropy this means that positivity is built into the divergence, so one projects onto affine marginal constraints without separately handling non-negativity.

Linear tilts and Gibbs references. The following proposition explains why adding a linear cost to a Bregman penalty merely shifts the reference point in dual coordinates. The usual Gibbs–KL reformulation is the entropy specialization.

Proposition 24 (Linear tilts of Bregman penalties). *Let Φ be differentiable and strictly convex, and let D_Φ be its Bregman divergence. Fix a reference point Q in the interior of the domain. Assume that there exists Q^C such that*

$$\nabla\Phi(Q^C) = \nabla\Phi(Q) - C/\varepsilon.$$

Then, for all P in the domain,

$$\langle P, C \rangle + \varepsilon D_\Phi(P|Q) = \varepsilon D_\Phi(P|Q^C) + \text{cst},$$

where the constant does not depend on P .

Proof. Subtract the two Bregman divergences:

$$D_\Phi(P|Q^C) - D_\Phi(P|Q) = \langle \nabla\Phi(Q) - \nabla\Phi(Q^C), P \rangle + \text{cst}.$$

Using $\nabla\Phi(Q) - \nabla\Phi(Q^C) = C/\varepsilon$ and multiplying by ε gives the claim. $\square$

For the negative entropy $\Phi(P) = \sum_{i,j} P_{i,j} \log P_{i,j}$, one has $D_\Phi = \text{KL}$. Taking $Q = a \otimes b$ gives the tilted reference

$$K_{a,b}^\varepsilon := (a \otimes b) \odot e^{-C/\varepsilon}.$$

Thus

$$\langle P, C \rangle + \varepsilon \text{KL}(P|a \otimes b) = \varepsilon \text{KL}(P|K_{a,b}^\varepsilon) + \text{cst}.$$

On the transport polytope, scaling $K_{a,b}^\varepsilon$ is equivalent to scaling the Gibbs kernel $K = e^{-C/\varepsilon}$ because the factors a_i and b_j can be absorbed into the Sinkhorn scalings.

This shows that the unique solution P_ε of (22) is a projection onto $U(a, b)$ of the tilted Gibbs reference

$$P_\varepsilon = \text{Proj}_{U(a,b)}^{\text{KL}}(K_{a,b}^\varepsilon) := \underset{P \in U(a,b)}{\text{argmin}} \text{KL}(P|K_{a,b}^\varepsilon). \quad (34)$$

Cyclic projection convergence. The general convergence mechanism is the classical one of Bregman [31].

Proposition 25 (Cyclic Bregman projections on affine constraints). *Let Φ be a Legendre strictly convex generator on a finite-dimensional convex domain, and let $\mathcal{C}_1, \mathcal{C}_2$ be affine constraint sets whose intersection meets the domain. Define*

$$P_{k+1} = \text{Proj}_{\mathcal{C}_2}^{D_\Phi} \text{Proj}_{\mathcal{C}_1}^{D_\Phi}(P_k),$$

starting from an interior point P_0 , and assume the projections are well-defined. Then P_k converges to the Bregman projection of P_0 onto $\mathcal{C}_1 \cap \mathcal{C}_2$. In particular, the KL case converges for positive affine marginal constraints.

Proof. We first prove the Pythagorean identity used by the projection argument. For three interior points one has the Bregman three-point formula

$$D_\Phi(Q|P) = D_\Phi(Q|P^+) + D_\Phi(P^+|P) + \langle \nabla\Phi(P^+) - \nabla\Phi(P), Q - P^+ \rangle.$$

If $P^+ = \text{Proj}_{\mathcal{C}}^{D_\Phi}(P)$ and $\mathcal{C}$ is affine, the first-order optimality condition for minimizing $R \mapsto D_\Phi(R|P)$ over $R \in \mathcal{C}$ is

$$\langle \nabla \Phi(P^+) - \nabla \Phi(P), R - P^+ \rangle = 0 \quad \forall R \in \mathcal{C},$$

because $R - P^+$ ranges over the tangent linear space of $\mathcal{C}$. Taking $R = Q \in \mathcal{C}$ cancels the last term and gives

$$D_\Phi(Q|P) = D_\Phi(Q|P^+) + D_\Phi(P^+|P) \quad \forall Q \in \mathcal{C}.$$

Let $(Z_\ell)_\ell$ be the half-step sequence obtained by alternating projections onto $\mathcal{C}_1$ and $\mathcal{C}_2$, so that $Z_{2k} = P_k$ and $Z_{2k+2} = P_{k+1}$. Fix $Q \in \mathcal{C}_1 \cap \mathcal{C}_2$. Applying the identity at each half-step gives

$$D_\Phi(Q|Z_\ell) - D_\Phi(Q|Z_{\ell+1}) = D_\Phi(Z_{\ell+1}|Z_\ell) \geq 0.$$

Thus $D_\Phi(Q|Z_\ell)$ is decreasing and the series $\sum_\ell D_\Phi(Z_{\ell+1}|Z_\ell)$ is finite. In finite dimension, the iterates remain in the Bregman sublevel set $\{Z : D_\Phi(Q|Z) \leq D_\Phi(Q|Z_0)\}$; for a Legendre generator this set stays in the domain and is compact under the stated well-definedness assumptions. Hence cluster points exist. Since the projection drops tend to zero and Φ is strictly convex on compact subsets of the domain, $\|Z_{\ell+1} - Z_\ell\| \rightarrow 0$. Every cluster point of the even subsequence is therefore also a cluster point of the adjacent odd subsequence, and because these two subsequences lie alternately in the closed affine sets $\mathcal{C}_1$ and $\mathcal{C}_2$, every cluster point belongs to $\mathcal{C}_1 \cap \mathcal{C}_2$.

Let $\bar{P}$ be such a cluster point. For each half-step, the dual displacement

$$\nabla \Phi(Z_{\ell+1}) - \nabla \Phi(Z_\ell)$$

belongs to the normal space of the affine set onto which one projects. Telescoping and using the convergence of Z_ℓ gives

$$\nabla \Phi(\bar{P}) - \nabla \Phi(P_0) \in N_{\mathcal{C}_1} + N_{\mathcal{C}_2} = N_{\mathcal{C}_1 \cap \mathcal{C}_2},$$

where the last equality uses that the sets are affine. This is precisely the first-order optimality condition for minimizing $R \mapsto D_\Phi(R|P_0)$ over $R \in \mathcal{C}_1 \cap \mathcal{C}_2$. Thus $\bar{P}$ is the Bregman projection of P_0 onto the intersection. Strict convexity gives uniqueness of this minimizer, hence all cluster points coincide and the whole sequence converges. The KL statement is obtained by choosing the negative entropy generator. $\square$

Denoting

$$\mathcal{C}_a^1 := \{P ; P \mathbf{1}_m = a\} \quad \text{and} \quad \mathcal{C}_b^2 := \{P ; P^\top \mathbf{1}_n = b\}$$

the rows and columns constraints, one has $U(a, b) = \mathcal{C}_a^1 \cap \mathcal{C}_b^2$. One can use KL, or more generally Bregman, iterative projections [31, 150, 151]

$$P^{(\ell+1)} := \text{Proj}_{\mathcal{C}_a^1}^{\text{KL}}(P^{(\ell)}) \quad \text{and} \quad P^{(\ell+2)} := \text{Proj}_{\mathcal{C}_b^2}^{\text{KL}}(P^{(\ell+1)}). \quad (35)$$

Since the sets $\mathcal{C}_a^1$ and $\mathcal{C}_b^2$ are affine, Proposition 25 applies with $P_0 = K_{a,b}^\varepsilon$ and shows convergence to the solution of (34).

Row and column scalings. The two projectors are simple to compute since they correspond to scaling respectively the rows and the columns, as explained in this proposition.

Proposition 26 (KL projections are scalings). *One has*

$$\text{Proj}_{\mathcal{C}_a^1}^{\text{KL}}(P) = \text{diag} \left(\frac{a}{P \mathbf{1}_m} \right) P \quad \text{and} \quad \text{Proj}_{\mathcal{C}_b^2}^{\text{KL}}(P) = P \text{diag} \left(\frac{b}{P^\top \mathbf{1}_n} \right).$$

Proof. Consider the problem along each row or column vector to impose a fixed sum $s \in \mathbb{R}_+$

$$\min_p \{ \text{KL}(p|q) ; \langle p, \mathbf{1} \rangle = s \}.$$

The Lagrange multiplier equation for this problem reads

$$\log(p/q) + \lambda \mathbf{1} = 0 \implies p = uq \quad \text{where} \quad u = e^{-\lambda} > 0.$$

The constraint $\langle p, \mathbf{1} \rangle = s$ is equivalent to $\langle uq, \mathbf{1} \rangle = s$, i.e. $u = s / \sum_i q_i$, which gives the desired scaling formula $p = sq / \sum_i q_i$. $\square$

These iterations are equivalent to Sinkhorn iterations (26) since defining

$$P^{(2\ell)} := \text{diag}(u^{(\ell)}) K \text{diag}(v^{(\ell)}),$$

one has

$$\begin{aligned} P^{(2\ell+1)} &:= \text{diag}(u^{(\ell+1)}) K \text{diag}(v^{(\ell)}) \\ \text{and } P^{(2\ell+2)} &:= \text{diag}(u^{(\ell+1)}) K \text{diag}(v^{(\ell+1)}) \end{aligned}$$

In practice, however, one should prefer using (26), which only requires manipulating scaling vectors and multiplying by a Gibbs kernel, and can often be accelerated when the kernel has separable, sparse, low-rank or geometric structure.

Such a convergence analysis using Bregman projection is of limited interest because it only works in finite dimension. For instance, the linear convergence speed one can obtain with these analyses (because the objective is strongly convex) degrades with the dimension and with ε . Section 8.2 gives a more robust dual analysis: the constants are expressed through the oscillation of the potentials and the cost range, and the resulting $O(1/k)$ estimate explains what can be guaranteed before any asymptotic linear regime becomes visible. It is also possible to decay ε during the iterates to improve the speed and rely on multiscale strategies in low dimensions.

4.6 Hilbert Metric View of Sinkhorn

Hilbert's projective metric gives a complementary convergence mechanism. Instead of following objective values, it measures distances between positive scaling vectors modulo global multiplication. Positive kernels are contractions in this geometry, yielding a global linear convergence statement.

As initially explained by [67], the global convergence analysis of Sinkhorn is greatly simplified using Hilbert projective metric on $\mathbb{R}_{+,*}^n$ (positive vectors), defined as

$$\forall (u, u') \in (\mathbb{R}_{+,*}^n)^2, \quad d_{\mathcal{H}}(u, u') := \|\log(u) - \log(u')\|_V \quad (36)$$

where the variation semi-norm is

$$\|z\|_V = \max(z) - \min(z).$$

Proposition 27 (Hilbert metric on the projective cone). *The function $d_{\mathcal{H}}$ defines a complete distance on the projective cone $\mathbb{R}_{+,*}^n / \sim$, where $u \sim u'$ means that $u = su'$ for some $s > 0$.*

Proof. The map $u \mapsto \log u$ identifies $\mathbb{R}_{+,*}^n / \sim$ with the quotient vector space $\mathbb{R}^n / \text{Span}(\mathbf{1}_n)$, because multiplying u by $s > 0$ adds the constant vector $\log(s)\mathbf{1}_n$. The variation semi-norm $\|z\|_V = \max_i z_i - \min_i z_i$ vanishes exactly on constant vectors, so it induces a norm on this quotient. Symmetry, the triangle inequality and separation therefore follow from the corresponding norm properties. Completeness follows because $\mathbb{R}^n / \text{Span}(\mathbf{1}_n)$ is finite-dimensional and all finite-dimensional normed spaces are complete. $\square$

It was introduced independently by [23] and [152] to provide quantitative proofs of the Perron-Frobenius theorem (convergence of iterations of positive matrices). Sinkhorn should be viewed as a nonlinear generalization of Perron-Frobenius.

Theorem 3 (Birkhoff contraction theorem). *Let $K \in \mathbb{R}_{+,*}^{n \times m}$, then for $(v, v') \in (\mathbb{R}_{+,*}^m)^2$*

$$d_{\mathcal{H}}(Kv, Kv') \leq \lambda(K) d_{\mathcal{H}}(v, v') \text{ where } \begin{cases} \lambda(K) := \frac{\sqrt{\eta(K)} - 1}{\sqrt{\eta(K)} + 1} < 1 \\ \eta(K) := \max_{i,j,k,\ell} \frac{K_{i,k} K_{j,\ell}}{K_{j,k} K_{i,\ell}}. \end{cases}$$

Proof. We recall the finite-dimensional Birkhoff–Hopf estimate. For a positive linear map A on a cone, define its projective diameter

$$\Delta(A) := \sup_{u,v>0} d_{\mathcal{H}}(Au, Av).$$

Then

$$d_{\mathcal{H}}(Au, Av) \leq \tanh(\Delta(A)/4) d_{\mathcal{H}}(u, v).$$

Indeed, after quotienting by positive scalings, write $r_k = u_k/v_k$ and normalize so that $e^{-h/2} \leq r_k \leq e^{h/2}$, where $h = d_{\mathcal{H}}(u, v)$. The ratio between two coordinates of Au/Av is a quotient of two weighted averages of the numbers r_k . A two-point extremal argument shows that the largest possible contraction is obtained when the mass of the two weights is placed on the two endpoints $e^{-h/2}$ and $e^{h/2}$; the cross-ratio bound defining $\Delta(A)$ then gives

$$d_{\mathcal{H}}(Au, Av) \leq 2 \log \frac{e^{\Delta(A)/4} e^{h/2} + e^{-\Delta(A)/4} e^{-h/2}}{e^{\Delta(A)/4} e^{-h/2} + e^{-\Delta(A)/4} e^{h/2}} \leq \tanh(\Delta(A)/4) h.$$

For the matrix K , its projective diameter is

$$\Delta(K) = \log \eta(K), \quad \eta(K) = \max_{i,j,k,\ell} \frac{K_{i,k} K_{j,\ell}}{K_{j,k} K_{i,\ell}}.$$

Therefore $\tanh(\Delta(K)/4) = (\sqrt{\eta(K)} - 1)/(\sqrt{\eta(K)} + 1)$, which is the claimed contraction factor. $\square$

This result extends to arbitrary convex cones and affine mappings from the cone to its interior.

The following theorem proved by [67], makes use of this Theorem 3 to show the linear convergence of Sinkhorn’s iterations.

Theorem 4 (Linear convergence of Sinkhorn). *One has $(u^{(\ell)}, v^{(\ell)}) \rightarrow (u^*, v^*)$ and*

$$d_{\mathcal{H}}(u^{(\ell)}, u^*) = O(\lambda(K)^{2\ell}), \quad d_{\mathcal{H}}(v^{(\ell)}, v^*) = O(\lambda(K)^{2\ell}). \quad (37)$$

One also has

$$d_{\mathcal{H}}(u^{(\ell)}, u^*) \leq \frac{d_{\mathcal{H}}(P^{(\ell)} \mathbf{1}_m, \mathbf{a})}{1 - \lambda(K)} \quad \text{and} \quad d_{\mathcal{H}}(v^{(\ell)}, v^*) \leq \frac{d_{\mathcal{H}}(P^{(\ell), \top} \mathbf{1}_n, \mathbf{b})}{1 - \lambda(K)}, \quad (38)$$

where we denoted $P^{(\ell)} := \text{diag}(u^{(\ell)}) K \text{diag}(v^{(\ell)})$. Lastly, one has

$$\|\log(P^{(\ell)}) - \log(P^*)\|_{\infty} \leq d_{\mathcal{H}}(u^{(\ell)}, u^*) + d_{\mathcal{H}}(v^{(\ell)}, v^*) \quad (39)$$

where P^ is the unique solution of (22).*

Proof. Notice that for any $(v, v') \in (\mathbb{R}_{+,*}^m)^2$, one has

$$d_{\mathcal{H}}(v, v') = d_{\mathcal{H}}(v/v', \mathbf{1}_m) = d_{\mathcal{H}}(\mathbf{1}_m/v, \mathbf{1}_m/v'),$$

since indeed $d_{\mathcal{H}}(a/v, a/v') = d_{\mathcal{H}}(v, v')$. This shows that

$$d_{\mathcal{H}}(u^{(\ell+1)}, u^*) = d_{\mathcal{H}}\left(\frac{\mathbf{a}}{Kv^{(\ell)}}, \frac{\mathbf{a}}{Kv^*}\right) = d_{\mathcal{H}}(Kv^{(\ell)}, Kv^*) \leq \lambda(K) d_{\mathcal{H}}(v^{(\ell)}, v^*).$$

where we used Theorem 3. This shows (37). One also has, using the triangular inequality

$$\begin{aligned} d_{\mathcal{H}}(u^{(\ell)}, u^*) &\leq d_{\mathcal{H}}(u^{(\ell+1)}, u^{(\ell)}) + d_{\mathcal{H}}(u^{(\ell+1)}, u^*) \leq d_{\mathcal{H}}\left(\frac{a}{Kv^{(\ell)}}, u^{(\ell)}\right) + \lambda(K)d_{\mathcal{H}}(u^{(\ell)}, u^*) \\ &= d_{\mathcal{H}}\left(a, u^{(\ell)} \odot (Kv^{(\ell)})\right) + \lambda(K)d_{\mathcal{H}}(u^{(\ell)}, u^*), \end{aligned}$$

which gives the first part of (38) since $u^{(\ell)} \odot (Kv^{(\ell)}) = P^{(\ell)}\mathbf{1}_m$ (the second one being similar). The proof of (39) follows from [67, Lemma 3] $\square$

The bound (38) shows that some error measures on the marginal constraints violation, for instance, $\|P^{(\ell)}\mathbf{1}_m - a\|_1$ and $\|P^{(\ell)\top}\mathbf{1}_n - b\|_1$, are useful stopping criteria to monitor the convergence. This theorem shows that the Sinkhorn algorithm converges linearly, but the worst-case rate becomes exponentially bad as $\varepsilon \rightarrow 0$, since the global contraction factor is controlled by a term of the form $e^{-\|c\|_{\infty}/\varepsilon}$. In practice, one often observes a much better local linear regime after enough iterations. The same Hilbert-metric mechanism extends beyond finite matrices to positive integral operators under suitable compactness and positivity assumptions. An important limitation of this analysis is that it requires a uniformly bounded cost and a kernel bounded away from degeneracy; Gaussian distributions with quadratic cost therefore require a different approach.

5 Dual Problem

Duality turns the transport problem into a search for potentials rather than couplings. This section explains why potentials certify optimality, how c -transforms regularize them, and why the quadratic case reveals convex analysis behind Brenier maps. Linear-programming duality gives the discrete picture [20], while the continuous form is one of the central theorems of OT [170, 154].

5.1 Discrete dual

The discrete dual gives finite-dimensional certificates of optimality. Its complementary slackness conditions identify where an optimal coupling can put mass.

The Kantorovich problem (16) is a linear program so that one can equivalently compute its value by solving a dual linear program.

Proposition 28 (Discrete Kantorovich duality). *One has*

$$L_C(a, b) = \max_{(f, g) \in R(a, b)} \langle f, a \rangle + \langle g, b \rangle \quad (40)$$

where the set of admissible potentials is

$$R(a, b) := \{(f, g) \in \mathbb{R}^n \times \mathbb{R}^m ; \forall (i, j) \in \llbracket n \rrbracket \times \llbracket m \rrbracket, f \oplus g \leq C\} \quad (41)$$

Proof. For the sake of completeness, let us derive this dual problem using Lagrangian duality. The Lagrangian associated to (16) reads

$$\min_{P \geq 0} \max_{(f, g) \in \mathbb{R}^n \times \mathbb{R}^m} \langle C, P \rangle + \langle a - P\mathbf{1}_m, f \rangle + \langle b - P^\top\mathbf{1}_n, g \rangle. \quad (42)$$

For a linear program, if the primal constraint set is non-empty, one can always exchange the min and the max and get the same value. We thus consider

$$\max_{(f, g) \in \mathbb{R}^n \times \mathbb{R}^m} \langle a, f \rangle + \langle b, g \rangle + \min_{P \geq 0} \langle C - f\mathbf{1}_m^\top - \mathbf{1}_n g^\top, P \rangle.$$

We conclude by remarking that

$$\min_{P \geq 0} \langle Q, P \rangle = \begin{cases} 0 & \text{if } Q \geq 0 \\ -\infty & \text{otherwise} \end{cases}$$

so that the constraint reads $C - f\mathbf{1}_m^\top - \mathbf{1}_n g^\top = C - f \oplus g \geq 0$. $\square$

The primal-dual optimality relation for the Lagrangian (42) allows locating the support of the optimal transport plan

$$\text{Supp}(P) \subset \{(i, j) \in \llbracket n \rrbracket \times \llbracket m \rrbracket ; f_i + g_j = C_{i,j}\}. \quad (43)$$

Figure 22 shows these finite-dimensional certificates on a one-dimensional quadratic problem. The potentials are not transport maps themselves; rather, their contact set with the cost matrix is where an optimal coupling is allowed to put mass.

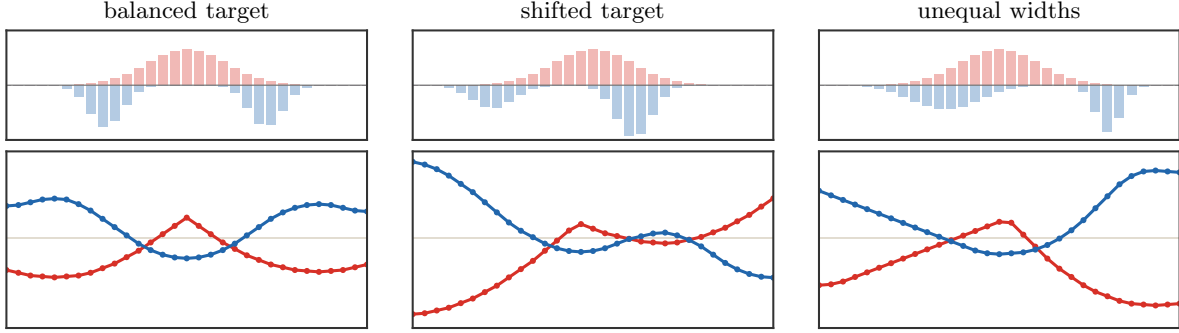


Figure 22: Discrete Kantorovich dual potentials for the quadratic cost $C_{i,j} = |x_i - y_j|^2$. In each panel the upper strip shows the fixed source histogram in red and the target histogram in blue. The lower strip shows an optimal pair of dual vectors (f, g) , with gauge chosen so that $\langle f, a \rangle = 0$. Complementary slackness states that mass can be transported only through entries where $f_i + g_j = C_{i,j}$.

The formulation (40) shows that $(a, b) \mapsto L_C(a, b)$ is a convex function (as a supremum of linear functions). From the primal problem (16), one also sees that $C \mapsto L_C(a, b)$ is concave.

5.2 Auction Algorithm and Dual Prices

The assignment algorithms mentioned in Section 1 become more transparent once one has dual variables. The auction algorithm is a dual price method: it updates target prices and maintains an approximate complementary-slackness certificate. The small tolerance ε removes ties, stabilizes the price updates and gives a quantitative optimality certificate [17, 18, 123].

Consider the square assignment problem with costs $C_{i,j}$ and rewrite it as the profit maximization problem with $a_{i,j} = -C_{i,j}$. The auction algorithm keeps prices p_j on the target points and a partial assignment. For an unassigned source i , define the best and second-best reduced profits

$$v_i = \max_j (a_{i,j} - p_j), \quad j_i \in \arg\max_j (a_{i,j} - p_j), \quad w_i = \max_{j \neq j_i} (a_{i,j} - p_j).$$

Source i bids for j_i and increases its price by the gap to the second-best target, plus a margin:

$$p_{j_i} \leftarrow p_{j_i} + v_i - w_i + \varepsilon.$$

The target j_i is then assigned to i , and its previous owner, if any, becomes unassigned. The iteration stops when all sources are assigned.

For fixed prices p , eliminating the bidder utilities u_i in the dual minimization gives the convex objective

$$D(p) = \sum_j p_j + \sum_i \max_j (a_{i,j} - p_j),$$

which comes from the dual constraints $u_i + p_j \geq a_{i,j}$. The auction update should thus be viewed as a price-adjustment method on this nonsmooth dual landscape, rather than as a generic gradient step: the proof of correctness is through approximate complementary slackness.

Definition 10 (ε -complementary slackness). *An assignment σ and prices p satisfy ε -complementary slackness if, for every source i ,*

$$a_{i,\sigma(i)} - p_{\sigma(i)} \geq \max_j (a_{i,j} - p_j) - \varepsilon.$$

Proposition 29 (Auction optimality certificate). *If a complete assignment σ satisfies ε -complementary slackness, then it is $n\varepsilon$ -optimal for the profit maximization problem, or equivalently $n\varepsilon$ -optimal for the original cost minimization problem. If all costs are integers and $\varepsilon < 1/n$, then σ is optimal.*

Proof. Let τ be any assignment. By ε -complementary slackness,

$$a_{i,\tau(i)} - p_{\tau(i)} \leq \max_j (a_{i,j} - p_j) \leq a_{i,\sigma(i)} - p_{\sigma(i)} + \varepsilon.$$

Summing over i cancels prices, because both σ and τ are permutations:

$$\sum_i a_{i,\tau(i)} \leq \sum_i a_{i,\sigma(i)} + n\varepsilon.$$

Thus no assignment has profit more than $n\varepsilon$ above that of σ . Since $a = -C$, the same statement says that the cost of σ is at most $n\varepsilon$ above the minimum cost. If the costs are integers, all assignment costs are integers; a gap strictly smaller than one therefore forces the gap to be zero. $\square$

During the bidding process the last bid made by a source makes its chosen target better, up to the margin ε , than all alternatives. Subsequent price increases can only make targets less attractive, and one checks by induction that currently assigned pairs satisfy ε -complementary slackness. The standard finite-termination proof normalizes prices by subtracting their minimum, observes that each bid increases one target price by at least ε , and bounds the normalized price spread in terms of the range of the profits; see [18, 123] for the full bound and implementation details.

Remark 7 (ε -scaling and relation with Sinkhorn). In practice one starts with a coarse ε and repeatedly decreases it, warm-starting the prices and assignment. This ε -scaling strategy is a homotopy method: large ε regularizes the combinatorial problem by enforcing a visible margin between the best and second-best reduced profits, while small ε recovers the exact dual certificate. If one wants a continuous-optimization analogy, the margin is closer to an exact-penalty or proximal continuation parameter than to a literal quadratic penalty.

Sinkhorn scaling plays a parallel role for entropic OT. There, the hard minimum in the dual c -transform is replaced by a soft minimum, or log-sum-exp, with temperature ε ; in the auction algorithm, the hard maximum is kept but the complementary slackness condition is relaxed by ε . Both methods therefore use an ε -controlled dual continuation, and both recover the unregularized transport certificate as $\varepsilon \rightarrow 0$ under the usual assumptions. The outputs are different: Sinkhorn produces dense entropic couplings, whereas auction keeps a sparse assignment throughout.

5.3 General formulation

The continuous dual is the analytic counterpart of the discrete linear program. It uses continuous potentials because measures are probed through integration.

To extend this primal-dual construction to arbitrary measures, it is important to realize that measures are naturally paired in duality with continuous functions, using the pairing $\langle f, \alpha \rangle := \int f d\alpha$.

Proposition 30 (Kantorovich duality). *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact metric spaces and that $c \in \mathcal{C}(\mathcal{X} \times \mathcal{Y})$. Then*

$$\mathcal{L}_c(\alpha, \beta) = \max_{(f,g) \in \mathcal{R}(c)} \int_{\mathcal{X}} f(x) d\alpha(x) + \int_{\mathcal{Y}} g(y) d\beta(y), \quad (44)$$

where the set of admissible dual potentials is

$$\mathcal{R}(c) := \{(f, g) \in \mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y}) ; \forall (x, y), f(x) + g(y) \leq c(x, y)\}. \quad (45)$$

Here, (f, g) is a pair of continuous functions, often called “Kantorovich potentials”. The same formula extends under the usual lower-semicontinuity and integrability assumptions, replacing maxima by suprema when dual optimizers need not exist.

Proof. Weak duality is immediate: if $f(x) + g(y) \leq c(x, y)$ and $\pi \in \mathcal{U}(\alpha, \beta)$, then

$$\int f d\alpha + \int g d\beta = \int (f(x) + g(y)) d\pi(x, y) \leq \int c d\pi.$$

Taking the supremum over admissible potentials and the infimum over couplings gives “ $\leq$ ”.

For the reverse inequality, view the primal problem as a linear program over the locally convex space of Radon measures, paired with $\mathcal{C}(\mathcal{X} \times \mathcal{Y})$. The affine map $\pi \mapsto (\pi_1, \pi_2)$ is continuous for the weak topology, the feasible set is non-empty because it contains $\alpha \otimes \beta$, and the cost is continuous and bounded on compact sets. The separating-hyperplane theorem applied to the convex set of attainable triples

$$\left\{ (\pi_1, \pi_2, \int c d\pi + r) : \pi \geq 0, r \geq 0 \right\}$$

gives a continuous affine separator, hence functions (f, g) and a scalar multiplier which can be normalized so that $f \oplus g \leq c$. The separating inequality then states that the supremum over such potentials is at least the primal value. This proves equality. The same argument is the infinite-dimensional analogue of the finite linear-programming proof in Proposition 28. $\square$

The discrete case (40) corresponds to the dual vectors being samples of the continuous potentials, *i.e.* $(f_i, g_j) = (f(x_i), g(y_j))$. The primal-dual optimality conditions allow for tracking the support of the optimal plan, and (43) is generalized as

$$\text{Supp}(\pi) \subset \{(x, y) \in \mathcal{X} \times \mathcal{Y} ; f(x) + g(y) = c(x, y)\}. \quad (46)$$

For the one-dimensional quadratic cost, the continuous potentials can be read from the monotone map $T = F_\beta^{-1} \circ F_\alpha$: on the active graph, $f'(x) = 2(x - T(x))$ and $g = f^c$.

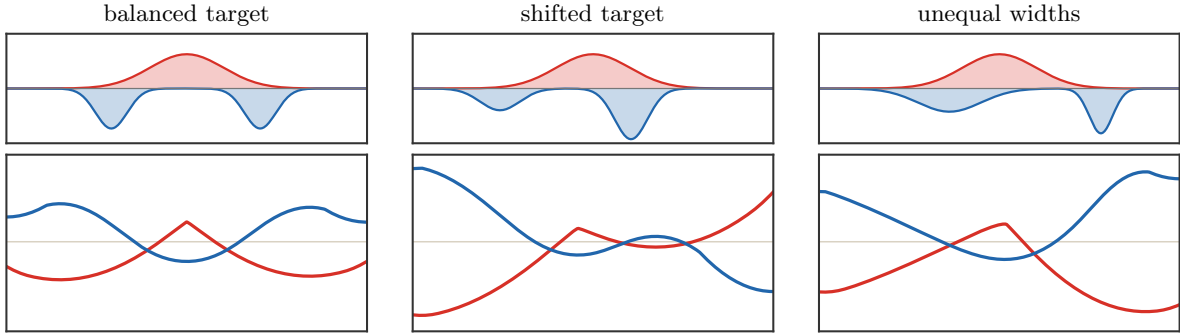


Figure 23: Continuous Kantorovich potentials for the same source and target families as Figure 22. The upper strips show the source density α in red and the target density β in blue. The lower strips show potentials f and $g = f^c$ for the quadratic cost $c(x, y) = |x - y|^2$, with the same gauge convention as in the discrete figure. The equality set $f(x) + g(y) = c(x, y)$ contains the monotone transport graph.

Note that in contrast to the primal problem (15), showing the existence of solutions to (44) is non-trivial, because the constraint set $\mathcal{R}(c)$ is not compact and the objective is not coercive. Using the machinery of c -transforms detailed in Section 5.4, one can show that optimal (f, g) are necessarily Lipschitz regular, which enables the replacement of the constraint by a compact one.

5.4 c -transforms

The c -transform is the operation that improves potentials without changing feasibility. It is both a proof device for dual attainment and the route from duality to Brenier's convex potentials.

Best-response potentials and the c -transform. Keeping a dual potential f fixed, one can maximize in closed form over the second potential in the dual problem (44), which leads to consider

$$\sup_{g \in \mathcal{C}(\mathcal{Y})} \left\{ \int g d\beta ; \forall (x, y), g(y) \leq c(x, y) - f(x) \right\}.$$

The constraint can be replaced by

$$\forall y \in \mathcal{Y}, \quad g(y) \leq f^c(y)$$

Definition 11 (c -transform). *For a function $f : \mathcal{X} \rightarrow \mathbb{R} \cup \{-\infty\}$, its c -transform is*

$$\forall y \in \mathcal{Y}, \quad f^c(y) := \inf_{x \in \mathcal{X}} c(x, y) - f(x). \quad (47)$$

For a function $g : \mathcal{Y} \rightarrow \mathbb{R} \cup \{-\infty\}$, the $\bar{c}$ -transform associated with $\bar{c}(y, x) = c(x, y)$ is

$$\forall x \in \mathcal{X}, \quad g^{\bar{c}}(x) := \inf_{y \in \mathcal{Y}} c(x, y) - g(y).$$

Since β is positive, the maximization of $\int g d\beta$ is thus achieved at those functions such that $g = f^c$ on the support of β , which means β -almost everywhere.

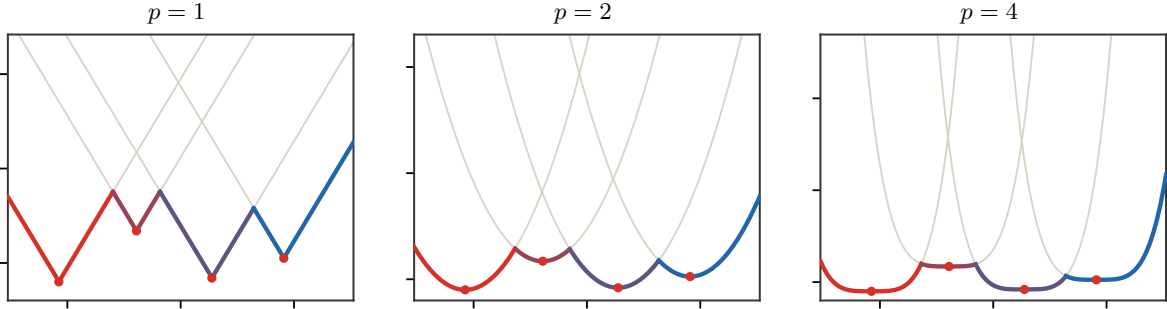


Figure 24: Discrete c -transform as a lower envelope for costs $c_p(x, y) = |x - y|^p$. The red circles are four source atoms x_i with potential values f_i ; the gray curves are the translated functions $y \mapsto c_p(x_i, y) - f_i$; the colored curve is their lower envelope $f^c(y) = \min_i c_p(x_i, y) - f_i$. This is the semi-discrete situation where $\mathcal{X}$ is finite, equivalently the source measure α is discrete; Section 6 studies the complementary case where the eliminated potential is supported on finitely many target atoms.

The failure of alternate optimization. A crucial property of the Legendre transform is that $f^{***} = f^*$, and that f^{**} is the convex envelope of f (the largest convex function below f). These properties carry over for the more general setting of c -transforms. The required convex-analytic background is standard in Rockafellar's theory [148].

Proposition 31 (Algebra of c -transforms). *The following identities, in which the inequality sign between vectors should be understood elementwise, hold, denoting $f^{c\bar{c}} := (f^c)^{\bar{c}}$:*

$$(i) \ f \leq f' \Rightarrow f^c \geq f'^c, \quad (ii) \ f^{c\bar{c}} \geq f, \quad (iii) \ g^{\bar{c}c} \geq g, \quad (iv) \ f^{c\bar{c}c} = f^c.$$

Proof. The first inequality (i) follows from the definition of c -transforms (because of the $-$ sign). To prove (ii), expanding the definition of $f^{c\bar{c}}$ we have

$$(f^{c\bar{c}})(x) = \min_y c(x, y) - f^c(y) = \min_y c(x, y) - \min_{x'} (c(x', y) - f(x')).$$

Now, since $-\min_{x'} (c(x', y) - f(x')) \geq -(c(x, y) - f(x))$, we recover

$$(f^{c\bar{c}})(x) \geq \min_y c(x, y) - c(x, y) + f(x) = f(x).$$

The relation $g^{\bar{c}c} \geq g$ is obtained in the same way. Now, to prove (iv), we first apply (ii) and then (i) with $f' = f^{c\bar{c}}$ to have $f^c \geq f^{c\bar{c}c}$. Then we apply (iii) to $g = f^c$ to obtain $f^c \leq f^{c\bar{c}c}$. $\square$

This invariance property shows that one can “improve” only once the dual potential this way. Indeed, starting from any pair (f, g) , one obtains the following iterates by alternating maximization

$$(f, g) \mapsto (f, f^c) \mapsto (f^{c\bar{c}}, f^c) \mapsto (f^{c\bar{c}}, f^{c\bar{c}c}) = (f^{c\bar{c}}, f^c) \dots \quad (48)$$

so that one reaches a stationary point. This failure is the classical behavior of alternating maximization on a non-smooth problem, where the non-smooth part of the functional (here the constraint) mixes the two variables. The workaround is to introduce smoothing, which is the classical method of augmented Lagrangian, and that we will develop here using entropic regularization, which corresponds to Sinkhorn’s algorithm.

For the bilinear cost $c(x, y) = -xy$ on a compact interval, the c -concave functions are ordinary concave functions and $f^{c\bar{c}}$ is the smallest concave majorant of f . In that model case, a hard transform removes non-concave oscillations in one closure step rather than producing a gradual ascent.

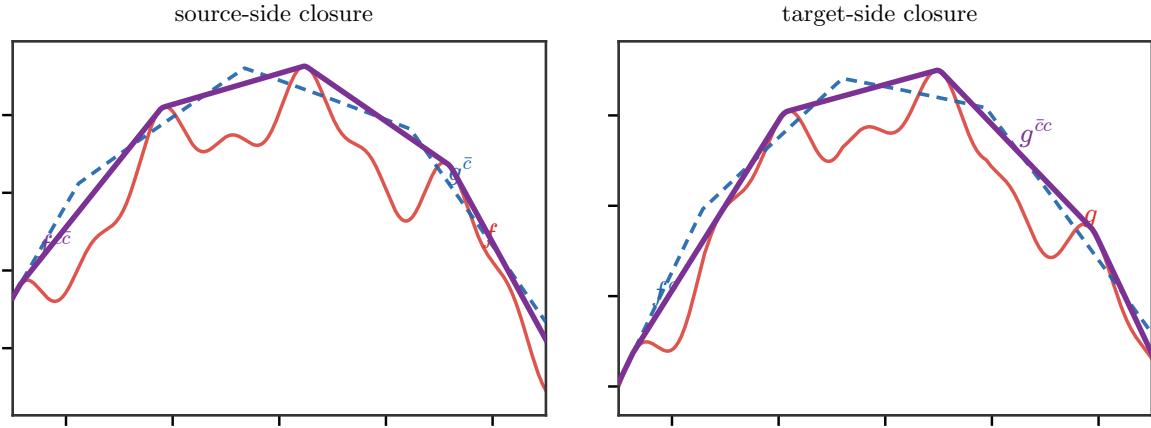


Figure 25: Hard c -transforms for the bilinear cost $c(x, y) = -xy$. Red curves are deliberately non-concave initial potentials f and g . Violet curves are the double-transform closures $f^{c\bar{c}}$ and $g^{\bar{c}c}$, which are concave majorants. Dashed blue curves are the one-sided best responses $g^{\bar{c}}$ and f^c , displayed after a harmless vertical gauge shift. The figure illustrates why exact alternating best responses are useful for dual certificates but do not give the smooth iterative dynamics later provided by entropic regularization.

6 Semi-discrete and $\mathcal{W}_1$

This section focuses on two computationally useful degeneracies of the dual problem. Semi-discrete OT turns a continuous-to-discrete map into finite-dimensional geometry, while $\mathcal{W}_1$ replaces convex potentials by Lipschitz functions and flow fields. The material connects computational geometry [11, 120, 121] with the Kantorovich–Rubinstein and Beckmann formulations [97, 13].

6.1 Semi-dual

The semi-dual eliminates one potential by an exact c -transform. It keeps concavity while removing explicit inequality constraints.

From the dual problem (44), that we write as

$$\sup_{f,g \in \mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y})} \mathcal{E}(f,g)$$

where $\mathcal{E}$ takes into account the constraints one can “marginalize” it with respect to g by minimizing over it to obtain the following “semi-dual” problem

$$\sup_{f \in \mathcal{C}(\mathcal{X})} \tilde{\mathcal{E}}(f) := \mathcal{E}(f, f^c) = \sup_g \mathcal{E}(f, g) = \int_{\mathcal{X}} f d\alpha + \int_{\mathcal{Y}} f^c d\beta. \quad (49)$$

Note that marginalizing a concave problem retains concavity so that $\tilde{\mathcal{E}}$ is still concave. The major advantage of this new “semi-dual” problem is that it is unconstrained, which allows the use of simpler optimization algorithms, as we will now see.

6.2 Semi-discrete

The semi-discrete case is where dual potentials become weights of Laguerre cells. This gives both geometry and algorithms for quantization and density fitting.

Discrete target and Laguerre cells. A case of particular interest is when $\beta = \sum_j b_j \delta_{y_j}$ is discrete (of course the same construction applies if α is discrete by exchanging the role of α, β). One can adapt the definition of the $\bar{c}$ transform (47) to this setting by restricting the minimization to the support $(y_j)_j$ of β ,

$$\forall g \in \mathbb{R}^m, \forall x \in \mathcal{X}, \quad g^{\bar{c}}(x) := \min_{j \in \llbracket m \rrbracket} c(x, y_j) - g_j. \quad (50)$$

This transform maps a vector g to a continuous function $g^{\bar{c}} \in \mathcal{C}(\mathcal{X})$. Note that this definition coincides with (47) when imposing that the space $\mathcal{X}$ is equal to the support of β .

Crucially, using the discrete $\bar{c}$ -transform, when β is a discrete measure, yields a finite-dimensional optimization,

$$\mathcal{L}_c(\alpha, \beta) = \max_{g \in \mathbb{R}^m} \mathcal{E}(g) := \int_{\mathcal{X}} g^{\bar{c}}(x) d\alpha(x) + \sum_j g_j b_j. \quad (51)$$

The Laguerre cells associated to the dual weights g

$$\mathbb{L}_j(g) := \{x \in \mathcal{X} ; \forall j' \neq j, c(x, y_j) - g_j \leq c(x, y_{j'}) - g_{j'}\}$$

induce a disjoint decomposition of $\mathcal{X} = \bigcup_j \mathbb{L}_j(g)$. When g is constant, the Laguerre cells decomposition corresponds to the Voronoi diagram partition of the space. For quadratic costs, varying the dual weights moves the walls between adjacent cells while keeping them parallel; this is the geometric mechanism by which the cell masses are adjusted.

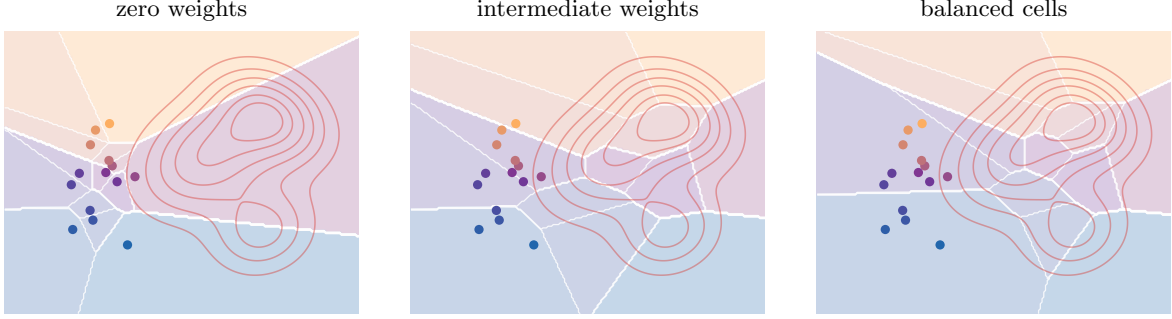


Figure 26: Laguerre cells for semi-discrete quadratic transport. The red contours show a continuous source density α given by a three-component Gaussian mixture on the right. The colored circular sites are the atoms of the discrete target β sampled from a compact Gaussian cloud on the left; each site color matches its Laguerre cell. Starting from ordinary Voronoi cells, semi-dual weight updates deform the cells so that the α -mass captured by each cell approaches the prescribed target mass.

Mass balance and quantization. This allows one to conveniently rewrite the minimized energy as

$$\mathcal{E}(g) = \sum_{j=1}^m \int_{\mathbb{L}_j(g)} (c(x, y_j) - g_j) d\alpha(x) + \langle g, b \rangle. \quad (52)$$

The following proposition provides a formula for the gradient of this concave function.

Proposition 32 (Gradient of the semi-discrete dual). *If α has a density with respect to Lebesgue measure and if c is smooth away from the diagonal, then $\mathcal{E}$ is differentiable and*

$$\forall j \in \llbracket m \rrbracket, \quad \nabla \mathcal{E}(g)_j = b_j - \int_{\mathbb{L}_j(g)} d\alpha.$$

Proof. One has

$$\mathcal{E}(g + \varepsilon \delta_j) - \mathcal{E}(g) - \varepsilon \left(b_j - \int_{\mathbb{L}_j(g)} d\alpha \right) = \sum_k \int_{\mathbb{L}_k(g + \varepsilon \delta_j)} c(x, y_k) d\alpha(x) - \int_{\mathbb{L}_k(g)} c(x, y_k) d\alpha(x).$$

Most of the terms on the right-hand side vanish (because most of the Laguerre cells associated with $g + \varepsilon \delta_j$ are equal to those of g) and the only terms remaining correspond to neighboring cells (j, k) such that $\mathbb{L}_j(g) \cap \mathbb{L}_k(g) \neq \emptyset$ (for the cost $\|x - y\|^2$ and $g = 0$ this forms the Delaunay triangulation). On these pairs, the right integral differs on a volume of the order of ε (since α has a density), and the function being integrated only varies on the order of ε (since the cost is smooth). So the right-hand side is of the order of ε^2 . $\square$

The first order optimality condition shows that to solve the dual semi-discrete problem, one needs to select the weights g to drive the Laguerre cell in a configuration such that $\int_{\mathbb{L}_j(g)} d\alpha = b_j$, i.e. each cell should capture the correct amount of mass. In this case, the optimal transport T such that $T_{\#}\alpha = \beta$ (which exists and is unique according to Brenier's theorem if α has a density) is piecewise constant and maps $x \in \mathbb{L}_j(g)$ to y_j .

The same cell geometry also underlies quantization. If the target locations y_j are not fixed, one can minimize $\int \min_j \|x - y_j\|^2 d\alpha(x)$ by Lloyd iterations: assign each point x to its nearest site, then replace each site by the centroid of its cell.



Figure 27: Lloyd quantization for the same continuous density and initial sites as Figure 26. The red contours show the density α , while the colored disks are the current codepoints and have the same colors as their Voronoi cells. The iterations move the initially left-located sites toward the high-density region and reshape the cells according to centroidal Voronoi geometry.

$\mathcal{W}_1$ on Euclidean spaces In the special case of Euclidean spaces $\mathcal{X} = \mathcal{Y} = \mathbb{R}^d$, using $c(x, y) = \|x - y\|$, the global Lipschitz constraint in the Kantorovich–Rubinstein formula can be made local as a uniform bound on the gradient of f ,

$$\mathcal{W}_1(\alpha, \beta) = \sup_f \left\{ \int_{\mathbb{R}^d} f(d\alpha - d\beta) ; \|\nabla f\|_\infty \leq 1 \right\}. \quad (53)$$

Here the constraint $\|\nabla f\|_\infty \leq 1$ signifies that the norm of the gradient of f at any point x is upper bounded by 1, $\|\nabla f(x)\|_2 \leq 1$ for any x .

Considering the dual problem to (53), denoting $\xi := \alpha - \beta$, and using that

$$\iota_{\|\cdot\|_{\mathbb{R}^d} \leq 1}(u) = \max_v \langle u, v \rangle - \|v\|_{\mathbb{R}^d}$$

one has a maximization on flow vector fields $s : \mathbb{R}^d \rightarrow \mathbb{R}^d$

$$\begin{aligned} \mathcal{W}_1(\alpha, \beta) &= \sup_f \inf_{s(x) \in \mathbb{R}^d} \int_{\mathbb{R}^d} f d\xi - \int \langle \nabla f(x), s(x) \rangle dx + \int \|s(x)\|_{\mathbb{R}^d} dx \\ &= \inf_{s(x) \in \mathbb{R}^d} \int \|s(x)\|_{\mathbb{R}^d} dx + \sup_f \int f(x) (d\xi - \operatorname{div}(s) dx) \end{aligned}$$

one obtains an optimization problem under a fixed divergence constraint

$$\mathcal{W}_1(\alpha, \beta) = \inf_s \left\{ \int_{\mathbb{R}^d} \|s(x)\|_{\mathbb{R}^d} dx ; \operatorname{div}(s) = \alpha - \beta \right\}, \quad (54)$$

which is often called the Beckmann formulation [13]. Here the vectorial function $s(x) \in \mathbb{R}^2$ can be interpreted as a flow field, describing locally the movement of mass. Outside the support of the two input measures, $\operatorname{div}(s) = 0$, which is the conservation of mass constraint. Once properly discretized using finite elements, Problems (53) and (54) become a nonsmooth convex optimization problem.

The previous formulations (53) and (54) of $\mathcal{W}_1$ can be generalized to the setting where $\mathcal{X}$ is a Riemannian manifold, i.e. $c(x, y) = d(x, y)$ where d is the associated geodesic distance (and then for smooth manifolds, the gradient and divergence should be understood as differential operators on manifolds).

Proposition 33 ($\mathcal{W}_1$ and Beckmann flow on a graph). *Let $G = (V, E)$ be a connected finite graph with positive edge lengths $(\ell_e)_{e \in E}$. The graph geodesic distance is*

$$d_G(i, j) = \min_{\gamma: i \rightsquigarrow j} \sum_{e \in \gamma} \ell_e.$$

For two probability vectors a, b on V , set $r = a - b$ and orient each edge $e = (i, j)$. If

$$(\nabla_G f)_e = f_j - f_i, \quad \text{div}_G = -\nabla_G^*$$

are the finite-difference gradient and its negative adjoint, then a positive flow on the oriented edge $i \rightarrow j$ has positive divergence at i and negative divergence at j . With this convention,

$$\mathcal{W}_{1,G}(a, b) = \max_f \left\{ \sum_{i \in V} f_i r_i ; |f_i - f_j| \leq \ell_e \quad \forall e = (i, j) \right\} = \min_m \left\{ \sum_{e \in E} \ell_e |m_e| ; \text{div}_G m = r \right\}.$$

The vector m_e is an oriented edge flow, and the constraint $\text{div}_G m = r$ is conservation of mass at each vertex.

Proof. The edge constraint $|f_i - f_j| \leq \ell_e$ implies, by summing along paths, that $|f_i - f_j| \leq d_G(i, j)$ for all vertices. Conversely, any 1-Lipschitz function for d_G satisfies the edge constraints because each edge is a path of length ℓ_e . The first equality is therefore the Kantorovich–Rubinstein formula on the metric space (V, d_G) .

For the second equality, write the graph Beckmann problem and dualize its equality constraint with a potential f :

$$\inf_m \sum_e \ell_e |m_e| + \sup_f \sum_i f_i (r_i - (\text{div}_G m)_i).$$

Using $\text{div}_G = -\nabla_G^*$, the coupling term is $\sum_e m_e (\nabla_G f)_e$. The minimization over each scalar flow m_e is finite exactly when $|(\nabla_G f)_e| \leq \ell_e$, and is then equal to zero. The dual problem is therefore precisely the graph Lipschitz dual above. Strong duality holds because this is a finite-dimensional linear program with a non-empty feasible set: connectedness and $\sum_i r_i = 0$ allow the signed surplus to be routed along paths. This proves the graph Beckmann formula. $\square$

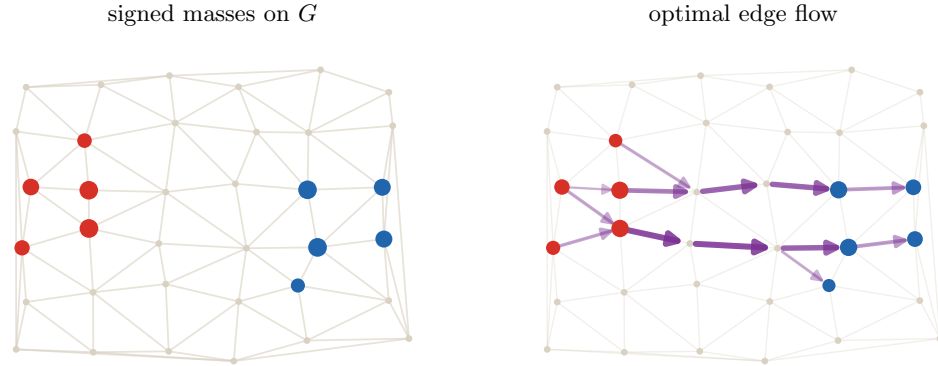


Figure 28: Graph Beckmann formulation of $\mathcal{W}_1$ on a Delaunay graph $G = (V, E)$ with 36 vertices. Red and blue disks encode the positive and negative parts of $r = \alpha - \beta$, localized on five vertices each. The violet arrows display the optimal signed edge flow m : orientation gives the sign, width is proportional to $\sqrt{|m_e|}$, and the flow satisfies the conservation constraint $\text{div}_G m = r$.

This graph formulation is the transshipment version of $\mathcal{W}_1$. It is the natural discrete analogue of (54): gradients are edge differences, divergences are incidence-matrix balances, and geodesic distance is the shortest-path length. It can be solved by min-cost flow methods on sparse graphs, and entropic or KL-projection variants lead to flow-Sinkhorn algorithms for graph $\mathcal{W}_1$ [13, 136].

7 Divergences and Dual Norms

This section compares OT with divergence-based and adversarial ways of measuring discrepancy. The main stake is topological: φ -divergences are cheap but strong, while dual norms and GAN objectives can be weak enough to compare singular measures. The discussion connects classical information divergences [48, 3] with modern integral probability metrics and generative modeling [165, 80, 10].

7.1 Dual norms (Integral Probability Metrics)

Dual norms generalize the $\mathcal{W}_1$ test-function principle. They are useful in statistics because they compare distributions by restricting the discriminator class.

Integral probability metrics. Formulation (53) is a special case of a dual norm. A dual norm is a convenient way to design “weak” norms that can deal with arbitrary measures. For a symmetric convex set B of measurable functions, one defines

$$\|\alpha\|_B := \sup_f \left\{ \int_{\mathcal{X}} f(x) d\alpha(x) ; f \in B \right\}. \quad (55)$$

These dual norms are often called “integral probability metrics”; see [166, 165, 167].

Example 2 (Total variation). The total variation norm (Example 6) is a dual norm associated with the whole space of continuous functions

$$B = \{f \in \mathcal{C}(\mathcal{X}) ; \|f\|_{\infty} \leq 1\}.$$

Total variation is the canonical nontrivial example of a discrepancy that is both a φ -divergence and a dual norm; see [165].

Example 3 ($\mathcal{W}_1$ norm). $\mathcal{W}_1$ as defined in (53), is a special case of dual norm (55), using

$$B = \{f ; \text{Lip}(f) \leq 1\}$$

the set of 1-Lipschitz functions.

Example 4 (Flat norm and Dudley metric). If the set B is bounded and separates measures, then $\|\cdot\|_B$ is a norm on the whole space $\mathcal{M}(\mathcal{X})$ of finite measures. This is not the case of $\mathcal{W}_1$, which is only defined for α such that $\int_{\mathcal{X}} d\alpha = 0$ (otherwise $\|\alpha\|_B = +\infty$). This can be alleviated by imposing a bound on the value of the potential f , in order to define for instance the flat norm,

$$B = \{f ; \text{Lip}(f) \leq 1 \quad \text{and} \quad \|f\|_{\infty} \leq 1\}. \quad (56)$$

It metrizes the weak convergence on the whole space $\mathcal{M}(\mathcal{X})$. The finite-dimensional version is obtained from the usual $\mathcal{W}_1$ dual linear program by adding the box constraints $|f_k| \leq 1$. The flat norm is sometimes called the “Kantorovich–Rubinstein” norm [89] and has been used as a fidelity term for inverse problems in imaging [107]. The flat norm is similar to the Dudley metric, which uses

$$B = \{f ; \|\nabla f\|_{\infty} + \|f\|_{\infty} \leq 1\}.$$

Kernel mean embeddings. Another important family of dual norms is obtained from reproducing kernel Hilbert spaces.

Definition 12 (Kernel MMD norm). *Let $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be a continuous positive definite kernel and let $\mathcal{H}$ be its reproducing kernel Hilbert space. The maximum mean discrepancy associated with k is*

$$\text{MMD}_k(\alpha, \beta) := \sup_{\|f\|_{\mathcal{H}} \leq 1} \int_{\mathcal{X}} f d(\alpha - \beta).$$

Equivalently, if $\mu_{\alpha} = \int k(x, \cdot) d\alpha(x) \in \mathcal{H}$ is the kernel mean embedding, then

$$\text{MMD}_k(\alpha, \beta) = \|\mu_{\alpha} - \mu_{\beta}\|_{\mathcal{H}}.$$

Proposition 34 (Universal kernels metrize weak convergence). *Assume that $\mathcal{X}$ is compact and that the RKHS generated by the continuous kernel k is dense in $\mathcal{C}(\mathcal{X})$ for the uniform norm. Then*

$$\text{MMD}_k(\alpha_n, \alpha) \rightarrow 0 \iff \alpha_n \rightharpoonup \alpha$$

for probability measures on $\mathcal{X}$.

Proof. If $\text{MMD}_k(\alpha_n, \alpha) \rightarrow 0$, then integrals of all RKHS functions converge. For any $h \in \mathcal{C}(\mathcal{X})$ and any $\eta > 0$, choose $g \in \mathcal{H}$ with $\|h - g\|_\infty \leq \eta$. Since α_n and α are probabilities,

$$\left| \int h d(\alpha_n - \alpha) \right| \leq 2\eta + \left| \int g d(\alpha_n - \alpha) \right|,$$

and the last term tends to zero. This proves weak convergence. Conversely, if $\alpha_n \rightharpoonup \alpha$, then $\alpha_n \otimes \alpha_n$, $\alpha_n \otimes \alpha$ and $\alpha \otimes \alpha$ converge weakly on the compact product space. Applying this to the continuous bounded function k in the identity

$$\text{MMD}_k(\alpha_n, \alpha)^2 = \iint k d\alpha_n d\alpha_n - 2 \iint k d\alpha_n d\alpha + \iint k d\alpha d\alpha$$

gives convergence to zero. $\square$

We refer to [16, 93, 156] for more details on RKHS functional spaces.

Remark 8 (Universal kernels). According to Proposition 34, the MMD norm $\|\cdot\|_k$ metrizes weak convergence if the span of the dual ball B is dense in the space of continuous functions $\mathcal{C}(\mathcal{X})$. This means that finite sums of the form $\sum_{i=1}^n a_i k(x_i, \cdot)$ (for arbitrary choices of n and points $(x_i)_i$) are dense in $\mathcal{C}(\mathcal{X})$ for the uniform norm $\|\cdot\|_\infty$. For translation-invariant kernels over $\mathcal{X} = \mathbb{R}^d$, $k(x, y) = k_0(x - y)$, this is equivalent to having a non-vanishing Fourier transform, $\hat{k}_0(\omega) > 0$.

In the special case where α is a discrete measure, one thus has the simple expression

$$\|\alpha\|_k^2 = \sum_{i=1}^n \sum_{i'=1}^n a_i a_{i'} k_{i,i'} = \langle \mathbf{ka}, \mathbf{a} \rangle \quad \text{where} \quad k_{i,i'} := k(x_i, x_{i'}).$$

In particular, when $\alpha = \sum_{i=1}^n a_i \delta_{x_i}$ and $\beta = \sum_{i=1}^n b_i \delta_{x_i}$ are supported on the same set of points, $\|\alpha - \beta\|_k^2 = \langle \mathbf{k}(\mathbf{a} - \mathbf{b}), \mathbf{a} - \mathbf{b} \rangle$, so that $\|\cdot\|_k$ is a Euclidean norm (proper if k is positive definite, degenerate otherwise if k is semidefinite) on the simplex Σ_n . To compute the discrepancy between two discrete measures, one can use

$$\|\alpha - \beta\|_k^2 = \sum_{i,i'} a_i a_{i'} k(x_i, x_{i'}) + \sum_{j,j'} b_j b_{j'} k(y_j, y_{j'}) - 2 \sum_{i,j} a_i b_j k(x_i, y_j). \quad (57)$$

7.2 Linear Optimal Transport

Linear OT replaces a nonlinear transport distance by a Hilbert norm between maps. It is not a dual norm on measures, but it belongs naturally in this chapter because, once a reference measure is fixed, measures are embedded into a linear Hilbert space, as in kernel mean embeddings. This tangent-linear viewpoint is useful when many nearby distributions must be compared cheaply.

Fix a reference absolutely continuous measure ρ and let T_α be the Brenier map pushing ρ to α . The linear OT embedding is

$$\alpha \mapsto T_\alpha - \text{Id} \in L^2(\rho; \mathbb{R}^d), \quad \text{LOT}_\rho(\alpha, \beta) = \|T_\alpha - T_\beta\|_{L^2(\rho)}. \quad (58)$$

If one of the two targets equals the reference, the linearized distance is exact: for instance, $\text{LOT}_\rho(\rho, \alpha) = \|T_\alpha - \text{Id}\|_{L^2(\rho)} = \mathcal{W}_2(\rho, \alpha)$. For two arbitrary targets it is instead a tangent-space approximation of the Wasserstein geometry [172]. For a family $(\alpha_s)_s$ with weights $(\lambda_s)_s$, the linearized barycenter is obtained by averaging maps,

$$\bar{T} = \sum_s \lambda_s T_{\alpha_s}, \quad \bar{\alpha}_{\text{LOT}} = \bar{T}_\# \rho.$$

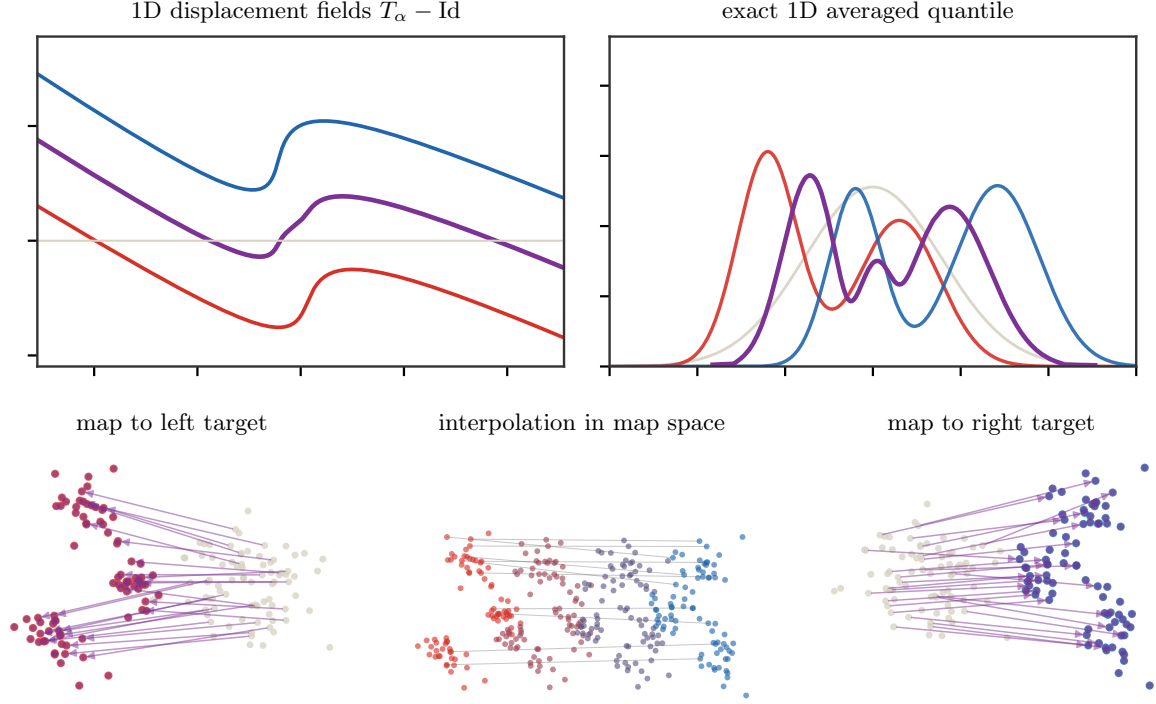


Figure 29: Linear OT coordinates. Fixing a reference measure ρ turns each target into the displacement field $T_\alpha - \text{Id}$. In one dimension this is exactly the quantile parametrization of $\mathcal{W}_2$, so averaging displacement fields gives the true Wasserstein barycenter. In two dimensions the same construction averages Brenier maps from a central Gaussian reference toward two multimodal targets; it is a tangent approximation and should not be confused with the generally nonlinear Wasserstein barycenter.

This is exact in one dimension, where quantile functions linearize $\mathcal{W}_2$, and it is especially useful when many barycenters with changing weights must be evaluated quickly.

Proposition 35 (Local stability of linear OT). *Assume that the measures are supported on a fixed convex compact set, with densities bounded above and below, and that the Brenier maps from ρ are regular. Then, for α, β in a sufficiently small regular neighborhood of ρ ,*

$$\mathcal{W}_2(\alpha, \beta) \leq \text{LOT}_\rho(\alpha, \beta) \quad \text{and} \quad \text{LOT}_\rho(\alpha, \beta) \leq C \mathcal{W}_2(\alpha, \beta)^\eta$$

for constants $C > 0$ and $\eta \in (0, 1]$ depending on regularity.

Proof. The first inequality is immediate: $(T_\alpha, T_\beta)_\# \rho$ is a feasible coupling between α and β . The reverse local estimate is a standard stability statement for the Monge–Ampère equation under the stated regularity assumptions: changes in the target measure control changes in the Brenier potential in Hölder norms, hence control $T_\alpha - T_\beta$ in $L^2(\rho)$. In simple one-dimensional settings, quantile functions make this exact with $\eta = 1$. In several dimensions one should not read the statement as a global Lipschitz estimate in $\mathcal{W}_2$. Quantitative stability results for semi-discrete and Monge–Ampère maps give Hölder exponents depending on the dimension, density bounds, support geometry and regularity; see for instance the estimates of Mérigot, Delalande and Chazal [122]. Under stronger smooth perturbations of uniformly convex smooth densities, elliptic regularity can give Lipschitz dependence in stronger function norms, but converting those controls to Wasserstein perturbations generally loses powers. $\square$

7.3 φ -divergences

This subsection develops divergences based on pointwise density ratios. They are computationally simple and statistically classical, but they do not see small spatial displacements between singular measures.

Definition by density ratios. We now consider a radically different class of methods to compare distributions, which are simpler to compute ($O(n)$ for discrete distributions) but never metrize weak-* convergence. Note that yet another way is possible, using Bregman divergence, which might metrize weak-* convergence when the associated entropy function is weak-* regular.

Definition 13 (Entropy function). *A function $\varphi : \mathbb{R} \rightarrow \mathbb{R} \cup \{\infty\}$ is an entropy function if it is lower semicontinuous, convex, $\text{dom } \varphi \subset [0, \infty[$, and satisfies the feasibility condition $\text{dom } \varphi \cap (0, +\infty) \neq \emptyset$. The speed of growth of φ at ∞ is described by*

$$\varphi'_\infty = \lim_{x \rightarrow +\infty} \varphi(x)/x \in \mathbb{R} \cup \{\infty\}.$$

If $\varphi'_\infty = \infty$, then φ grows faster than any linear function and φ is said to be *superlinear*. Any entropy function φ induces a φ -divergence (also known as Ciszár divergence [48, 3] or f -divergence) as follows.

Definition 14 (φ -Divergences). *Let φ be an entropy function. For $\alpha, \beta \in \mathcal{M}(\mathcal{X})$, let $\frac{d\alpha}{d\beta}\beta + \alpha^\perp$ be the Lebesgue decomposition¹ of α with respect to β . The divergence $\mathcal{D}_\varphi$ is defined by*

$$\mathcal{D}_\varphi(\alpha|\beta) := \int_{\mathcal{X}} \varphi\left(\frac{d\alpha}{d\beta}\right) d\beta + \varphi'_\infty \alpha^\perp(\mathcal{X}) \quad (59)$$

if α, β are nonnegative and ∞ otherwise.

The additional term $\varphi'_\infty \alpha^\perp(\mathcal{X})$ in (59) is the recession contribution of the perspective functional. It gives the weak-* lower-semicontinuous extension of the density-ratio integral when singular mass appears. This is essential for entropies with linear growth at infinity, such as the absolute value (63) defining the TV norm. If φ has superlinear growth, e.g. the usual entropy (62), then $\varphi'_\infty = +\infty$ so that $\mathcal{D}_\varphi(\alpha|\beta) = +\infty$ if α does not have a density with respect to β .

In the discrete setting, assuming

$$\alpha = \sum_i a_i \delta_{x_i} \quad \text{and} \quad \beta = \sum_i b_i \delta_{x_i} \quad (60)$$

are supported on the same set of n points $(x_i)_{i=1}^n \subset \mathcal{X}$, (59) defines a divergence on Σ_n

$$\mathcal{D}_\varphi(a|b) = \sum_{i \in \text{Supp}(b)} \varphi\left(\frac{a_i}{b_i}\right) b_i + \varphi'_\infty \sum_{i \notin \text{Supp}(b)} a_i, \quad (61)$$

where $\text{Supp}(b) := \{i \in \llbracket n \rrbracket ; b_i \neq 0\}$.

Proposition 3 (Basic properties of φ -divergences). *If φ is an entropy function, then $\mathcal{D}_\varphi$ is jointly 1-homogeneous, convex and weak-* lower semicontinuous in (α, β) .*

Proof. One defines the associated perspective function

$$\forall (u, v) \in (\mathbb{R}_+)^2, \quad \psi(u, v) = \begin{cases} \varphi(u/v)v & \text{if } v \neq 0, \\ u\varphi'_\infty & \text{if } v = 0 \end{cases}$$

¹The Lebesgue decomposition theorem asserts that, given β , α admits a unique decomposition as the sum of two measures $\alpha^{\text{ac}} + \alpha^\perp$ such that α^{ac} is absolutely continuous with respect to β and $\alpha^\perp$ and β are singular.

The joint 1-homogeneity follows from the definition of this perspective. We prove convexity in the discrete case, where

$$D_\varphi(\mathbf{a}|\mathbf{b}) = \sum_i \psi(a_i, b_i),$$

and it is enough to show that ψ is convex on $(\mathbb{R}_+)^2$. We first prove this on $\mathbb{R}_+ \times \mathbb{R}_+^*$; the extension to $v = 0$ follows by lower semicontinuity of the recession value $u\varphi'_\infty$. Indeed, for any $\lambda \in [0, 1]$, $\tau = 1 - \lambda$

$$\varphi\left(\frac{\tau u_1 + \lambda u_2}{\tau v_1 + \lambda v_2}\right)(\tau v_1 + \lambda v_2) = \varphi\left(\frac{\tau v_1}{\tau v_1 + \lambda v_2} \frac{u_1}{v_1} + \frac{\lambda v_2}{\tau v_1 + \lambda v_2} \frac{u_2}{v_2}\right)(\tau v_1 + \lambda v_2) \leq \tau \varphi\left(\frac{u_1}{v_1}\right) v_1 + \lambda \varphi\left(\frac{u_2}{v_2}\right) v_2.$$

In the general measure case, weak-* lower semicontinuity is the standard lower-semicontinuity theorem for convex integral functionals with recession extension; in the discrete case it is immediate from the lower semicontinuity of ψ . $\square$

The following proposition records when $\mathcal{D}_\varphi$ is nonnegative.

Proposition 4 (Non-negativity of φ -divergences). *Assume that φ is normalized by $\varphi(1) = 0$. For probability distributions $(\alpha, \beta) \in \mathcal{M}_+^1(\mathcal{X})$, one has $\mathcal{D}_\varphi(\alpha|\beta) \geq 0$. If φ is strictly convex, then one has $\mathcal{D}_\varphi(\alpha|\beta) = 0$ if and only if $\alpha = \beta$. This property extends to arbitrary distributions $(\alpha, \beta) \in \mathcal{M}_+(\mathcal{X})$ if one furthermore imposes that $\varphi \geq 0$.*

Proof. Let $m = \alpha + \beta$ and write $a = \frac{d\alpha}{dm}$ and $b = \frac{d\beta}{dm}$. Using the perspective function ψ from the previous proof,

$$\mathcal{D}_\varphi(\alpha|\beta) = \int \psi(a, b) dm.$$

Since α and β are probabilities, $\int a dm = \int b dm = 1$. Jensen's inequality and $\psi(1, 1) = \varphi(1) = 0$ give

$$\mathcal{D}_\varphi(\alpha|\beta) \geq \psi\left(\int a dm, \int b dm\right) = 0.$$

If φ is strictly convex, equality in Jensen forces $a = b$ m -almost everywhere, hence $\alpha = \beta$. In the general non-probability case, if $\varphi \geq 0$ then the divergence is positive by construction. $\square$

Classical examples and topology.

Example 5 (Kullback–Leibler divergence). The Kullback–Leibler divergence $\text{KL} := \mathcal{D}_{\varphi_{\text{KL}}}$, also known as the relative entropy, was already introduced in (33) and (27). It is the divergence associated to the Shannon–Boltzmann entropy function φ_{KL} , given by

$$\varphi_{\text{KL}}(s) = \begin{cases} s \log(s) - s + 1 & \text{for } s > 0, \\ 1 & \text{for } s = 0, \\ +\infty & \text{otherwise.} \end{cases} \quad (62)$$

Example 6 (Total variation). The total variation distance $\text{TV} := \mathcal{D}_{\varphi_{\text{TV}}}$ is the divergence associated to

$$\varphi_{\text{TV}}(s) = \begin{cases} |s - 1| & \text{for } s \geq 0, \\ +\infty & \text{otherwise.} \end{cases} \quad (63)$$

It actually defines a norm on the full space of measures $\mathcal{M}(\mathcal{X})$ where

$$\text{TV}(\alpha|\beta) = \|\alpha - \beta\|_{\text{TV}}, \quad \text{where} \quad \|\alpha\|_{\text{TV}} = |\alpha|(\mathcal{X}) = \int_{\mathcal{X}} d|\alpha|(x). \quad (64)$$

If α has a density ρ_α on $\mathcal{X} = \mathbb{R}^d$, then the TV norm is the L^1 norm on functions, $\|\alpha\|_{\text{TV}} = \int_{\mathcal{X}} |\rho_\alpha(x)| dx = \|\rho_\alpha\|_{L^1}$. If α is discrete as in (60), then the TV norm is the ℓ^1 norm of vectors in $\mathbb{R}^n$, $\|\alpha\|_{\text{TV}} = \sum_i |a_i| = \|\mathbf{a}\|_{\ell^1}$.

Remark 9 (Strong vs. weak topology). The total variation norm (64) defines the so-called “strong” topology on the space of measures. On a compact domain $\mathcal{X}$ of radius R , one has

$$\mathcal{W}_1(\alpha, \beta) \leq R \|\alpha - \beta\|_{\text{TV}}$$

so that this strong notion of convergence implies the weak convergence metrized by Wasserstein distances. The converse is, however, not true, since δ_x does not converge strongly to δ_y if $x \rightarrow y$ (note that $\|\delta_x - \delta_y\|_{\text{TV}} = 2$ if $x \neq y$). A chief advantage is that $\mathcal{M}_+^1(\mathcal{X})$ (once again on a compact ground space $\mathcal{X}$) is compact for the weak topology so that from any sequence of probability measures $(\alpha_k)_k$, one can always extract a converging subsequence, which makes it a suitable space for several optimization problems.

Main families of φ -divergences. Several classical divergences fit in the same template. The power-divergence family

$$\varphi_\gamma(s) = \frac{s^\gamma - \gamma s + \gamma - 1}{\gamma(\gamma - 1)} \quad (\gamma \neq 0, 1)$$

interpolates between the Pearson χ^2 divergence at $\gamma = 2$, the Hellinger-type behavior at $\gamma = 1/2$, and, by taking limits, the KL divergence as $\gamma \rightarrow 1$ and the reverse KL or Burg entropy $\varphi_0(s) = -\log s + s - 1$ as $\gamma \rightarrow 0$. The Hellinger divergence is often written separately with $\varphi_H(s) = (\sqrt{s} - 1)^2$, giving $\mathfrak{h}(\alpha, \beta) = \|\sqrt{\rho_\alpha} - \sqrt{\rho_\beta}\|_{L^2}$ when both measures have densities. The Jensen–Shannon divergence is the symmetrized and bounded KL-to-the-mixture divergence

$$\text{JS}(\alpha, \beta)^2 = \frac{1}{2} \text{KL}\left(\alpha \middle| \frac{\alpha + \beta}{2}\right) + \frac{1}{2} \text{KL}\left(\beta \middle| \frac{\alpha + \beta}{2}\right),$$

and is generated by a bounded entropy equivalent to $\varphi_{\text{JS}}(s) = s \log s - (s + 1) \log((s + 1)/2)$ up to an irrelevant affine term. Total variation corresponds to the non-smooth entropy $\varphi_{\text{TV}}(s) = |s - 1|$ and is exceptional because it is both a φ -divergence and an integral probability metric.

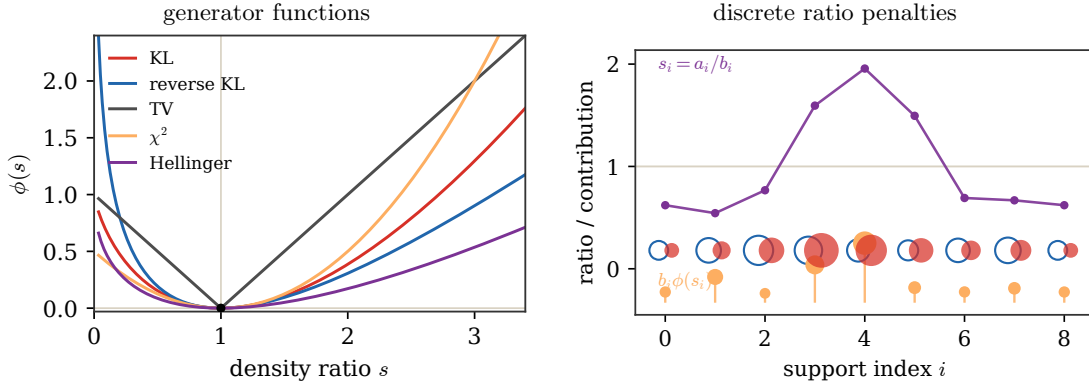


Figure 30: φ -divergences through density ratios. The left panel shows normalized generators for common divergences as functions of $s = d\alpha/d\beta$; all curves vanish at $s = 1$ up to affine normalization. The right panel shows the discrete formula $D_\varphi(a|b) = \sum_i b_i \varphi(a_i/b_i)$: hollow blue circles encode b_i , filled red circles encode a_i , the violet curve gives the ratios a_i/b_i , and orange lollipops show local KL-type contributions.

Remark 10 (φ -divergences versus Bregman divergences). Except for KL-type entropies, φ -divergences should not be confused with Bregman divergences. A φ -divergence compares measures pointwise through the density ratio $d\alpha/d\beta$ and is invariant under measurable changes of variables. A Bregman divergence is generated by a convex functional on a linear space and compares two points through first-order Taylor error. KL is special because the integral entropy $\alpha \mapsto \int \rho \log \rho$ produces a Bregman divergence whose density-ratio form is also a φ -divergence.

Variational dual formula.

Proposition 5 (Dual expression). *A φ -divergence can be expressed using the Legendre transform*

$$\varphi^{*,\geq 0}(s) := \sup_{t \in \mathbb{R}^+} st - \varphi(t)$$

(notice that we restrict the function to the positive real) of φ as

$$\mathcal{D}_\varphi(\alpha|\beta) = \sup_{f:\mathcal{X} \rightarrow \mathbb{R}} \int_{\mathcal{X}} f(x) d\alpha(x) - \int_{\mathcal{X}} \varphi^{*,\geq 0}(f(x)) d\beta(x). \quad (65)$$

which equivalently reads that the Legendre transform of $\mathcal{D}_\varphi(\cdot|\beta)$ reads

$$\forall f \in \mathcal{C}(\mathcal{X}), \quad \mathcal{D}_\varphi^*(f|\beta) = \int_{\mathcal{X}} \varphi^{*,\geq 0}(f(x)) d\beta(x). \quad (66)$$

Proof. We first consider the superlinear case $\varphi'_\infty = +\infty$, so that $\mathcal{D}_\varphi(\alpha|\beta) = +\infty$ if α does not have a density $\rho \geq 0$ with respect to β , $d\alpha = \rho d\beta$. Thus the Legendre-Fenchel transform of $\mathcal{D}_\varphi(\cdot|\beta)$ reads

$$\begin{aligned} \mathcal{D}_\varphi^*(f|\beta) &= \sup_{\rho \geq 0} \int_{\mathcal{X}} f(x) \rho(x) d\beta(x) - \int_{\mathcal{X}} \varphi(\rho(x)) d\beta(x) \\ &= \int_{\mathcal{X}} \sup_{\rho(x) \geq 0} (f(x) \rho(x) - \varphi(\rho(x))) d\beta(x) = \int_{\mathcal{X}} \varphi^{*,\geq 0}(f(x)) d\beta(x). \end{aligned}$$

Fenchel–Moreau then gives the displayed dual expression. For a general entropy, the same argument is applied to the perspective with its recession term; the singular part is exactly encoded by the effective domain of $\varphi^{*,\geq 0}$. $\square$

7.4 GANs via Duality

GANs fit naturally into the dual viewpoint: the discriminator is a parameterized potential and the generator moves a reference measure. This subsection first explains the original divergence-based GAN objective, then contrasts it with integral probability metrics such as MMD and Wasserstein distances.

The goal is to fit a generative parametric model $\alpha_\theta = g_{\theta,\#}\zeta$ to empirical data $\beta = \frac{1}{m} \sum_j \delta_{y_j}$, where $\zeta \in \mathcal{M}_+^1(\mathcal{Z})$ is a fixed density over the latent space and $g_\theta : \mathcal{Z} \rightarrow \mathcal{X}$ is the generator, often a neural network.

Divergence-based adversarial losses. Any φ -divergence can be written in adversarial form through the dual formula (65):

$$\min_\theta \mathcal{D}_\varphi(\alpha_\theta|\beta) = \min_\theta \sup_f \int_{\mathcal{X}} f(x) d\alpha_\theta(x) - \mathcal{D}_\varphi^*(f|\beta) = \min_\theta \sup_f \int_{\mathcal{X}} f(g_\theta(z)) d\zeta(z) - \frac{1}{m} \sum_j \varphi^*(f(y_j)).$$

Replacing the unrestricted potential f by a neural network f_ξ gives a saddle problem

$$\min_\theta \max_\xi \int_{\mathcal{Z}} f_\xi(g_\theta(z)) d\zeta(z) - \frac{1}{m} \sum_j \varphi^*(f_\xi(y_j)).$$

The original vanilla GAN [80] is this construction for the Jensen–Shannon generator discussed above,

$$\varphi_{\text{JS}}(s) = s \log s - (s+1) \log \frac{s+1}{2}, \quad \varphi_{\text{JS}}^*(u) = -\log(2 - e^u), \quad u < \log 2,$$

up to affine normalizations and the usual reparametrization of the potential by a discriminator with values in $(0, 1)$. In practice the min–max problem is solved by alternating stochastic gradient descent/ascent on (θ, ξ) . Unlike the convex-concave variational formula, the neural parametrization is non-convex in θ and non-concave in ξ , which explains the instability and mode-collapse pathologies of divergence-based GAN training. These losses estimate density ratios, which is statistically meaningful when the measures overlap but can saturate when the model and data are mutually singular; for example, the Jensen–Shannon divergence is maximal for disjoint supports.

Dual norms and integral probability metrics. Instead of a density-ratio divergence, one can minimize a dual norm (55), also called an integral probability metric,

$$\min_{\theta} \|\alpha_{\theta} - \beta\|_B = \min_{\theta} \sup_{f \in B} \int_{\mathcal{X}} f(x) d(\alpha_{\theta} - \beta)(x) = \min_{\theta} \sup_{f \in B} \int_{\mathcal{Z}} f(g_{\theta}(z)) d\zeta - \frac{1}{m} \sum_j f(y_j).$$

MMD-GANs take B to be a unit ball in an RKHS [60]; Wasserstein GANs take B to be a Lipschitz ball, following Kantorovich–Rubinstein duality [10, 68]. The advantage of such choices is topological: for bounded continuous RKHS balls, or for bounded Lipschitz balls on compact spaces, the resulting objective is weak-* continuous. It can therefore compare singular empirical and generated measures through test functions, instead of requiring pointwise density ratios. The price is that the discriminator class must be controlled geometrically, either by a kernel norm, a Lipschitz constraint or a related regularization. For instance, Wasserstein GANs originally used weight clipping, constraining $\|\xi\|_{\infty} \leq 1$ as a proxy for enforcing $f_{\xi} \in B = \{f; \text{Lip}(f) \leq 1\}$. This parameter set is both smaller than the true Lipschitz ball and non-convex, so clipping should be understood as a practical heuristic rather than a faithful implementation of the Kantorovich–Rubinstein dual constraint.

8 Advanced Topics on Entropic Regularization

This section collects the refinements that make entropic OT practical and statistically meaningful. It develops the dual smooth problem, revisits convergence, introduces the debiased Sinkhorn divergence and explains why regularization improves sample complexity. These topics connect algorithm design [53, 35] with modern statistical uses of regularized OT [76, 64].

8.1 Dual of Sinkhorn

The dual point of view replaces couplings by potentials and soft c -transforms. It is the right formulation for stabilized implementations and differentiation.

Discrete dual. The following proposition details the dual problem associated to (22).

Proposition 36 (Dual of entropic OT). *One has*

$$L_C^{\varepsilon}(\mathbf{a}, \mathbf{b}) = \max_{\mathbf{f} \in \mathbb{R}^n, \mathbf{g} \in \mathbb{R}^m} \langle \mathbf{f}, \mathbf{a} \rangle + \langle \mathbf{g}, \mathbf{b} \rangle - \varepsilon \sum_{i,j} \exp\left(\frac{f_i + g_j - C_{i,j}}{\varepsilon}\right) a_i b_j + \varepsilon. \quad (67)$$

The optimal $(\mathbf{f}, \mathbf{g})$ are linked to scalings $(\mathbf{u}, \mathbf{v})$ appearing in (23) through

$$\mathbf{u}_i = a_i e^{f_i/\varepsilon} \quad \text{and} \quad \mathbf{v}_j = b_j e^{g_j/\varepsilon}. \quad (68)$$

Proof. We introduce Lagrange multipliers and consider

$$\min_{\mathbf{P} \geq 0} \max_{\mathbf{f}, \mathbf{g}} \langle \mathbf{C}, \mathbf{P} \rangle + \varepsilon \text{KL}(\mathbf{P} | \mathbf{a} \otimes \mathbf{b}) + \langle \mathbf{a} - \mathbf{P} \mathbf{1}, \mathbf{f} \rangle + \langle \mathbf{b} - \mathbf{P}^{\top} \mathbf{1}, \mathbf{g} \rangle.$$

Finite-dimensional convex duality allows us to exchange the minimum over $\mathbf{P}$ with the maximum over $(\mathbf{f}, \mathbf{g})$, giving

$$\max_{\mathbf{f}, \mathbf{g}} \langle \mathbf{f}, \mathbf{a} \rangle + \langle \mathbf{g}, \mathbf{b} \rangle + \varepsilon \min_{\mathbf{P} \geq 0} \left(\text{KL}(\mathbf{P} | \mathbf{a} \otimes \mathbf{b}) - \left\langle \frac{\mathbf{f} \oplus \mathbf{g} - \mathbf{C}}{\varepsilon}, \mathbf{P} \right\rangle \right) = \langle \mathbf{f}, \mathbf{a} \rangle + \langle \mathbf{g}, \mathbf{b} \rangle - \varepsilon \text{KL}^* \left(\frac{\mathbf{f} \oplus \mathbf{g} - \mathbf{C}}{\varepsilon} | \mathbf{a} \otimes \mathbf{b} \right).$$

One concludes by using (66) for $\varphi(r) = r \log(r) - r + 1$

$$\text{KL}^*(H | \mathbf{a} \otimes \mathbf{b}) = \sum_{i,j} \varphi^*(H_{i,j}) a_i b_j.$$

Indeed, the scalar maximization

$$\varphi^*(s) = \sup_{r \geq 0} \{rs - r \log r + r - 1\}$$

has first-order condition $s - \log r = 0$, hence $r = e^s$ and $\varphi^*(s) = e^s - 1$. $\square$

Discrete soft c -transforms. Since the dual problem (67) is smooth and concave, one can perform alternating block maximization. For a fixed g , maximizing with respect to f leads to the following equation after zeroing the derivative with respect to f :

$$a_i - e^{\frac{f_i}{\varepsilon}} a_i \sum_j \exp\left(\frac{g_j - C_{i,j}}{\varepsilon}\right) b_j = 0$$

which leads to the explicit solution

$$f_i = -\varepsilon \log \sum_j \exp\left(\frac{g_j - C_{i,j}}{\varepsilon}\right) b_j.$$

We conveniently introduce the soft-min operator of some vector $h \in \mathbb{R}^m$

$$\min_b^\varepsilon(h) := -\varepsilon \log \sum_j e^{-h_j/\varepsilon} b_j$$

which is a smooth approximation of the minimum operator, and the optimal f for a fixed g is computed by a soft version of the c -transform

$$f_i = \min_b^\varepsilon(C_{i,\cdot} - g). \quad (69)$$

In a similar way, the optimal g for a fixed f is

$$g_j = \min_a^\varepsilon(C_{\cdot,j} - f). \quad (70)$$

Exponentiating these iterations recovers exactly the Sinkhorn algorithm. These iterations, however, become unstable for small ε . To apply the algorithm in this regime, one needs to stabilize it using the celebrated log-sum-exp trick. This follows from noticing that, similarly to the minimum operator, one has

$$\min_b^\varepsilon(h - \text{cst}) = \min_b^\varepsilon(h) - \text{cst}$$

and to replace the computation of $\min_b^\varepsilon(h)$ by its stabilized version (equal when using infinite precision computation) $\min_b^\varepsilon(h - \min(h)) + \min(h)$.

Other regularizers. The exponential form of Sinkhorn scalings is specific to KL regularization. Other convex penalties on the density of P with respect to $a \otimes b$ lead to the same marginal constraints but to different pointwise transfer laws. For an entropy $h : \mathbb{R}_+ \rightarrow \mathbb{R} \cup \{+\infty\}$ and $\lambda > 0$, consider

$$\min_{P \in U(a,b)} \langle C, P \rangle + \lambda \sum_{i,j} a_i b_j h\left(\frac{P_{i,j}}{a_i b_j}\right),$$

assuming for simplicity that $a_i, b_j > 0$.

Proposition 37 (Pointwise laws for entropy-regularized plans). *Let P be an optimal plan for the preceding problem and let (f, g) be dual multipliers for the two marginal constraints. With*

$$r_{i,j} := \frac{P_{i,j}}{a_i b_j}, \quad s_{i,j} := \frac{f_i + g_j - C_{i,j}}{\lambda},$$

one has $s_{i,j} \in \partial h(r_{i,j})$. In the smooth interior this reads $r_{i,j} = (h')^{-1}(s_{i,j})$. In particular,

$$\begin{aligned} h(r) = r \log r - r + 1 &\Rightarrow r_{i,j} = e^{s_{i,j}}, \\ h(r) = r - \log r - 1 &\Rightarrow r_{i,j} = (1 - s_{i,j})^{-1} \quad (s_{i,j} < 1), \\ h(r) = \frac{1}{2}(r - 1)^2 &\Rightarrow r_{i,j} = (1 + s_{i,j})_+. \end{aligned}$$

Proof. Introduce Lagrange multipliers in the form

$$\langle C, P \rangle + \lambda \sum_{i,j} a_i b_j h(r_{i,j}) + \langle a - P \mathbf{1}, f \rangle + \langle b - P^\top \mathbf{1}, g \rangle.$$

The stationarity condition with respect to $P_{i,j}$ gives

$$0 \in C_{i,j} - f_i - g_j + \lambda \partial h(r_{i,j}),$$

which is the announced inclusion. The three displayed formulas follow by inverting h' for KL and Burg. For the quadratic penalty, the unconstrained solution is $1 + s_{i,j}$ and the constraint $r_{i,j} \geq 0$ projects it to its positive part. $\square$

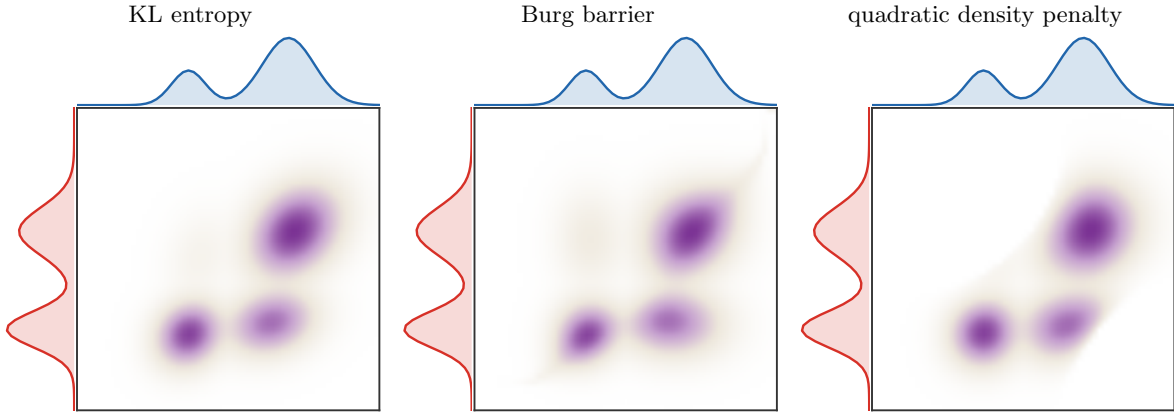


Figure 31: Effect of the regularizing entropy on the coupling. The red and blue side curves are fixed one-dimensional Gaussian-mixture marginals. The central matrix displays the square root of the coupling density on a common color scale. KL regularization gives the usual diffuse positive plan, the Burg barrier keeps a positive but differently tailed support, and the quadratic density penalty can set entries exactly to zero through its positive-part law.

Remark 11 (Gaussian rates). Because the Sinkhorn iterates remain quadratic, Gaussian Sinkhorn reduces to a finite-dimensional nonlinear iteration on means and covariance parameters. In the centered one-dimensional case with both marginals equal to $\mathcal{N}(0, 1)$ and kernel $K_\varepsilon(x, y) = e^{-(x-y)^2/\varepsilon}$, write one scaling as $v_k(y) = \exp(-q_k y^2 + \text{cst})$. The next row scaling is again Gaussian, with quadratic coefficient

$$F(q_k) = \frac{1}{\varepsilon} - \frac{1}{\varepsilon^2(q_k + \frac{1}{2} + \frac{1}{\varepsilon})}.$$

One full Sinkhorn cycle is therefore $q_{k+1} = F(F(q_k))$. If $q_\star = F(q_\star)$ and $B_\star = q_\star + \frac{1}{2} + \frac{1}{\varepsilon}$, then

$$B_\star^2 - \left(\frac{1}{2} + \frac{2}{\varepsilon}\right) B_\star + \frac{1}{\varepsilon^2} = 0, \quad F'(q_\star) = \frac{1}{\varepsilon^2 B_\star^2}.$$

The local linear rate for one full cycle is thus $(F'(q_\star))^2$. This explicit scalar formula shows the same qualitative behavior as the general Gaussian analysis of Chizat, Delalande and Vaškevičius [45]: the rate improves when ε is large and deteriorates as ε becomes small, where the entropic coupling approaches a deterministic Brenier map.

8.2 Convergence Analysis

This subsection revisits Sinkhorn convergence through three complementary lenses. Hilbert’s metric gives a clean contraction, Fortet’s order argument gives qualitative fixed-point convergence, and robust Bregman estimates give a non-asymptotic $O(1/k)$ dual-gap bound. The first two viewpoints explain asymptotic convergence mechanisms, while the last one is often the most useful for explicit complexity estimates.

Hilbert-metric linear convergence. The Hilbert metric convergence result is equivalent to convergence on the dual potentials (f, g) according to the variation norms. For bounded cost c (e.g. on compact spaces),

$$\|f_k - f^*\|_V = O(\lambda^k) \quad \text{and} \quad \|g_k - g^*\|_V = O(\lambda^k)$$

$$\left\| \log \frac{d\pi_k}{d\pi^*} \right\|_\infty = \|(f_k - f^*) \oplus (g_k - g^*)\|_\infty \leq \|f_k - f^*\|_V + \|g_k - g^*\|_V$$

where the contraction ratio is the Birkhoff factor of the Gibbs kernel $K_\varepsilon = e^{-c/\varepsilon}$. Namely, with $\eta = \eta(K_\varepsilon)$ as in Theorem 3 and $R := \sup c - \inf c$,

$$\lambda = \frac{\sqrt{\eta} - 1}{\sqrt{\eta} + 1} \leq \tanh(R/(2\varepsilon)) < 1.$$

One also has the following bounds

$$\|f_k - f^*\|_V \leq \frac{\left\| \log \frac{d\pi_{k,1}}{d\alpha} \right\|_\infty}{1 - \lambda}$$

which can be used to provide a posterior estimate of the rate of convergence and serve as a stopping criterion.

Order-monotone convergence. There is another, older way to understand convergence, going back to Fortet’s proof of the Schrödinger system [65, 63, 110]. It does not primarily estimate a contraction factor. Instead, it uses the order structure of the soft transforms.

Proposition 38 (Monotone fixed-point route to Sinkhorn convergence). *Let c be bounded and continuous on compact spaces, and define the double Sinkhorn map, normalized by subtracting its α -mean,*

$$\mathcal{A}(f) := (f^{c,\varepsilon})^{\bar{c},\varepsilon} - \int (f^{c,\varepsilon})^{\bar{c},\varepsilon} d\alpha.$$

The map $\mathcal{A}$ is order preserving modulo additive constants. If an initialization is chosen so that $f_0 \leq \mathcal{A}(f_0)$ after normalization, then the iterates $f_{k+1} = \mathcal{A}(f_k)$ increase pointwise up to constants, remain uniformly bounded in oscillation, and converge to a fixed point. A convenient practical choice is $f_0 = 0$, followed if necessary by subtracting a sufficiently large constant so that the subsolution inequality holds. This fixed point is the entropic potential, hence the associated Sinkhorn scalings converge.

Proof. The soft c -transform is order reversing: if $g \leq g'$, then

$$-\varepsilon \log \int e^{(g-c)/\varepsilon} d\beta \geq -\varepsilon \log \int e^{(g'-c)/\varepsilon} d\beta.$$

The composition of two order-reversing transforms is therefore order preserving. The transform also commutes with additive constants in the projective sense, which is why one fixes a normalization. Starting from a subsolution gives $f_0 \leq f_1$, and order preservation gives $f_k \leq f_{k+1}$ for all k . Soft-transform oscillation bounds, controlled by $\sup c - \inf c$, prevent escape to infinity after normalization. Monotone pointwise convergence, compactness of equicontinuous soft transforms, and continuity of $\mathcal{A}$ then give a fixed point. Uniqueness of entropic potentials up to constants, Proposition 19 and the dual uniqueness statement above identify this fixed point with the Sinkhorn solution. $\square$

This proof is qualitative rather than quantitative, but it is conceptually useful: Sinkhorn is not only alternating projection or projective contraction; it is also a monotone fixed-point iteration on potentials once constants are quotiented out.

Robust sublinear Bregman rate. Sinkhorn is cyclic coordinate ascent on the smooth dual problem and, equivalently, alternating KL projection on the two marginal constraint sets. We denote the continuous entropic dual objective by

$$\mathcal{D}_\varepsilon(f, g) := \int f d\alpha + \int g d\beta - \varepsilon \int \left(e^{(f \oplus g - c)/\varepsilon} - 1 \right) d\alpha \otimes d\beta. \quad (71)$$

The following statement records the rate that is most useful for complexity estimates: before entering a possible linear regime, the dual objective gap decreases at order $1/k$. We state it for balanced entropic OT, which is the specialization needed here; related robust Bregman-projection rates are developed in [136, 4, 59], and statistical consequences of entropic smoothing are analyzed in [76, 22].

Proposition 39 (Pinsker inequality). *If $p, q \in \Sigma_n$, then*

$$\|p - q\|_1^2 \leq 2 \text{KL}(p|q).$$

Proof. It is enough to prove the scalar inequality

$$u \log u - u + 1 \geq \frac{(u - 1)^2}{2(u + 1)} \quad u \geq 0.$$

Indeed, after bringing the right-hand side to the left, the resulting function h satisfies $h(1) = h'(1) = 0$ and, for $u > 0$,

$$h''(u) = \frac{1}{u} - \frac{4}{(u + 1)^3} = \frac{(u + 1)^3 - 4u}{u(u + 1)^3} \geq 0,$$

so h is convex and minimized at $u = 1$. Multiplying by q_i and summing with $u = p_i/q_i$ gives, first when all $q_i > 0$ and then by approximation in general,

$$\text{KL}(p|q) \geq \frac{1}{2} \sum_i \frac{(p_i - q_i)^2}{p_i + q_i}.$$

Cauchy–Schwarz yields

$$\|p - q\|_1^2 = \left(\sum_i \frac{|p_i - q_i|}{\sqrt{p_i + q_i}} \sqrt{p_i + q_i} \right)^2 \leq \left(\sum_i \frac{(p_i - q_i)^2}{p_i + q_i} \right) \left(\sum_i (p_i + q_i) \right),$$

and $\sum_i (p_i + q_i) = 2$. □

Proposition 40 (A compact $O(1/k)$ dual rate). *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact, that c is bounded, and write $R = \sup c - \inf c$. Let (f_k, g_k) be Sinkhorn dual iterates normalized by $\int f_k d\alpha = 0$, and let*

$$\Delta_k := \mathcal{D}_\varepsilon(f^*, g^*) - \mathcal{D}_\varepsilon(f_k, g_k)$$

be the dual suboptimality gap for the entropic dual objective (71). Then there exists a numerical constant C such that

$$\Delta_k \leq \frac{CR^2}{\varepsilon(k + 1)}.$$

Proof. The proof uses three elementary ingredients. First, the soft c -transform bounds the oscillation of every normalized iterate and every normalized optimum:

$$\|f_k\|_V + \|g_k\|_V + \|f^*\|_V + \|g^*\|_V \leq CR.$$

Here $\|h\|_V := \sup h - \inf h$ is the variation seminorm, the same projective norm used in Hilbert’s metric in Section 4.6. It is natural because dual potentials can be shifted by constants without changing the coupling.

Second, each Sinkhorn half-step is an exact KL projection. The Pythagorean identity for KL projections gives the ascent identity

$$\mathcal{D}_\varepsilon(f_{k+1}, g_{k+1}) - \mathcal{D}_\varepsilon(f_k, g_k) = \varepsilon [\text{KL}(\pi^* | \pi_k) - \text{KL}(\pi^* | \pi_{k+1})],$$

where $\pi_k = e^{(f_k \oplus g_k - c)/\varepsilon} \alpha \otimes \beta$ and π^* is the optimal entropic coupling. The KL drop controls the marginal residuals through Pinsker's inequality, Proposition 39. Third, convexity of the exponential dual objective gives a one-step estimate of the form

$$\Delta_k^2 \leq \frac{CR^2}{\varepsilon} (\mathcal{D}_\varepsilon(f_{k+1}, g_{k+1}) - \mathcal{D}_\varepsilon(f_k, g_k)).$$

This is the usual Bregman-projection estimate: the dual gap is controlled by the product of a bounded dual radius and the marginal residual corrected by the next projection, and the residual squared is controlled by the KL drop. Summing the reciprocal inequality obtained from the last display yields

$$\frac{1}{\Delta_{k+1}} - \frac{1}{\Delta_k} \geq \frac{\varepsilon}{CR^2},$$

and therefore $\Delta_k \leq CR^2/(\varepsilon(k+1))$. □

Corollary 1 (Approximating unregularized OT by regularized dual costs). *Consider discrete histograms $a \in \Sigma_n$, $b \in \Sigma_m$ and a finite cost matrix C . Let $\mathcal{D}_{\varepsilon,k}$ be the KL-normalized entropic dual value after k Sinkhorn cycles, and let Δ_k be its dual gap. Define the entropy-corrected lower bound*

$$L_{\varepsilon,k} := \mathcal{D}_{\varepsilon,k} - \varepsilon H(a) - \varepsilon H(b), \quad H(a) = - \sum_i a_i \log a_i.$$

Then

$$0 \leq L_C(a, b) - L_{\varepsilon,k} \leq \varepsilon \log(nm) + \Delta_k.$$

Consequently, choosing $\varepsilon \leq \delta/(2 \log(nm))$ and running Sinkhorn until $\Delta_k \leq \delta/2$ gives a δ -accurate lower bound on the unregularized OT value. Under Proposition 40, the intermediate condition is $k+1 \geq 2CR^2/(\varepsilon\delta)$. With the above choice of ε , it is sufficient to take

$$k+1 \geq \frac{4CR^2 \log(nm)}{\delta^2},$$

hence $k = O(R^2 \log(nm)/\delta^2)$, up to constants and logarithmic stabilization factors.

Proof. The KL-normalized objective differs from the entropy convention (22) by the constant $\varepsilon H(a) + \varepsilon H(b)$ on the transport polytope, because

$$\text{KL}(P | a \otimes b) = -H(P) + H(a) + H(b).$$

Let E_ε be the optimum of the entropy-regularized objective $\langle P, C \rangle - \varepsilon H(P)$. Since $0 \leq H(P) \leq \log(nm)$ for any coupling matrix,

$$L_C(a, b) - \varepsilon \log(nm) \leq E_\varepsilon \leq L_C(a, b).$$

The corrected iterate satisfies $L_{\varepsilon,k} = E_\varepsilon - \Delta_k$, which gives the displayed value bound. The final iteration estimate follows by combining $\Delta_k \leq CR^2/(\varepsilon(k+1))$ with the target $\Delta_k \leq \delta/2$. □

The same identity also yields computable stopping diagnostics. The KL drops are exactly marginal defects: after a row update the row marginal is correct and the remaining drop is measured by the column marginal, and conversely after a column update. Thus marginal violations monitor both feasibility and the remaining dual gap, up to the bounded-radius constant above.

8.3 Sinkhorn Divergences

Sinkhorn divergences remove the entropic self-bias while retaining smoothness. They interpolate between OT-like geometry and kernel-like norms, which explains their statistical behavior.

Entropic bias. A major issue with the value of the Sinkhorn problem (32) is that $\mathcal{L}_c^\varepsilon(\alpha, \beta) > 0$. In particular,

$$\alpha_\varepsilon = \operatorname{argmin}_\beta \mathcal{L}_c^\varepsilon(\alpha, \beta)$$

does not satisfy $\alpha_\varepsilon = \alpha$ unless $\varepsilon = 0$. The following proposition shows that the bias induced by this entropic regularization dominates the large ε limit.

Proposition 41 (Large-temperature entropic bias). *Assume that c is bounded and continuous. Then $\mathcal{L}_c^\varepsilon(\alpha, \beta) \rightarrow \iint c(x, y) d\alpha(x) d\beta(y)$ as $\varepsilon \rightarrow +\infty$.*

Proof. Let $(f_\varepsilon, g_\varepsilon)$ be optimal dual potentials, normalized by $\int g_\varepsilon d\beta = 0$. The soft c -transform equation gives

$$f_\varepsilon(x) = -\varepsilon \log \int \exp\left(\frac{g_\varepsilon(y) - c(x, y)}{\varepsilon}\right) d\beta(y).$$

For bounded c , the oscillations of normalized entropic potentials are bounded uniformly in ε by the oscillation of c . Hence the log-sum-exp expansion is uniform:

$$f_\varepsilon(x) = - \int (g_\varepsilon(y) - c(x, y)) d\beta(y) + O(\varepsilon^{-1}) = \int c(x, y) d\beta(y) + O(\varepsilon^{-1}).$$

At optimality the exponential penalty in the dual integrates to zero, so

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) = \int f_\varepsilon d\alpha + \int g_\varepsilon d\beta = \iint c(x, y) d\alpha(x) d\beta(y) + O(\varepsilon^{-1}).$$

This proves the limit. □

So in the large ε limit, $\mathcal{L}_c^\varepsilon$ behaves like an inner product and not like a norm. For instance, in the case

$$\alpha_\varepsilon \rightarrow \min_\beta \left\langle \int c(x, \cdot) d\alpha(x), \beta \right\rangle = \delta_{y^*(\alpha)} \quad \text{where} \quad y^*(\alpha) = \operatorname{argmin}_y \int c(x, y) d\alpha(x).$$

For instance, when $c(x, y) = \|x - y\|^2$ then α_ε collapses towards a Dirac located at the mean $\int x d\alpha(x)$ of α .

Sinkhorn divergences. The usual way to go from an inner product to a norm is to use the polarization formula. We thus define the debiased Sinkhorn divergence

$$\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) := \mathcal{L}_c^\varepsilon(\alpha, \beta) - \frac{1}{2} \mathcal{L}_c^\varepsilon(\alpha, \alpha) - \frac{1}{2} \mathcal{L}_c^\varepsilon(\beta, \beta). \quad (72)$$

Although this formula is a debiasing by self-costs, its non-negativity is not automatic from the definition; Proposition 43 proves it below by a kernel Cauchy–Schwarz argument.

We first record a fundamental lemma: at an optimal dual pair, the exponential regularization term integrates to zero, so the entropic cost can be read directly from the potentials. The same cancellation holds along Sinkhorn iterations after each exact block update.

Lemma 3 (Entropic dual cost at optimum). *Let $(f_{\alpha, \beta}, g_{\alpha, \beta})$ be optimal dual potentials, normalized arbitrarily. One has*

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) = \langle f_{\alpha, \beta}, \alpha \rangle + \langle g_{\alpha, \beta}, \beta \rangle. \quad (73)$$

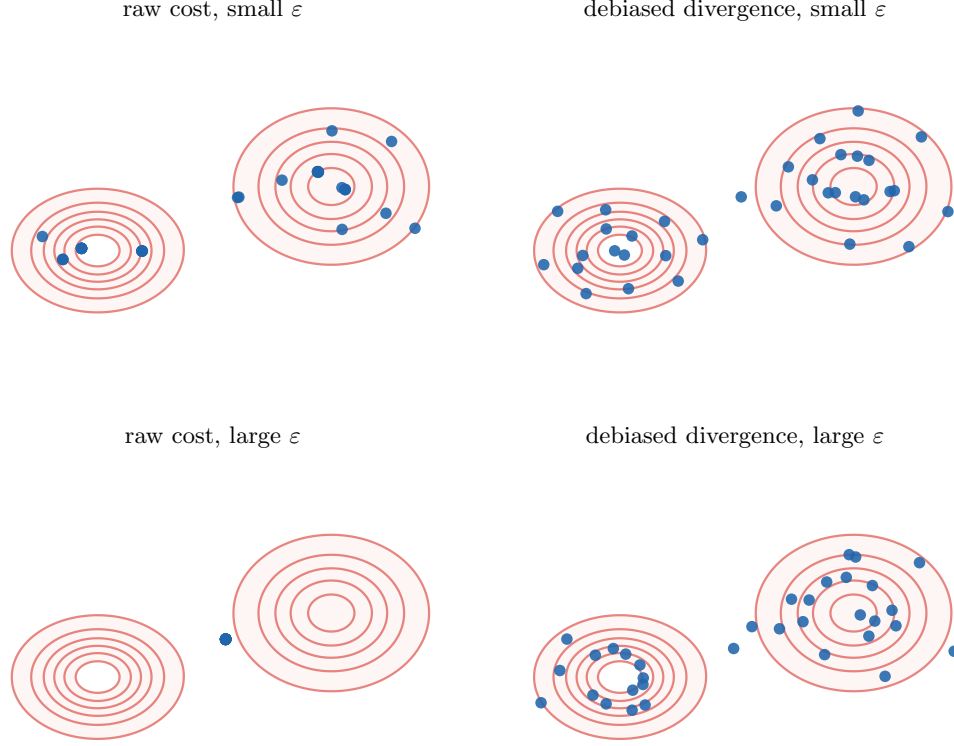


Figure 32: Debiating by point optimization. The red level sets show a fixed two-Gaussian target density β , while the blue atoms are an optimized empirical measure α_n . For small ε , the raw entropic cost and the Sinkhorn divergence both keep the two modes. For large ε , minimizing the raw cost collapses the atoms toward the barycenter predicted by the large-temperature bias, whereas the self-cost subtraction in (72) keeps a bimodal cloud.

Proof. We first notice that at optimality, the relation

$$f_{\alpha,\beta} = -\varepsilon \log \int_{\mathcal{Y}} e^{\frac{g_{\alpha,\beta}(y) - c(x,y)}{\varepsilon}} d\beta(y)$$

after taking the exponential, equivalently reads

$$1 = \int_{\mathcal{Y}} e^{\frac{f_{\alpha,\beta}(x) + g_{\alpha,\beta}(y) - c(x,y)}{\varepsilon}} d\beta(y) \implies \int_{\mathcal{X} \times \mathcal{Y}} \left(e^{\frac{f_{\alpha,\beta} \oplus g_{\alpha,\beta} - c}{\varepsilon}} - 1 \right) d(\alpha \otimes \beta) = 0.$$

Plugging this in formula (71), one obtains the result. $\square$

We first record the two limiting regimes of this debiased quantity.

Proposition 42 (Asymptotics of Sinkhorn divergences). *Assume that the two measures are supported on the same space and that c is bounded, continuous, nonnegative and satisfies $c(x, x) = 0$. One has $\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \rightarrow \mathcal{L}_c(\alpha, \beta)$ when $\varepsilon \rightarrow 0$ and*

$$\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \rightarrow \frac{1}{2} \int -cd(\alpha - \beta) \otimes d(\alpha - \beta) \quad \text{when } \varepsilon \rightarrow +\infty.$$

Proof. For discrete measures, the convergence is already proved in (30); we now give a general treatment. **Case $\varepsilon \rightarrow 0$.** The first limit follows from the standard Γ -convergence argument for entropic optimal transport: the entropy term is lower semicontinuous along weakly converging couplings, while any finite-cost coupling can be approximated by couplings with finite entropy. Since $c(x, x) = 0$, the two self-costs in the debiased expression converge to zero, and the cross term converges to $\mathcal{L}_c(\alpha, \beta)$.

Case $\varepsilon \rightarrow +\infty$. We denote by $(f_\varepsilon, g_\varepsilon)$ optimal dual potentials. After normalizing them and using boundedness of c , their oscillations stay uniformly bounded, so the following expansion is uniform. The optimality condition on f_ε (equivalently the Sinkhorn fixed point on f_ε) reads

$$\begin{aligned} f_\varepsilon &= -\varepsilon \log \int \exp \left(\frac{g_\varepsilon(y) - c(\cdot, y)}{\varepsilon} \right) d\beta(y) = -\varepsilon \log \int \left(1 + \frac{g_\varepsilon(y) - c(\cdot, y)}{\varepsilon} + o(1/\varepsilon) \right) d\beta(y) \\ &= -\varepsilon \log \left(1 + \frac{1}{\varepsilon} \int (g_\varepsilon(y) - c(\cdot, y)) d\beta(y) + o(1/\varepsilon) \right) = - \int g_\varepsilon d\beta + \int c(\cdot, y) d\beta(y) + o(1). \end{aligned}$$

Plugging this relation in the dual expression (73)

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) = \int f_\varepsilon d\alpha + \int g_\varepsilon d\beta = \iint c(x, y) d\alpha(x) d\beta(y) + o(1).$$

Applying this expansion to (α, β) , (α, α) and (β, β) gives

$$\tilde{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \rightarrow \int c d\alpha \otimes d\beta - \frac{1}{2} \int c d\alpha \otimes d\alpha - \frac{1}{2} \int c d\beta \otimes d\beta = -\frac{1}{2} \int c d(\alpha - \beta) \otimes d(\alpha - \beta).$$

□

In the case where $-c$ defines a conditionally positive definite kernel, $\tilde{\mathcal{L}}_c^\varepsilon(\alpha, \beta)$ converges to the square of a Hilbertian kernel norm. A typical example is $c(x, y) = \|x - y\|^p$ for $0 < p < 2$, which corresponds to the energy-distance kernel. This kernel norm is the dual of a homogeneous Sobolev norm.

We now show that this debiased Sinkhorn divergence is positive.

Proposition 43 (Non-negativity of Sinkhorn divergences). *If $k(x, y) = e^{-c(x, y)/\varepsilon}$ is positive definite, then $\tilde{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \geq 0$.*

Proof. In the following, we denote by $(f_{\alpha, \beta}, g_{\alpha, \beta})$ optimal dual potentials for the dual Schrödinger problem between α and β . We denote by $f_{\alpha, \alpha} = g_{\alpha, \alpha}$ (one can assume they are equal by symmetry) the solution for the problem between α and itself. Using the suboptimal function $(f_{\alpha, \alpha}, g_{\beta, \beta})$ in the dual maximization problem, and using relation (73) for the simplified expression of the dual cost, one obtains

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) \geq \langle f_{\alpha, \alpha}, \alpha \rangle + \langle g_{\beta, \beta}, \beta \rangle - \varepsilon \langle e^{\frac{f_{\alpha, \alpha} \oplus g_{\beta, \beta} - c}{\varepsilon}} - 1, \alpha \otimes \beta \rangle$$

But one has $\langle f_{\alpha, \alpha}, \alpha \rangle = \frac{1}{2} \mathcal{L}_c^\varepsilon(\alpha, \alpha)$, and similarly for β , so that the previous inequality equivalently reads

$$\frac{1}{\varepsilon} \tilde{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \geq 1 - \langle e^{\frac{f_{\alpha, \alpha} \oplus g_{\beta, \beta} - c}{\varepsilon}}, \alpha \otimes \beta \rangle = 1 - \langle \tilde{\alpha}, \tilde{\beta} \rangle_k$$

where $\tilde{\alpha} = e^{f_{\alpha, \alpha}/\varepsilon} \alpha$, $\tilde{\beta} = e^{f_{\beta, \beta}/\varepsilon} \beta$ and we introduced the inner product (which is valid because k is positive definite) $\langle \tilde{\alpha}, \tilde{\beta} \rangle_k := \int k(x, y) d\tilde{\alpha}(x) d\tilde{\beta}(y)$. One notes that the Sinkhorn fixed point equation, once exponentiated, reads $e^{f_{\alpha, \alpha}/\varepsilon} \odot [k(\tilde{\alpha})] = 1$ and hence

$$\|\tilde{\alpha}\|_k^2 = \langle k(\tilde{\alpha}), \tilde{\alpha} \rangle = \langle e^{f_{\alpha, \alpha}/\varepsilon} \odot k(\tilde{\alpha}), \alpha \rangle = \langle 1, \alpha \rangle = 1$$

and similarly $\|\tilde{\beta}\|_k^2 = 1$. Therefore, by Cauchy–Schwarz, one has $1 - \langle \tilde{\alpha}, \tilde{\beta} \rangle_k \geq 0$. □

Remark 12 (Strict positivity). Under additional assumptions on the kernel, one can furthermore show that $\tilde{\mathcal{L}}_c^\varepsilon(\alpha, \beta) = 0$ implies $\alpha = \beta$, and that this debiased divergence metrizes convergence in law.

8.4 Sample Complexity

This subsection explains why regularization can improve statistical rates. The message is that a biased but smooth distance may be preferable to exact OT in high dimension.

The sample complexity of unregularized OT suffers from the curse of dimensionality. Entropic regularization changes this picture: for a fixed $\varepsilon > 0$, Sinkhorn divergences have parametric $n^{-1/2}$ statistical rates, although the constant deteriorates when $\varepsilon \rightarrow 0$ [76, 22]. Related two-sample-testing viewpoints are developed in [145], and the large- ε kernel limit connects to classical MMD tests [83]. This improvement should be balanced against the regularization bias, which vanishes only when ε is sent to zero.

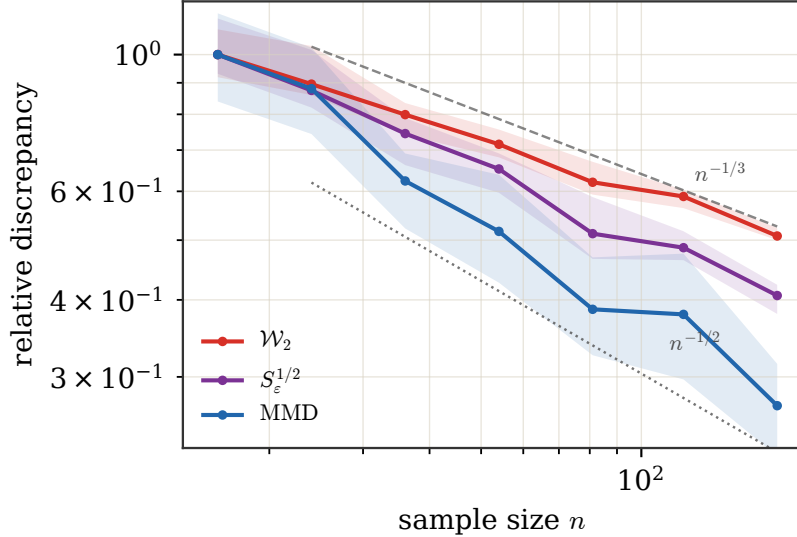


Figure 33: Empirical fluctuation in dimension three. For each sample size n , two independent empirical measures α_n and α'_n are drawn from the same standard Gaussian law in $\mathbb{R}^3$. The curves show the median of $D(\alpha_n, \alpha'_n)$, normalized by its value at the smallest displayed n , with interquartile bands. Exact OT follows the slower dimension-dependent scale, while MMD and the fixed- ε Sinkhorn divergence behave closer to the parametric $n^{-1/2}$ guide. This is a statistical illustration, not a solver benchmark.

Proposition 44 (Empirical OT has dimension-dependent rates). *Let α be a probability distribution with a density bounded above and below on $[0, 1]^d$, and let $\hat{\alpha}_n = n^{-1} \sum_{i=1}^n \delta_{X_i}$ be its empirical measure. For $d > 2p$, the typical scale of empirical OT is*

$$\mathbb{E} \mathcal{W}_p(\hat{\alpha}_n, \alpha) \asymp n^{-1/d}.$$

The exponent changes in low dimension, but the important message is that exact OT deteriorates with the ambient dimension.

Proof. We recall the standard multiscale argument, suppressing constants [57, 66, 173]. For the upper bound, partition $[0, 1]^d$ into dyadic cubes. At scale 2^{-j} , the empirical mass fluctuation over the cells is of order $n^{-1/2} 2^{jd/2}$, while moving this excess mass inside cells costs 2^{-j} . Summing the multiscale contributions up to the scale where the expected number of samples per cell is of order one gives 2^{-J} with $2^{Jd} \simeq n$, hence $n^{-1/d}$. For the lower bound, a packing argument shows that a positive fraction of cells at the critical scale are empty or overfull with non-negligible probability, forcing mass to travel at least a distance comparable to $n^{-1/d}$. The matching upper and lower bounds give the displayed rate. $\square$

Proposition 45 (MMD has a parametric empirical rate). *Let k be a bounded positive definite kernel with RKHS $\mathcal{H}_k$, and define*

$$\text{MMD}_k(\alpha, \beta) = \left\| \int k(x, \cdot) d(\alpha - \beta)(x) \right\|_{\mathcal{H}_k}.$$

If $\hat{\alpha}_n$ is the empirical measure of n i.i.d. samples from α , then

$$\mathbb{E} \text{MMD}_k(\hat{\alpha}_n, \alpha)^2 = \frac{1}{n} \left(\mathbb{E} k(X, X) - \mathbb{E} k(X, X') \right),$$

where X, X' are independent with law α . In particular, if $k(x, x) \leq \kappa^2$, then $\mathbb{E} \text{MMD}_k(\hat{\alpha}_n, \alpha) \leq \kappa/\sqrt{n}$.

Proof. Let $\Phi(x) = k(x, \cdot)$ be the feature map and $m_\alpha = \mathbb{E}\Phi(X)$. Then

$$\text{MMD}_k(\hat{\alpha}_n, \alpha) = \left\| \frac{1}{n} \sum_{i=1}^n (\Phi(X_i) - m_\alpha) \right\|_{\mathcal{H}_k}.$$

Independence cancels the cross terms after taking the squared norm and expectation, giving

$$\mathbb{E} \text{MMD}_k(\hat{\alpha}_n, \alpha)^2 = \frac{1}{n} \mathbb{E} \|\Phi(X) - m_\alpha\|_{\mathcal{H}_k}^2 = \frac{1}{n} \left(\mathbb{E} k(X, X) - \mathbb{E} k(X, X') \right).$$

The last bound follows from Jensen's inequality and $k(x, x) \leq \kappa^2$. $\square$

Proposition 46 (Sinkhorn divergences interpolate the rates). *Assume that α and β are supported in a compact subset of $\mathbb{R}^d$ and that the cost is smooth. For fixed $\varepsilon > 0$, debiased Sinkhorn divergences satisfy representative empirical bounds of the form*

$$\mathbb{E} \left| \bar{\mathcal{L}}_c^\varepsilon(\hat{\alpha}_n, \hat{\beta}_m) - \bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \right| \leq C_{c,d} \varepsilon^{-d/2} \left(\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}} \right),$$

up to constants and exponents depending on the precise smoothness class and support diameter. Thus regularization removes the $n^{-1/d}$ curse for fixed ε , while the prefactor deteriorates as $\varepsilon \rightarrow 0$.

Proof. The proof follows the empirical-process argument of [76]. By the envelope theorem, the fluctuation of $\mathcal{L}_c^\varepsilon$ with respect to its first marginal is controlled by the class of entropic dual potentials. The soft c -transform smooths these potentials at spatial scale $\sqrt{\varepsilon}$ for a quadratic-type cost. Covering a bounded d -dimensional domain at this scale gives an effective complexity of order $\varepsilon^{-d/2}$. Standard Rademacher or Dudley entropy bounds then give an empirical-process fluctuation of order $\varepsilon^{-d/2}/\sqrt{n}$ for each marginal. Applying the same estimate to the three terms defining the debiased divergence gives the stated bound. $\square$

Remark 13 (No free lunch when approximating exact OT). The parametric rate in Proposition 46 holds for fixed ε . If the goal is to approximate the unregularized OT value, one must also account for the regularization bias. In a typical bounded-cost finite-dimensional regime,

$$\left| \bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) - \mathcal{L}_c(\alpha, \beta) \right| \leq C\varepsilon, \quad \mathbb{E} \left| \bar{\mathcal{L}}_c^\varepsilon(\hat{\alpha}_n, \hat{\beta}_n) - \bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \right| \leq C_{c,d} \varepsilon^{-d/2} n^{-1/2}.$$

Balancing the two terms gives $\varepsilon \simeq n^{-1/(d+2)}$ and total error of order $n^{-1/(d+2)}$. Equivalently, target accuracy η requires choosing $\varepsilon \simeq \eta$ and $n \simeq \eta^{-(d+2)}$ samples under this bound. Thus entropic smoothing improves the statistical behavior at fixed scale, but approximating exact OT still forces a bias-variance tradeoff whose exponent deteriorates with dimension.

9 Generalized Wasserstein Distances

The first family of extensions keeps the idea of a distance between measures, but changes the geometry used to compare them. The variants below relax mass conservation, reduce high-dimensional transport to one-dimensional projections, or replace the trace quadratic cost by spectral gauges and robust projected viewpoints. They are useful when standard $\mathcal{W}_p$ is too rigid or too expensive, while still preserving a metric interpretation.

9.1 Unbalanced OT

Unbalanced OT allows mass creation and destruction by penalizing marginal mismatch. This is essential when histograms are not normalized, when observations contain outliers, or when only part of the source should match the target [111, 47, 46].

Relaxed formulation. For nonnegative measures $(\alpha, \beta) \in \mathcal{M}_+(\mathcal{X}) \times \mathcal{M}_+(\mathcal{Y})$, a generic relaxed formulation is

$$\text{UW}_c(\alpha, \beta) = \inf_{\pi \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y})} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) + \mathcal{D}_{\psi_1}(\pi_1 | \alpha) + \mathcal{D}_{\psi_2}(\pi_2 | \beta), \quad (74)$$

where ψ_1, ψ_2 are convex entropy functions. Exact conservation $(\pi_1, \pi_2) = (\alpha, \beta)$ is replaced by a cost for changing the marginals. Writing $\psi_s = \tau \bar{\psi}_s$ exposes the relaxation scale:

$$\text{UW}_{c, \tau}(\alpha, \beta) = \inf_{\pi \geq 0} \int c d\pi + \tau \mathcal{D}_{\bar{\psi}_1}(\pi_1 | \alpha) + \tau \mathcal{D}_{\bar{\psi}_2}(\pi_2 | \beta).$$

Large τ makes marginal mismatch expensive and approaches balanced OT when the total masses are compatible. Small τ makes it cheaper to destroy source mass and create target mass instead of transporting it over long distances.

Homogeneous viewpoint. The conic formulation of unbalanced OT is obtained by rewriting the marginal penalties as a local cost. For positive density ratios r, s , set

$$L_c(r, s) := c + r\psi_1(1/r) + s\psi_2(1/s). \quad (75)$$

Its positively homogeneous perspective is

$$H_c(r, s) := \inf_{\theta > 0} \theta L_c(r/\theta, s/\theta). \quad (76)$$

The associated homogeneous unbalanced value can be written formally as

$$\text{HW}_c(\alpha, \beta) = \inf_{\pi \geq 0} \int H_{c(x, y)} \left(\frac{d\pi_1}{d\alpha}(x), \frac{d\pi_2}{d\beta}(y) \right) d\pi(x, y), \quad (77)$$

with the usual recession terms on the singular parts. This perspective is the right one for proving metric properties.

Theorem 5 (Cone formulation of unbalanced OT). *Assume $\mathcal{X} = \mathcal{Y}$, $\psi_1 = \psi_2$, and choose the ground cost and exponent so that*

$$\Delta((x, r), (y, s)) := H_{c(x, y)}(r^p, s^p)^{1/p}$$

is a distance on the cone $\mathfrak{C}[\mathcal{X}]$. Then $\text{UW}_c = \text{HW}_c$ and the corresponding cone OT value $\text{CW}_c^{1/p}$ is a distance between nonnegative measures.

Proof. The equality $\text{UW}_c = \text{HW}_c$ follows from the perspective relaxation: disintegrating a feasible plan with respect to its marginals turns the marginal penalties into the local expression L_c , and optimizing over a common scaling variable gives H_c without changing the infimum. Lifting each point x to a cone point (x, r) encodes the transported amount as the radial coordinate. The cone marginal constraints recover α and β after integrating r^p and s^p . If Δ is a distance, the cone formulation is ordinary Wasserstein transport on a metric cone, hence its p th root satisfies symmetry, separation and the triangle inequality. $\square$

Entropic KL relaxation. For discrete histograms $\mathbf{a}, \mathbf{b}$ and a cost matrix $\mathbf{C}$, the KL-penalized entropic problem reads

$$\min_{\mathbf{P} \geq 0} \langle \mathbf{C}, \mathbf{P} \rangle + \varepsilon \text{KL}(\mathbf{P} | \mathbf{a} \otimes \mathbf{b}) + \tau \text{KL}(\mathbf{P} \mathbf{1} | \mathbf{a}) + \tau \text{KL}(\mathbf{P}^\top \mathbf{1} | \mathbf{b}).$$

With $K_{i,j} = e^{-C_{i,j}/\varepsilon} a_i b_j$ and $\rho = \tau/(\tau + \varepsilon)$, the generalized Sinkhorn scaling is

$$u_i \leftarrow \left(\frac{a_i}{(Kv)_i} \right)^\rho, \quad v_j \leftarrow \left(\frac{b_j}{(K^\top u)_j} \right)^\rho, \quad \mathbf{P} = \text{diag}(\mathbf{u}) \mathbf{K} \text{diag}(\mathbf{v}).$$

The exponent $\rho < 1$ is the visible difference with balanced Sinkhorn: marginal corrections are damped because violating the marginals is allowed.

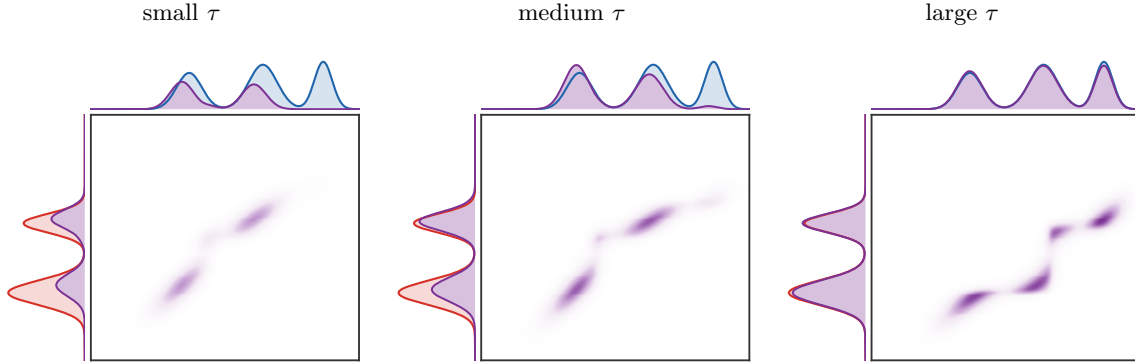


Figure 34: KL unbalanced OT on one-dimensional Gaussian-mixture densities. The central matrix is the transported coupling. On the left, the red curve is the prescribed source marginal and the violet curve is the transported source marginal; the red gap is destroyed mass. On the top, the blue curve is the prescribed target marginal and the violet curve is the transported target marginal; the blue gap is created mass. Increasing τ makes marginal mismatch more expensive, so more mass is transported, including toward the far right target mode.

The entropy used in the marginal relaxation also changes the qualitative behavior. A KL penalty leads to smooth multiplicative rescaling. The reverse-KL, or Burg, penalty blows up when a transported marginal vanishes where the prescribed marginal is positive, so it discourages complete deletion of small modes. Total variation has a linear kink and behaves closer to partial transport: mass is either kept active or created and destroyed at nearly constant marginal price.

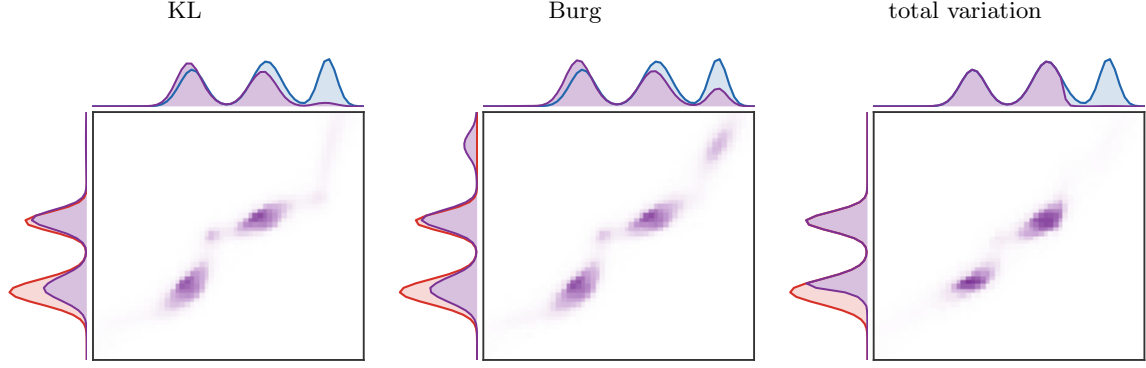


Figure 35: Effect of the marginal divergence in unbalanced entropic OT. The geometric cost, entropic plan regularization ε , and relaxation strength τ are fixed; only the marginal penalty changes. KL allows smooth mass variation, Burg keeps transported marginals from vanishing on prescribed modes, and total variation gives a sharper active-mass selection. The side plots use the same convention as Figure 34.

9.2 Sliced Wasserstein Distances

Sliced Wasserstein replaces one hard high-dimensional OT problem by many one-dimensional OT problems. For $\theta \in \mathbb{S}^{d-1}$, let $P_\theta(x) = \langle \theta, x \rangle$ be the projection on direction θ . The sliced distance is

$$\text{SW}_p(\alpha, \beta)^p := \int_{\mathbb{S}^{d-1}} \mathcal{W}_p((P_\theta)_\# \alpha, (P_\theta)_\# \beta)^p d\sigma(\theta), \quad (78)$$

where σ is the uniform probability measure on the sphere. This construction is computationally attractive because each projected problem is solved by sorting or quantiles [142, 30, 101].

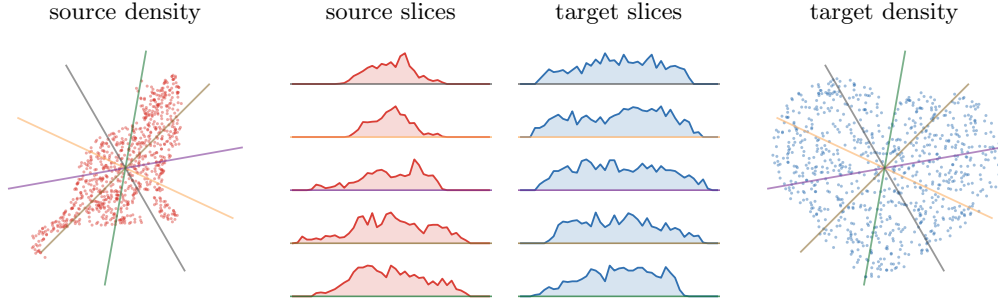


Figure 36: Sliced Wasserstein projections between two planar densities. Five fixed directions are drawn on the source and target point clouds. For each direction, the middle panels show the one-dimensional projected histograms $(P_\theta)_\# \alpha$ and $(P_\theta)_\# \beta$. Sliced OT averages one-dimensional Wasserstein discrepancies over many such directions, replacing a difficult planar comparison by a collection of sorted one-dimensional comparisons.

Definition 15 (Sliced variants). *The max-sliced distance replaces the average in (78) by a supremum,*

$$\text{MaxSW}_p(\alpha, \beta)^p := \sup_{\theta \in \mathbb{S}^{d-1}} \mathcal{W}_p((P_\theta)_\# \alpha, (P_\theta)_\# \beta)^p.$$

The same definitions can be made with projections on k -dimensional subspaces, giving $\text{SW}_{p,k}$ and $\text{MaxSW}_{p,k}$. Min-sliced lifted plans keep the projection direction but lift the induced one-dimensional matching back to the ambient space, giving an interpretable upper-bound construction for $\mathcal{W}_2$.

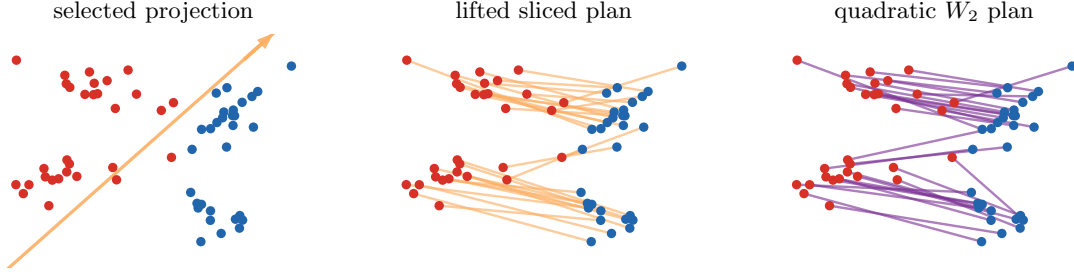


Figure 37: Lifted min-sliced plan. A one-dimensional direction is selected by a small deterministic sweep, then red and blue atoms are sorted after projection and matched in that order. The middle panel lifts this one-dimensional matching back to the plane; it is a feasible coupling but not the same object as the quadratic W_2 matching shown on the right. This illustrates why sliced constructions are computationally light and interpretable, while losing some of the geometry of the full transport problem.

9.3 Spectral and Robust Wasserstein Distances

For quadratic cost, an OT coupling has a second-moment displacement matrix

$$M_\pi := \int (x - y)(x - y)^\top d\pi(x, y) \in \mathbb{S}_+^d.$$

Standard $\mathcal{W}_2^2$ corresponds to the trace gauge $M \mapsto \text{tr}(M)$. More generally, let $\gamma : \mathbb{S}_+^d \rightarrow \mathbb{R}_+$ be a monotone gauge, meaning that it is convex, positively homogeneous, vanishes only at 0, and is monotone for the Loewner order: $0 \preceq M \preceq N$ implies $\gamma(M) \leq \gamma(N)$. Define

$$\mathcal{W}_\gamma(\alpha, \beta)^2 := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \gamma(M_\pi). \quad (79)$$

The trace gauge gives back $\mathcal{W}_2$. The spectral gauge $\gamma(M) = \lambda_{\max}(M)$ instead measures the worst transported variance direction.

The polar set of the gauge is

$$\mathcal{B}_\gamma := \{A \succeq 0; \langle A, M \rangle \leq \gamma(M) \text{ for all } M \succeq 0\}, \quad (80)$$

so that $\gamma(M) = \sup_{A \in \mathcal{B}_\gamma} \langle A, M \rangle$. For $A \succeq 0$, introduce the projected quadratic transport cost

$$\mathcal{W}_{2,A}(\alpha, \beta)^2 := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int \langle A, (x - y)(x - y)^\top \rangle d\pi(x, y). \quad (81)$$

Proposition 47 (Robust form of spectral Wasserstein). *Assume the measures have compact support and $\mathcal{B}_\gamma$ is compact. Then*

$$\mathcal{W}_\gamma(\alpha, \beta) = \sup_{A \in \mathcal{B}_\gamma} \mathcal{W}_{2,A}(\alpha, \beta).$$

In particular, if $a \text{tr}(M) \leq \gamma(M) \leq b \text{tr}(M)$ on $\mathbb{S}_+^d$ for some $0 < a \leq b < \infty$, then $\mathcal{W}_\gamma$ is a distance and is equivalent to $\mathcal{W}_2$.

Proof. Using the support-function representation of γ ,

$$\mathcal{W}_\gamma(\alpha, \beta)^2 = \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \sup_{A \in \mathcal{B}_\gamma} \int \langle A, (x - y)(x - y)^\top \rangle d\pi.$$

The coupling set is compact and convex for compactly supported measures, $\mathcal{B}_\gamma$ is compact and convex, and the displayed functional is bilinear in (π, A) . Sion's minimax theorem exchanges the infimum and supremum

and gives the robust formula. Each $\mathcal{W}_{2,A}$ is a pseudometric, so the supremum is a metric as soon as the polar family detects every nonzero displacement. The two trace bounds give

$$\sqrt{a} \mathcal{W}_2(\alpha, \beta) \leq \mathcal{W}_\gamma(\alpha, \beta) \leq \sqrt{b} \mathcal{W}_2(\alpha, \beta),$$

which proves separation and equivalence. $\square$

The Paty–Cuturi subspace robust distance [132] is recovered by choosing the polar family $\mathcal{B}$ to be rank- k orthogonal projectors. Its convex hull is $\{A ; 0 \preceq A \preceq I, \text{tr}(A) = k\}$, and the associated support function is the Ky Fan gauge γ_k . Thus $\mathcal{W}_{\gamma_k}$ is the convexified spectral counterpart of $\text{SRW}_{2,k}$, while $\text{SRW}_{2,k}$ keeps the original non-convex rank constraint. For $k = 1$, $\gamma_1(M) = \lambda_{\max}(M)$ and $\mathcal{B}_{\gamma_1} = \{A \succeq 0 ; \text{tr}(A) \leq 1\}$. This top-eigenvalue spectral Wasserstein geometry is connected to Muon-type spectral dynamics in [135].

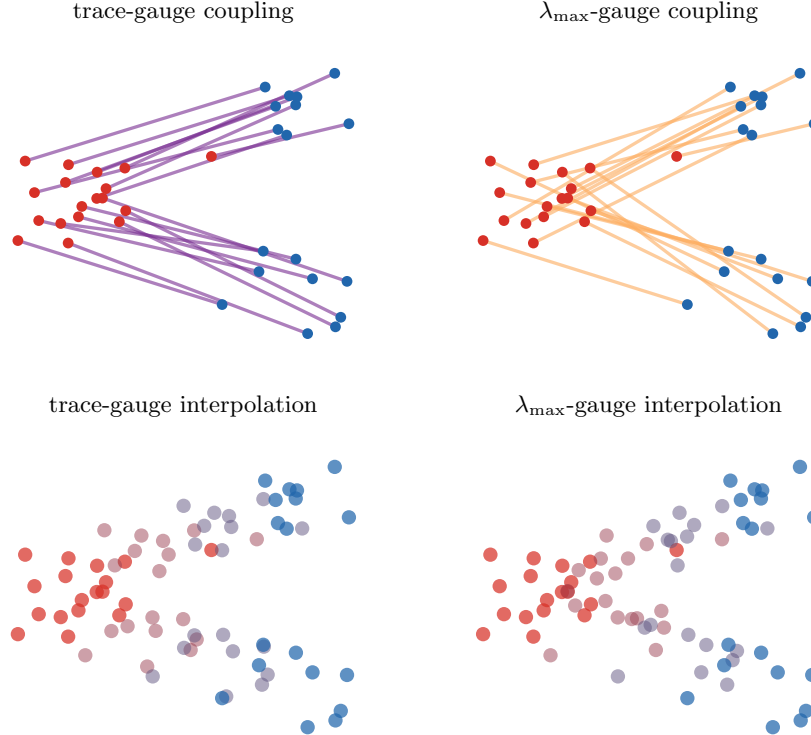


Figure 38: Trace and spectral gauges for displacement covariances. The trace gauge minimizes the average squared displacement and gives the usual quadratic transport plan. The $\lambda_{\max}$ gauge penalizes the worst projected displacement variance; the displayed plan is obtained by a small finite-direction robust linear program. The bottom row shows the corresponding displacement interpolations, colored from red to blue. The figure is illustrative: it visualizes how changing the gauge changes the coupling and the induced interpolation geometry.

10 Generalized OT Problems

The second family changes the optimization problem rather than only the ground distance. Barycenters average several measures, multi-marginal OT couples many measures at once, inverse OT learns the cost from observed transport, and weak OT allows randomized conditional responses. These formulations remain close to Kantorovich linear programming, but the object being optimized is richer than a single two-marginal coupling.

10.1 OT Barycenters

Barycenters ask how to average probability measures rather than points. This subsection explains the variational definition, the special closed forms in one dimension and for Gaussians, and the entropic algorithms used in practice.

Fréchet means. For discrete input histograms $\{b_s\}_{s=1}^S$, with $b_s \in \Sigma_{n_s}$, and weights $\lambda \in \Sigma_S$, a Wasserstein barycenter can be computed by minimizing

$$\min_{a \in \Sigma_n} \sum_{s=1}^S \lambda_s L_{C_s}(a, b_s), \quad (82)$$

where the cost matrices $C_s \in \mathbb{R}^{n \times n_s}$ are prescribed.

This barycenter problem was originally introduced by [1] following earlier ideas of [36]. For the quadratic cost on $\mathcal{X} = \mathbb{R}^d$, their theory gives existence and uniqueness when at least one input measure is absolutely continuous, and more generally under hypotheses ensuring that the relevant optimal maps are well defined. Discrete existence, consistency and fixed-point constructions are further studied in [7, 5, 106].

Given a set of input measures $(\beta_s)_s$ defined on some space $\mathcal{X}$, the barycenter problem becomes

$$\min_{\alpha \in \mathcal{M}_+^1(\mathcal{X})} \sum_{s=1}^S \lambda_s \mathcal{L}_c(\alpha, \beta_s). \quad (83)$$

Remark 14 (Two measures give a Wasserstein geodesic). For $S = 2$, $c(x, y) = \|x - y\|^2$ and weights $(1 - t, t)$, the barycenter is the point at time t on the Wasserstein geodesic between β_0 and β_1 . If T is the Brenier map from β_0 to β_1 , this barycenter is $((1 - t)\text{Id} + tT)_\# \beta_0$, the McCann interpolation detailed in Section 13.2. If no Monge map is available, the same construction uses an optimal coupling π and the interpolation map $(x, y) \mapsto (1 - t)x + ty$, giving $((1 - t)x + ty)_\# \pi$.

Example 7 (Dirac inputs). Problem (83) generalizes the computation of barycenters of points $(x_s)_{s=1}^S \in \mathcal{X}^S$ to arbitrary measures. Indeed, if $\beta_s = \delta_{x_s}$ is a single Dirac mass, then a solution to (83) is δ_{x^*} where x^* is a Fréchet mean of the points $(x_s)_s$.

Remark 15 (Mean of a quadratic barycenter). For $c(x, y) = \|x - y\|^2$, the mean of the barycenter α^* is necessarily the barycenter of the means,

$$\int_{\mathcal{X}} x d\alpha^*(x) = \sum_s \lambda_s \int_{\mathcal{X}} x d\beta_s(x).$$

Indeed, the squared Wasserstein distance decomposes into a squared distance between means plus a centered Wasserstein term. Minimizing the resulting quadratic function of the barycenter mean gives the displayed identity. If the input measures have compact support, the usual multi-marginal barycentric construction also gives a barycenter supported in the convex hull of the input supports.

Gaussian Case Gaussian barycenters show that the same separation as in the Gaussian Wasserstein formula (12) persists: means average linearly, while covariances average according to the Bures–Wasserstein geometry.

Remark 16 (Gaussian barycenters). The barycenter of Gaussian measures is Gaussian. In one dimension, it is obtained by averaging the means and the standard deviations, so the barycenter variance is the square of this averaged standard deviation. In higher dimensions, the covariance Σ minimizes the Bures objective

$$\Sigma \mapsto \sum_s \lambda_s \mathcal{B}(\Sigma, \Sigma_s)^2,$$

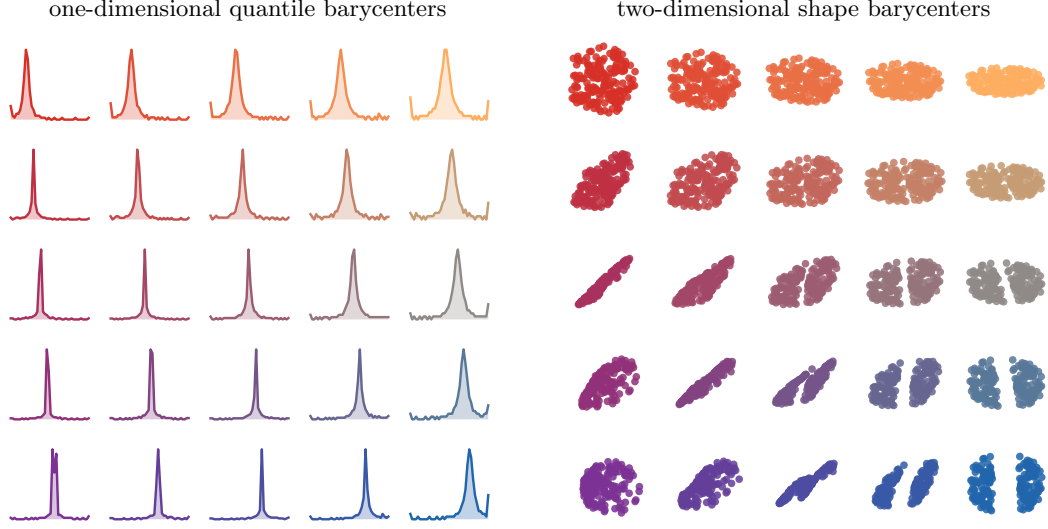


Figure 39: Wasserstein barycenter grids for four corner measures. The left panel uses the one-dimensional formula $Q_{u,v} = \sum_{i,j} \lambda_{ij}(u,v)Q_{ij}$ and displays the densities reconstructed from the averaged quantiles. The right panel uses four deformation maps from a common reference cloud to simple shapes and averages these maps with the same bilinear weights, which is the discrete Monge analogue of a barycentric map. Colors interpolate between the four corners.

and equivalently solves the fixed-point equation

$$\Sigma = \sum_s \lambda_s \left(\Sigma^{1/2} \Sigma_s \Sigma^{1/2} \right)^{1/2}.$$

This is the covariance analogue of the usual Euclidean barycenter equation: the mean part averages linearly, while the covariance part averages through the Bures–Wasserstein geometry [5, 21].

Corollary 2 (Gaussian and discrete barycenters). *Quadratic Wasserstein barycenters of Gaussian measures are Gaussian. If the input measures are discrete, then there exists a barycenter supported on the set of weighted averages $\sum_s \lambda_s x_{s,i_s}$ of one support point from each input; in particular, if the s -th input has n_s atoms, a barycenter exists with at most $\prod_s n_s$ atoms.*

Proof. Let the input Gaussians have means $\mathbf{m}_s$ and covariances Σ_s . For any candidate barycenter α with mean $\mathbf{m}$ and covariance Σ , Gelbrich’s inequality [75] gives

$$\mathcal{W}_2^2(\alpha, \beta_s) \geq \|\mathbf{m} - \mathbf{m}_s\|^2 + \mathcal{B}(\Sigma, \Sigma_s)^2,$$

with equality for the Gaussian law with mean $\mathbf{m}$ and covariance Σ . Therefore the barycenter objective is bounded below by a function depending only on $(\mathbf{m}, \Sigma)$, and this lower bound is attained by the Gaussian measure with the minimizing mean and covariance. Hence at least one barycenter is Gaussian, and uniqueness in the usual nondegenerate setting gives the Gaussian barycenter mentioned above. For discrete inputs, the multi-marginal formulation uses the barycentric map

$$B(x_1, \dots, x_S) = \sum_s \lambda_s x_s.$$

Any multi-marginal optimizer is supported on the finite product of the input supports, and B maps this product to at most $\prod_s n_s$ points. $\square$

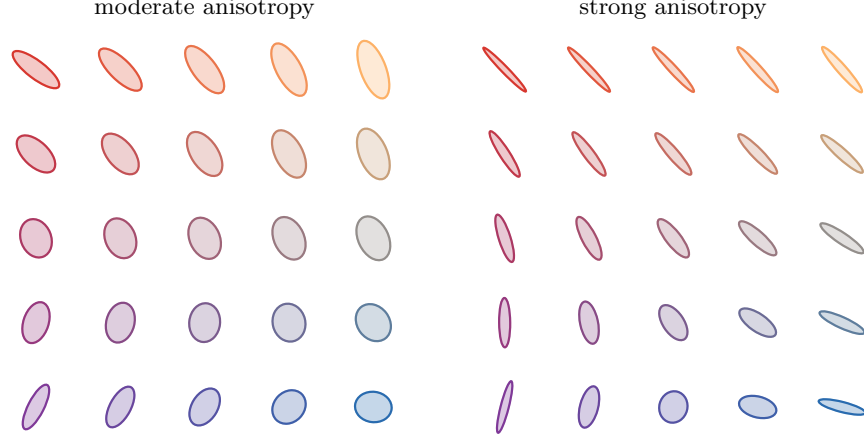


Figure 40: Bures–Wasserstein barycenters of centered Gaussian covariance matrices. Each panel shows a 5×5 grid of barycenter ellipses for four corner covariances, without separate input panels: the corner ellipses are the four input covariances themselves. The right grid uses more anisotropic inputs, making the nonlinear rotation and scaling of covariance barycenters more visible.

Entropic regularization of multi-marginal. As in the two-marginal case, adding an entropic penalty with respect to the product measure $\alpha_1 \otimes \cdots \otimes \alpha_S$ leads to scaling algorithms:

$$\inf_{\pi \in \mathcal{U}(\alpha_1, \dots, \alpha_S)} \int c d\pi + \varepsilon \text{KL}(\pi | \alpha_1 \otimes \cdots \otimes \alpha_S).$$

The optimizer has the generalized Gibbs form

$$d\pi^*(x_1, \dots, x_S) = \exp\left(\frac{\sum_s f_s(x_s) - c(x_1, \dots, x_S)}{\varepsilon}\right) \prod_s d\alpha_s(x_s),$$

and generalized Sinkhorn iterations alternately update one potential f_s so that the s -th marginal is correct. The bottleneck is that a naive tensor representation has size $\prod_s n_s$ in the discrete case. Practical barycenter solvers therefore exploit separability of the cost, low-rank structure, convolutional kernels, or a fixed barycenter support.

10.2 Metric learning and inverse OT

This final subsection points to inverse problems where the ground cost itself is learned. Such problems are typically bilevel and non-convex, but OT provides useful gradients with respect to the cost.

Metric learning and derivatives of OT OT is convex with respect to the measure and concave with respect to the cost. Ground-metric learning was explicitly studied in [51], and it connects to the broader metric-learning literature [104, 14].

Proposition 48 (Derivative with respect to the cost). *In the discrete setting, assume that the optimal coupling for $L_C(a, b)$ is unique and denote it by $P^*(C)$. Then $C \mapsto L_C(a, b)$ is differentiable at C and*

$$\nabla_C L_C(a, b) = P^*(C).$$

Proof. The value is the minimum of affine functions of C ,

$$L_C(a, b) = \min_{P \in \mathcal{U}(a, b)} \langle C, P \rangle.$$

The envelope theorem, or equivalently Danskin’s theorem, states that the subdifferential with respect to C is the convex hull of the optimal couplings. If the optimizer is unique, this subdifferential is the singleton $\{P^*(C)\}$, hence the value is differentiable with the displayed gradient. $\square$

Thus, if the cost is parameterized as C_θ , gradients of losses involving OT values are obtained by back-propagating through the inner product $\langle P^*(C_\theta), \partial_\theta C_\theta \rangle$. The difficulty is not differentiating a solved OT problem, but learning a cost for which the resulting matching has the desired semantic behavior; this is a bilevel and usually non-convex optimization problem.

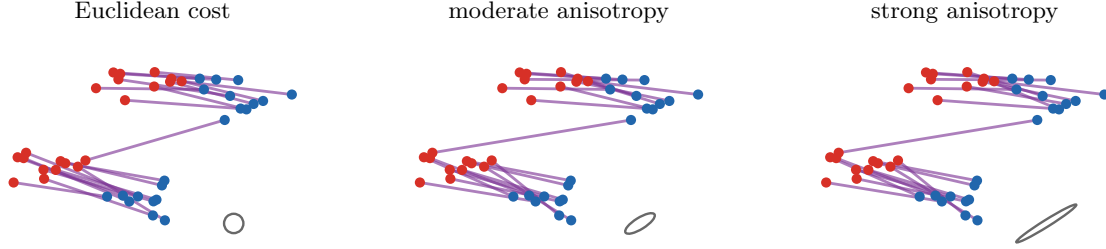


Figure 41: Changing the ground metric changes the optimal coupling. The same red and blue empirical measures are matched with $c_A(x, y) = (x - y)^\top A(x - y)$ for the Euclidean metric and two increasingly anisotropic Mahalanobis metrics. The small gray ellipse shows the unit ball of the metric: directions in which the ellipse is elongated are cheaper, and this deforms the transport segments selected by the OT plan.

Inverse Optimal Transport Inverse OT asks for a ground cost that explains observed matchings or flows as optimal transport plans. In its most direct form, one observes a plan $\hat{\pi}$ with marginals (α, β) and seeks a cost c such that $\hat{\pi}$ is optimal for

$$\inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int c(x, y) d\pi(x, y).$$

This is ill-posed without structure: adding potentials $u(x) + v(y)$ to a cost does not change the set of optimal couplings, and many costs can rationalize the same sparse plan.

A useful statistical methodology is to measure the suboptimality of the observed plan through a Fenchel–Young loss. Write the score as $s = -c$ and define the convex regularized prediction value

$$G_\varepsilon(s) = \sup_{\pi \in \mathcal{U}(\alpha, \beta)} \int s d\pi - \varepsilon \text{KL}(\pi | \alpha \otimes \beta).$$

The Fenchel–Young loss

$$\mathcal{L}_\varepsilon(c; \hat{\pi}) = G_\varepsilon(-c) + G_\varepsilon^*(\hat{\pi}) + \int c d\hat{\pi}$$

is nonnegative by Fenchel’s inequality and vanishes exactly when $\hat{\pi} \in \partial G_\varepsilon(-c)$, i.e. when $\hat{\pi}$ satisfies the regularized optimality conditions for c . Sharpened Fenchel–Young losses for inverse problems over measures and inverse entropic/unbalanced OT are developed in [9]; curvature and identifiability of inverse OT with respect to the cost are studied in [140]. Entropic regularization is important here because it makes the forward map smoother and provides gradients with respect to c , at the price of a bias that must be controlled statistically.

In practice, one restricts the cost to a finite-dimensional model class, often affine:

$$C_\theta = \sum_{r=1}^R \theta_r C^{(r)}, \quad \theta \in \Theta,$$

where Θ is convex and the matrices $C^{(r)}$ encode features, graph distances or a Mahalanobis parameterization. This viewpoint appears in low-rank and sparse inverse OT models [58, 8] and in convex formulations for learning OT costs from observed plans [115, 140].

Proposition 49 (Convex dual-gap formulation of inverse OT). *Let $\hat{P} \in \mathcal{U}(\mathbf{a}, \mathbf{b})$ be an observed coupling and let C_θ depend affinely on $\theta \in \Theta$, where Θ is convex. The condition that $\hat{P}$ is optimal for the cost C_θ is equivalent to the existence of dual potentials (f, g) such that*

$$f_i + g_j \leq (C_\theta)_{i,j} \quad \text{and} \quad \sum_{i,j} \hat{P}_{i,j} ((C_\theta)_{i,j} - f_i - g_j) = 0.$$

Consequently, for a convex regularizer R , the noisy inverse problem can be relaxed as the convex program

$$\min_{\theta \in \Theta, f, g} R(\theta) + \lambda \sum_{i,j} \hat{P}_{i,j} ((C_\theta)_{i,j} - f_i - g_j) \quad \text{subject to} \quad f_i + g_j \leq (C_\theta)_{i,j} \quad \forall i, j. \quad (84)$$

Proof. For a fixed cost C_θ , Kantorovich duality gives

$$\min_{P \in \mathcal{U}(\mathbf{a}, \mathbf{b})} \langle C_\theta, P \rangle = \max_{f_i + g_j \leq (C_\theta)_{i,j}} \langle f, \mathbf{a} \rangle + \langle g, \mathbf{b} \rangle.$$

Since $\hat{P}$ has marginals $(\mathbf{a}, \mathbf{b})$, every dual feasible pair satisfies

$$\langle C_\theta, \hat{P} \rangle - \langle f, \mathbf{a} \rangle - \langle g, \mathbf{b} \rangle = \sum_{i,j} \hat{P}_{i,j} ((C_\theta)_{i,j} - f_i - g_j) \geq 0.$$

This nonnegative quantity is exactly the primal-dual gap of $\hat{P}$. It vanishes if and only if $\hat{P}$ reaches the dual value and is therefore optimal. If C_θ is affine and Θ and R are convex, the constraints and objective in (84) are convex, proving the relaxation claim. $\square$

The formulation (84) is useful because it avoids differentiating through a forward OT solver: it learns a cost by making the observed plan satisfy complementary slackness. In statistical settings, $\hat{P}$ is only partially observed or noisy, so one adds sparsity, low-rank, smoothness or metric constraints to select a meaningful cost [58, 8]. For entropic OT, the optimality condition becomes smoother:

$$\hat{P}_{i,j} \approx a_i b_j \exp \left(\frac{f_i + g_j - (C_\theta)_{i,j}}{\varepsilon} \right),$$

which leads to likelihood-based or KL-based convex objectives when C_θ is affine, and connects inverse OT with generalized Sinkhorn iterations and transport-regularized inverse problems [98, 115]. Neural parameterizations of C_θ are more flexible but reintroduce non-convexity; the convex formulation above is the clean mathematical baseline.

10.3 Weak Optimal Transport

Weak OT relaxes the cost so that it depends on the conditional distribution of destinations rather than only on pointwise pairs. It is useful when a source point is allowed to choose a randomized response and the model only penalizes some aggregate of that response, such as its conditional mean. The framework was introduced through general transport costs and weak transport inequalities in [81]; existence, duality and optimality conditions on Polish spaces are developed in [12].

Given $\pi \in \mathcal{U}(\alpha, \beta)$, disintegrate it as $\pi(dx, dy) = \pi_x(dy)\alpha(dx)$. For a weak cost $C : \mathcal{X} \times \mathcal{P}(\mathcal{Y}) \rightarrow \mathbb{R} \cup \{+\infty\}$, the weak OT value is

$$\mathcal{W}_C(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int C(x, \pi_x) d\alpha(x). \quad (85)$$

The classical Kantorovich problem is recovered when $C(x, \nu) = \int c(x, y) d\nu(y)$, because the objective then becomes $\int c(x, y) d\pi(x, y)$. The genuinely weak behavior starts when C is nonlinear in ν .

Proposition 50 (Weak Kantorovich duality). *Assume that $\mathcal{X}, \mathcal{Y}$ are compact metric spaces and that $C(x, \nu)$ is lower semicontinuous, bounded from below and convex in ν , with the standard qualification assumptions ensuring Fenchel–Rockafellar duality. For $g \in \mathcal{C}(\mathcal{Y})$ define the weak C -transform*

$$g^C(x) := \inf_{\nu \in \mathcal{P}(\mathcal{Y})} \left\{ C(x, \nu) - \int g(y) d\nu(y) \right\}.$$

Then

$$\mathcal{W}_C(\alpha, \beta) = \sup_{g \in \mathcal{C}(\mathcal{Y})} \left\{ \int g^C(x) d\alpha(x) + \int g(y) d\beta(y) \right\}.$$

When $C(x, \nu) = \int c(x, y) d\nu(y)$, this reduces to the usual Kantorovich dual with $g^C(x) = \inf_y (c(x, y) - g(y))$.

Proof. For any coupling π and any $g \in \mathcal{C}(\mathcal{Y})$, the definition of g^C gives

$$C(x, \pi_x) \geq g^C(x) + \int g(y) d\pi_x(y).$$

After integration with respect to α , the second term becomes $\int g d\beta$ because the second marginal of π is β . This proves weak duality.

For the reverse inequality, consider the convex minimization over probability kernels $x \mapsto \pi_x$ with the affine constraint $\int \pi_x d\alpha(x) = \beta$. Fenchel–Rockafellar duality gives a continuous Lagrange multiplier g for this marginal constraint. Minimizing the Lagrangian over each conditional law gives exactly the pointwise term $g^C(x)$, while the multiplier contributes $\int g d\beta$. The compactness, lower semicontinuity, convexity and qualification assumptions ensure no duality gap. This is the weak-cost analogue of Proposition 28. $\square$

Proposition 51 (Barycentric weak transport is weaker than $\mathcal{W}_2$). *Let $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$ and define*

$$C_{\text{bar}}(x, \nu) = \|x - \int y d\nu(y)\|^2.$$

Then

$$\mathcal{W}_{C_{\text{bar}}}(\alpha, \beta) \leq \mathcal{W}_2^2(\alpha, \beta).$$

Proof. Let π be any coupling and disintegrate it as $\pi_x \alpha$. By Jensen’s inequality,

$$\|x - \int y d\pi_x(y)\|^2 \leq \int \|x - y\|^2 d\pi_x(y).$$

Integrating in x gives $\int C_{\text{bar}}(x, \pi_x) d\alpha(x) \leq \int \|x - y\|^2 d\pi(x, y)$. Taking the infimum over π proves the claim. $\square$

The barycentric cost is the canonical example to keep in mind: it only asks the conditional mean of Y given $X = x$ to be close to x , and ignores the conditional variance. This connects weak OT with martingale transport, Strassen-type convex-order constraints, barycentric projections and learning problems where conditional averages are meaningful objects.

11 Beyond Comparing Measures

The last group leaves the setting of scalar measures on a common ambient space. Vector and tensor-valued OT transports matrix-valued mass with internal degrees of freedom, Gromov–Wasserstein compares metric-measure spaces without a prescribed correspondence, and quantum OT replaces scalar couplings by positive operators. In each case, the transport plan also has to encode structure carried by the support, the fibers or the non-commutative state space.

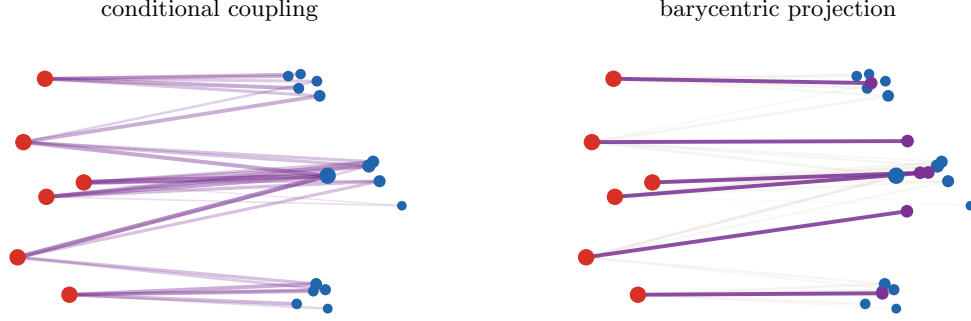


Figure 42: Weak barycentric transport on a small discrete coupling. The left panel shows the full conditional laws: each red source atom splits its mass among several blue target atoms, with segment thickness proportional to transported mass. The right panel collapses each conditional law π_x to its barycenter $\bar{y}(x) = \int y d\pi_x(y)$, shown in violet. The barycentric weak cost only sees the red-to-violet displacement, and therefore ignores the conditional spread around each barycenter.

11.1 Vector and Tensor-Valued Measures

Scalar OT transports a nonnegative density. In imaging, diffusion tensor imaging, spectral analysis and quantum-inspired models, the mass attached to a point can instead be a vector, a covariance matrix or a positive semidefinite tensor. A convenient finite-dimensional model is a matrix-valued measure A on $\mathcal{X}$ with values in $\mathbb{H}_m^+$, the cone of $m \times m$ positive semidefinite Hermitian matrices. If A has a density, we write $A(x) \succeq 0$ and interpret $\text{tr } A(x)$ as the local amount of mass while the normalized matrix describes an internal state. Static matrix-valued Monge–Kantorovich problems and dual test-function metrics were developed in [128, 95, 129]; dynamic non-commutative versions are closely related to the matrix transport models of [43, 41] and to non-commutative Wasserstein geometries for density matrices [34, 137].

A Benamou–Brenier-type construction introduces, in addition to the matrix density $A_t(x)$, a spatial matrix flux $M_t(x) = (M_{t,1}, \dots, M_{t,d})$ and sometimes an internal flux $Q_t(x) = (Q_{t,r})_r$ that rotates or mixes the matrix fibers. A schematic matrix-perspective action, written in the simplest left-perspective convention, is

$$\mathcal{W}_{\text{mv}}^2(A_0, A_1) = \inf_{A, M, Q} \int_0^1 \int_{\mathcal{X}} \left[\sum_{\ell=1}^d \text{tr} (M_{t,\ell}^* A_t^\dagger M_{t,\ell}) + \lambda \sum_r \text{tr} (Q_{t,r}^* A_t^\dagger Q_{t,r}) \right] dx dt \quad (86)$$

subject to the endpoint constraints and to a continuity equation

$$\partial_t A_t + \nabla_x \cdot M_t + \nabla_L^* Q_t = 0. \quad (87)$$

Here $A^\dagger$ denotes the Moore–Penrose inverse, with the usual perspective convention: the action is finite only when the fluxes are supported in the range of A . In genuinely non-commutative settings, one often replaces the displayed left-perspective expression by a symmetrized left-right perspective; this choice is part of the definition of the metric and is what ensures convexity and compatibility with adjoints. The operator ∇_L is a non-commutative gradient generated by a finite family of matrices $(L_r)_r$, for instance $\nabla_L X = (L_r X - X L_r)_r$, and ∇_L^* is its adjoint. The spatial term moves matrix-valued mass through $\mathcal{X}$; the internal term changes the fiber state without spatial displacement. If $m = 1$ and $Q = 0$, (86) reduces to the scalar Benamou–Brenier action of Section 12. If $\mathcal{X}$ is a single point, only the internal non-commutative transport remains, giving a dynamic geometry on positive matrices.

The important point is not the particular choice of ∇_L , but the variational grammar. One chooses admissible spatial and internal directions, writes a linear continuity equation for the matrix density, and minimizes a convex matrix-perspective kinetic energy. Different choices of L_r , symmetrization conventions and boundary conditions give different tensor-valued Wasserstein metrics. These models are useful when

the transported object carries orientation, color covariance, spectral content or a quantum state, and they provide the dynamic counterpart of the finite-dimensional quantum couplings described below.

11.2 Gromov–Wasserstein

Gromov–Wasserstein compares spaces through their internal distance structures rather than through a fixed ambient ground cost. This is the right extension for graphs, shapes and point clouds whose points are not pre-aligned.

Discrete formulation. Optimal transport needs a ground cost C to compare histograms (a, b) , and thus cannot be used directly if the histograms are not defined on the same underlying space, or if one cannot pre-register these spaces to define a ground cost. To address this issue, one can instead use a weaker requirement: two matrices $D \in \mathbb{R}^{n \times n}$ and $D' \in \mathbb{R}^{m \times m}$ are available and represent relationships between the points on which the histograms are defined. A typical scenario is when these matrices are powers of distance matrices. The Gromov–Wasserstein problem reads

$$\text{GW}((a, D), (b, D'))^p := \min_{P \in \mathcal{U}(a, b)} \mathcal{E}_{D, D'}(P) := \sum_{i, j, i', j'} \Delta(D_{i, i'}, D'_{j, j'})^p P_{i, j} P_{i', j'}, \quad (88)$$

where $p \geq 1$ and Δ is a distance on $\mathbb{R}$, typically $\Delta(u, v) = |u - v|$. This is a non-convex quadratic problem over the transport polytope. In the uniform case with the additional hard-assignment constraint $m = n$ and P a permutation matrix, it becomes a Quadratic Assignment Problem (QAP) [113]; this restricted graph-matching form is already NP-hard in full generality. The relaxed coupling formulation used in GW can therefore be read as a soft graph-matching model [114].

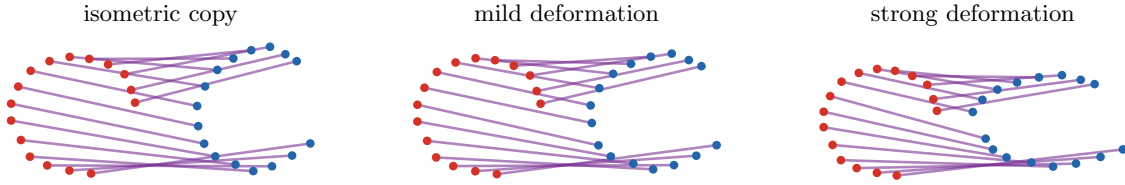


Figure 43: Gromov–Wasserstein correspondences under increasing deformation. The red and blue point clouds are not compared through an ambient Euclidean cross-cost; instead, the GW coupling compares their internal pairwise distances. A perfectly isometric copy admits a clean structural match, while mild and strong deformations progressively bend the correspondence.

When the matrices D, D' are genuine distance matrices, the general construction below shows that GW satisfies the triangle inequality and defines a distance between metric spaces equipped with a probability distribution, up to measure-preserving isometries. This distance was introduced and studied in detail by Memoli in [119]. An in-depth mathematical exposition (in particular, its geodesic structure and gradient flows) is given in [168]. See also [155] for applications in computer vision. Its relation to Hausdorff and Gromov–Hausdorff distances is discussed at the end of this subsection.

General setting. The general setting corresponds to computing couplings between metric measure spaces $\mathbb{X} = (\mathcal{X}, d_{\mathcal{X}}, \alpha)$ and $\mathbb{Y} = (\mathcal{Y}, d_{\mathcal{Y}}, \beta)$ where $(d_{\mathcal{X}}, d_{\mathcal{Y}})$ are distances and (α, β) are measures on their respective spaces. The natural setting is that of Polish metric spaces; compactness is often assumed in this section to avoid existence and integrability issues. One defines

$$\text{GW}(\mathbb{X}, \mathbb{Y})^p := \min_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{\mathcal{X}^2 \times \mathcal{Y}^2} \Delta(d_{\mathcal{X}}(x, x'), d_{\mathcal{Y}}(y, y'))^p d\pi(x, y) d\pi(x', y'). \quad (89)$$

Proposition 52 (Euclidean GW is controlled by Wasserstein). *Let α, β be probability measures on $\mathbb{R}^d$, equipped with the Euclidean distance, and take $\Delta(u, v) = |u - v|$ in (89). Then*

$$\mathcal{GW}((\mathbb{R}^d, \|\cdot\|, \alpha), (\mathbb{R}^d, \|\cdot\|, \beta)) \leq 2\mathcal{W}_p(\alpha, \beta).$$

Proof. Let π be any coupling between α and β . For two independent pairs $(X, Y), (X', Y') \sim \pi$, the reverse triangle inequality gives

$$\|X - X'\| - \|Y - Y'\| \leq \|X - Y\| + \|X' - Y'\|.$$

Taking the L^p norm and using Minkowski gives a bound by $2(\int \|x - y\|^p d\pi)^{1/p}$. Optimizing over π proves the claim. $\square$

In the following, one says that $\mathbb{X}$ and $\mathbb{Y}$ are isometric mm-spaces if there exists $\varphi : \mathcal{X} \rightarrow \mathcal{Y}$ such that $\varphi_\# \alpha = \beta$ and which is a bijective isometry between the supports of α and β , i.e. $d_{\mathcal{Y}}(\varphi(x), \varphi(x')) = d_{\mathcal{X}}(x, x')$ on the support of α .

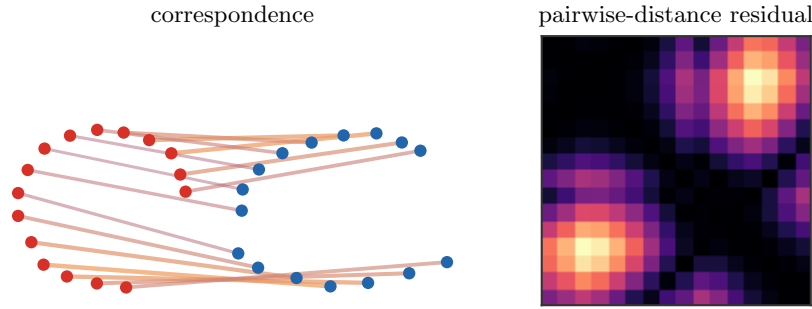


Figure 44: Local distortion in a non-isometric GW match. The left panel colors transport segments by the average residual induced by the displayed hard correspondence. The right panel shows the pairwise-distance residual matrix $|d_{\mathcal{X}}(x_i, x_{i'}) - d_{\mathcal{Y}}(y_{\sigma(i)}, y_{\sigma(i')})|$, which is the local contribution minimized by the discrete GW objective for this correspondence.

Adding features. Fused Gromov–Wasserstein augments the structural term with a feature transport cost. In the discrete case, given a cross-feature cost $M \in \mathbb{R}^{n \times m}$ and a parameter $\eta \in [0, 1]$, one minimizes

$$\min_{P \in \mathcal{U}(\mathbf{a}, \mathbf{b})} (1 - \eta) \sum_{i,j} M_{ij} P_{ij} + \eta \sum_{i,j,i',j'} \Delta(D_{ii'}, D'_{jj'})^p P_{ij} P_{i'j'}.$$

The first term compares node attributes in the usual OT sense, while the second compares the intrinsic geometry. The endpoints $\eta = 0$ and $\eta = 1$ recover feature-only OT and pure GW respectively; intermediate values trade attribute matching against structural matching. This is useful when the two spaces have both distances and features, and these two sources of information may disagree.

11.3 Quantum Optimal Transport

Quantum OT replaces scalar probability vectors by density matrices. In finite dimension, let $\mathbb{H}_n$ be the Hermitian matrices and let $\mathbb{H}_n^{+,1}$ be the positive semidefinite Hermitian matrices with trace one. A coupling between $A \in \mathbb{H}_n^{+,1}$ and $B \in \mathbb{H}_m^{+,1}$ is a positive semidefinite matrix $T \in \mathbb{H}_{nm}^+$ whose partial traces are A and B :

$$\text{Tr}_B(T) = A, \quad \text{Tr}_A(T) = B, \quad (90)$$

where the partial traces are characterized by

$$\text{tr}(F \text{Tr}_B T) = \text{tr}((F \otimes \text{Id}_m)T), \quad \text{tr}(G \text{Tr}_A T) = \text{tr}((\text{Id}_n \otimes G)T)$$

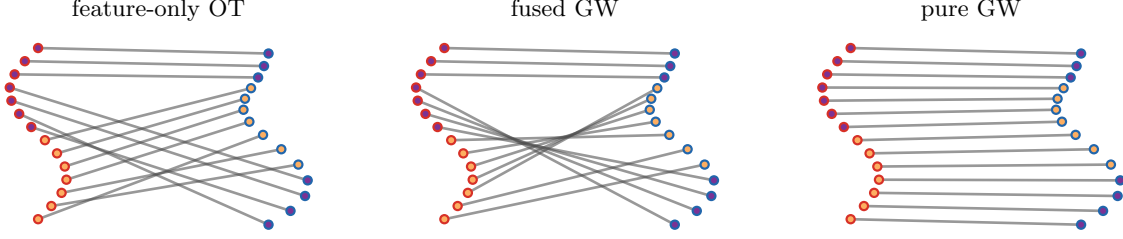


Figure 45: Feature information and intrinsic geometry in fused Gromov–Wasserstein. Small inner disks encode binary node features. Feature-only OT follows the attributes even when this crosses the shape structure, pure GW follows the intrinsic ordering, and fused GW balances the feature term with the pairwise-distance distortion.

for all $F \in \mathbb{H}_n$ and $G \in \mathbb{H}_m$. Given a Hermitian cost observable $C \in \mathbb{H}_{nm}$, the finite-dimensional quantum OT problem is the semidefinite program

$$\text{QOT}_C(A, B) := \min_{T \succeq 0} \{ \text{tr}(CT) : \text{Tr}_B(T) = A, \text{Tr}_A(T) = B \}. \quad (91)$$

This is the direct non-commutative analogue of a Kantorovich coupling: nonnegativity becomes positive semidefiniteness and marginal constraints become partial-trace constraints.

Proposition 53 (Quantum Kantorovich duality). *For $A \in \mathbb{H}_n^{+,1}$ and $B \in \mathbb{H}_m^{+,1}$, the dual of (91) is*

$$\text{QOT}_C(A, B) = \max_{F \in \mathbb{H}_n, G \in \mathbb{H}_m} \{ \text{tr}(FA) + \text{tr}(GB) : F \otimes \text{Id}_m + \text{Id}_n \otimes G \preceq C \}. \quad (92)$$

If A and B are positive definite, strong duality follows directly from Slater’s condition; the semidefinite case follows by restriction to the supports of A and B or by approximation.

Proof. Introduce Hermitian Lagrange multipliers F and G for the two marginal constraints. The Lagrangian is

$$\text{tr}(CT) + \text{tr}(F(A - \text{Tr}_B T)) + \text{tr}(G(B - \text{Tr}_A T)) = \text{tr}(FA) + \text{tr}(GB) + \text{tr}((C - F \otimes \text{Id}_m - \text{Id}_n \otimes G)T),$$

where (90) was used in the last equality. Minimizing over $T \succeq 0$ gives a finite lower bound if and only if $C - F \otimes \text{Id}_m - \text{Id}_n \otimes G \succeq 0$, in which case the infimum in T is 0. This gives the dual program. When $A, B \succ 0$, the coupling $A \otimes B$ is strictly feasible, so Slater’s theorem gives equality of primal and dual values and dual attainment. The general finite-dimensional semidefinite case is obtained by approximation $A_\delta = (1 - \delta)A + \delta \text{Id}_n/n$, $B_\delta = (1 - \delta)B + \delta \text{Id}_m/m$ and by compactness, or equivalently by reducing to the supports of A and B . $\square$

The dual potentials have the usual scalar gauge freedom: replacing (F, G) by $(F + t\text{Id}_n, G - t\text{Id}_m)$ leaves both the constraint and the value unchanged because $\text{tr}(A) = \text{tr}(B) = 1$.

Entropic regularization and Bregman iterations. As in scalar OT, one obtains a smoother problem by adding the von Neumann entropy

$$H(T) = \text{tr}(T(\log T - \text{Id}_{nm})), \quad \nabla H(T) = \log T,$$

with the convention $0 \log 0 = 0$. For $\varepsilon > 0$ define

$$\text{QOT}_C^\varepsilon(A, B) = \min_{T \succeq 0} \{ \text{tr}(CT) + \varepsilon H(T) : \text{Tr}_B(T) = A, \text{Tr}_A(T) = B \}. \quad (93)$$

This is the non-commutative analogue of entropic OT [137, 39]: the Shannon entropy of a coupling is replaced by the trace entropy of a density matrix.

Proposition 54 (Entropic quantum OT duality). *Assume $A \succ 0$, $B \succ 0$ and $\varepsilon > 0$. Then (93) has a unique positive minimizer. Its dual is*

$$\text{QOT}_C^\varepsilon(A, B) = \max_{F \in \mathbb{H}_n, G \in \mathbb{H}_m} \left\{ \text{tr}(FA) + \text{tr}(GB) - \varepsilon \text{tr} \exp \left(\frac{F \otimes \text{Id}_m + \text{Id}_n \otimes G - C}{\varepsilon} \right) \right\}. \quad (94)$$

At optimality, primal and dual variables are linked by the Gibbs formula

$$T_e(F, G) = \exp \left(\frac{F \otimes \text{Id}_m + \text{Id}_n \otimes G - C}{\varepsilon} \right), \quad (95)$$

with $\text{Tr}_B(T_e) = A$ and $\text{Tr}_A(T_e) = B$.

Proof. The feasible set is compact and nonempty, and it contains the positive definite point $A \otimes B$. The trace entropy is strictly convex on positive semidefinite matrices, hence the regularized primal has a unique minimizer. Slater's condition justifies the Lagrange dual computation. The Fenchel identity

$$\sup_{T \succeq 0} \text{tr}(YT) - \varepsilon H(T) = \varepsilon \text{tr} \exp(Y/\varepsilon)$$

is the matrix analogue of the scalar exponential conjugacy. Applying it to the Lagrangian of (93), with $Y = F \otimes \text{Id}_m + \text{Id}_n \otimes G - C$, gives (94). The stationarity condition of this Fenchel identity gives (95); differentiating the dual objective with respect to F and G yields the two marginal equations. $\square$

Writing $K = \exp(-C/\varepsilon)$, the objective in (93) differs by a constant from ε times the quantum KL divergence

$$D_H(T|K) = \text{tr}(T(\log T - \log K) - T + K).$$

The exact quantum analogue of Sinkhorn is an implicit alternating Bregman projection scheme onto the affine marginal sets

$$\mathcal{M}_A = \{T \succeq 0 : \text{Tr}_B(T) = A\}, \quad \mathcal{M}_B = \{T \succeq 0 : \text{Tr}_A(T) = B\}.$$

Proposition 55 (Exact Bregman projections). *Assume $A, B \succ 0$ and let $K = \exp(-C/\varepsilon)$. The minimizer of (93) is equivalently the minimizer of $D_H(T|K)$ over $\mathcal{M}_A \cap \mathcal{M}_B$. Moreover, if a current positive definite matrix has Gibbs form $T_e(F, G)$, then its Bregman projection onto $\mathcal{M}_A$ has the form $T_e(F^+, G)$, where F^+ is chosen so that $\text{Tr}_B T_e(F^+, G) = A$. The projection onto $\mathcal{M}_B$ is analogous. Thus, when each one-block marginal equation is solved exactly, alternating Bregman projections are equivalent to alternating block maximization of the dual (94).*

Proof. Since $\log K = -C/\varepsilon$, the identity

$$\text{tr}(CT) + \varepsilon H(T) = \varepsilon D_H(T|K) - \varepsilon \text{tr}(K)$$

holds, so the primal minimizer is the constrained Bregman projection of K up to an additive constant. For the projection of a positive definite matrix S onto $\mathcal{M}_A$, the affine set contains the positive definite point $A \otimes \text{Id}_m/m$; the entropy derivative $\log T - \log S$ is singular at the boundary, so the projection lies in the interior of the positive cone. Its Lagrangian first variation is

$$\log T - \log S - \Lambda \otimes \text{Id}_m = 0$$

for a Hermitian multiplier Λ . Hence $T = \exp(\log S + \Lambda \otimes \text{Id}_m)$. If $S = T_e(F, G)$, this is again of the form $T_e(F + \varepsilon \Lambda, G)$. The multiplier is fixed by the marginal equation $\text{Tr}_B(T) = A$. The same argument applies to $\mathcal{M}_B$. Finally, the first-order optimality condition for maximizing (94) over one block is exactly the corresponding marginal equation, so the Bregman and block-dual views coincide. $\square$

In the diagonal case this proposition gives the usual multiplicative Sinkhorn updates. In the non-commutative case, however, the exact block equations

$$\mathrm{Tr}_B T_e(F, G) = A, \quad \mathrm{Tr}_A T_e(F, G) = B$$

do not admit scalar division formulas, because the exponential of $F \otimes \mathrm{Id}_m + \mathrm{Id}_n \otimes G - C$ cannot be separated unless the local potential $F \otimes \mathrm{Id}_m + \mathrm{Id}_n \otimes G$ commutes with the cost C . When all matrices are diagonal in the same basis, commutativity restores the scalar form $T_e = \mathrm{diag}(u)K\mathrm{diag}(v)$ and the marginal equations reduce to the usual Sinkhorn divisions.

Gurvits scaling and quantum Sinkhorn. The algorithm often called quantum Sinkhorn comes from the operator-scaling literature of Gurvits and subsequent developments [85, 86, 77, 74]. It replaces the true Gibbs coupling (95) by the symmetric factorization

$$T_s(F, G) = \exp\left(\frac{Z}{2\varepsilon}\right) \exp(-C/\varepsilon) \exp\left(\frac{Z}{2\varepsilon}\right) = (U \otimes V)K(U \otimes V), \quad Z = F \otimes \mathrm{Id}_m + \mathrm{Id}_n \otimes G, \quad (96)$$

where $U = \exp(F/(2\varepsilon))$, $V = \exp(G/(2\varepsilon))$ and $K = \exp(-C/\varepsilon)$. If $[Z, C] = 0$, then $T_s(F, G) = T_e(F, G)$; otherwise this is a Strang-type symmetric surrogate.

Fix a Choi convention and let $\mathcal{K} : \mathbb{H}_m \rightarrow \mathbb{H}_n$ be the completely positive map represented by the positive Choi matrix K ; let $\mathcal{K}^*$ be its Hilbert–Schmidt adjoint. Up to the transpose dictated by the chosen Choi convention, the marginal equations for the symmetric coupling take the operator-scaling form

$$U \mathcal{K}(V^2) U = A, \quad V \mathcal{K}^*(U^2) V = B.$$

They can be enforced by the explicit congruence normalizations

$$\begin{aligned} R_V &= \mathcal{K}(V^2), \quad U \leftarrow R_V^{-1/2} (R_V^{1/2} A R_V^{1/2})^{1/2} R_V^{-1/2}, \\ S_U &= \mathcal{K}^*(U^2), \quad V \leftarrow S_U^{-1/2} (S_U^{1/2} B S_U^{1/2})^{1/2} S_U^{-1/2}. \end{aligned} \quad (97)$$

These inverse square roots are well-defined when $K \succ 0$ and $U, V, A, B \succ 0$. This is Gurvits/operator scaling with prescribed targets; when all matrices are diagonal it reduces to classical Sinkhorn scaling, and when the targets are proportional to identities it matches the usual bistochastic operator-scaling normalization, up to the conventional trace normalization.

It is important not to identify (97) with the exact Bregman scheme for (93). The exact Bregman step would enforce the marginals of $T_e(F, G) = \exp((Z - C)/\varepsilon)$ and would be a block maximization of the true concave dual (94). Gurvits scaling instead enforces the marginals of the surrogate

$$T_s = \exp\left(\frac{Z}{2\varepsilon}\right) \exp(-C/\varepsilon) \exp\left(\frac{Z}{2\varepsilon}\right).$$

The two coincide in the commuting/diagonal regime, but in general the Baker–Campbell–Hausdorff commutator terms do not vanish. The Gurvits iteration should therefore be understood as a tractable symmetric operator-scaling approximation to entropic Q–OT, not as the literal alternating KL projection algorithm.

Remark 17 (Operator-valued couplings). The same definitions extend formally from matrices to separable Hilbert spaces by replacing density matrices with positive trace-class operators of trace one, observables with bounded self-adjoint operators and (90) with partial traces defined by duality against local bounded observables. If $\Pi(A, B)$ denotes positive trace-class operators with partial traces A and B , a bounded cost observable C gives the problem $\inf_{T \in \Pi(A, B)} \mathrm{Tr}(CT)$. For unbounded positive costs one must define the energy through the quadratic form or spectral truncations, and in the entropic case one must ensure that the Gibbs operator $\exp(-C/\varepsilon)$ is trace class and that the partial traces of the candidate coupling are well-defined. The matrix formulas above are therefore the clean finite-dimensional core; the operator version adds domain and compactness assumptions rather than a different algebraic structure.

12 Dynamic Optimal Transport

Optimal transport becomes especially powerful once distances between measures are seen as actions of moving mass. This section first develops the dynamic language: continuity equations describe admissible measure evolutions, while the Benamou–Brenier formula identifies $\mathcal{W}_2$ with a least-action principle. These ideas prepare the gradient-flow and generative-model sections that follow.

12.1 Evolutions over the Space of Measures

We start with the continuity equation because it is the common language for particles, densities and weak measure evolutions. It also makes precise which velocity fields actually move mass.

Lagrangian and Eulerian descriptions. We consider the evolution $t \mapsto \alpha_t \in \mathcal{P}(\mathbb{R}^d)$. Such an evolution can be described in a “Lagrangian” way as the advection of particles along a (time-dependent) vector field $v_t(x)$ in $\mathbb{R}^d$. At the particle level, this advection is governed by

$$\frac{dx(t)}{dt} = v_t(x(t)), \quad (98)$$

and we write T_t for the associated flow map, so that $T_t(x(0)) = x(t)$. The advected measure is then $\alpha_t = (T_t)_\# \alpha_0$. For discrete measures, $\alpha_t = \frac{1}{n} \sum_{i=1}^n \delta_{x_i(t)}$, meaning each $x_i(t)$ solves (98).

In the Eulerian description, the same motion is written directly on the evolving measure: the particle ODE becomes the PDE

$$\frac{\partial \alpha_t}{\partial t} + \operatorname{div}(v_t \alpha_t) = 0. \quad (99)$$

This PDE is often referred to as the advection equation, the continuity equation, or Liouville’s equation when operating over a phase space. For singular measures, and in particular for empirical measures, it is understood in the distributional sense:

$$\frac{d}{dt} \int_{\mathbb{R}^d} \varphi(x) d\alpha_t(x) = \int_{\mathbb{R}^d} \langle \nabla \varphi(x), v_t(x) \rangle d\alpha_t(x) \quad (100)$$

for every smooth compactly supported test function φ . This weak form is exactly what is obtained by differentiating $\frac{1}{n} \sum_i \varphi(x_i(t))$ when the measure is empirical.

Benamou–Brenier action. The dynamic formulation of the quadratic Wasserstein distance is

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{\substack{\partial_t \alpha_t + \operatorname{div}(\alpha_t v_t) = 0 \\ \alpha_{t=0} = \alpha_0, \alpha_{t=1} = \alpha_1}} \int_0^1 \int_{\mathbb{R}^d} \|v_t(x)\|^2 d\alpha_t(x) dt. \quad (101)$$

For a prescribed smooth curve $(\alpha_t)_t$, the tangent velocity of minimal kinetic energy is the solution of the constrained least-squares problem

$$v_t = \operatorname{argmin}_v \left\{ \int \|v(x)\|^2 d\alpha_t(x) ; \partial_t \alpha_t + \operatorname{div}(\alpha_t v) = 0 \right\}. \quad (102)$$

If $\alpha_t = \rho_t dx$ is a smooth positive density, this minimizer is a gradient field $v_t = \nabla \varphi_t$ where

$$-\operatorname{div}(\rho_t \nabla \varphi_t) = \partial_t \rho_t. \quad (103)$$

While this is not the focus of this chapter, let us note that although the formulation (101) is not jointly convex in (α_t, v_t) , for density curves $\alpha_t = \rho_t dx$ the change of variable $m_t := \rho_t v_t$ gives the convex perspective form

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{\substack{\partial_t \rho_t + \operatorname{div} m_t = 0 \\ \rho_{t=0} dx = \alpha_0, \rho_{t=1} dx = \alpha_1}} \int_0^1 \int_{\mathbb{R}^d} \frac{\|m_t(x)\|^2}{\rho_t(x)} dx dt,$$

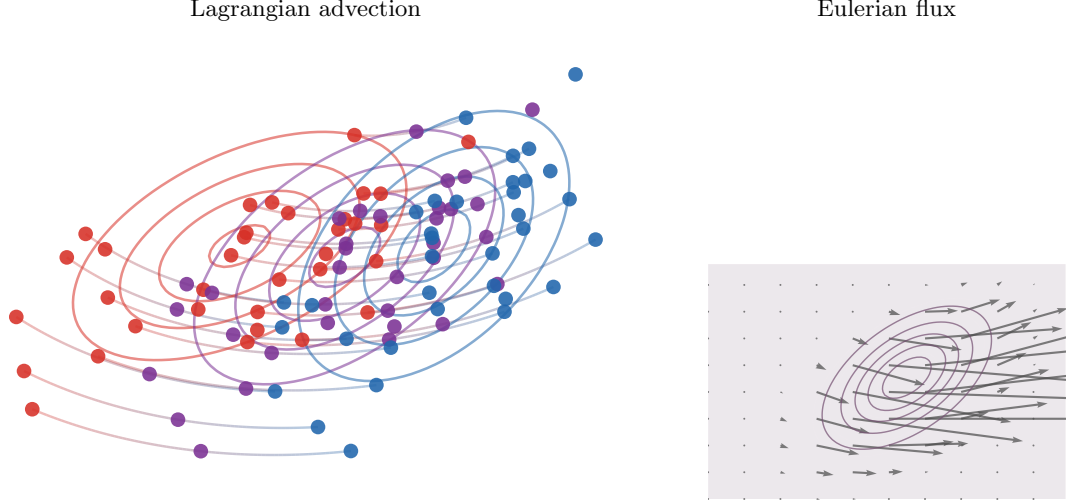


Figure 46: Two views of the continuity equation. The left panel advects particles and Gaussian density contours under a smooth affine vector field, showing the Lagrangian flow $x'(t) = v(x(t))$. The right panel shows the Eulerian flux $m_t = \rho_t v_t$ at an intermediate time, where $\alpha_t = \rho_t dx$: arrows are weighted by the local density, so the velocity contributes to the PDE only where mass is present.

with the usual convention that the integrand is 0 when $(\rho_t, m_t) = (0, 0)$ and $+\infty$ when $\rho_t = 0$ but $m_t \neq 0$. This convex reformulation enables geodesic interpolation by convex optimization once the domain is discretized.



Figure 47: Benamou–Brenier geodesic between two sampled silhouettes. A discrete quadratic OT plan between the cat and two-disks point clouds induces the McCann interpolation $Z_t = (1 - t)X + tY$, which is the Lagrangian realization of the least-action solution. The left panel shows kernel-density contours along this geodesic, and the right panel displays midpoint particles with their constant characteristic velocities $Y - X$.

Remark 18 (Path-space formulation). Let $\mathcal{S} = C([0, 1]; \mathbb{R}^d)$ be the space of continuous paths endowed with the uniform topology. For $t \in [0, 1]$ define the evaluation map

$$P_t : \mathcal{S} \rightarrow \mathbb{R}^d, \quad P_t(\gamma) = \gamma(t).$$

The Benamou–Brenier cost admits the equivalent formulation over probability measures concentrated on absolutely continuous paths,

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{M \in \mathcal{P}(\mathcal{S})} \left\{ \int_{\mathcal{S}} \int_0^1 |\dot{\gamma}(t)|^2 dt dM(\gamma) : (P_0)_\# M = \alpha_0, (P_1)_\# M = \alpha_1 \right\}.$$

If α_0 has a density, the minimizer M^* is unique. Its time marginals reproduce the optimal curve: $\alpha_t = (P_t)_\# M^*$ for all t . Furthermore, for a.e. t , denoting $Q_t(\gamma) = \dot{\gamma}(t)$ on the absolutely continuous paths where

this derivative is defined, the conditional law of the velocity is deterministic:

$$(P_t, Q_t)_\# M^*(dx, dq) = \alpha_t(dx) \delta_{v_t^*(x)}(dq),$$

where v_t^* is the optimal velocity field in the Benamou–Brenier formulation. Hence M^* concentrates on straight-line geodesics and, for a.e. t , assigns exactly one direction at α_t -a.e. spatial point.

12.2 Extensions of the Dynamic Formulation

The same variational grammar extends beyond the quadratic Wasserstein distance. One changes either the kinetic exponent, the mobility or the balance equation, while keeping a continuity-type constraint and a convex perspective action.

Remark 19 (Generalized Benamou–Brenier distances). The dynamic formulation is not specific to $\mathcal{W}_2$. For measures with finite p -th moments and $p > 1$, one has the analogous action formula

$$\mathcal{W}_p^p(\alpha_0, \alpha_1) = \inf_{\substack{\partial_t \alpha_t + \nabla \cdot (\alpha_t v_t) = 0 \\ \alpha_{t=0} = \alpha_0, \alpha_{t=1} = \alpha_1}} \int_0^1 \int_{\mathbb{R}^d} |v_t(x)|^p d\alpha_t(x) dt.$$

When $\alpha_t = \rho_t dx$ and $m_t = \rho_t v_t$, this becomes the convex perspective action

$$\int_0^1 \int_{\mathbb{R}^d} \frac{|m_t(x)|^p}{\rho_t(x)^{p-1}} dx dt, \quad \partial_t \rho_t + \nabla \cdot m_t = 0,$$

with the usual convention that the integrand is 0 if $(\rho, m) = (0, 0)$ and $+\infty$ if $\rho = 0$ but $m \neq 0$.

A second class of variants changes the mobility of the medium: the quadratic action $|m|^2/\rho$ is replaced by $|m|^2/\theta(\rho)$ for a suitable concave mobility θ . Under appropriate structural assumptions, this produces transport metrics adapted to nonlinear diffusions and finite-volume discretizations [56]. On finite graphs and Markov chains, the analogous action uses an edge mobility, often the logarithmic mean of the endpoint densities, and leads to discrete Wasserstein geometries [116, 124]. A third class allows creation and destruction of mass by replacing the continuity equation by a balance equation and penalizing both transport and growth. A representative quadratic action is

$$\partial_t \rho_t + \nabla \cdot m_t = s_t, \quad \int_0^1 \int \left(\frac{|m_t|^2}{\rho_t} + \kappa^2 \frac{s_t^2}{\rho_t} \right) dx dt,$$

with the same perspective convention. This leads, up to normalization conventions, to the Wasserstein–Fisher–Rao or Hellinger–Kantorovich geometry for unbalanced measures [111, 47, 46]. These extensions keep the same variational grammar as Benamou–Brenier: a balance equation, an action density, and geodesics obtained by minimizing an integrated kinetic cost.

13 Wasserstein Gradient Flows

Once $\mathcal{W}_2$ is a dynamic metric, one can run gradient descent directly on the space of measures. This section derives the formal Wasserstein gradient, explains the JKO minimizing-movement scheme, records the role of geodesic convexity in convergence, and then applies the same calculus to mean-field neural-network training.

13.1 Minimizing Movements and Wasserstein Gradients

This first subsection explains how a variational implicit-Euler step on measures gives rise, in the small-step limit, to a continuity equation driven by the Wasserstein gradient of the energy.

We now consider a function $f(\alpha)$ and seek a minimizing evolution $(\alpha_t)_t$. The general strategy of minimizing movement over a metric space is to construct a discrete-time evolution using an implicit Euler scheme:

$$\alpha_{t+\tau} := \arg \min_{\alpha} \frac{1}{2\tau} \mathcal{W}_2(\alpha_t, \alpha)^2 + f(\alpha). \quad (104)$$

Euclidean gradient flows. If we restrict (104) to finite dimensions and assume $\alpha_t = \delta_{x(t)}$ and $\alpha = \delta_x$ (single Dirac measures), this matches the implicit Euler scheme:

$$x(t + \tau) := \arg \min_x \frac{1}{2\tau} \|x - x(t)\|^2 + h(x),$$

where $h(x) = f(\delta_x)$. Its solution is formally given by the implicit Euler formula:

$$x(t + \tau) = (\text{Id} + \tau \nabla h)^{-1}(x(t)).$$

In contrast, the explicit Euler scheme is:

$$x(t + \tau) = (\text{Id} - \tau \nabla h)(x(t)) = x(t) - \tau \nabla h(x(t)).$$

Both schemes converge as $\tau \rightarrow 0$ to:

$$\dot{x}(t) = -\nabla h(x(t)). \quad (105)$$

Wasserstein gradient formula. The implicit Euler scheme has the advantage that it does not require h or f to be smooth. For f , this is crucial to handle evolutions over measures that may have densities, atoms or other singular parts.

As $\tau \rightarrow 0$, under certain conditions on f , (104) defines a continuous evolution $t \mapsto \alpha_t$. As discussed earlier, this evolution can be described as a Lagrangian evolution (98). A key point is that it provides an explicit vector field v_t (depending on α_t), denoted as $\nabla_{\mathcal{W}} f(\alpha)$, called the Wasserstein gradient. In the weak sense, α_t satisfies:

$$\frac{\partial \alpha_t}{\partial t} + \text{div}(-\nabla_{\mathcal{W}} f(\alpha_t) \alpha_t) = 0. \quad (106)$$

Here, $\nabla_{\mathcal{W}} f(\alpha_t)(x) \in \mathbb{R}^d$ is a vector field and is a gradient (similar to the computation in (102)), computable as:

$$\nabla_{\mathcal{W}} f(\alpha) = \nabla_x \varphi, \quad \text{where } \varphi := \delta f(\alpha).$$

The function $\delta f(\alpha) \in \mathcal{C}(\mathbb{R}^d)$ is known as the first variation or Fréchet (directional) derivative, satisfying for any $\beta \in \mathcal{P}(\mathbb{R}^d)$:

$$f((1 - \tau)\alpha + \tau\beta) = f(\alpha + \tau\rho) = f(\alpha) + \tau \int [\delta f(\alpha)](x) d\rho(x) + o(\tau),$$

where $\rho = \beta - \alpha$ is a signed measure of total mass zero.

Proposition 6 (Formal Wasserstein gradient). *Assume that f admits a smooth first variation $\delta f(\alpha)$ and that α has a smooth positive density. For infinitesimal perturbations generated by a velocity field v through $(\text{Id} + \tau v)_\# \alpha$, the differential of f is*

$$\frac{d}{d\tau} \Big|_{\tau=0} f((\text{Id} + \tau v)_\# \alpha) = \int \langle \nabla \delta f(\alpha)(x), v(x) \rangle d\alpha(x).$$

Hence, for the Riemannian metric $\|v\|_{L^2(\alpha)}^2 = \int \|v\|^2 d\alpha$, the Wasserstein gradient is the vector field

$$\nabla_{\mathcal{W}} f(\alpha) = \nabla \delta f(\alpha).$$

Proof. The push-forward expansion gives, in the sense of distributions,

$$(\text{Id} + \tau v)_\# \alpha = \alpha - \tau \text{div}(\alpha v) + o(\tau).$$

Using the definition of the first variation,

$$f((\text{Id} + \tau v)_\# \alpha) = f(\alpha) - \tau \int \delta f(\alpha) \text{div}(\alpha v) dx + o(\tau).$$

An integration by parts, with either compact support or vanishing boundary flux, gives

$$-\int \delta f(\alpha) \operatorname{div}(\alpha v) dx = \int \langle \nabla \delta f(\alpha), v \rangle d\alpha.$$

By definition of the Riesz representative for the $L^2(\alpha)$ metric, this representative is $\nabla \delta f(\alpha)$. $\square$

The Wasserstein gradient-flow viewpoint already appears in John D. Lafferty’s PhD work, published as “The Density Manifold and Configuration Space Quantization”, under the name “density manifold”. It was then systematically developed by Otto, who exposed the formal Riemannian structure of this space [131]. Rigorous metric-space treatments and numerical JKO schemes can be found in [6, 15, 133, 71].

From the JKO step to the velocity field. A first-order expansion of the JKO step explains why (106) uses the vector field $\nabla_{\mathcal{W}} f(\alpha)$. Write (104) as a minimization over displacement fields v such that $\alpha = (\operatorname{Id} + \tau v)_\# \alpha_t$:

$$\min_v \frac{1}{2\tau} \tau^2 \|v\|_{L^2(\alpha_t)}^2 + f((\operatorname{Id} + \tau v)_\# \alpha_t).$$

Then we perform a first-order Taylor expansion of this formulation using

$$(\operatorname{Id} + \tau v)_\# \alpha_t = \alpha_t - \tau \operatorname{div}(v \alpha_t) + o(\tau)$$

$$\begin{aligned} f((\operatorname{Id} + \tau v)_\# \alpha_t) &= f(\alpha_t) - \tau \int \delta f(\alpha_t) \operatorname{div}(v \alpha_t) dx + o(\tau) \\ &= f(\alpha_t) + \tau \int \langle \nabla_x \delta f(\alpha_t)(x), v(x) \rangle d\alpha_t(x) + o(\tau) \end{aligned}$$

to obtain the following first-order expansion in τ of the problem minimized in (104)

$$\min_v f(\alpha_t) + \tau \int \left[\frac{1}{2} \|v(x)\|^2 + \langle \nabla_{\mathcal{W}} f(\alpha_t)(x), v(x) \rangle \right] d\alpha_t(x) + o(\tau).$$

The pointwise minimizer is $v = -\nabla_{\mathcal{W}} f(\alpha_t)$, which gives the velocity in the continuity equation. We now detail examples of such Wasserstein gradient flows.

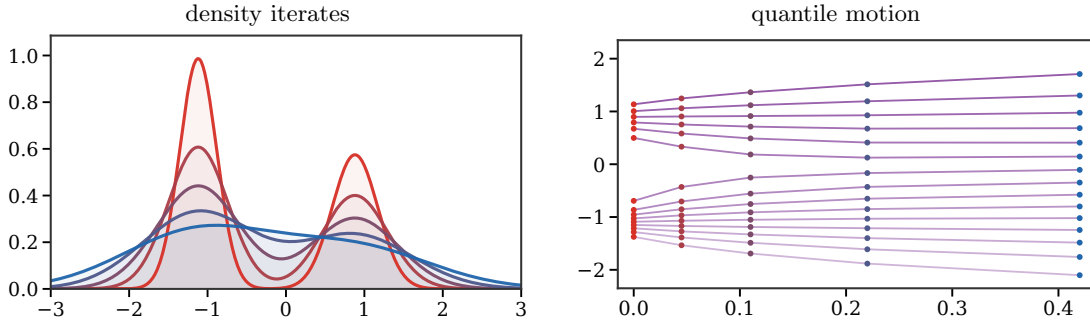


Figure 48: JKO minimizing movements for the entropy flow in one dimension. The left panel displays successive implicit-Euler minimizers for the heat equation, colored from red to blue. The right panel tracks inverse CDF values $Q_t(s) = F_t^{-1}(s)$ for selected probability levels s , giving a Lagrangian view of the proximal movement in Wasserstein space.

Discrete evolutions. If $f(\alpha)$ can be evaluated on discrete distributions and $\nabla_{\mathcal{W}}$ is continuous in this case, the flow (106) maintains the number of Dirac masses, $\alpha_t = \frac{1}{n} \sum_i \delta_{x_i(t)}$. The particles $X(t) := (x_i(t))_i$ evolve according to a system of coupled ODEs:

$$\dot{x}_i(t) = -n \nabla_{x_i} F(X(t)), \quad (107)$$

where $F(X) := f\left(\frac{1}{n} \sum_i \delta_{x_i}\right)$ and the factor n comes from the empirical Wasserstein metric $\frac{1}{n} \sum_i \|\dot{x}_i\|^2$.

Linear Functionals. The simplest example of flows is for linear functions

$$f(\alpha) = \int h(x) d\alpha(x). \quad (108)$$

Here, $\delta f(\alpha) = h$ is a fixed function (independent of α). The flow (106) becomes:

$$\frac{\partial \alpha_t}{\partial t} + \operatorname{div}(-\nabla h \alpha_t) = 0.$$

This implies particles move independently according to the usual gradient flow (105).

Shannon Neg-Entropy. A very different behavior is obtained by considering functionals that require α_t to have a density, the canonical example being Shannon neg-entropy

$$f(\alpha) = \int \log\left(\frac{d\alpha}{dx}(x)\right) d\alpha(x). \quad (109)$$

Here, $\delta f(\alpha) = \log\left(\frac{d\alpha}{dx}\right)$ up to an additive constant, so $\nabla_{\mathcal{W}} f(\alpha) = \frac{\nabla \alpha}{\alpha}$ (often called the score). The flow (106) becomes the heat equation:

$$\partial_t \alpha_t = \Delta \alpha_t.$$

Other entropy functionals lead to nonlinear diffusion equations; finite-volume and particle discretizations are discussed in [38, 78, 116, 61]. For example, a generalized entropy of the form

$$f(\alpha) = \int g\left(\frac{d\alpha}{dx}\right) dx, \quad (110)$$

for a scalar convex function g , leads to nonlinear diffusions in the smooth-density regime:

$$\frac{\partial \alpha_t}{\partial t} = \Delta(P(\alpha_t)),$$

where the pressure P satisfies $P'(s) = sg''(s)$. For example, $g(s) = s \log(s)$ gives $P(s) = s$ and recovers (109), while $g(s) = s^m/(m-1)$, $m > 1$, gives $P(s) = s^m$ up to an additive constant and yields the porous-medium equation. A celebrated theorem by McCann [117] states that an internal energy of the form (110), for $g : \mathbb{R}^+ \rightarrow \mathbb{R} \cup \{+\infty\}$ with $g(0) = 0$, is geodesically convex on $\mathcal{P}(\mathbb{R}^d)$ when g is convex and the map $r \mapsto r^d g(r^{-d})$ is convex and nonincreasing on $(0, +\infty)$. Examples of such functions are $g(s) = s^q$ for $q > 1$ and Shannon entropy $g(s) = s \log(s)$. By contrast, $g(s) = -\log(s)$, associated with the reverse KL divergence, does not satisfy this displacement-convexity criterion.

Interaction Energies. In a similar spirit, to obtain nonlinear evolutions without requiring the measure to have density, one can consider

$$f(\alpha) := \iint k(x, y) d\alpha(x) d\alpha(y). \quad (111)$$

For a symmetric kernel k :

$$\delta f(\alpha)(x) = 2 \int k(x, y) d\alpha(y), \quad \nabla_{\mathcal{W}} f(\alpha)(x) = 2 \int \nabla_x k(x, y) d\alpha(y).$$

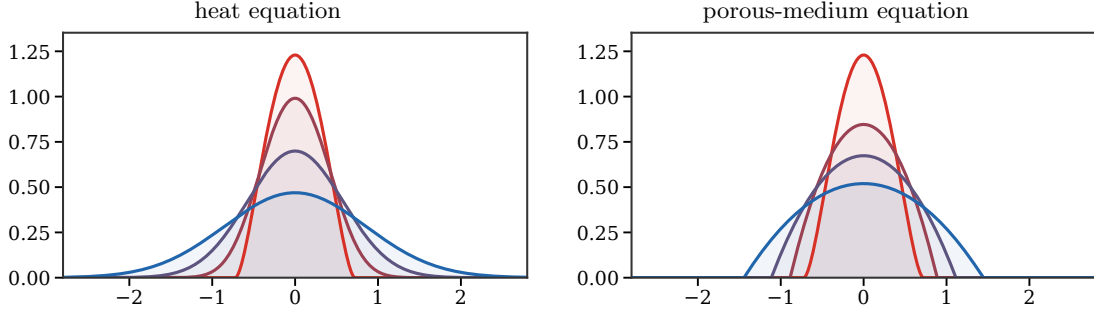


Figure 49: Two entropy-driven Wasserstein gradient flows from the same compact initial density. The heat flow is generated by Shannon entropy $g(\rho) = \rho \log \rho$ and instantly develops Gaussian tails. The porous-medium flow uses the power entropy $g(\rho) = \rho^m / (m - 1)$ with $m = 2$, hence $\partial_t \rho = \Delta(\rho^2)$; it spreads more slowly and keeps a compact free boundary.

For $\alpha_0 = \frac{1}{n} \sum_i \delta_{x_i}$, the flow (106) implies particles $(x_i(t))_i$ obey:

$$\dot{x}_i(t) = -\frac{2}{n} \sum_j \nabla k(x_i(t), x_j(t)).$$

If k is positive definite and one minimizes the squared MMD distance to a teacher distribution β , then

$$\|\alpha - \beta\|_k^2 = \iint k d\alpha d\alpha - 2 \int \left(\int k(x, y) d\beta(y) \right) d\alpha(x) + \text{constant}.$$

Thus MMD training energies are exactly an interaction energy plus a linear potential; the teacher distribution appears through the potential $x \mapsto -2 \int k(x, y) d\beta(y)$.

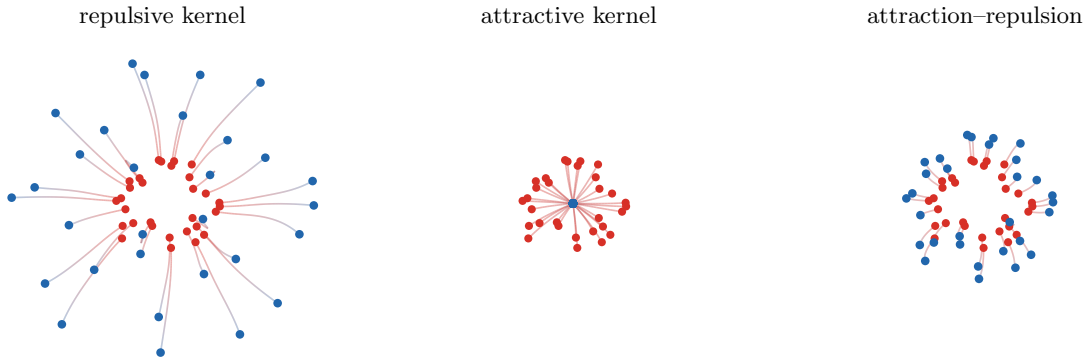


Figure 50: Interaction-energy particle flows for three choices of k . A positive Gaussian kernel $k(x, y) = \exp(-\|x - y\|^2 / (2\sigma^2))$ produces short-range repulsion under Wasserstein descent; changing its sign produces attraction and collapse; adding a quadratic long-range attraction to the repulsive kernel yields a balanced attraction–repulsion dynamics. In all cases the particles follow the coupled ODE structure of (107).

13.2 Geodesic Convexity and Convergence

Geodesic convexity is the convexity notion adapted to Wasserstein geometry. It is the condition that turns the formal gradient-flow calculus into a convergence theory.

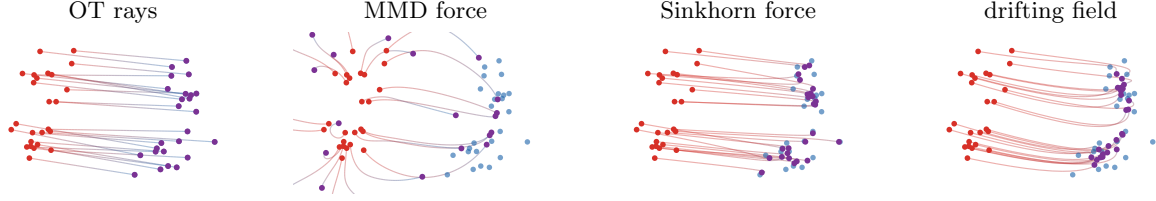


Figure 51: Particle trajectories induced by different discrepancy geometries. The red particles and blue target cloud are the same in all panels. Straight OT displacement produces rays from an optimal matching; an MMD-type witness field gives smoother nonlocal forces; the Sinkhorn-divergence force is an entropic, debiased transport attraction; and the normalized drifting field combines attraction to data with self-repulsion. The figure is qualitative: it compares geometric behavior, not solver performance.

Geodesics and convexity. A constant-speed $\mathcal{W}_2$ geodesic between α_0 and α_1 is obtained from any optimal coupling $\pi^* \in \mathcal{U}(\alpha_0, \alpha_1)$ by the McCann interpolation

$$\alpha_t = ((1-t)P_0 + tP_1)_\# \pi^*, \quad t \in [0, 1],$$

where $P_0(x, y) = x$ and $P_1(x, y) = y$. If the optimal plan is induced by a Brenier map T , this reduces to $((1-t)\text{Id} + tT)_\# \alpha_0$. The coupling formula is important because geodesics exist even when no Monge map exists, for instance when a Dirac mass must split.

Definition 16 (Geodesic convexity). *A functional f on $\mathcal{P}_2(\mathbb{R}^d)$ is geodesically convex if for every $\mathcal{W}_2$ geodesic $(\alpha_t)_t$,*

$$f(\alpha_t) \leq (1-t)f(\alpha_0) + tf(\alpha_1).$$

It is λ -geodesically convex if the right-hand side is improved by $-\frac{\lambda}{2}t(1-t)\mathcal{W}_2^2(\alpha_0, \alpha_1)$.

Proposition 56 (Basic geodesically convex energies). *The following formal statements hold on $\mathcal{P}_2(\mathbb{R}^d)$.*

1. *If h is convex, then $\alpha \mapsto \int h d\alpha$ is geodesically convex; if h is λ -strongly convex, it is λ -geodesically convex.*
2. *If $W(x-y)$ is convex as a function of the displacement, then $\alpha \mapsto \frac{1}{2} \iint W(x-y) d\alpha(x) d\alpha(y)$ is geodesically convex.*
3. *Shannon entropy $\alpha \mapsto \int \rho \log \rho dx$ is geodesically convex.*
4. *The relative entropy $\text{KL}(\alpha|\gamma)$ with $d\gamma = e^{-V} dx/Z$ is λ -geodesically convex when V is λ -strongly convex.*

Proof. Along a Monge geodesic $X_t = (1-t)X_0 + tX_1$, convexity of h gives $h(X_t) \leq (1-t)h(X_0) + th(X_1)$, and strong convexity gives the additional quadratic term; integrating proves the first claim. The interaction claim follows similarly by applying convexity of W to pairwise differences $X_t - X'_t = (1-t)(X_0 - X'_0) + t(X_1 - X'_1)$ and integrating over two independent copies. The entropy claim is McCann's displacement convexity theorem; at the density level it follows from the concavity of the Jacobian determinant under the interpolation of optimal maps. Finally, $\text{KL}(\alpha|\gamma) = \int \rho \log \rho dx + \int V d\alpha + \text{constant}$, so it is the sum of displacement-convex entropy and a λ -geodesically convex linear potential. $\square$

Convergence of the flow. In general, analyzing (106) is delicate. The cleanest case is when f is geodesically convex in the sense above. This condition is the Wasserstein analogue of convexity in Euclidean gradient descent.

Proposition 57 (Energy decay for convex Wasserstein flows). *Assume formally that f is geodesically convex, admits a smooth first variation, and has a minimizer α^* . Let $(\alpha_t)_t$ be a smooth solution of the Wasserstein gradient flow*

$$\partial_t \alpha_t + \operatorname{div}(\alpha_t v_t) = 0, \quad v_t = -\nabla_{\mathcal{W}} f(\alpha_t).$$

Then

$$\frac{d}{dt} f(\alpha_t) = - \int \|\nabla_{\mathcal{W}} f(\alpha_t)(x)\|^2 d\alpha_t(x) \leq 0.$$

If T_t is the optimal map from α_t to α^* , then

$$f(\alpha_t) - f(\alpha^*) \leq -\frac{d}{dt} \frac{1}{2} \mathcal{W}_2^2(\alpha_t, \alpha^*),$$

and consequently

$$f(\alpha_t) - f(\alpha^*) \leq \frac{\mathcal{W}_2^2(\alpha_0, \alpha^*)}{2t}.$$

If f is λ -geodesically convex with $\lambda > 0$, then

$$f(\alpha_t) - f(\alpha^*) \leq e^{-2\lambda t} (f(\alpha_0) - f(\alpha^*)).$$

Proof. The chain rule and Proposition 6 give

$$\frac{d}{dt} f(\alpha_t) = \int \langle \nabla_{\mathcal{W}} f(\alpha_t)(x), v_t(x) \rangle d\alpha_t(x) = - \int \|\nabla_{\mathcal{W}} f(\alpha_t)(x)\|^2 d\alpha_t(x).$$

Geodesic convexity along the geodesic $((1-s)\operatorname{Id} + sT_t)_\# \alpha_t$ gives

$$f(\alpha^*) - f(\alpha_t) \geq \int \langle \nabla_{\mathcal{W}} f(\alpha_t)(x), T_t(x) - x \rangle d\alpha_t(x).$$

Since $v_t = -\nabla_{\mathcal{W}} f(\alpha_t)$, this reads

$$f(\alpha_t) - f(\alpha^*) \leq \int \langle v_t(x), T_t(x) - x \rangle d\alpha_t(x).$$

The standard first-variation formula for the squared Wasserstein distance gives

$$\frac{d}{dt} \frac{1}{2} \mathcal{W}_2^2(\alpha_t, \alpha^*) = \int \langle x - T_t(x), v_t(x) \rangle d\alpha_t(x),$$

which proves the differential inequality. Integrating it from 0 to t and using the monotonicity of $s \mapsto f(\alpha_s)$ gives

$$t(f(\alpha_t) - f(\alpha^*)) \leq \int_0^t (f(\alpha_s) - f(\alpha^*)) ds \leq \frac{1}{2} \mathcal{W}_2^2(\alpha_0, \alpha^*).$$

If f is λ -geodesically convex, the Wasserstein analogue of strong convexity gives the slope inequality

$$\int \|\nabla_{\mathcal{W}} f(\alpha_t)\|^2 d\alpha_t \geq 2\lambda (f(\alpha_t) - f(\alpha^*)).$$

Combining it with the energy dissipation identity yields

$$\frac{d}{dt} (f(\alpha_t) - f(\alpha^*)) \leq -2\lambda (f(\alpha_t) - f(\alpha^*)),$$

and Gronwall's lemma gives the exponential rate. $\square$

Proposition 58 (Convex examples covered by the theory). *The hypotheses of Proposition 57 are satisfied in the following standard cases, at least at the formal smooth level used in this section.*

1. For the linear energy $f(\alpha) = \int h d\alpha$, geodesic convexity holds when h is convex. If h is λ -strongly convex, then f is λ -geodesically convex and the flow enjoys the exponential rate of Proposition 57.
2. For the interaction energy $f(\alpha) = \frac{1}{2} \iint W(x - y) d\alpha(x) d\alpha(y)$, geodesic convexity holds when W is convex and even. This covers repulsive or attractive pairwise models whose displacement cost has no non-convex wells.
3. The Shannon entropy $f(\alpha) = \int \rho \log \rho dx$ and, more generally, McCann displacement-convex internal energies generate diffusion-type Wasserstein gradient flows.
4. If $\gamma = Z^{-1} e^{-V} dx$ and V is λ -strongly convex, then the relative entropy $\text{KL}(\alpha|\gamma)$ is λ -geodesically convex. Its flow is the Fokker-Planck equation with invariant law γ .

Proof. Let $(\alpha_t)_t$ be the McCann interpolation between α_0 and α_1 , written with an optimal coupling as $X_t = (1 - t)X_0 + tX_1$. For a linear energy, Jensen's inequality gives

$$h(X_t) \leq (1 - t)h(X_0) + th(X_1),$$

and the strong convexity version gives the additional term $-\frac{\lambda}{2}t(1 - t)\|X_0 - X_1\|^2$. Integrating over the optimal coupling proves geodesic convexity and λ -geodesic convexity.

For interaction energies, use two independent copies of the optimal coupling. The pairwise displacement evolves as

$$X_t - X'_t = (1 - t)(X_0 - X'_0) + t(X_1 - X'_1).$$

Convexity of W gives the convexity inequality after integration over the product coupling. Evenness of W ensures that the interaction is symmetric in the two particles and matches the usual factor $1/2$ in (111).

The entropy claim is McCann's displacement-convexity theorem. For smooth positive densities and Brenier maps, it follows from the change-of-variables formula and the concavity of the determinant along positive matrices; the general statement is obtained by approximation. Finally,

$$\text{KL}(\alpha|\gamma) = \int \rho \log \rho dx + \int V d\alpha + \log Z,$$

so it is the sum of the displacement-convex entropy and the λ -geodesically convex linear potential generated by V . Proposition 57 then applies to all four cases. $\square$

Outside this convex regime, such as in the mean-field neural-network example below, the flow may still be informative but its convergence analysis requires problem-specific arguments.

13.3 Training Two-Layer MLPs as Wasserstein Flows

Mean-field limits recast the training of wide neural networks as transport of a distribution of neurons. This subsection shows how the particle ODE of gradient descent becomes a Wasserstein flow in parameter space.

We use $z \in \mathbb{R}^d$ for the input data and $y \in \mathbb{R}^{d'}$ for the label. A neuron is a particle

$$x = (u, v) \in \mathbb{R}^d \times \mathbb{R}^{d'},$$

where u is the inner weight and v is the outer vector weight. For a scalar nonlinearity σ , define the vector-valued feature

$$\psi(x, z) = v \sigma(\langle u, z \rangle) \in \mathbb{R}^{d'}.$$

The width- n network and its mean-field version are

$$G_X(z) = \frac{1}{n} \sum_{i=1}^n \psi(x_i, z), \quad G_\alpha(z) = \int \psi(x, z) d\alpha(x), \quad \alpha = \frac{1}{n} \sum_i \delta_{x_i}.$$

This formulation removes the artificial ordering of neurons and allows α to be a continuous distribution of infinitely many neurons.

Let ρ be a probability distribution on data-label pairs $(z, y) \in \mathbb{R}^d \times \mathbb{R}^{d'}$. The population risk is

$$f(\alpha) = \int \ell(G_\alpha(z), y) d\rho(z, y),$$

and the empirical risk is the special case $\rho = \rho_N := N^{-1} \sum_{k=1}^N \delta_{(z_k, y_k)}$. Since $\alpha \mapsto G_\alpha$ is linear, f is convex as a function of α whenever $\ell(\cdot, y)$ is convex. For the empirical neuron law $\alpha_X = n^{-1} \sum_i \delta_{x_i}$, the Wasserstein metric induces on particles the rescaled metric $n^{-1} \sum_i \|\dot{x}_i\|^2$. The corresponding particle flow is

$$\dot{x}_i = -n \nabla_{x_i} F(X), \quad F(X) = f\left(\frac{1}{n} \sum_i \delta_{x_i}\right),$$

which is the gradient flow of $F(X) = f(\alpha_X)$ for this Wasserstein particle metric, equivalently Euclidean gradient descent with the time scale multiplied by n . It gives a particle discretization of (106).

Assume that ℓ is differentiable in its first variable. The first variation is

$$\delta f(\alpha)(x) = \int \langle \nabla_1 \ell(G_\alpha(z), y), \psi(x, z) \rangle d\rho(z, y), \quad (112)$$

and the Wasserstein gradient in parameter space is

$$\nabla_{\mathcal{W}} f(\alpha)(x) = \nabla_x \delta f(\alpha)(x) = \int [D_x \psi(x, z)]^\top \nabla_1 \ell(G_\alpha(z), y) d\rho(z, y).$$

For the squared Euclidean loss $\ell(s, y) = \frac{1}{2} \|s - y\|^2$, the energy is the sum of a quadratic interaction and a linear potential:

$$f(\alpha) = \frac{1}{2} \iint k(x, x') d\alpha(x) d\alpha(x') + \frac{1}{2} \int \|y\|^2 d\rho(z, y), \quad (113)$$

with

$$k(x, x') = \int \langle \psi(x, z), \psi(x', z) \rangle d\rho(z, y), \quad g(x) = - \int \langle y, \psi(x, z) \rangle d\rho(z, y). \quad (114)$$

Thus

$$\delta f(\alpha)(x) = \int k(x, x') d\alpha(x') + g(x), \quad \nabla_{\mathcal{W}} f(\alpha)(x) = \int \nabla_x k(x, x') d\alpha(x') + \nabla_x g(x).$$

These kernels are generally not convex in the particle variable, so the geodesic-convex convergence theory above does not apply directly.

The mean-field description of two-layer training was developed in several works, including [44, 118]. The distinctive contribution of Chizat and Bach is a global-convergence analysis for positively homogeneous networks without adding an explicit regularizer or relying on noisy SGD to create a Laplacian term. The following formal statement isolates the core mechanism and ignores the technical issues due to ReLU non-smoothness, support propagation and compactness.

Proposition 59 (Formal global optimality for two-homogeneous mean-field flows). *Assume that the feature is positively two-homogeneous in the neuron variable,*

$$\psi(\lambda x, z) = \lambda^2 \psi(x, z) \quad (\lambda > 0),$$

and that $f(\alpha) = J(G_\alpha)$ with J convex and differentiable as a functional of the predictor. Let α be a smooth stationary point of the Wasserstein flow, so that $\nabla_x \delta f(\alpha)(x) = 0$ on $\text{supp}(\alpha)$. Assume also full directional support: for every nonzero direction ω , the support of α intersects the ray $\{\lambda \omega : \lambda > 0\}$. Then α is a global minimizer of f over the mean-field model class.

neuron trajectories

directional concentration

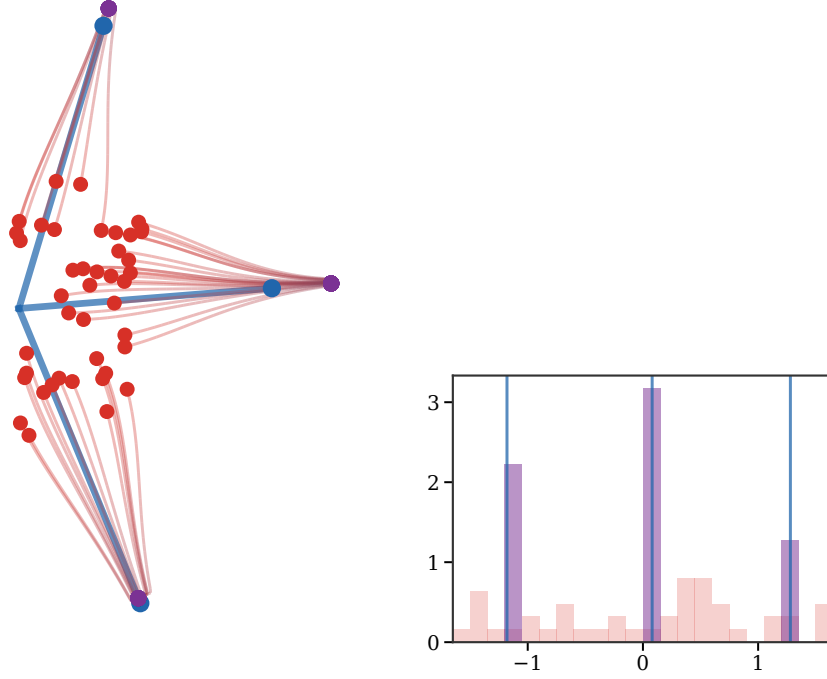


Figure 52: Mean-field training of a homogeneous two-layer model as transport in neuron space. Blue rays represent teacher directions or active homogeneous features. Red neurons move as particles under a Wasserstein gradient flow, rotating and changing radius until their empirical support concentrates near the useful rays. The right panel shows the same effect in angular coordinates: the initial red directional distribution becomes concentrated around the blue teacher rays. This illustrates why support propagation in parameter space is central to the homogeneous mean-field analysis.

Proof. Write

$$h_\alpha(x) = \delta f(\alpha)(x) = \langle \nabla J(G_\alpha), \psi(x, \cdot) \rangle_\rho.$$

By two-homogeneity of ψ , one has $h_\alpha(\lambda x) = \lambda^2 h_\alpha(x)$. Normalize a nonzero direction ω and choose $r_\omega > 0$ with $r_\omega \omega \in \text{supp}(\alpha)$. Stationarity gives a zero radial derivative at this point:

$$0 = \left. \frac{d}{dr} h_\alpha(r\omega) \right|_{r=r_\omega} = 2r_\omega h_\alpha(\omega).$$

Hence $h_\alpha(\omega) = 0$ for every direction ω , and by homogeneity $h_\alpha(x) = 0$ for every x .

For any competitor β , convexity of J gives

$$f(\beta) - f(\alpha) \geq \int h_\alpha(x) d(\beta - \alpha)(x) = 0.$$

Thus no competitor has smaller risk. The rigorous theorem replaces the full directional support assumption by propagation and overparameterization hypotheses ensuring that a negative descent direction would be present in the support and would contradict stationarity. $\square$

14 General Models via Transportation

The preceding gradient-flow calculus is variational. Modern machine-learning models often use the same transportation language more broadly: one may prescribe an interpolation and regress its velocity, fit a one-step generator to a descent field, or view network depth as a continuous transport of token measures. The examples below separate what is genuinely a Wasserstein gradient flow from what is a transportation dynamics with a useful geometric interpretation.

14.1 Generative Models via Flow Matching

Flow matching constructs a generative map by learning the velocity field of an interpolation. The key computational insight is that a constrained continuity-equation problem can be trained by an unconstrained regression.

Generative models aim to build a transportation map T between a reference distribution α (typically an isotropic Gaussian) and the target data distribution β . Since such reference measures are non-atomic, a measurable map with $T_\# \alpha = \beta$ exists on standard Borel spaces, for instance by identifying both probability spaces with the unit interval and using a quantile-type rearrangement. This abstract existence statement is much weaker than having an explicit and numerically stable construction of T . Optimal transport is one approach to achieving this, but it is computationally expensive and raises questions about how to estimate it from samples. A recent idea, first introduced in diffusion models [92, 164] and later systematically developed in flow matching by Yaron Lipman and collaborators [112], is to obtain T by integrating a time-dependent vector field v_t . This vector field v_t is obtained by constructing an interpolation α_t and then finding v_t using the least-squares formula (103). As we will explain, for a specific class of interpolation (obtained by a parametric push-forward), this v_t can be obtained by avoiding explicitly inverting a Laplacian and instead computing a simple conditional expectation. This conditional expectation can itself be estimated by solving another least-squares problem, but this time unconstrained, making the estimation feasible from finite samples of α and β .

Stochastic interpolant. We assume that α_t is defined via a “projection” (in a loose sense) of a latent distribution $\pi \in \mathcal{P}(\mathbb{R}^{d'})$, using an operator $P_t : \mathbb{R}^{d'} \rightarrow \mathbb{R}^d$ where $d' \gg d$, i.e.

$$\forall t \in [0, 1], \quad \alpha_t := (P_t)_\# \pi. \quad (115)$$

A common case is $d' = 2d$. We write $(x, y) \in \mathbb{R}^{d'} = \mathbb{R}^d \times \mathbb{R}^d$ and assume $P_0(x, y) = x$ and $P_1(x, y) = y$, so that π is a probabilistic coupling between α_0 and α_1 , i.e. π has marginals (α_0, α_1) . For instance, one can use $\pi = \alpha_0 \otimes \alpha_1$, the trivial coupling. An even more special case is the linear interpolation $P_t(x, y) = (1-t)x + ty$. More complex constructions are also possible, provided that sampling from π remains simple; stochastic interpolants provide a general framework unifying deterministic flows and diffusions [2].

If $\pi = \alpha \otimes \beta$ and $\alpha = \frac{1}{n} \sum_i \delta_{x_i}$, $\beta = \frac{1}{m} \sum_j \delta_{y_j}$, then α_t consists of $n \times m$ Dirac masses

$$\alpha_t = \frac{1}{nm} \sum_{i,j} \delta_{P_t(x_i, y_j)}.$$

If $\pi = (\text{Id}, T)_\# \alpha$ is a Brenier-type coupling, then $\alpha_t = ((1-t)\text{Id} + tT)_\# \alpha$ is the so-called McCann OT interpolation.

Flow matching formula. This interpolation is not directly useful for sampling from β , but it can be used to define a flow field v_t so that the Eulerian advection equation (99) holds. This flow field is computed by solving an unconstrained least-squares problem, or equivalently, it is a conditional expectation.

Proposition 60 (Flow matching vector field). *For each fixed t , assume $\partial_t P_t \in L^2(\pi; \mathbb{R}^d)$. The minimizer over measurable fields $v_t : \mathbb{R}^d \rightarrow \mathbb{R}^d$ of the flow-matching problem*

$$\min_{v_t} \int_{\mathbb{R}^d} \|v_t(P_t(u)) - [\partial_t P_t](u)\|^2 d\pi(u). \quad (116)$$

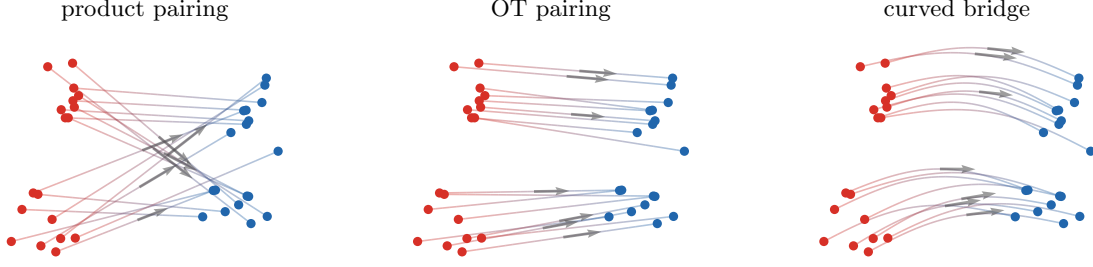


Figure 53: Flow matching interpolants between the same empirical source and target measures. A product-style random pairing produces crossing paths, an OT pairing gives direct displacement rays, and a curved bridge changes the path geometry while keeping the same endpoints. Gray arrows mark representative midpoint velocities $\partial_t P_t$.

is, α_t -almost everywhere, the conditional expectation

$$v_t(z) = \mathbb{E}_{u \sim \pi}([\partial_t P_t](u) \mid z = P_t(u)). \quad (117)$$

The pair (α_t, v_t) satisfies the continuity equation (99).

Proof. The minimizer is the orthogonal projection in $L^2(\pi; \mathbb{R}^d)$ of the random velocity $\partial_t P_t(u)$ onto the closed subspace of functions that depend on u only through $P_t(u)$. This projection is the conditional expectation (117). Equivalently, for every test vector field m ,

$$\int \langle m(z), v_t(z) \rangle d\alpha_t(z) = \int \langle m(P_t(u)), [\partial_t P_t](u) \rangle d\pi(u). \quad (118)$$

Let φ be a smooth scalar test function. Since $\alpha_t = (P_t)_\# \pi$,

$$\frac{d}{dt} \int \varphi(z) d\alpha_t(z) = \int \langle \nabla \varphi(P_t(u)), [\partial_t P_t](u) \rangle d\pi(u).$$

Using (118) with $m = \nabla \varphi$ gives

$$\frac{d}{dt} \int \varphi(z) d\alpha_t(z) = \int \langle \nabla \varphi(z), v_t(z) \rangle d\alpha_t(z),$$

which is the weak form of $\partial_t \alpha_t + \operatorname{div}(\alpha_t v_t) = 0$. $\square$

The solution of (116) is the conditional expectation of the velocities $\partial_t P_t$. Intuitively, $v_t(z)$ is the average velocity of all trajectories passing through z . Numerically, $(x, t) \rightarrow v_t(x)$ can be parameterized by a neural network (e.g., a U-Net for vision tasks) and estimated using stochastic gradient descent on the objective in (116). For the exact field v_t , integrating the ODE $\dot{x} = v_t(x)$ defines a transport map T_t . If v_t is regular enough, or more generally if the continuity equation has a unique solution for this velocity, then $(T_t)_\# \alpha_0 = \alpha_t$. Thus the same interpolation as (115) is represented by a deterministic flow rather than by the original coupling. The sampling procedure consists in first drawing $X_0 \sim \alpha$, and then integrating the ODE $\dot{X}_t = v_t(X_t)$ starting with $X_{t=0} = X_0$. In the ideal exact-field limit, the resulting $X_{t=1}$ is distributed according to $\alpha_1 = \beta$.

Connection with diffusion models. In the special case where $P_t(x, y) = (1 - t)x + ty$ is a linear interpolation and $\pi = \alpha \otimes \beta$, the curve α_t is a convolution of rescaled versions of α_0 and α_1 . The flow-matching problem (116) becomes

$$\min_{(v_t)_t} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|v_t((1 - t)x + ty) - (y - x)\|^2 d\alpha_0(x) d\alpha_1(y).$$

When one endpoint is an isotropic Gaussian, this construction is closely related to the probability-flow formulation of diffusion models, up to the usual change of time parametrization. The next two propositions are written in the noising direction, from a data law α to a Gaussian; reversing time gives the corresponding sampling flow. They also give an explicit closed form for v_t and show that it is a gradient field. In this setting, v_t is also the solution of the constrained least-squares problem (103). The regression (116) is computationally simpler because the continuity equation has already been enforced by the chosen interpolant. To prove this, we rely on Tweedie's formula, which expresses the optimal Gaussian denoiser through the score, i.e. the gradient of the log-density.

Proposition 7 (Tweedie identity). *Let W be a random vector in $\mathbb{R}^d$ with density β . For $\sigma > 0$, observe*

$$Z = W + \sigma \varepsilon, \quad \text{where } \varepsilon \sim \mathcal{N}(0, I_d) \text{ is independent of } W.$$

Denote by

$$\beta_\sigma = \beta * \mathcal{N}(0, \sigma^2 I_d)$$

the density of Z . Then

$$\mathbb{E}[W \mid Z = z] = z + \sigma^2 \nabla \log \beta_\sigma(z) \quad \text{for all } z \in \mathbb{R}^d.$$

Proof. Bayes' rule gives the conditional density $p_{W|Z}(w \mid z) = \frac{\beta(w) \varphi_\sigma(z - w)}{\beta_\sigma(z)}$ with φ_σ the $\mathcal{N}(0, \sigma^2 I_d)$ density. Hence

$$\mathbb{E}[W \mid Z = z] = \frac{1}{\beta_\sigma(z)} \int_{\mathbb{R}^d} w \beta(w) \varphi_\sigma(z - w) dw.$$

Differentiating the Gaussian convolution under the integral sign and using $\nabla_z \varphi_\sigma(z - w) = -\sigma^{-2}(z - w) \varphi_\sigma(z - w)$ yields

$$\nabla_z \beta_\sigma(z) = \int \beta(w) \nabla_z \varphi_\sigma(z - w) dw = -\sigma^{-2} \left(z - \mathbb{E}[W \mid Z = z] \right) \beta_\sigma(z).$$

Rearranging finishes the proof. $\square$

Proposition 8 (Gaussian-endpoint flow-matching field). *Let $X \sim \alpha$ and $Y \sim \mathcal{N}(0, I_d)$ be independent. For $t \in (0, 1)$ set*

$$Z_t = (1 - t)X + tY, \quad \alpha_t = \text{Law}(Z_t).$$

The regression minimizer $v^ : \mathbb{R}^d \times (0, 1) \rightarrow \mathbb{R}^d$ of*

$$\min_v \int_0^1 \iint_{\mathbb{R}^d \times \mathbb{R}^d} |y - x - v((1 - t)x + ty, t)|^2 d\alpha(x) d\mathcal{N}(y) dt$$

is

$$v^*(x, t) = -\frac{1}{1 - t} x - \frac{t}{1 - t} \nabla \log \alpha_t(x) \quad (x \in \mathbb{R}^d, t \in (0, 1)).$$

In particular, for each $t \in (0, 1)$ this field is a gradient field,

$$v^*(\cdot, t) = -\nabla \left(\frac{\|\cdot\|^2}{2(1 - t)} + \frac{t}{1 - t} \log \alpha_t \right).$$

Proof. Fix $t \in (0, 1)$ and write $W = (1 - t)X$, $\sigma = t$, so that $Z_t = W + \sigma Y$ matches the setting of Proposition 7. Conditional expectations satisfy $v^*(z, t) = \mathbb{E}[Y - X \mid Z_t = z] = \frac{1}{t} \mathbb{E}[Z_t - W \mid Z_t = z] - \frac{1}{1 - t} \mathbb{E}[W \mid Z_t = z]$. Applying Proposition 7 to $\mathbb{E}[W \mid Z_t = z]$ and noting $\mathbb{E}[Y \mid Z_t = z] = -t \nabla \log \alpha_t(z)$ gives the claimed formula. $\square$

The same probability-flow intuition is visible in two dimensions. When the data law is a Gaussian mixture, the noising density remains a Gaussian mixture and its score can be evaluated explicitly. This makes it possible to draw backward trajectories without training a neural network.

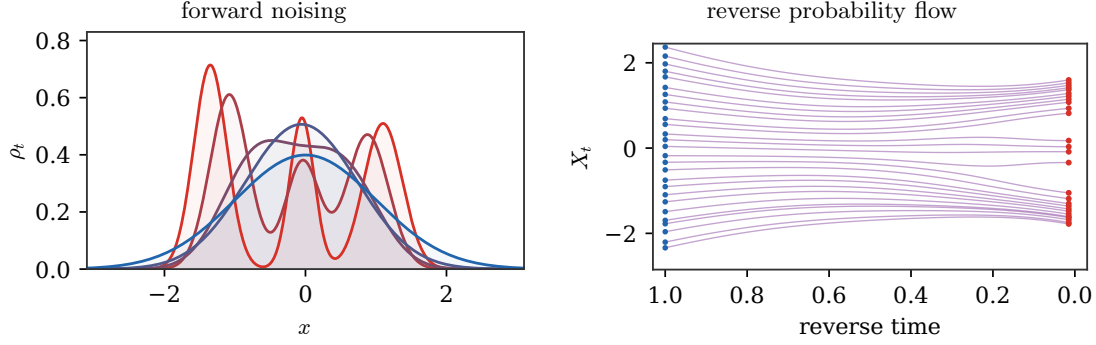


Figure 54: One-dimensional diffusion bridge for a Gaussian-mixture data law. The forward path $Z_t = (1-t)X + tY$ smooths the red data density toward a blue Gaussian endpoint. Reversing the probability-flow ODE transports a denser set of blue noise samples back toward the data modes, making the splitting of trajectories across mixture components visible.

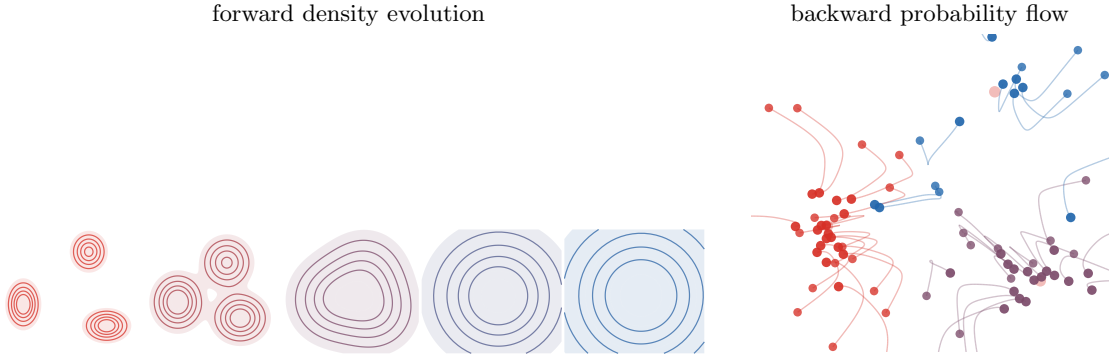


Figure 55: Two-dimensional diffusion bridge from a three-component Gaussian mixture to an isotropic Gaussian. The left panel shows the forward noising densities of $Z_t = (1-t)X + tY$. The right panel integrates the probability-flow ODE backward from near the Gaussian endpoint; trajectories split toward the three red data modes while keeping the smooth, non-straight geometry characteristic of diffusion-type samplers.

When is the induced map optimal? Integrating the learned velocity gives a deterministic map from α_0 to α_1 , but this map is not automatically the Brenier optimal map. It is optimal only in special cases where the accumulated flow remains the gradient of a convex potential. For Gaussian marginals, the flow is affine. If the covariance matrices co-diagonalize, the affine map is symmetric positive definite in the common eigenbasis, hence it coincides with the Gaussian Brenier map. More generally, non-commuting covariance dynamics or mixture-like paths can create rotational or shearing components, and the final map need not be optimal even if every instantaneous velocity looks natural. This distinction is closely related to counterexamples showing that flow maps associated with Fokker–Planck or diffusion-type evolutions do not in general provide optimal transport maps [105]. In particular, starting from an isotropic Gaussian does not by itself guarantee optimality once the target distribution is non-Gaussian; additional structural assumptions on the path or on the coupling are needed.

14.2 One-Step Generative Models

One-step generative models try to keep the geometric training principle of flows while removing the expensive multi-step integration at sampling time. The idea is to evolve the model distribution during

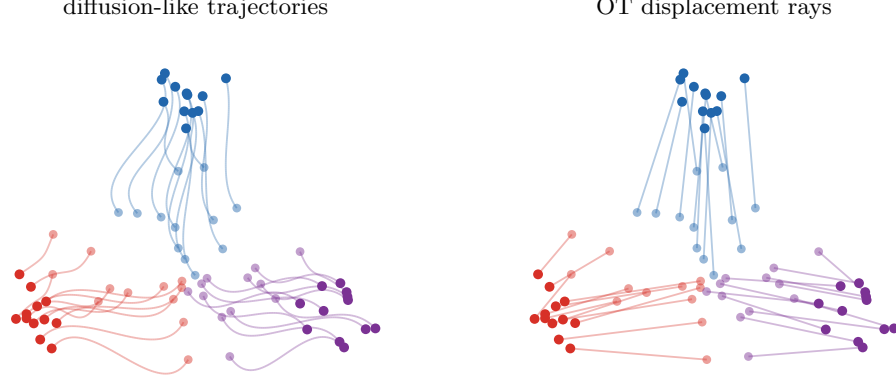


Figure 56: Diffusion-style sampling trajectories compared with OT rays in a small two-dimensional example. The initial particles are sampled near the center and each trajectory is colored, including its initial disk, by the final target cluster. Curved multi-step trajectories can approach modes through non-straight paths, while the OT map uses straight displacement rays selected by the quadratic matching.

training, but to store the final evolution in a single generator evaluation.

Training a one-step flow. Let ζ be a simple latent distribution and let $\alpha_\theta = (G_\theta)_\# \zeta$ be the model distribution. Assume that the target data distribution is β . A Wasserstein-flow construction chooses a discrepancy

$$\mathcal{E}_\beta(\alpha),$$

for instance a smoothed $\text{KL}(\alpha|\beta)$, an MMD/IPM loss, or the debiased Sinkhorn divergence $\tilde{\mathcal{L}}_\epsilon^\varepsilon(\alpha, \beta)$ introduced in Section 8.3. The associated formal descent is

$$\partial_t \mu_t + \text{div}(\mu_t w_t) = 0, \quad w_t(x) = -\nabla_{\delta_\alpha} \mathcal{E}_\beta(\mu_t)(x). \quad (119)$$

Instead of integrating (119) at inference time, one fits a parametric residual field U_η along the current model distribution:

$$\min_\eta \int_0^1 \int \|U_\eta(t, x) - w_t(x)\|^2 d\mu_t(x) dt. \quad (120)$$

In a particle or generator implementation, the learned residual is then used to update the current generator by

$$\alpha_\theta^+ = (\text{Id} + \tau U_\eta)_\# \alpha_\theta, \quad \text{or equivalently} \quad G_\theta^+(z) = G_\theta(z) + \tau U_\eta(G_\theta(z)).$$

After many training updates, the accumulated generator is evaluated once at test time. This is the organizing principle behind recent one-step methods based on Wasserstein gradient flows: W-Flow uses such a construction with the Sinkhorn divergence as a tractable global discrepancy [88], while drifting methods evolve the generated distribution during training through a fitted vector field and also admit one-step inference [55]. The gradient-flow interpretation of drifting models, and its relation to KL, MMD, sliced-Wasserstein and Sinkhorn-type discrepancies, is analyzed in [84, 91]. These ideas are also connected to the Sinkhorn-type normalization dynamics used to model attention in Sinkformers [153].

Normalized drifting fields. Drifting methods need not start from an exact Wasserstein gradient. They often prescribe a normalized attraction-minus-repulsion field and then regress this field in $L^2(\mu_t)$. A simple continuous version uses a positive kernel $K_\varepsilon(x, y)$ and defines, for any measure ν ,

$$B_\varepsilon[\nu](x) := \frac{\int (y - x) K_\varepsilon(x, y) d\nu(y)}{\int K_\varepsilon(x, y) d\nu(y)}. \quad (121)$$

For the Gaussian kernel $K_\varepsilon(x, y) = \exp(-\|x - y\|^2/(2\varepsilon))$, this normalized field is a score of a smoothed density:

$$B_\varepsilon[\nu](x) = \varepsilon \nabla \log \left(\int K_\varepsilon(x, y) d\nu(y) \right). \quad (122)$$

The drifting velocity is then

$$u_t(x) = B_\varepsilon[\beta](x) - B_\varepsilon[\mu_t](x) = \varepsilon \nabla \log \frac{\int K_\varepsilon(x, y) d\beta(y)}{\int K_\varepsilon(x, y) d\mu_t(y)}. \quad (123)$$

The first term pulls samples toward data, while the second term corrects self-attraction and prevents all particles from collapsing onto the same high-density region. Sinkhorn drifting replaces these one-sided kernel normalizations by two-sided entropic OT couplings, so that the cross and self terms are normalized by Sinkhorn scaling rather than by a single denominator [91].

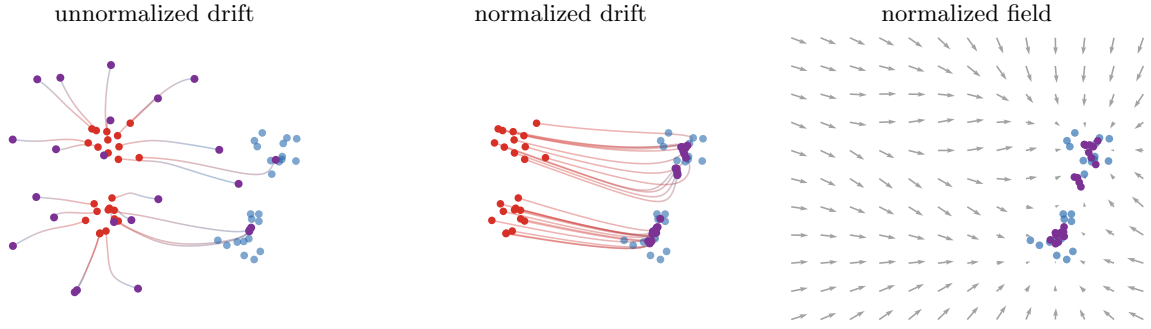


Figure 57: Drifting fields for a small particle generator. The unnormalized Laplacian-kernel drift has weak long-range attraction and can leave particles away from the data modes. The normalized drift uses the local average $B_\varepsilon[\beta] - B_\varepsilon[\mu_t]$, so particles reach the blue modes while the self-normalized correction prevents collapse. The right panel displays the normalized vector field around an intermediate violet model distribution.

Proposition 61 (Drifting as a time-dependent Wasserstein gradient). *Let μ_t be a smooth curve of positive densities and let $u_t = \nabla \varphi_t$ be a smooth time-dependent gradient field. Define the semi-relaxed functional*

$$\mathcal{R}_t(\alpha|\mu_t) := - \int \varphi_t(x) d\alpha(x) + \int \varphi_t(x) d\mu_t(x). \quad (124)$$

Here μ_t and φ_t are frozen when taking the first variation with respect to the first argument α . Then the continuity equation

$$\partial_t \mu_t + \operatorname{div}(\mu_t u_t) = 0$$

is the formal Wasserstein gradient descent of the time-dependent functional $\alpha \mapsto \mathcal{R}_t(\alpha|\mu_t)$.

Proof. Since μ_t and φ_t are fixed in the variation with respect to α , the first variation is

$$\delta_\alpha \mathcal{R}_t(\alpha|\mu_t)(x) = -\varphi_t(x).$$

By Proposition 6,

$$\nabla_{\mathcal{W}} \mathcal{R}_t(\alpha|\mu_t) = \nabla \delta_\alpha \mathcal{R}_t(\alpha|\mu_t) = -\nabla \varphi_t = -u_t.$$

The Wasserstein gradient-descent velocity is the negative of this gradient, namely u_t . Substituting this velocity in the continuity equation gives the claimed flow. $\square$

Example 8 (Kernel drifting as a semi-relaxed divergence). For the Gaussian-kernel drift (123), set

$$\varphi_t(x) = \varepsilon \log \frac{\int K_\varepsilon(x, y) d\beta(y)}{\int K_\varepsilon(x, y) d\mu_t(y)}.$$

Then $u_t = \nabla \varphi_t$, so Proposition 61 shows that kernel drifting is the Wasserstein gradient descent of

$$\mathcal{R}_t^{\text{drift}}(\alpha|\mu_t) = \varepsilon \int \log \frac{\int K_\varepsilon(x, y) d\mu_t(y)}{\int K_\varepsilon(x, y) d\beta(y)} d\alpha(x) + \text{constant}.$$

It is “semi-relaxed” because the current model μ_t is used to build the potential, but it is not varied inside the denominator when computing the first variation in α .

Remark 20 (General fields and projection onto gradients). A general regressed field b_t is not necessarily a Wasserstein gradient, since Wasserstein tangent vectors are represented by gradient fields modulo $L^2(\mu_t)$ -null directions. The gradient component is obtained by the weighted projection

$$\nabla \varphi_t = \operatorname{argmin}_{\nabla \varphi} \int \|\nabla \varphi(x) - b_t(x)\|^2 d\mu_t(x).$$

One may first normalize b_t pointwise, for instance by $b_t/(\|b_t\| + \eta)$, or globally by $\|b_t\|_{L^2(\mu_t)}$, before this projection. Proposition 61 then applies to the projected field. Non-gradient components can still be useful in a parametric model, but they are not descent directions of a scalar functional for the $\mathcal{W}_2$ Riemannian metric.

14.3 Evolution in Depth of Transformers

Deep residual architectures can be read as time discretizations of ODEs or PDEs. For transformers, the transported objects are token measures and the velocity is induced by attention.

We consider very deep transformers, focusing on a single-head attention mechanism for simplicity while ignoring MLP layers, layer normalization, causality, and masking. This framework is best suited to modeling encoders and vision transformers.

Attention as a context-dependent velocity. After tokenization, embedding, and positional encoding, each input (from a set of tokens) is represented as a point cloud $(x_i)_{i=1}^n$ of n points in the space of vectorized tokens. An attention layer with skip connection and rescaling by $1/T$ (where T is the depth) defines a transformation of the tokens:

$$x_i \mapsto x_i + \frac{1}{T} \sum_j \frac{e^{\langle Qx_i, Kx_j \rangle} Vx_j}{\sum_\ell e^{\langle Qx_i, Kx_\ell \rangle}},$$

where $\theta = (K, Q, V)$ are the parameters of the attention layer, represented by three matrices.

Token measure evolution. To handle an arbitrary number of tokens, we define $\alpha = \frac{1}{n} \sum_i \delta_{x_i}$ as the empirical measure of tokens and rewrite the transformer mapping as:

$$x_i \mapsto x_i + \frac{1}{T} \Gamma_\theta[\alpha](x_i),$$

where

$$\Gamma_\theta[\alpha](x) := \frac{\int e^{\langle Qx, Ky \rangle} Vy d\alpha(y)}{\int e^{\langle Qx, Kz \rangle} d\alpha(z)}.$$

At the level of the token distribution, the layer pushes α forward by the “in-context” mapping $\Gamma_{\theta_t}[\alpha]$, which depends on the context α , the tokens, and the depth-dependent parameters θ_t . Denoting $t \in [0, 1]$ as the depth and $\tau = 1/T$ as the step size, this gives:

$$\alpha_{t+\tau} = (\text{Id} + \tau \Gamma_{\theta_t}[\alpha_t])_\# \alpha_t.$$

As $\tau \rightarrow 0$, this converges formally to the conservation equation

$$\partial_t \alpha_t + \operatorname{div}(\alpha_t \Gamma_{\theta_t}[\alpha_t]) = 0.$$

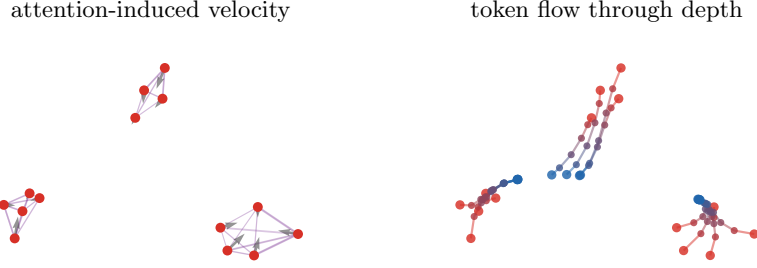


Figure 58: Attention as a transportation dynamics on token measures. The schematic uses the attractive setup $Q = K = I$ and velocity $W(X)X - X$, where $W(X)$ is the softmax attention matrix. The left panel shows one layer: faint violet edges encode attention weights and gray arrows show the residual velocity applied to red tokens. The right panel follows the same tokens through residual depth, colored from red to blue, and shows local clustering induced by attention.

Gradient structure and limitations. When the token space has dimension d and the query/key space has dimension r , take $Q, K \in \mathbb{R}^{r \times d}$ and $V \in \mathbb{R}^{d \times d}$. If $V = Q^\top K$, the field $\Gamma_\theta[\alpha]$ is a gradient vector field in the token variable. Indeed, define the log-partition potential

$$\Phi_\alpha(x) = \int \exp(\langle Qx, Ky \rangle) d\alpha(y), \quad U_\alpha(x) = \log \Phi_\alpha(x).$$

Then

$$\nabla_x U_\alpha(x) = \frac{\int Q^\top Ky \exp(\langle Qx, Ky \rangle) d\alpha(y)}{\int \exp(\langle Qx, Kz \rangle) d\alpha(z)} = \Gamma_\theta[\alpha](x).$$

This is an instantaneous gradient in x . It is not, however, the gradient of the first variation of a fixed functional of α , because the potential U_α itself depends on the current measure through the same attention normalization. Thus the PDE is generally a transportation dynamics, not a Wasserstein gradient flow. This formulation is related to continuous-depth neural networks [40] and was introduced for attention dynamics in the Sinkformers paper [153], building on the transformer architecture of [169]. It models deep transformers as PDEs on token measures. The key challenge is to understand training, which corresponds to optimizing the time-dependent parameters $(\theta_t)_t$.

14.4 Flows over the Gaussian Manifold

Gaussian measures provide a useful testing ground for the preceding dynamics. They are not invariant under a general Wasserstein gradient flow: a nonlinear velocity usually creates non-Gaussian densities immediately. The useful substitute is to either identify affine velocities, which exactly preserve Gaussianity, or to project the dynamics onto the Gaussian manifold. In both cases the measure PDE reduces to matrix ODEs for the mean and covariance. This viewpoint is emphasized in the survey [134] and is useful for comparing diffusion paths, Wasserstein gradient flows, drifting fields and transformer-type dynamics.

For constrained gradient flows on this family, the covariance equation is the finite-dimensional Bures–Wasserstein gradient flow on positive definite matrices. Thus Gaussian closure is not just a computational shortcut: it is the restriction of Wasserstein geometry to the Gaussian submanifold, where affine gradient fields encode tangent vectors.

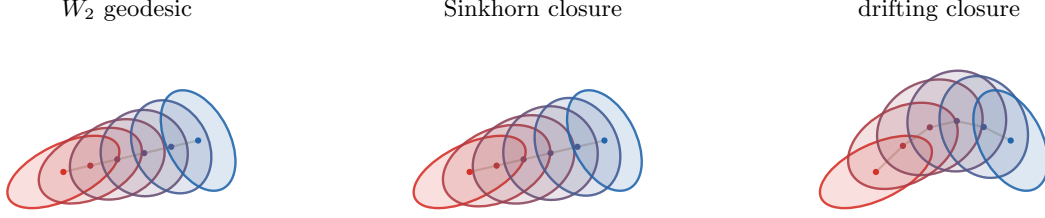


Figure 59: Gaussian closures of transport dynamics between two overlapping anisotropic Gaussians. The left panel is the exact W_2 Gaussian geodesic. The middle panel shows a regularized Sinkhorn-style closure, where the same mean path is accompanied by inflated intermediate covariances. The right panel shows a drifting-style closure with a curved mean path and moment-matched covariance ellipses. These finite-dimensional pictures keep only means and covariances, and therefore discard higher-order shape information that would be created by a genuinely nonlinear velocity field.

Gaussianity preservation. The first question is invariance: one wants a simple criterion ensuring that the continuity equation does not leave the finite-dimensional Gaussian family.

Proposition 62 (Affine velocities preserve Gaussianity). *Let $\alpha_t = \mathcal{N}(\mathbf{m}_t, \Sigma_t)$, with Σ_t positive definite, solve the continuity equation with an affine velocity*

$$v_t(x) = b_t + A_t(x - \mathbf{m}_t).$$

Then α_t remains Gaussian and its moments solve

$$\dot{\mathbf{m}}_t = b_t, \quad \dot{\Sigma}_t = A_t \Sigma_t + \Sigma_t A_t^\top.$$

Conversely, any smooth Gaussian curve with positive definite covariance can be generated by such an affine velocity. If one wants the velocity to be a Wasserstein tangent gradient, one chooses the unique symmetric solution of the Lyapunov equation

$$A_t \Sigma_t + \Sigma_t A_t = \dot{\Sigma}_t.$$

Proof. Let X_t follow the characteristic ODE $\dot{X}_t = b_t + A_t(X_t - \mathbf{m}_t)$. This linear ODE maps Gaussian random variables to Gaussian random variables. Taking expectation gives $\dot{\mathbf{m}}_t = b_t$. Writing $\tilde{X}_t = X_t - \mathbf{m}_t$, one has $\dot{\tilde{X}}_t = A_t \tilde{X}_t$, hence

$$\dot{\Sigma}_t = \frac{d}{dt} \mathbb{E}(\tilde{X}_t \tilde{X}_t^\top) = A_t \Sigma_t + \Sigma_t A_t^\top.$$

For the converse, set $b_t = \dot{\mathbf{m}}_t$ and choose any matrix A_t satisfying the covariance equation. Since Σ_t is positive definite, the Lyapunov map $A \mapsto A \Sigma_t + \Sigma_t A$ is invertible on symmetric matrices, which gives the unique symmetric choice when a gradient velocity is required. In that case v_t is the gradient of the quadratic potential $x \mapsto \langle b_t, x \rangle + \langle A_t(x - \mathbf{m}_t), x - \mathbf{m}_t \rangle / 2$. $\square$

Constrained evolution on the Gaussian manifold. For non-affine velocities, the finite-dimensional substitute is to project the Wasserstein dynamics onto the Gaussian manifold.

Let

$$\mathcal{G} = \{\mathcal{N}(\mathbf{m}, \Sigma) : \mathbf{m} \in \mathbb{R}^d, \Sigma \succ 0\}$$

be the Gaussian submanifold of $\mathcal{P}_2(\mathbb{R}^d)$. The Wasserstein gradient of a functional constrained to a smooth submanifold $\mathcal{M} \subset \mathcal{P}_2$ is defined as the Riesz representative of the differential restricted to tangent velocities of $\mathcal{M}$. Equivalently, it is the small-step limit of the constrained JKO scheme

$$\alpha^{k+1} \in \operatorname{argmin}_{\alpha \in \mathcal{M}} \frac{1}{2\tau} \mathcal{W}_2^2(\alpha, \alpha^k) + f(\alpha).$$

For $\mathcal{M} = \mathcal{G}$, tangent velocities are affine gradient fields $v(x) = b + A(x - \mathbf{m})$ with $A = A^\top$. The constrained gradient is therefore the $L^2(\mathcal{N}(\mathbf{m}, \Sigma))$ projection of the ambient Wasserstein gradient onto this finite-dimensional affine space, whenever the ambient gradient exists.

Proposition 63 (Gaussian-constrained Wasserstein gradients). *Let f be a smooth functional and assume that its restriction to nondegenerate Gaussian measures can be written as*

$$f(\mathcal{N}(\mathbf{m}, \Sigma)) = F(\mathbf{m}, \Sigma).$$

Then the Wasserstein gradient constrained to the Gaussian family is the affine vector field

$$v_F(x) = \nabla_{\mathbf{m}} F(\mathbf{m}, \Sigma) + 2\nabla_{\Sigma} F(\mathbf{m}, \Sigma)(x - \mathbf{m}),$$

where $\nabla_{\Sigma} F$ denotes the symmetric matrix derivative. Equivalently, v_F is the $L^2(\mathcal{N}(\mathbf{m}, \Sigma))$ projection of the ambient Wasserstein gradient onto affine gradient fields, whenever the ambient gradient exists. Hence the gradient descent flow constrained to Gaussian measures satisfies

$$\dot{\mathbf{m}}_t = -\nabla_{\mathbf{m}} F(\mathbf{m}_t, \Sigma_t), \quad \dot{\Sigma}_t = -2(\Sigma_t \nabla_{\Sigma} F(\mathbf{m}_t, \Sigma_t) + \nabla_{\Sigma} F(\mathbf{m}_t, \Sigma_t) \Sigma_t), \quad (125)$$

and the descent velocity is affine.

Proof. Test the functional along a Gaussian tangent vector, represented by an affine gradient field

$$v(x) = b + A(x - \mathbf{m})$$

with A symmetric. The induced first-order variations are $\dot{\mathbf{m}} = b$ and $\dot{\Sigma} = A\Sigma + \Sigma A$. Therefore

$$dF(\mathbf{m}, \Sigma)[b, A\Sigma + \Sigma A] = \langle \nabla_{\mathbf{m}} F, b \rangle + \text{tr}(\nabla_{\Sigma} F(A\Sigma + \Sigma A)).$$

Since A , Σ and $\nabla_{\Sigma} F$ are symmetric, the second term equals

$$2 \text{tr}(\nabla_{\Sigma} F A \Sigma) = \int \langle 2\nabla_{\Sigma} F(x - \mathbf{m}), A(x - \mathbf{m}) \rangle d\mathcal{N}(\mathbf{m}, \Sigma)(x).$$

Together with the mean term, this gives

$$dF(\mathbf{m}, \Sigma)[\dot{\mathbf{m}}, \dot{\Sigma}] = \int \langle v_F(x), v(x) \rangle d\mathcal{N}(\mathbf{m}, \Sigma)(x)$$

for all affine gradient fields v . This identifies the constrained Wasserstein gradient in the induced $L^2(\alpha)$ metric, or equivalently the projection of the ambient gradient when it exists. Substituting the descent velocity $-v_F$ in Proposition 62 gives (125). $\square$

This proposition should be read as the organizing rule for Gaussian closures: once the scalar energy has been reduced to a function of $(\mathbf{m}, \Sigma)$, its constrained Wasserstein gradient is automatically affine and the covariance follows the Bures-type ODE (125). When the first variation of f is quadratic, this constrained gradient coincides with the full Wasserstein gradient.

Gaussian-preserving gradient flows. The next examples show that many familiar energies already have affine Wasserstein gradients on Gaussian inputs, so their full flow remains inside the Gaussian family.

Example 9 (Gaussian energies and affine gradients). Proposition 63 turns many standard energies into explicit affine fields:

- *Quadratic potential energy.* If

$$f(\alpha) = \int \left(\frac{1}{2} x^\top H x + \langle \ell, x \rangle \right) d\alpha(x), \quad H = H^\top,$$

then

$$\nabla_{\mathcal{W}} f(\alpha)(x) = Hx + \ell = (H\mathbf{m} + \ell) + H(x - \mathbf{m}).$$

This is the Gaussian form of transport under a quadratic confinement.

- *Quadratic interaction energy.* If

$$f(\alpha) = \frac{1}{4} \iint (x - y)^\top G(x - y) d\alpha(x) d\alpha(y), \quad G = G^\top,$$

then $F(\mathbf{m}, \Sigma) = \frac{1}{2} \text{tr}(G\Sigma)$ and

$$\nabla_{\mathcal{W}} f(\alpha)(x) = G(x - \mathbf{m}).$$

The mean is unchanged and the covariance contracts or expands according to the signs of G .

- *Relative entropy to a Gaussian.* For $\bar{\alpha} = \mathcal{N}(\bar{\mathbf{m}}, \bar{\Sigma})$,

$$f(\alpha) = \text{KL}(\alpha | \bar{\alpha})$$

has

$$\nabla_{\mathcal{W}} f(\alpha)(x) = \bar{\Sigma}^{-1}(\mathbf{m} - \bar{\mathbf{m}}) + (\bar{\Sigma}^{-1} - \Sigma^{-1})(x - \mathbf{m}).$$

The descent equations are the Ornstein–Uhlenbeck moment equations

$$\dot{\mathbf{m}}_t = -\bar{\Sigma}^{-1}(\mathbf{m}_t - \bar{\mathbf{m}}), \quad \dot{\Sigma}_t = 2\text{Id} - \bar{\Sigma}^{-1}\Sigma_t - \Sigma_t\bar{\Sigma}^{-1}.$$

- *Squared Wasserstein distance to a Gaussian.* For

$$f(\alpha) = \frac{1}{2} \mathcal{W}_2^2(\alpha, \bar{\alpha}), \quad \bar{\alpha} = \mathcal{N}(\bar{\mathbf{m}}, \bar{\Sigma}),$$

the Gaussian Brenier map $T_{\alpha \rightarrow \bar{\alpha}}$ is affine,

$$T_{\alpha \rightarrow \bar{\alpha}}(x) = \bar{\mathbf{m}} + M(x - \mathbf{m}), \quad M = \Sigma^{-1/2}(\Sigma^{1/2}\bar{\Sigma}\Sigma^{1/2})^{1/2}\Sigma^{-1/2}.$$

Hence

$$\nabla_{\mathcal{W}} f(\alpha)(x) = x - T_{\alpha \rightarrow \bar{\alpha}}(x) = (\mathbf{m} - \bar{\mathbf{m}}) + (\text{Id} - M)(x - \mathbf{m}),$$

and descent moves each Gaussian infinitesimally along the Bures–Wasserstein geodesic toward $\bar{\alpha}$.

- *Gaussian-only losses.* Sliced SW_2^2 losses to a Gaussian, Gaussian Sinkhorn divergences, and any smooth closed formula depending only on $(\mathbf{m}, \Sigma)$ fit the same constrained-gradient template:

$$v_F(x) = \nabla_{\mathbf{m}} F + 2\nabla_{\Sigma} F(x - \mathbf{m}).$$

For Gaussian Sinkhorn divergences this finite-dimensional flow is studied in [90].

Not every PDE preserves Gaussianity exactly. For instance, Wasserstein flows of relative Fisher information, related to quantum-drift or higher-order diffusion equations, typically require a Gaussian projection to close on $(\mathbf{m}, \Sigma)$. Such projected closures are still useful: they expose the finite-dimensional dynamics predicted by a variational model and make it easy to compare variational flows with non-variational affine dynamics such as drifting fields or the Gaussian transformer closure below.

Proposition 64 (Centered Gaussian covariance catalogue). *Let $\gamma = \mathcal{N}(0, \text{Id})$ and let $\mu_t = \mathcal{N}(0, C_t)$ with $C_t \succ 0$. For the normalizations displayed below, the Wasserstein descent constrained to the centered Gaussian manifold satisfies $\dot{C}_t = h(C_t)$, with*

$$\begin{aligned} \text{KL}(\mu | \gamma) &: h(C) = 2(\text{Id} - C), \\ \frac{1}{2} \mathcal{I}(\mu | \gamma) &: h(C) = 2(C^{-1} - C), \\ \mathcal{W}_2^2(\mu, \gamma) &: h(C) = 4(C^{1/2} - C), \\ \text{MMD}_k^2(\mu, \gamma), \quad k(x, y) = \langle x, y \rangle^2 &: h(C) = 8(C - C^2), \\ S_\varepsilon(\mu, \gamma) &: h(C) = 4 \left(C + \frac{\varepsilon^2}{16} \text{Id} \right)^{1/2} - 2 \left(C^2 + \frac{\varepsilon^2}{16} \text{Id} \right)^{1/2} - 2C - \frac{\varepsilon}{2} \text{Id}, \\ \text{SW}_2^2(\mu, \gamma) &: h(C) = V(C)C + CV(C), \end{aligned}$$

where S_ε is the debiased Sinkhorn divergence for the quadratic cost $\|x - y\|^2$ and KL regularization strength ε , and

$$V(C) = 2 \int_{\mathbb{S}^{d-1}} \left(\frac{1}{\sqrt{\theta^\top C \theta}} - 1 \right) \theta \theta^\top d\sigma(\theta)$$

for the normalized spherical measure σ . Here

$$\mathcal{I}(\mu|\gamma) = \int |\nabla \log \rho(x) + x|^2 \rho(x) dx \quad (\mu = \rho dx).$$

Thus the unhalved Fisher divergence has right-hand side $4(C^{-1} - C)$. Multiplying any of these energies by a constant simply rescales the corresponding right-hand side.

Proof. Each row is obtained by identifying the affine descent velocity $v(x) = M_C x$ generated by the corresponding Gaussian-constrained calculation and then applying Proposition 62, which gives $\dot{C} = M_C C + C M_C^\top$. For $\text{KL}(\cdot|\gamma)$, the Fokker–Planck velocity is $(C^{-1} - \text{Id})x$, hence $\dot{C} = 2(\text{Id} - C)$. For the Fisher row, the restriction of $\frac{1}{2}\mathcal{I}$ to centered Gaussians is

$$\frac{1}{2} (\text{tr}(C) + \text{tr}(C^{-1}) - 2d).$$

Using Proposition 63 gives the descent velocity $(C^{-2} - \text{Id})x$, hence $\dot{C} = 2(C^{-1} - C)$. This row should be read as a Gaussian projected closure of the fourth-order Fisher flow.

For $\mathcal{W}_2^2(\cdot, \gamma)$, the Brenier map from $\mathcal{N}(0, C)$ to γ is $C^{-1/2}x$, so the descent velocity for the unhalved squared distance is $2(C^{-1/2} - \text{Id})x$, giving $4(C^{1/2} - C)$. For the polynomial MMD row, centered Gaussians satisfy $\text{MMD}_k^2(\mu, \gamma) = \|C - \text{Id}\|_F^2$; the first variation is quadratic and its descent velocity is $4(\text{Id} - C)x$, giving $8(C - C^2)$.

Gaussian Sinkhorn dual potentials are quadratic, so the velocity is again linear; differentiating the closed Gaussian formula yields the displayed spectral expression. The square roots are spectral functions of C , hence commute with C , which is why the covariance ODE closes as a matrix function of C alone. For sliced Wasserstein, each one-dimensional projection is a Gaussian transport with velocity $2((\theta^\top C \theta)^{-1/2} - 1)(\theta, x)\theta$; averaging these velocities over $\mathbb{S}^{d-1}$ gives $v(x) = V(C)x$ and thus $\dot{C} = V(C)C + CV(C)$. $\square$

Example 10 (Linear mean-field networks as cross-moment flows). In the two-layer model above, take the linear activation $\sigma(s) = s$, so that

$$\psi((u, v), z) = v \langle u, z \rangle.$$

The predictor is the linear map

$$G_{\alpha_t}(z) = Q_t z, \quad Q_t = \int v u^\top d\alpha_t(u, v) \in \mathbb{R}^{d' \times d}.$$

Thus the energy depends on the neuron law only through the cross moment Q_t , a subblock of the raw second moment of $x = (u, v)$. For square loss, set

$$S = \int z z^\top d\rho(z, y), \quad R = \int y z^\top d\rho(z, y), \quad H_t = Q_t S - R.$$

The first variation is

$$\delta f(\alpha_t)(u, v) = \langle H_t, v u^\top \rangle = v^\top H_t u.$$

Hence the particle velocity in parameter space is linear:

$$-\nabla_{(u, v)} \delta f(\alpha_t)(u, v) = - \begin{pmatrix} 0 & H_t^\top \\ H_t & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}.$$

Therefore a Gaussian law of neurons remains Gaussian. Its mean and covariance follow Proposition 62, with a matrix depending only on the current cross moment Q_t , a raw second-moment subblock that becomes a covariance subblock when the neuron law is centered. This exact closure is special to the linear activation; for nonlinear activations, Gaussian closures are usually projections rather than invariant families.

Non-variational Gaussian-preserving flows. The last examples are not ordinary gradient flows of a fixed scalar energy on the full Wasserstein space. They preserve Gaussianity because the prescribed velocity field is affine when evaluated on Gaussian measures.

Example 11 (Flow matching and diffusion paths between Gaussians). Consider a prescribed Gaussian interpolation $\alpha_t = \mathcal{N}(\mathbf{m}_t, \Sigma_t)$. Proposition 62 shows that an exact flow-matching velocity can be taken affine:

$$v_t(x) = \dot{\mathbf{m}}_t + A_t(x - \mathbf{m}_t), \quad A_t \Sigma_t + \Sigma_t A_t = \dot{\Sigma}_t.$$

In the isotropic case $\Sigma_t = s_t^2 \text{Id}$, this reduces to the transparent formula

$$v_t(x) = \dot{\mathbf{m}}_t + \frac{\dot{s}_t}{s_t}(x - \mathbf{m}_t).$$

For instance, the diffusion noising path

$$X_t = a_t X_0 + \sigma_t Z, \quad Z \sim \mathcal{N}(0, \text{Id}),$$

has $\mathbf{m}_t = a_t \mathbf{m}_0$ and $\Sigma_t = a_t^2 \Sigma_0 + \sigma_t^2 \text{Id}$. Thus, in the Gaussian case, diffusion paths and flow-matching paths reduce to the same mean-covariance bookkeeping, although the corresponding training objectives are different.

Example 12 (Gaussian kernel drifting). Let the target be $\beta = \mathcal{N}(\bar{\mathbf{m}}, \bar{\Sigma})$ and assume $\mu_t = \mathcal{N}(\mathbf{m}_t, \Sigma_t)$. For the Gaussian kernel

$$K_\varepsilon(x, y) = \exp(-\|x - y\|^2 / (2\varepsilon)),$$

the normalized field (121) satisfies

$$B_\varepsilon[\mu_t](x) = -\varepsilon(\Sigma_t + \varepsilon \text{Id})^{-1}(x - \mathbf{m}_t).$$

Thus the drifting velocity (123) is affine and preserves Gaussianity. With

$$A_t = (\Sigma_t + \varepsilon \text{Id})^{-1}, \quad \bar{A} = (\bar{\Sigma} + \varepsilon \text{Id})^{-1},$$

the ODE is

$$\dot{\mathbf{m}}_t = \varepsilon \bar{A}(\bar{\mathbf{m}} - \mathbf{m}_t), \quad \dot{\Sigma}_t = \varepsilon((A_t - \bar{A})\Sigma_t + \Sigma_t(A_t - \bar{A})).$$

This finite-dimensional model explains the stabilizing role of the self-normalized repulsion term in drifting: without it, the covariance equation loses the $A_t \Sigma_t + \Sigma_t A_t$ contribution.

Example 13 (Gaussian closure of attention dynamics). For the transformer PDE, assume $\alpha = \mathcal{N}(\mathbf{m}, \Sigma)$. Since exponential tilting preserves Gaussianity,

$$\frac{\int e^{\langle Qx, Ky \rangle} y \, d\alpha(y)}{\int e^{\langle Qx, Kz \rangle} d\alpha(z)} = \mathbf{m} + \Sigma K^\top Qx.$$

Therefore

$$\Gamma_\theta[\alpha](x) = V\mathbf{m} + V\Sigma K^\top Qx$$

is affine. The Gaussian token law is preserved and satisfies

$$\dot{\mathbf{m}}_t = (V_t + V_t \Sigma_t K_t^\top Q_t) \mathbf{m}_t, \quad \dot{\Sigma}_t = B_t \Sigma_t + \Sigma_t B_t^\top, \quad B_t = V_t \Sigma_t K_t^\top Q_t.$$

When $V_t = Q_t^\top K_t$, the matrix $B_t = Q_t^\top K_t \Sigma_t K_t^\top Q_t$ is symmetric positive semidefinite, matching the gradient-field case mentioned above. This closure is not a convergence theorem for trained transformers. It is instead a tractable model of how attention can shear, amplify or contract a cloud of tokens through its covariance.

Conclusion

Optimal transport is useful because it keeps several viewpoints active at once. It is a linear program over couplings, a duality theory for potentials, a geometry on probability measures, a source of PDEs and gradient flows, and a computational toolbox built around linear programming, Sinkhorn scaling and low-dimensional projections. These viewpoints reinforce each other: Brenier maps explain geodesics, geodesic convexity explains convergence of flows, entropic regularization turns transport into scalable differentiable losses, and dual norms or sliced variants reveal what is gained and lost when OT is simplified.

For machine learning, this interplay is especially stimulating. Modern generative modeling, diffusion and flow matching, inverse problems, robust optimization, and even continuous-depth views of transformers all ask for ways to move, compare or learn distributions in high dimension. These applications do not merely consume existing OT theory; they create difficult mathematical questions because empirical measures are singular, dimensions are large, models are parametrized and non-convex, and computational approximations are inseparable from statistical error. The strength of OT is precisely that it gives a common language for these tensions, while still leaving enough open structure for new mathematics to be needed.

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A Notation Table

This appendix collects the main notation used throughout the book. The last column points to the first section, equation, definition, proposition or theorem where the notation is defined or first used in a mathematically meaningful way.

Notation	Meaning	First reference
Ambient spaces, measures and elementary objects		
$\mathbb{R}^d$	Euclidean ambient space.	Section 2.1
$\mathcal{X}, \mathcal{Y}$	Source and target spaces.	Eq. (6)
$\mathcal{M}(\mathcal{X})$	Finite signed Radon measures on $\mathcal{X}$.	Section 2.1
$\mathcal{M}_+(\mathcal{X}), \mathcal{M}_+^1(\mathcal{X})$	Positive finite measures and probability measures.	Section 2.1
$\mathcal{P}(\mathcal{X}), \mathcal{P}_p(\mathcal{X})$	Probability measures, with finite p -moment for $\mathcal{P}_p$.	Section 3.1
Σ_n	Probability simplex of histograms of length n .	Section 2.1
δ_x	Dirac mass at x .	Eq. (2)
α, β, γ	Source, target and auxiliary probability measures.	Eq. (6)
ρ_α	Density of α with respect to a reference measure.	Section 2.1
$d\alpha, dx$	Integration against α and against Lebesgue measure.	Section 2.1
$\mathbb{E}$	Expectation of a random variable.	Section 2.1
$\text{supp}(\pi)$	Topological support of a measure.	Definition 1
$\text{Supp}(\mathbf{b})$	Index support of a histogram.	Eq. (61)
$\mathcal{C}(\mathcal{X})$	Continuous real-valued functions on $\mathcal{X}$.	Section 2.1
$\ \cdot\ $	Euclidean norm or the norm indicated by a subscript.	Section 1
$\langle \cdot, \cdot \rangle$	Euclidean/Frobenius pairing or measure-function pairing.	Section 2.1
Discrete matching and discrete Kantorovich OT		
$(x_i)_i, (y_j)_j$	Source and target point clouds.	Eq. (1)
$\mathbf{C} = (\mathbf{C}_{i,j})$	Cost matrix between source and target points.	Eq. (1)
$\sigma \in \text{Perm}(n)$	Permutation encoding a one-to-one matching.	Eq. (1)
$\mathbf{P}_\sigma, \text{Perm}_n$	Permutation matrix and the set of all such matrices.	Figure 9
$\mathcal{B}_n$	Bistochastic matrix polytope.	Section 4
$\mathbf{a}, \mathbf{b}$	Discrete probability histograms.	Eq. (13)
$\mathbf{P}$	Discrete transport/coupling matrix.	Eq. (13)
$\mathbf{U}(\mathbf{a}, \mathbf{b})$	Polytope of discrete couplings with marginals $\mathbf{a}, \mathbf{b}$.	Eq. (13)
$\mathbf{1}_n, \mathbf{P}^\top$	All-ones vector and transpose of $\mathbf{P}$.	Eq. (13)
$\mathbf{L}_\mathbf{C}(\mathbf{a}, \mathbf{b})$	Discrete Kantorovich optimal value with cost $\mathbf{C}$.	Eq. (16)
$\mathbf{D}$	Ground distance matrix for discrete Wasserstein distances.	Eq. (18)
$\mathbf{W}_p(\mathbf{a}, \mathbf{b})$	Discrete p -Wasserstein distance.	Eq. (18)
Monge maps, one-dimensional OT and Gaussians		
T, T	Transport map.	Eq. (6)
$T_\# \alpha$	Push-forward of α by T .	Definition 3
Id	Identity map.	Definition 3
$\tilde{\mathcal{W}}_p$	Directed Monge transport distance.	Eq. (7)
$\nabla \varphi$	Brenier map for quadratic cost.	Theorem 1
$\mathcal{C}_\alpha$	Cumulative distribution function of a 1-D measure.	Eq. (8)
$\mathcal{C}_\alpha^{-1}$	Quantile function of a 1-D measure.	Eq. (9)

Notation	Meaning	First reference
$\mathcal{N}(\mathbf{m}, \Sigma)$	Gaussian law with mean $\mathbf{m}$ and covariance Σ .	Eq. (10)
$\mathbf{m}_\alpha, \Sigma_\alpha$	Mean and covariance of a Gaussian measure α .	Eq. (12)
$\mathcal{B}(\Sigma_\alpha, \Sigma_\beta)$	Bures covariance distance.	Eq. (11)
$\text{tr}(\Sigma)$	Trace of a matrix.	Eq. (12)
Continuous Kantorovich OT and Wasserstein distances		
π	Coupling or transport plan.	Definition 5
$\mathcal{U}(\alpha, \beta)$	Set of couplings between α and β .	Eq. (14)
$\mathcal{L}_c(\alpha, \beta)$	Kantorovich optimal value with ground cost c .	Eq. (15)
d	Ground distance on the underlying metric space.	Eq. (19)
$\mathcal{W}_p(\alpha, \beta)$	p -Wasserstein distance.	Eq. (19)
$\mathcal{W}_\infty(\alpha, \beta)$	Worst-displacement Wasserstein distance.	Eq. (21)
Γ	c -cyclically monotone subset of $\mathcal{X} \times \mathcal{Y}$.	Definition 6
ρ	Glued or composed coupling.	Lemma 2
$\rightharpoonup$	Weak* convergence of measures.	Definition 7
$\text{TV}, \ \cdot\ _{\text{TV}}$	Total variation divergence/norm.	Section 2.1
Duality, transforms and weak norms		
f, g	Discrete dual potentials.	Eq. (40)
f, g	Continuous dual potentials.	Eq. (44)
$R(a, b)$	Feasible set of discrete dual potentials.	Eq. (41)
$\mathcal{R}(c)$	Feasible set of continuous dual potentials.	Eq. (45)
f^c, g^c	c -transform of a potential.	Definition 11
$\text{Lip}(f)$	Lipschitz constant of f .	Eq. (53)
$\mathcal{W}_1$	Kantorovich–Rubinstein distance/norm.	Eq. (53)
$\mathcal{W}_{1,G}$	Graph Wasserstein-1/transshipment distance.	Proposition 33
$d_G, \nabla_G, \text{div}_G$	Graph geodesic distance, gradient and divergence.	Proposition 33
$\ \cdot\ _B$	Dual norm induced by a discriminator class B .	Eq. (55)
$\mathcal{H}, k$	Reproducing kernel Hilbert space and its kernel.	Definition 12
MMD_k	Maximum mean discrepancy/kernel norm for k .	Definition 12
$\mathcal{D}_\varphi, D_\varphi$	Continuous and discrete φ -divergences.	Eq. (59)
φ'_∞	Recession slope of an entropy function.	Definition 13
φ^*	Legendre transform of φ .	Eq. (66)
KL, KL	Continuous and discrete Kullback–Leibler divergences.	Eq. (33)
$\mathfrak{h}$	Hellinger divergence/distance.	Section 7.3
JS	Jensen–Shannon divergence.	Section 7.3
Entropic regularization and Sinkhorn algorithms		
ε	Entropic regularization strength.	Eq. (22)
$H(P)$	Shannon–Boltzmann entropy of a matrix.	Eq. (22)
$L_C^\varepsilon(a, b)$	Discrete entropic OT value.	Eq. (22)
$\mathcal{L}_c^\varepsilon(\alpha, \beta)$	Continuous entropic OT value.	Eq. (32)
K	Gibbs kernel $e^{-C/\varepsilon}$.	Eq. (23)
u, v	Left and right Sinkhorn scalings.	Eq. (23)
$\text{diag}(u)K\text{diag}(v)$	Scaling form of the entropic coupling.	Eq. (4.5)
$\odot$	Entrywise product of vectors.	Eq. (25)
$u^{(\ell)}, u^{(\ell+1)}$	Current and next Sinkhorn iterates.	Eq. (26)
$d_{\mathcal{H}}$	Hilbert projective metric on positive vectors.	Eq. (36)
Proj^{KL}	KL/Bregman projection.	Eq. (34)

Notation	Meaning	First reference
$\tilde{\mathcal{L}}_c^\varepsilon(\alpha, \beta)$	Debiased Sinkhorn divergence.	Eq. (72)

Notation	Meaning	First reference
Extensions of OT		
ψ_1, ψ_2	Entropy functions penalizing marginal mismatch.	Eq. (74)
$UW_c, UW_{c,\tau}$	Relaxed unbalanced OT value with marginal penalties.	Eq. (74)
L_c	Reverse-formulation local unbalanced cost.	Eq. (75)
H_c	Homogeneous perspective of the local cost L_c .	Eq. (76)
HW	Homogeneous unbalanced formulation.	Eq. (77)
$\mathfrak{C}[\mathcal{X}]$	Cone over the metric space $\mathcal{X}$.	Section 9.1
CW	Cone formulation of unbalanced OT.	Theorem 5
β_s, λ_s	Input measures and weights in barycenter problems.	Eq. (83)
α^*	Optimal measure, often a barycenter.	Section 10.1
SW_p	Sliced Wasserstein distance.	Eq. (78)
$\mathbb{S}^{d-1}$	Unit sphere of projection directions.	Eq. (78)
P_θ	Projection on direction θ .	Eq. (78)
MaxSW_p	Max-sliced Wasserstein distance.	Definition 15
MSWGG_2	Min-sliced lifted-plan upper bound on $\mathcal{W}_2$.	Section 9.2
$SW_{p,k}, \text{MaxSW}_{p,k}$	Average and max Wasserstein distances over k -dimensional projections.	Section 9.2
$\mathcal{W}_\gamma$	Spectral Wasserstein distance associated with a matrix gauge γ .	Eq. (79)
$\mathcal{B}_\gamma$	Polar set defining the robust projected form of $\mathcal{W}_\gamma$.	Eq. (80)
$\mathcal{W}_{2,A}$	Quadratic Wasserstein pseudodistance after projection by $A^{1/2}$.	Eq. (81)
$\text{SRW}_{2,k}$	Paty–Cuturi subspace robust Wasserstein distance.	Section 9.3
LOT_ρ	Linear OT distance around reference ρ .	Eq. (58)
$\mathcal{W}_C$	Weak OT value with conditional-law cost C .	Eq. (85)
g^C	Weak C -transform in weak OT duality.	Proposition 50
D, D'	Intra-domain distance matrices in discrete GW.	Eq. (88)
Δ	Discrepancy between intra-domain distances.	Eq. (88)
GW	Discrete Gromov–Wasserstein cost.	Eq. (88)
$\mathbb{X}, \mathbb{Y}$	Metric-measure spaces.	Eq. (89)
$\mathcal{GW}$	Continuous Gromov–Wasserstein distance.	Eq. (89)
d_H, d_{GH}	Hausdorff and Gromov–Hausdorff distances.	Section 11.2
$\text{FGW}_{\lambda,p}$	Fused Gromov–Wasserstein distance.	Section 11.2
A_t, M_t, Q_t	Matrix-valued density, spatial flux and internal flux.	Eqs. (86), (87)
$\mathcal{W}_{mv}$	Dynamic matrix-valued Wasserstein-type cost.	Eq. (86)
∇_L, ∇_L^*	Non-commutative gradient and its adjoint.	Eq. (87)
$\mathbb{H}_n, \mathbb{H}_n^+, \mathbb{H}_n^{+,1}$	Hermitian matrices, positive semidefinite Hermitian matrices and density matrices.	Section 11.3
Tr_A, Tr_B	Partial traces of a bipartite matrix.	Eq. (90)
$\text{QOT}_C(A, B)$	Finite-dimensional quantum OT value with cost observable C .	Eq. (91)
$\text{QOT}_C^\varepsilon(A, B)$	Entropically regularized quantum OT value.	Eq. (93)
$T_e(F, G), T_s(F, G)$	Exact Gibbs coupling and symmetric Gurvits-scaling surrogate.	Eqs. (95), (96)
Dynamic OT and Wasserstein gradient flows		
α_t	Time-dependent curve of probability measures.	Eq. (99)
v_t	Eulerian velocity field transporting α_t .	Eq. (99)

Notation	Meaning	First reference
T_t	Lagrangian particle flow map.	Eq. (98)
P_t	Interpolant map in flow matching; later also path evaluation.	Eq. (115)
$\mathcal{W}_2^2$ via action	Benamou–Brenier dynamic formulation.	Eq. (101)
$\nabla_{\mathcal{W}} f(\alpha)$	Wasserstein gradient of a functional.	Proposition 6
$\delta f(\alpha)$	First variation of f at α .	Proposition 6
$\partial_t \alpha + \operatorname{div}(\alpha v) = 0$	Continuity equation.	Eq. (99)
$\alpha_{t+\tau}$	One JKO/minimizing-movement step.	Eq. (104)
$\mathcal{S} = C([0, 1]; \mathbb{R}^d)$	Path space in the superposition formulation.	Section 12